Foundations of Applied Dynamical Systems

A Concise First Graduate Course

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Preface

Introduction to the Foundations

Dynamical systems are ubiquitous in mathematics and beyond. They not only describe a vast range of natural and technical phenomena but also provide elegant techniques to prove results within multiple mathematical areas. The field has advanced rapidly in recent decades. One may even argue that newly emerging areas in science and engineering have shown an even greater need to describe nonlinear dynamical phenomena. As a consequence, teaching courses in dynamical systems has become more challenging. The amount of material that one could present has drastically grown. The time allocated for students to acquire essential foundations has often remained constant. With this series of books, I propose a significant restructuring and streamlining of the entire dynamics curriculum. There are two basic books:

(F1) Foundations of Applied Dynamical Systems:
Introduction to ODEs and Nonlinear Dynamics

(F2) Foundations of Applied Dynamical Systems:
A Concise First Graduate Course

These books form the foundation of an entire follow-up series of dynamical systems textbooks entitled "Topics in Applied Dynamical Systems: An Introduction to (...)" to be discussed below.

The books (F1)-(F2) aim to attract students and researchers from different fields. In particular, you can enter the material at various stages depending upon your background. (F1)-(F2) are based upon the observation that disciplinary boundaries are becoming harder to identify but mathematics is still the undisputed basic language we all have to learn. During the first years of undergraduate study, students in mathematics and various other disciplines eventually should learn about dynamical systems and their applications. Yet, the background provided in different fields can vary quite significantly. There are two natural entry points. One subset of students is rooted in a theoretical proof-based perspective common in mathematics, theoretical physics and adjacent disciplines. Another subset of students comes to dynamics from a more application- or data-driven perspective focusing on modeling, explicit solutions and numerical simulation. Yet another subset of students only arrives at dynamics at a much later point, e.g., during a master thesis period or during doctoral studies. Beyond students, also many researchers eventually realize the need to understand the field of dynamics much better (in which case, all I can say is "congratulations" that you had this reflection rather than brushing it aside). One has to find a good entry point for all groups. The books (F1)-(F2) aim to cater to all these groups. The basic principle is to condense to the essentials and at the same time provide modularity for each entry level. Achieving these goals simultaneously is quite a challenge.

The main principle I have applied to focus on the essential aspects is to fundamentally rethink how to structure dynamical systems as a subject. I believe that it is best to really target the generic system classes and phenomena first. In particular, this means:

(a) rely first upon sufficient smoothness, regularity and measurability;

(b) study generic effects that appear in open sets of dynamical systems;

(c) only introduce specific subclasses postponing their detailed analysis.

These changes have a drastic effect. For example, this means studying local stability, bifurcations, invariant manifolds, and related topics in generic smooth ordinary differential equations in detail, while postponing details about systems with symmetries, integrable systems, stochastic systems, complex-analytic maps, networks, etc. More precisely, the aspect (c) mentioned above is intended to avoid making too strong non-generic structural assumptions and only rely initially upon the fundamental tools from undergraduate mathematics (real analysis, linear algebra, basic geometry). Initially this may look to some experienced readers as too reductive, e.g., why not discuss details regarding Hamiltonian systems in full detail early on? The truth is that if one starts selecting one sub-area, then the bias is simply too large for a condensed introduction. Here is precisely where a sequence of follow-up textbooks comes in entitled "Topics in Applied Dynamical Systems: An Introduction to (...)", which we shall abbreviate as (Tx). These books precisely fill this gap and are going to allow lecturers and readers to select a suitable augmentation after having finished the relevant parts of (F1)-(F2). For example, there will be introductions to "Stochastic Dynamics" and "Network Dynamics", among many others in the (Tx) series.

Entry Path to this Book

If you are interested in really becoming an expert in nonlinear dynamics, you will need all the ideas and techniques from (F2). Of course, the level of (F2) is still significantly lower compared to research publications. However, it is often surprising how far many of the foundational tools can reach. Therefore, one natural way would be to linearly work through (F2). If you feel uncomfortable with the level of mathematical abstraction, then starting in (F1) is the right strategy. It is possible to skip some sections in (F2) that are a bit more specialized than others, and then return to them later. This is particularly advisable if you do not exactly know how much time you can devote to work through or teach from this book. Examples are Sections 6, 7, 11, 17, 19, and 23.

Regarding background, one should have mastered standard undergraduate analysis and linear algebra. In this respect, one notable global difference seems to be when students encounter manifolds. I usually teach an introduction to this topic within the standard second-semester undergraduate analysis course. However, this seems not entirely standard globally, so I have included some additional background on manifolds in (F2) as well as some basic ideas in (F1). A second topic, which is not uniformly taught is measure theory. It is only relevant for the last four sections of this book, which require measure theory at a decent level. In most cases, this material is covered in a second-semester or third-semester undergraduate class but backgrounds may vary so I have included a small measure theory introduction in (F2) to make the books self-contained. Hence, basic analysis and linear algebra suffice if you are willing to learn some additional concepts along the way. If you feel you have already all the essentials of (F1)-(F2) internalized, then you may want to consider going directly to your favorite topic (Tx) and just utilize (F1)-(F2) as basic references.

As for (F1), I have also tried the material in (F2) for over ten years in a dedicated master/graduate-level dynamical systems course. A good guideline was that I managed to present about four pages of material per ninety minutes at a rather relaxed pace on the blackboard. If you have less time available as a lecturer, one could also easily craft smaller courses, e.g., skipping some material on iterated maps and ergodic theory by focusing on ODEs, hyperbolicity, bifurcations, and related topics. Or alternatively, one could keep the ODE components smaller and focus more on the latter parts of the book on chaos, attractors, ergodicity, and related topics.

There are exercises included in two forms. First, there are small comments in the text, where steps are omitted that mark a point to reflect and calculate as a reader. Second, there are slightly more elaborate exercises at the end of each section, which could be used as homework or tutorial material. Although the entire material of the book is structured and tested to be used in a minimal blackboard environment, it is easy to add more modern elements these days. For example, if you prefer additional computational illustrations, then there is a guide to accomplish this in the appendix of book (F1) already, which effectively applies verbatim.

A really BIG THANK YOU!

Last but not least, it is now my great pleasure to thank everyone who has been around in the differential equations and dynamics courses that I taught on which the book (F2) is based. In fact, the book arose out of the need to integrate students at Technical University Munich (TUM) coming from different disciplines and backgrounds wanting to learn nonlinear dynamics. I have taught various courses covering (F2) for over ten years at TUM. I would like to thank my colleagues Luca Arcidiacono, Paolo Bernuzzi, Tobias Böhle, Gideon Chiusole, Daniel Karrasch, Christian Kluge, Iacopo Longo, and Anne Pein who brilliantly led the exercises classes in various courses. Regarding the figures for this book, I would like to thank Andreas Burkhart and Maximilian Kapsegger, who have very nicely converted many of my hand-drawn figures into electronic formats. Furthermore, I have received remarkable and invaluable direct didactic feedback from many amazing students at TUM. Without them the design, tuning, implementation, feedback and updating of this book would not have nearly reached the level it has now. More broadly, I am also extremely grateful to the entire dynamical systems community for gradually shaping my view of the field over several decades, and for providing feedback on this book and on the entire book series.

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1 Definitions & Examples

Definition 1.1. A **dynamical system** is a triplet $(\mathcal{X}, \mathcal{T}, \phi_t)$, where $\mathcal{X}$ is the **phase space** (or **state space**), $\mathcal{T}$ is the **time set** (or **time domain**), and $\phi_t : \mathcal{X} \to \mathcal{X}$ for $t \in \mathcal{T}$ is a **family of evolution operators** satisfying

(D1) $\phi_0 = \mathrm{Id}$,

(D2) $\phi_t \circ \phi_s = \phi_{t+s} \quad \forall t, s \in \mathcal{T}$.

We shall assume here that $\mathcal{X}$ is at least a topological space and restrict to the cases where $\mathcal{T} \subseteq \mathbb{R}$ or $\mathcal{T} \subseteq \mathbb{Z}$; see also Figure 1(a) for an illustration of Definition 1.1.

Example 1.2. Let $\mathcal{X} = \mathbb{R}^2$, $\mathcal{T} = \mathbb{R}$, $A \in \mathbb{R}^{2\times 2}$ and consider

$$\phi_t(x_0) = \mathrm{e}^{tA} x_0 = \sum_{k=0}^{\infty} \frac{(tA)^k}{k!} x_0 \qquad \text{for } x_0 \in \mathbb{R}^2. \tag{1.1}$$

The example (1.1) arises from solving a **linear ordinary differential equation (ODE)**

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x' = Ax, \qquad x = x(t),\ x(0) = x_0$$

by setting $\phi_t(x_0) = x(t) = \mathrm{e}^{tA} x_0$. A very simple nontrivial example in $\mathbb{R}^2$ arises if A is a diagonal matrix with $A_{ii} \neq 0$ for $i \in \{1, 2\}$, which has a simple visualization in the phase plane/space; see Figure 1(b). ♦

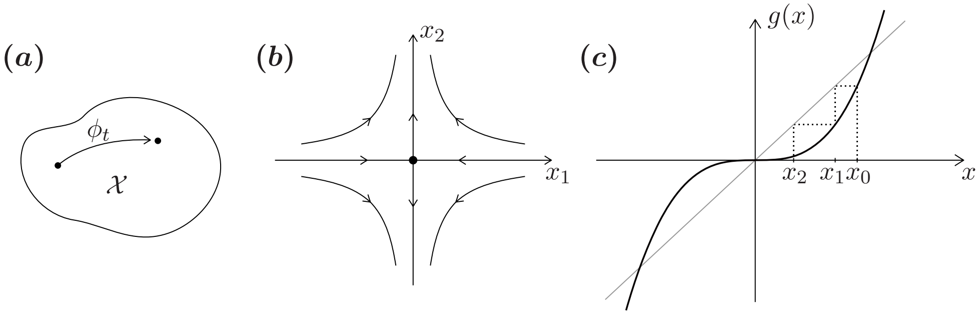

Figure 1: (a) General dynamical system on $\mathcal{X}$. (b) Linear ODE (1.1) with diagonal matrix $A \in \mathbb{R}^{2\times 2}$ and $A_{11} < 0$, $A_{22} > 0$. (c) Iterated cubic map (1.2) including cob-web diagram (dotted lines).

Example 1.3. Take $\mathcal{X} = \mathbb{R}$, $\mathcal{T} = \mathbb{N}_0 = \mathbb{N} \cup \{0\}$, and consider the mapping

$$x^3 =: g(x), \qquad g : \mathbb{R} \to \mathbb{R}, \tag{1.2}$$

and define

$$x_k := \phi_k(x_0) := \underbrace{(g \circ g \circ \cdots \circ g)}_{k\text{-times}}(x_0) =: g^k(x_0), \qquad \phi_0 = \mathrm{Id}, \tag{1.3}$$

where one has to be careful not to confuse k-fold composition superscripts with powers or differentiation. Since $x \mapsto x^3$ is invertible, we may extend the dynamical system to $\mathcal{T} = \mathbb{Z}$. A simple visualization of the map is obtained by the **cobweb construction** as shown in Figure 1(c). Instead of mapping back the value $g(x_k)$ from the vertical axis (representing the range) to the horizontal axis (representing the domain) to obtain x_{k+1} and then computing $g(x_{k+1})$, one alternates between the graph of g and the diagonal. ⧫

Definition 1.4. We call $(\mathcal{X}, \mathcal{T}, \phi_t)$ a **discrete-time** dynamical system (or simply an **iterated map**) if $\mathcal{T}$ has a bijection with $\mathbb{N}$ (e.g. $\mathcal{T} = \mathbb{N}_0, \mathbb{Z}$). If $\mathcal{T}$ is a connected subset of, or equal to, $\mathbb{R}$, we call it a **continuous-time** dynamical system.

Example 1.2 is continuous-time and Example 1.3 is discrete-time.

Definition 1.5. A dynamical system is called **invertible** if $\mathcal{T}$ is an additive group and ϕ_t is defined for all $t \in \mathcal{T}$, i.e., ϕ_{-t} is a single-valued map and $\phi_{-t} \circ \phi_t = \mathrm{Id}$. For the continuous-time case we also speak of **flows** if $\mathcal{T} = \mathbb{R}$ and **semiflows** if $\mathcal{T} = \mathbb{R}_+ = [0, \infty)$.

Example 1.6. Consider the **logistic map**

$$x_{k+1} = p x_k (1 - x_k) = g(x_k; p), \qquad p \in (0, 4], \tag{1.4}$$

where we view p as a parameter, which one checks (exercise!) to be well-defined; see Figure 2.

However, there exists an open interval $\mathcal{I} \subset [0, 1]$ such that the pre-image $g^{-1}(\mathcal{I})$ consists of two disconnected components. Indeed, we have

$$y = g(x; p) = px(1 - x) \qquad \Rightarrow \qquad x = g^{-1}(y) = \frac{1}{2} \pm \sqrt{\frac{1}{4} - \frac{y}{p}}$$

so the inverse is **multi-valued**. Multi-valued (or set-valued) dynamical systems will not be discussed here. ⧫

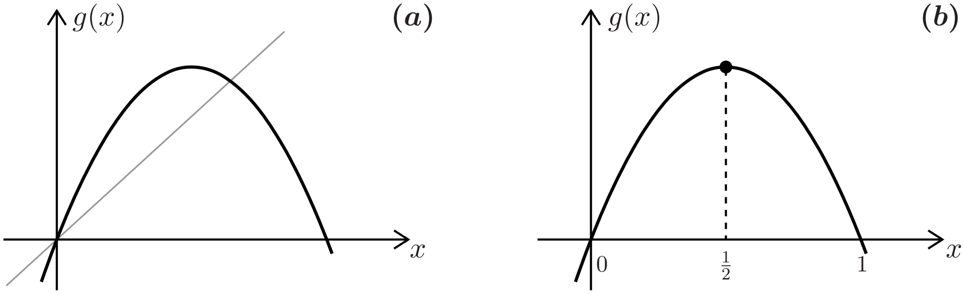

Figure 2: (a) Sketch of the logistic map (1.4) $g(x;p) =: g(x)$ for $p > 2$. (b) Logistic map; draw in $\mathcal{I}$ as discussed in the text (exercise!).

Recall the following variant of a classical theorem from a first ODE course:

Theorem 1.7. *(Existence, uniqueness and smooth dependence) Let $f : \mathbb{R}^d \to \mathbb{R}^d$ be smooth and consider the ODEs*

$$x' = f(x), \qquad x = x(t) \in \mathbb{R}^d, \qquad x(0) = x_0. \tag{1.5}$$

Then there exists a unique map $(t, x_0) \mapsto \phi_t(x_0)$, smooth in (t, x_0), such that

(T1) $\phi_0(x_0) = x_0$,

(T2) there exists a (maximal) interval $\mathcal{T} = (-t_1, t_2)$, $t_{1,2} = t_{1,2}(x_0) > 0$ such that for all $t \in \mathcal{T}$, $\phi_t(x_0) = x(t)$, i.e., the ODE (1.5) *is satisfied.*

The weakest form of Theorem 1.7 requires just Lipschitz continuity of f; see [40]. Yet, for this book, our main focus will not be regularity so we often just assume smoothness, and then the solution map $(t, x_0) \mapsto \phi_t(x_0)$ is also smooth. The differentiability of $\phi_t(x_0)$ can be refined [40] and a C^k solution map is obtained if the input **vector field** $f : \mathbb{R}^d \to \mathbb{R}^d$ is C^k. Recall that the proof of Theorem 1.7 proceeds via re-writing the ODEs in integral form, defining a suitable norm, and using the Banach fixed-point theorem; see [40]. Repeating the Banach fixed-point theorem is useful to realize that it makes a dynamical systems statement already:

Theorem 1.8. ***(Banach Fixed Point Theorem)*** *Let $(\mathcal{X}, d)$ be a nonempty complete metric space and assume $g : \mathcal{X} \to \mathcal{X}$ is a* ***contraction mapping***

$$d(g(x), g(y)) \leq \lambda d(x, y), \qquad \text{for some } \lambda \in [0, 1).$$

Then g admits a unique ***fixed point*** *$g(x_*) = x_*$.*

For the ODEs (1.5), if $\mathcal{T} = \mathbb{R}$, we have a flow.

Example 1.9. Not all ODEs define flows, just consider $x' = x^2$ with $x \in \mathbb{R}$. A standard calculation [40] using separation of variables shows that $\mathcal{T} \neq \mathbb{R}$

$$\int_{x_0}^{x(t)} \frac{1}{y^2}\,\mathrm{d}y = \int_0^t \,\mathrm{d}s$$

so we get the solution

$$\phi_t(x_0) = x(t) = \frac{1}{\frac{1}{x_0} - t},$$

which **blows up** in finite time. However, we may consider ϕ_t as a **local flow** on an open time interval. ◆

We remark that we always work with the usual metric induced by the Euclidean norm $|x| := |x|_2 := \sqrt{x_1^2 + \cdots + x_d^2}$ for ODEs until any further notice.

Definition 1.10. Suppose $(\mathbb{R}^d, \mathbb{R}, \phi_t)$ is a dynamical system. If $\phi_t(x)$ is **smooth** in (t, x), then we say that $(\mathcal{X}, \mathcal{T}, \phi_t)$ is a **smooth dynamical system**.

Remark: Similar definitions can be stated for C^k (i.e., k-times differentiable, $k \geq 1$), or just continuous C^0 dynamical systems. Usually we shall assume that the dynamical system is at least C^0; however, most of our examples are actually even smooth.

Our key goal is to analyze and classify possible dynamics.

Definition 1.11. An **orbit** (or **trajectory**) of a dynamical system starting at x_0 is the ordered set

$$\mathrm{Or}(x_0) := \{x \in \mathcal{X} : x = \phi_t(x_0) \text{ for some } t \in \mathcal{T} \text{ such that } \phi_t(x_0) \text{ is defined}\}.$$

Definition 1.12. A point $x_* \in \mathcal{X}$ is called a **steady state**, **equilibrium** or **fixed point** if $\phi_t(x_*) = x_*$ for all $t \in \mathcal{T}$, i.e., the orbit $\mathrm{Or}(x_*)$ is a singleton.

Fixed point is the more common name for discrete-time systems/maps while equilibrium point is used for continuous-time systems/flows.

Example 1.13. Let $p > 0$ be a parameter. Consider the planar **van der Pol** equation

$$\begin{aligned} x' &= y - \tfrac{x^3}{3} + x, \\ y' &= -px, \end{aligned} \tag{1.6}$$

An equilibrium of (1.6) must satisfy $y' = 0$ so $x = 0$, which implies using the first equation for x' that also $y = 0$. Therefore, $(x_*, y_*) = (0, 0)$ is the unique equilibrium point; see Figure 3. ◆

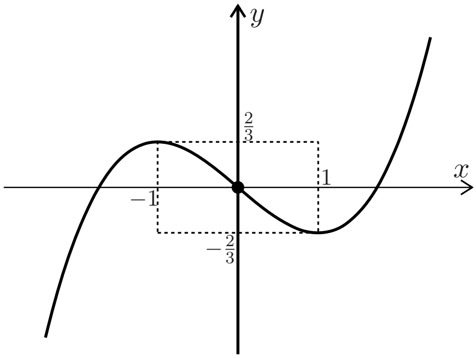

Figure 3: Phase space $\mathbb{R}^2 = \mathcal{X}$ of (1.6) including both nullclines as thick curves; cf. (1.7). Using the nullclines, it is relatively straightforward to sketch the flow (exercise!).

Definition 1.14. If we consider an ODE $x' = f(x)$, then the curves

$$\{x \in \mathbb{R}^d : f_k(x) = 0\}, \qquad k \in \{1, 2, \ldots, d\} \tag{1.7}$$

are called **nullclines**.

At an equilibrium point of an ODE, all nullclines intersect. A steady state is a particular case of the following class of sets.

Definition 1.15. A set $\mathcal{S} \subset \mathcal{X}$ is called **invariant** for ϕ_t if $x_0 \in \mathcal{S}$ implies $\phi_t(x_0) \in \mathcal{S}$ for all $t \in \mathcal{T}$. $\mathcal{S}$ is called **positively invariant** if $x_0 \in \mathcal{S}$ at $t = 0$ implies $\phi_t(x_0) \in \mathcal{S}$ for all $t > 0$; similarly one defines **negatively invariant**; see Figure 4(b).

Example 1.16. Let $\mathcal{T} = \mathbb{Z}$ and $\mathcal{X} = \Sigma_2$ be the space of all bi-infinite sequences of two symbols, say $\{0, 1\}$. A point $\omega \in \Sigma_2$ is given by

$$\omega = \{\ldots, \omega_{-2}, \omega_{-1}, \omega_0, \omega_1, \omega_2, \ldots\}$$

or just written as

$$\omega = \cdots \omega_{-2}\omega_{-1}\omega_0\omega_1\omega_2 \cdots .$$

Define the (one-step, left-)**shift map** $g : \Sigma_2 \to \Sigma_2$ by

$$g(\omega)_j = \omega_{j+1}, \tag{1.8}$$

i.e., we shift each entry one position to the left. Clearly, we have

$$\omega = \cdots 1111 \cdots \qquad \text{and} \qquad \omega = \cdots 0000 \cdots$$

are the only two fixed points. However, there are a lot more invariant sets, which are not fixed points. ♦

Definition 1.17. A **periodic orbit** (or **cycle**) is a non-equilibrium orbit γ such that each $x_0 \in \gamma$ satisfies

$$\phi_{t+T_0}(x_0) = \phi_t(x_0)$$

for some fixed $T_0 > 0$ and all $t \in \mathcal{T}$; see also Figure 4(a). We call the minimal such $T_0 > 0$ the **period** of γ.

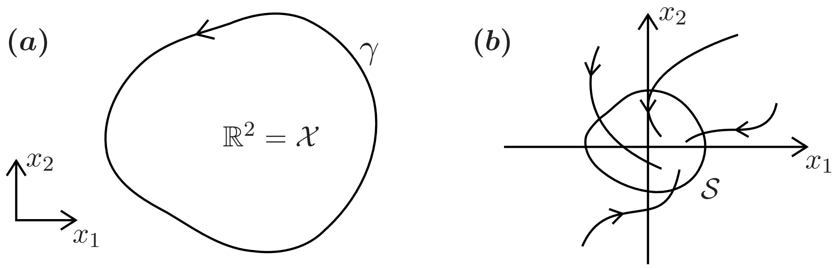

Figure 4: (a) Sketch of a periodic orbit for a planar ODE. (b) Sketch of a positively invariant set $\mathcal{S}$.

Example 1.18. Consider a circle $\mathcal{X} = \mathbb{S}^1 \simeq [0,1]/(0 \sim 1)$, $\mathcal{T} = \mathbb{N}_0$ and define the **(rational) circle rotation map** given by

$$x_{k+1} = g(x_k) := (x_k + r) \mod 1,$$

where we assume $r \in \mathbb{Q}$, $r = \frac{p}{q}$, $\gcd(p,q) = 1$. Then every point $x_0 \in \mathcal{X}$ lies in some periodic orbit. Indeed, consider

$$g^q(x_0) = \left(x_0 + q\frac{p}{q}\right) \mod 1 = x_0.$$

In fact, each point in $\mathcal{X}$ has period q but there are infinitely many distinct periodic orbits. ⧫

Definition 1.19. The **phase portrait** of a dynamical system is the partitioning of $\mathcal{X}$ into its orbits.

Ideally, we would be given a dynamical system and provide a full understanding of its phase portrait. Philosophy:

- $\approx$ *17th-century:* Given an ODE, compute its solutions! ("Newton")
- $\approx$ *19th-century:* Given an ODE, describe its phase portrait! ("Poincaré")
- $\approx$ *20th-century:* Given no equation, describe its phase portrait! ($\rightarrow$ modern dynamical systems)

Exercises for Section 1

E1.1 (a) Consider the ODE $x'(t) = Ax(t)$ with $x(0) = x_0 \in \mathbb{R}^d$ for $A \in \mathbb{R}^{d\times d}$. Prove that this ODE induces a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ for $\mathcal{X} = \mathbb{R}^d$, $\mathcal{T} = \mathbb{R}$ and $\phi_t(x) = \mathrm{e}^{tA}x$. (b) Now, let $A = A(t) \in \mathbb{R}^{d\times d}$ be time-dependent, i.e., consider $x'(t) = A(t)x(t)$ with initial condition $x(0) = x_0$. Does such an ODE in general induce a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ for $\mathcal{X} = \mathbb{R}^d$, $\mathcal{T} = \mathbb{R}$ and $\phi_t(x_0)$ being the solution to the ODE at time $t \in \mathcal{T}$?

E1.2 Decide whether the following are dynamical systems or not:

(a) $\mathcal{X} = \mathbb{R}^2, \quad \mathcal{T} = \mathbb{R}, \quad \phi_t(x) = \begin{pmatrix} \sin(t) & -\cos(t) \\ \cos(t) & \sin(t) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

(b) $\mathcal{X} = \mathbb{R}^2, \quad \mathcal{T} = \mathbb{R}, \quad \phi_t(x) = \mathrm{e}^{2t} \begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

(c) $\mathcal{X} = \mathbb{R}, \quad \mathcal{T} = \mathbb{Z}, \quad \phi_t(x) = a^t x + \frac{a^t - 1}{a - 1} \quad$ with $a \in \mathbb{R} \setminus \{1\}$

E1.3 (a) Let $(\mathcal{X}, \mathcal{T}, \phi_t)$ be an invertible dynamical system and $S \subset \mathcal{X}$ a positively (negatively) invariant set. Show that its complement $\mathcal{X} \setminus S$ is negatively (positively) invariant. (b) Let $A \in \mathbb{R}^{d\times d}$. Show that the (generalized) eigenspaces of the matrix A are invariant under the flow generated by $x' = Ax$.

E1.4 Consider the following system of differential equations

$$\begin{aligned} x' &= y \\ y' &= x + \cos(x) \end{aligned} \tag{1.9}$$

as well as the function

$$E : \mathbb{R}^2 \to \mathbb{R} , \quad E(x, y) = \tfrac{1}{2}y^2 - \tfrac{1}{2}x^2 - \sin(x) \ .$$

Show that the level sets $S_c = \{(x, y) \in \mathbb{R}^2 \mid E(x, y) = c\}$ of the function E are invariant under the flow of (1.9).

2 Stability

Definition 2.1. Let x_* be a steady state for $(\mathcal{X}, \mathcal{T}, \phi_t)$. Then x_* is called **(Lyapunov) stable** if for any given neighborhood $\mathcal{U} = \mathcal{U}(x_*)$ there exists another neighborhood $\mathcal{V}(x_*) = \mathcal{V} \subseteq \mathcal{U}$ such that

$$\phi_t(\mathcal{V}) \subseteq \mathcal{U} \qquad \text{for all } t \geq 0.$$

If stability does not hold, we say that x_* is **unstable**. x_* is called **asymptotically stable** if it is **stable** and there exists a neighborhood $\mathcal{U}$ such that

$$\lim_{t\to+\infty} \phi_t(x_0) = x_* \qquad \forall x_0 \in \mathcal{U}. \tag{2.1}$$

From now on, we shall always assume that $\mathcal{X}$ has a metric. Since most our examples are based in $\mathbb{R}^d$ with Euclidean norm $|\cdot|$, it is relevant to observe that the second requirement (2.1) for asymptotic stability can be written in the Euclidean norm setting as

$$\lim_{t\to+\infty} |\phi_t(x_0) - x_*| = 0 \qquad \forall x_0 \in \mathcal{U}.$$

Furthermore, we use **global asymptotic stability** to indicate $\mathcal{U} = \mathcal{X}$ and employ **local asymptotic stability** to emphasize that $\mathcal{U}$ may have to be chosen small enough.

Example 2.2. Consider the **harmonic oscillator**

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + x = x'' + x = 0,$$

which can be re-written as a first-order system, setting $x' = y$,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \tag{2.2}$$

The general solution is given by

$$(x(t), y(t)) = (c_1 \cos t + c_2 \sin t, c_2 \cos t - c_1 \sin t)$$

for constants $c_{1,2} \in \mathbb{R}$. One checks that $(x, y) = (0, 0)$ is an equilibrium point, which is Lyapunov stable, yet not asymptotically stable; see the phase portrait in Figure 5(a). ♦

We prove the following theorems for dynamical systems generated by an ODE

$$x' = f(x), \qquad x \in \mathbb{R}^d. \tag{2.3}$$

Analogous results for maps exist.

Definition 2.3. Let x_* be an equilibrium for (2.3) and $\mathcal{U}$ be a neighborhood of x_*. A function $L : \mathcal{U} \to \mathbb{R}$ is called a **Lyapunov function** if

(D1) L is continuous,

(D2) $L(x_*) = 0$ and $L(x) > 0$ for $x \neq x_*$,

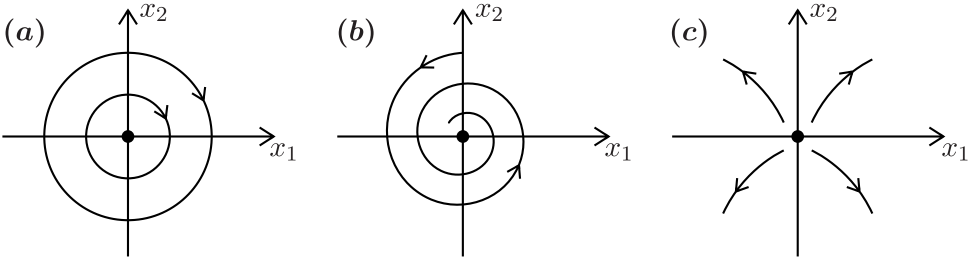

Figure 5: (a) Sketch of the phase portrait of (2.2); typical example of a center equilibrium at the origin. (b) Stable focus. (c) Unstable node.

(D3) L is decreasing along orbits, i.e.,

$$L(\phi_{t_0}(x_0)) \geq L(\phi_{t_1}(x_0)), \tag{2.4}$$

for $t_0 < t_1$ and $\phi_t(x_0) \in \mathcal{U} \setminus \{x_*\}$.

The Lyapunov function is called **strict** if equality in (2.4) never occurs; see Figure 6 for an illustration of the definition.

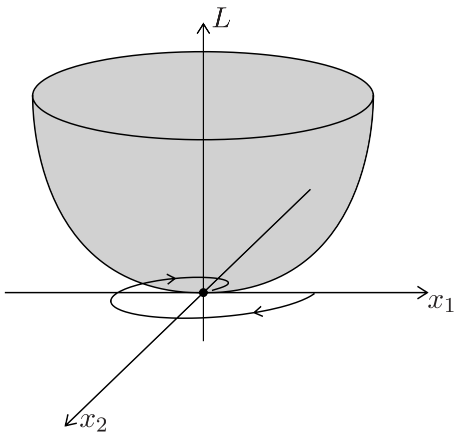

Figure 6: Sketch of a Lyapunov function $L : \mathbb{R}^2 \to \mathbb{R}$ (in gray). The flow has a stable focus and conditions (D1)-(D3) of Definition 2.3 can be checked.

Theorem 2.4 (Lyapunov Stability Criterion)**.** *Suppose x_* is an equilibrium and there exists a Lyapunov function L. Then x_* is stable.*

Proof. Let $\mathcal{S}_\delta$ be the maximal connected component of the sub-level set

$$\{x \in \mathcal{U} : L(x) \leq \delta\}$$

containing x_*. Inside any neighborhood $\mathcal{U}(x_*) = \mathcal{U}$, take a sufficiently small ball $\mathcal{B}(x_*;\delta)$, such that $\partial\mathcal{U} \cap \mathcal{B}(x_*;\delta) = \emptyset$. We claim that for every $\delta > 0$, there exists $\varepsilon > 0$ such that

$$\mathcal{S}_\varepsilon \subseteq \mathcal{B}(x_*;\delta) := \{x \in \mathbb{R}^d : |x - x_*| < \delta\} \tag{2.5}$$

To prove this claim, we argue by contradiction. Therefore, for every $n \in \mathbb{N}$, there exists $x_n \in \mathcal{S}_{1/n}$ such that $|x_n - x_*| \geq \delta$. Since $\mathcal{S}_{1/n}$ is connected, we can take x_n such that $|x_n - x_*| = \delta$. Since the sphere is compact, and all x_n lie in $\partial\mathcal{B}(x_*;\delta)$, we extract a convergent subsequence

$$\lim_{m\to\infty} x_{n_m} = y \qquad \Rightarrow \quad L(y) = \lim_{m\to\infty} L(x_{n_m}) = 0$$

since L is continuous by Definition 2.3(D1). Therefore, $y = x_*$ by Definition 2.3(D2) and this contradicts $|y - x_*| = \delta > 0$. Note that $\mathcal{S}_\varepsilon$ is now a closed set as it does not share boundary points with $\mathcal{U}$. We claim that $\mathcal{S}_\varepsilon$ is positively invariant. We argue by contradiction and assume there exists t_1 such that we leave $\mathcal{S}_\varepsilon$ at $\phi_{t_1}(x_0) =: x_1$. Since $\mathcal{S}_\varepsilon$ is closed it follows $x_1 \in \mathcal{U}$ and $x_1 \notin \partial\mathcal{U}$. There exists a ball

$$\mathcal{B}(x_1;r) \subseteq \mathcal{U}$$

of radius $r > 0$ such that $\phi_{t_1+\rho}(x_0) \in \mathcal{B}(x_1;r) \setminus \mathcal{S}_\varepsilon$ for sufficiently small $\rho > 0$. However, this implies

$$L(\phi_{t_1+\rho}(x_0)) > L(x_1), \tag{2.6}$$

since otherwise $\mathcal{S}_\varepsilon$ would not be the maximal connected component. The result (2.6) is a contradiction to Definition 2.3(D3), which concludes the proof. $\square$

Certain partial results on the existence of Lyapunov functions exist under the assumption that the equilibrium is asymptotically stable (exercise!).

Definition 2.5. Given any $x \in \mathcal{X}$, we define the **ω-limit set** of x by

$$\omega(x) := \left\{ y \in \mathcal{X} : \exists t_j \to +\infty \text{ such that } \lim_{j\to\infty} \phi_{t_j}(x) = y \right\}.$$

The **α-limit set** is given by

$$\alpha(x) := \left\{ y \in \mathcal{X} : \exists t_j \to -\infty \text{ such that } \lim_{j\to\infty} \phi_{t_j}(x) = y \right\}.$$

Theorem 2.6 (LaSalle's Invariance Principle)**.** *If a Lyapunov function L for an equilibrium x_* is not constant on any orbit lying entirely in $\mathcal{U}(x_*) \setminus \{x_*\}$, then x_* is asymptotically stable.*

Proof. For every x with $\phi_t(x) \in \mathcal{U}(x_*)$ for all $t \geq 0$, the limit

$$\lim_{t \to +\infty} L(\phi_t(x))$$

exists by monotonicity and since L is bounded from below. If $y \in \omega(x)$, it follows that

$$\lim_{j \to \infty} \phi_{t_j}(x) = y \qquad \text{for some sequence } t_j \to +\infty.$$

Therefore, we have the limit

$$\lim_{t \to +\infty} L(\phi_t(x)) = L(y)$$

Since $y \in \omega(x)$ and $\omega(x)$ is invariant (exercise!), it follows that $L(y)$ is constant on $\omega(x)$ so we must have $y = x_*$ since L is only constant on the orbit x_*. Therefore, $\omega(x) = \{x_*\}$ so x_* is asymptotically stable. $\square$

If a Lyapunov function is differentiable, we have Definition 2.3(D3) if and only if $L'(\phi_t(x)) \leq 0$, i.e.,

$$L'(\phi_t(x)) = \nabla L(\phi_t(x)) \cdot \phi_t'(x) = \nabla L(\phi_t(x)) \cdot f(\phi_t(x)) \leq 0.$$

The formula $\nabla L(x) \cdot f(x)$ is called the **Lie derivative** of L along f.

Definition 2.7. A function $g : \mathbb{R}^d \to \mathbb{R}$, for which the Lie derivative is zero on every orbit is called a **first integral** (or **constant of motion**).

Example 2.8. One easily checks that $L(x, y) = x^2 + y^2$ is a Lyapunov function for the harmonic oscillator (2.2). In this special case, since

$$\nabla L(x) \cdot f(x) = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \cdot \begin{pmatrix} y \\ -x \end{pmatrix} = 0$$

it follows that L is even a first integral. ♦

Example 2.9. Consider a particle at position $x = x(t) \in \mathbb{R}^d$ of mass 1 moving in a potential $V : \mathbb{R}^d \to \mathbb{R}$ according to Newton's law $x'' = -\nabla V(x)$. With $x' =: y$, we get

$$\begin{aligned} x' &= \qquad\; y = \;\; \partial_y H(x, y), \\ y' &= -\nabla V(x) = -\partial_x H(x, y), \end{aligned} \tag{2.7}$$

where $H : \mathbb{R}^{2d} \to \mathbb{R}$, $H(x, y) = \frac{1}{2}|y|^2 + V(x)$ is the **Hamiltonian** of the system and (2.7) is in the canonical form of **Hamilton's equations**. One checks (exercise!) that H is a first integral but in general not a Lyapunov function. ♦

A powerful alternative to Lyapunov functions, which are potentially applicable globally but are often difficult to find, are local linearization techniques. The reason, why we are often allowed to linearize will be covered in Section 4. First, let us quickly recall [40, Thm. 10.1] regarding linear systems stability.

Theorem 2.10 (Linear System Stability)**.** *Consider the ODE*

$$x' = Ax, \qquad A \in \mathbb{R}^{d\times d} \tag{2.8}$$

with equilibrium $x_* = 0$*.*

(T1) If all eigenvalues $\lambda_j = a_j + \mathrm{i}b_j$ *of* A *satisfy* $\mathrm{Re}(\lambda_j) = a_j < 0$*, then* 0 *is locally asymptotically stable.*

(T2) If there exists λ_j *with* $\mathrm{Re}(\lambda_j) > 0$ *then* 0 *is unstable.*

Proof. (Sketch) The detailed proof is given in [40, Sec. 10]. The idea is relatively easy: transform A to Jordan form, solve the ODE explicitly, and observe that solutions contain only terms of the form $(\cdots)\mathrm{e}^{t\mathrm{Re}(\lambda_j)}$, where $(\cdots)$ are polynomials, sines and cosines. Hence, the time asymptotic stability results easily follow. □

For planar systems, there is terminology that is also useful to recall (or learn quickly) as it helps to build geometric intuition; see also [40, Sec. 12].

Definition 2.11. Consider the linear system (2.8) for $d = 2$ and $\lambda_{1,2} = a_{1,2} + \mathrm{i}b_{1,2}$. Then $x_* = 0$ is called a

node if $b_{1,2} = 0$, $a_1a_2 > 0$;

saddle if $b_{1,2} = 0$, $a_1a_2 < 0$;

focus if $b_{1,2} \neq 0$, $a_{1,2} \neq 0$;

center if $b_{1,2} \neq 0$, $a_{1,2} = 0$;

see also Figure 5. The remaining cases are often just pre-fixed as **degenerate**, e.g., degenerate node if $b_{1,2} = 0$, and $a_1 = 0$ or $a_2 = 0$.

Additional natural pre-fixes apply to objects from Definition 2.11, e.g., if $a_1, a_2 < 0$ and $b_{1,2} = 0$, then $x_* = 0$ is a **stable node**; cf. Theorem 2.10. The notions also generalize to higher dimensions in a natural way, e.g., stable nodes are equilibria with negative real eigenvalues, saddles are equilibria with at least one stable and one unstable direction. In fact, the ideas of linear(ized) stability carry over to iterated maps. Consider the linear map

$$x \mapsto Ax \qquad x \in \mathbb{R}^d,\ x(0) = x_0.$$

Iterating yields $x_k = g^k(x_0) = A^k x_0$. Using Jordan canonical form $J = Q^{-1}AQ$ we have

$$\lim_{k\to\infty} x_k = Q\left(\lim_{k\to\infty} J^k\right) Q^{-1}x_0$$

converges to the fixed point $x_* = 0 \in \mathbb{R}^d$ if the eigenvalues μ_j of A satisfy $|\mu_j| < 1$. In this case, the eigenvalues are called **multipliers**; see also [40, Sec. 17] for additional concrete examples and motivation. In summary, there is a natural analogy for linear systems between ODEs and maps; see also Exercise E2.2 below. Regarding the spectrum, the main separating boundaries are the imaginary axis and the unit circle in the complex plane for flows and maps respectively.

Remark: Tacitly we introduced above the convention that 0 always takes the right dimensions as a scalar or vector to make sense of the used expression. Only if ambiguity could arise, then we explicitly write out a zero vector and indicate whether it is a column or row vector; by default, all vectors are column vectors and we use $(\cdot)^\top$ to turn them into column vectors if needed.

Exercises for Section 2

E2.1 (a) Find a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ – by means of an equation or a phase portrait – which has an unstable equilibrium x_*, that satisfies $\lim_{t\to\infty}\phi_t(x_0) = x_*$ for all $x_0 \in \mathcal{X}$. (b) Find another dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$, such that x_* is an asymptotically stable equilibrium, but there exists no $\epsilon > 0$ such that the ball $\mathcal{B}(x_*, \epsilon)$ is positively invariant.

E2.2 (a) Consider the dynamical system $D_1 = (\mathbb{R}^4, \mathbb{R}, \phi_t)$ that is induced by the linear differential equation $x' = Ax$, where

$$A = \begin{pmatrix} p-2 & 1 & 0 & 0 \\ -1 & p-2 & 0 & 0 \\ 0 & 0 & -p-1 & 4 \\ 0 & 0 & 0 & -2 \end{pmatrix},$$

and $p \in \mathbb{R}$ is a parameter. This dynamical system has an equilibrium at $x_* = 0$. (a) Determine the values of p such that the eigenvalues of A suffice to determine the stability of x_*; in these cases check when x_* is stable or unstable. (b) Now let $\tau > 0$ be fixed. Consider the dynamical system $D_2 = (\mathbb{R}^4, \tau\mathbb{N}_0, \tilde{\phi}_t)$, that is given by iterating the time-τ map ϕ_τ of D_1. Again, this dynamical system has an equilibrium at $x_* = 0$. Calculate the multipliers of this equilibrium and use them to decide when x_* is stable or unstable in D_2 (depending on p). (c) Relate the eigenvalues of A, which determine the stability of x_* in D_1, to the multipliers of x_* in D_2.

E2.3 Consider an example of a **nonlinear pendulum**

$$x'' + \sin(x) = 0\ , \quad x = x(t) \in \mathbb{R}. \tag{2.9}$$

(a) Show that (2.9) is a Hamiltonian system, i.e., it can be written in the form

$$\begin{aligned} x' &= \partial_y H(x,y), \\ y' &= -\partial_x H(x,y), \end{aligned}$$

for some $H \in C^1(\mathbb{R}^2,\mathbb{R})$. (b) Show that for every Hamiltonian system the Hamiltonian H is constant along trajectories, i.e. it is a first integral. (c) Find a suitable Lyapunov function to show that the equilibrium at 0 is stable. (d) Use the fact that the Hamiltonian H is a first integral to sketch the phase portrait.

E2.4 Next, consider a damped nonlinear pendulum, governed by the equation

$$x'' + x' + \sin(x) = 0. \tag{2.10}$$

(a) Prove that 0 is now asymptotically stable. (b) Prove that (2.10) is not a Hamiltonian system.

3 Stable/Unstable Manifolds

Before we can start discussing stable/unstable manifolds, we provide a brief introduction (and/or refresher) to elementary definitions from analysis or geometry/topology. Readers familiar with manifolds can skip ahead to equation (3.3).

Definition 3.1. A d-dimensional (topological) **manifold** $\mathcal{M}$ is a second-countable Hausdorff topological space such that each point $x \in \mathcal{M}$ has a neighborhood, which is locally homeomorphic to an open set in $\mathbb{R}^d$.

One may check that topological manifolds have an open covering $\{\mathcal{U}_\alpha\}$, for $\mathcal{U}_\alpha \subseteq \mathcal{M}$ of **charts**, i.e., homeomorphisms $h_\alpha : \mathcal{U}_\alpha \to \mathcal{V}_\alpha \subset \mathbb{R}^d$, where each $\mathcal{V}_\alpha$ is an open subset in $\mathbb{R}^d$. The collection $\{(\mathcal{U}_\alpha, h_\alpha)\}$ of charts is called an **atlas**; see Figure 7(a).

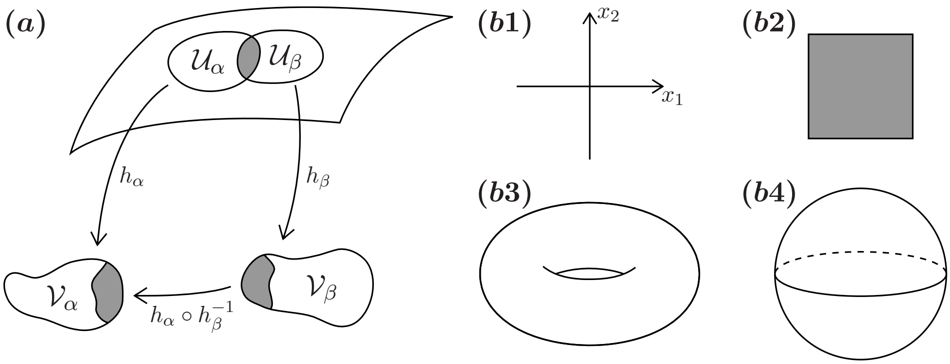

Figure 7: (a) Sketch of a manifold, charts, and transition maps. (b) Examples of two-dimensional manifolds: Euclidean space $\mathbb{R}^2$, a square in $\mathbb{R}^2$, a torus, and a sphere.

Definition 3.2. A d-dimensional C^k**-manifold** $\mathcal{M}$ is a topological manifold, where the atlas satisfies for $\mathcal{U}_\alpha \cap \mathcal{U}_\beta \neq \emptyset$ that the map

$$h_\alpha \circ h_\beta^{-1} : h_\beta(\mathcal{U}_\alpha \cap \mathcal{U}_\beta) \to h_\alpha(\mathcal{U}_\alpha \cap \mathcal{U}_\beta) \text{ is in } C^k, \tag{3.1}$$

and the atlas is maximal, i.e., if there exists another chart, which is compatible in sense of (3.1) with all other charts, then it is included in the atlas; see also Figure 7(a).

Remark: One may prove that it suffices to exhibit some C^k-atlas, where the smoothness condition for the transition maps is satisfied, to show that the manifold is C^k. Surprising fact: Any topological manifold for $d \leq 3$ can be given the structure of a smooth manifold but this fails for $d \geq 4$.

Example 3.3. Clearly, Euclidean space $\mathbb{R}^d$, an interval $(0,1)$ or the rectangle $(0,1) \times (0,1)$ are manifolds. The same holds for the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, $d \geq 1$ ($d = 1$ is just the circle), as well as unit sphere $\mathbb{S}^d = \{x \in \mathbb{R}^d : |x| = 1\}$. All these manifolds are C^k (they are even smooth $k = \infty$, and analytic $k = \omega$) and can be locally written as graphs of smooth functions (exercise!); see Figure 7(b). ♦

One naturally obtains the space $C^k(\mathcal{M}_1, \mathcal{M}_2)$ for maps between smooth manifolds by postulating that $f \in C^k$ if it is a smooth map upon application of coordinate charts.

Definition 3.4. A C^k**-curve** in $\mathcal{M}$ is a C^k-map $\gamma : (-s, s) \to \mathcal{M}$. The **tangent vector** to γ at $x = \gamma(0)$ is a linear map $v : C^1(\mathcal{M}, \mathbb{R}) \to \mathbb{R}$ defined by

$$v(f) := \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t))\bigg|_{t=0}.$$

The linear span of all tangent vectors at x is called the **tangent space** $\mathrm{T}_x\mathcal{M}$; see Figure 8.

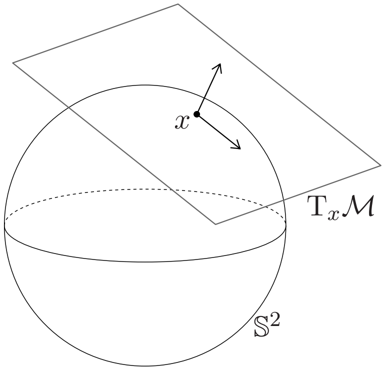

Figure 8: Two-dimensional sphere and tangent space at a point x.

In a chart $h : \mathcal{U} \subset \mathcal{M} \to \mathbb{R}^d$ with local coordinates $h(x) = (x_1, x_2, \ldots, x_d)$, we can view a tangent vector v as

$$v(f) = \sum_{i=1}^{d} v_i \left.\frac{\partial f}{\partial x_i}\right|_{h(x)} \qquad v_i := \left.\frac{\mathrm{d}x_i(\gamma(t))}{\mathrm{d}t}\right|_{t=0} \tag{3.2}$$

by the chain rule (exercise!) so v is just a vector in $\mathbb{R}^d$ with components v_i.

Example 3.5. Take $\mathcal{M} = \mathbb{S}^2$, then one checks that $\mathrm{T}_x\mathcal{M} \simeq \mathbb{R}^2$ for every x, i.e., we have just a tangent plane at every point. It holds in more generality (exercise!) that $\mathrm{T}_x\mathcal{M} \simeq \mathbb{R}^d$; see Figure 8. ♦

In summary, manifolds are geometric objects "locally equivalent to Euclidean space". They can be locally written as graphs of maps, when they are embedded into an ambient Euclidean space. An embedding into some Euclidean space is always possible (by the highly non-trivial **Whitney Embedding Theorem**). For this book, you can think of manifolds as embedded in Euclidean space. We now start the main dynamical constructions. As before, we mostly work in this section with dynamical systems generated by the ODE (2.3), i.e.,

$$x' = f(x), \qquad x \in \mathbb{R}^d. \tag{3.3}$$

Definition 3.6. Consider the ODE (3.3) with $f \in C^1$. An equilibrium x_* is called **hyperbolic** if the Jacobian $\mathrm{D}f(x_*)$ has no eigenvalues with zero real parts.

Definition 3.7. Consider an iterated map $g \in C^1$, $x \mapsto g(x)$. A fixed point x_* is called **hyperbolic** if the Jacobian $\mathrm{D}g(x_*)$ has no multiplier μ with $|\mu| = 1$.

Formulated concisely, hyperbolicity for steady states implies that eigenvalues are away from the imaginary axis and multipliers away from the unit circle. Suppose x_* is a hyperbolic equilibrium. Without loss of generality $x_* = 0$ (using a shift $\tilde{x} = x - x_*$ if necessary), so we may use Taylor's Theorem to write

$$x' = f(x) = \mathrm{D}f(0)x + R(x), \qquad R(x) = o(|x|) \qquad \text{as } |x| \to 0; \qquad (3.4)$$

we would have $R(x) = \mathcal{O}(|x|^2)$ for $f \in C^2$ or even $f \in C^\infty$, which is often just assumed in dynamical systems; here we shall often write the nonlinear terms just as $\mathcal{O}(|x|^2)$ with the understanding to be careful in the case of $f \in C^1$ and $f \notin C^2$. We expect the linear part $A = \mathrm{D}f(0)$, defining the **linearized system**

$$X' = \mathrm{D}f(0)X = AX, \qquad X \in \mathbb{R}^d$$

to dominate. Since 0 is hyperbolic, A is a hyperbolic matrix and there exists a splitting

$$\mathbb{R}^d = E^{\mathrm{s}}(0) \oplus E^{\mathrm{u}}(0)$$

where $E^{\mathrm{s}}(0)$ is the **stable eigenspace** and $E^{\mathrm{u}}(0)$ is the **unstable eigenspace**

$$\begin{aligned} E^{\mathrm{s}}(0) &= \{y \in \mathbb{R}^d : \mathrm{e}^{tA}y \to 0 \text{ as } t \to \infty\}, \\ E^{\mathrm{u}}(0) &= \{y \in \mathbb{R}^d : \mathrm{e}^{tA}y \to 0 \text{ as } t \to -\infty\}. \end{aligned}$$

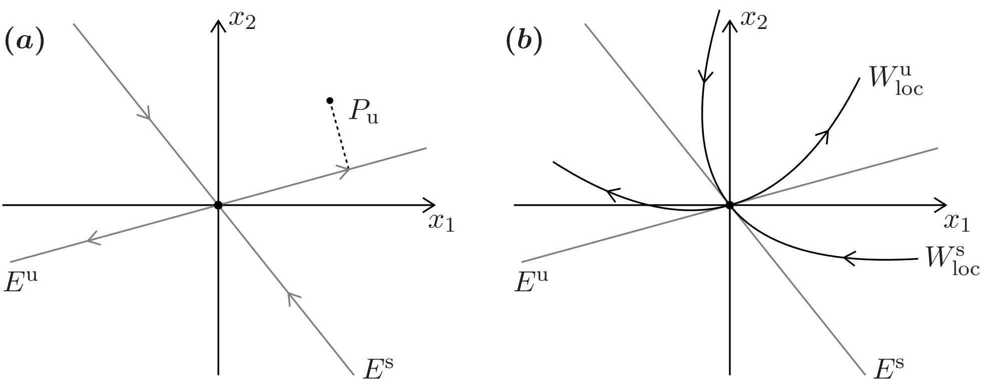

Figure 9: (a) Stable/unstable eigenspaces for a planar saddle. (b) Stable/unstable manifolds tangent to the eigenspaces.

The spaces $E^{\mathrm{s}} = E^{\mathrm{s}}(x_*)$, $E^{\mathrm{u}} = E^{\mathrm{u}}(x_*)$ for general equilibria are defined analogously and come with natural projections

$$P_{\mathrm{s}} : \mathbb{R}^d \to E^{\mathrm{s}} \qquad \text{and} \qquad P_{\mathrm{u}} : \mathbb{R}^d \to E^{\mathrm{u}}.$$

However, the nonlinear terms $\mathcal{O}(|x|^2)$ may locally deform the geometry of the linear spaces near x_*; see Figure 9.

Theorem 3.8 (**Stable/Unstable Manifolds (for flows)**). *Suppose the ODE* (2.3)*, $f \in C^1$, has a hyperbolic equilibrium x_* and $\mathrm{D}f(x_*)$ has k real-part negative and $d-k$ real-part positive eigenvalues with corresponding eigenspaces $E^{\mathrm{s}}(x_*)$ and $E^{\mathrm{u}}(x_*)$ for the linearized system. Then there exists a neighborhood $\mathcal{U}$ of x_* with local **stable** and **unstable manifolds** $W^{\mathrm{s}}_{\mathrm{loc}}(x_*)$ and $W^{\mathrm{u}}_{\mathrm{loc}}(x_*)$*

$$\begin{aligned} W^{\mathrm{s}}_{\mathrm{loc}}(x_*) &= \{y \in \mathcal{U} : \phi_t(y) \to x_* \text{ as } t \to \infty \text{ and } \phi_t(y) \in \mathcal{U}\ \forall t \geq 0\}, \\ W^{\mathrm{u}}_{\mathrm{loc}}(x_*) &= \{y \in \mathcal{U} : \phi_t(y) \to x_* \text{ as } t \to -\infty \text{ and } \phi_t(y) \in \mathcal{U}\ \forall t \leq 0\}. \end{aligned}$$

Furthermore, the stable/unstable manifolds are differentiable. $W^{\mathrm{s}}_{\mathrm{loc}}(x_*)$ and $W^{\mathrm{u}}_{\mathrm{loc}}(x_*)$ are tangent to $E^{\mathrm{s}}(x_*)$ and $E^{\mathrm{u}}(x_*)$ at x_*.

We carry out three steps, the first two are the most involved.

Proof. `Step 1:` We are going to prove the *existence* of $W^{\mathrm{s}}_{\mathrm{loc}}(x_*)$; the case $W^{\mathrm{u}}_{\mathrm{loc}}(x_*)$ can be obtained by time-reversal. Without loss of generality $x_* = 0$, consider (3.4), let $x(0) = x_0$, and define

$$x_{\mathrm{s,u}} := P_{\mathrm{s,u}} x_0, \qquad R_{\mathrm{s,u}}(x) := P_{\mathrm{s,u}} R(x), \qquad x_{\mathrm{s,u}}(t) := P_{\mathrm{s,u}} x(t).$$

The goal is to construct $W^{\mathrm{s}}_{\mathrm{loc}}(0)$ as a graph over $E^{\mathrm{s}}(0)$, i.e., formally

$$W^{\mathrm{s}}_{\mathrm{loc}}(0) := \{x = x_{\mathrm{s}} + x_{\mathrm{u}} \in \mathbb{R}^d \cap \mathcal{U} : x_{\mathrm{u}} = h(x_{\mathrm{s}})\}, \qquad h : \mathbb{R}^k \to \mathbb{R}^{d-k} \quad (3.5)$$

as shown in Figure 10(a).

There are three main techniques to construct h; here we use functional iteration (**Lyapunov-Perron method**). We re-write (3.4) using the **variation-of-constants formula**

$$x(t) = \mathrm{e}^{tA} x_0 + \int_0^t \mathrm{e}^{(t-r)A} R(x(r))\ \mathrm{d}r. \qquad (3.6)$$

$x(t)$ remains bounded only for certain $x_0 = x_{\mathrm{s}} + x_{\mathrm{u}}$. Hence, we project to the unstable part, multiply by e^{-tA} and re-arrange terms to obtain

$$x_{\mathrm{u}} = \mathrm{e}^{-tA} x_{\mathrm{u}}(t) - \int_0^t \mathrm{e}^{-rA} R_{\mathrm{u}}(x(r))\ \mathrm{d}r.$$

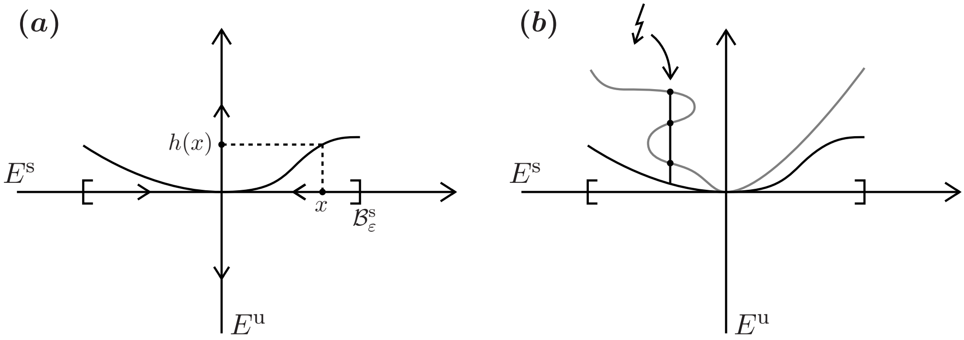

Figure 10: (a) Idea to construct $W^{\mathrm{s}}_{\mathrm{loc}}(0)$ as a graph with a sufficiently small (ball) domain $\mathcal{B}^{\mathrm{s}}_{\varepsilon} = \mathcal{B}(0;\varepsilon)$ within the stable eigenspace $E^{\mathrm{s}} = E^{\mathrm{s}}(0)$. (b) Potential obstacle for the alternative graph transform method mentioned after the functional iteration method; for the graph transform, one has to ensure that iterates remain graphs and do not become multi-valued.

Suppose $|x(t)|$ is bounded as $t \to +\infty$, then taking limits gives

$$x_{\mathrm{u}} = -\int_0^\infty \mathrm{e}^{-rA} R_{\mathrm{u}}(x(r))\ \mathrm{d}r. \tag{3.7}$$

Plugging (3.7) into (3.6) gives

$$x(t) = \mathrm{e}^{tA} x_{\mathrm{s}} + \int_0^t \mathrm{e}^{(t-r)A} R_{\mathrm{s}}(x(r))\ \mathrm{d}r - \int_t^\infty \mathrm{e}^{(t-r)A} R_{\mathrm{u}}(x(r))\ \mathrm{d}r. \tag{3.8}$$

Define $P(t) = P_{\mathrm{s}}$ for $t > 0$ and $P(t) = -P_{\mathrm{u}}$ for $t \leq 0$, then (3.8) is re-written as

$$x(t) = K(x)(t), \quad K(x)(t) = \mathrm{e}^{tA} x_{\mathrm{s}} + \int_0^\infty \mathrm{e}^{(t-r)A} P(t-r) R(x(r))\ \mathrm{d}r. \tag{3.9}$$

In particular, all bounded solutions satisfy the operator fixed-point equation (3.9). Suppose we could solve (3.9), denote its solution by $\psi(t, x_{\mathrm{s}})$, then the definition

$$h(y) := P_{\mathrm{u}} \psi(0, y)$$

is going to give $W^{\mathrm{s}}_{\mathrm{loc}}(0)$. Indeed, all forward-time bounded solutions staying near 0 must converge to 0 as can be shown using a direct approximation argument invoking linearization and hyperbolicity. Hence, it remains to solve (3.9). We introduce the Banach space $C_{\mathrm{b}}([0,\infty), \mathbb{R}^d)$ with norm

$$\|x\| = \sup_{t \geq 0} |x(t)|.$$

Since we work in some $\mathcal{U}$, assume $|x(t)| \leq \delta$ for some $\delta > 0$. Observe that R satisfies

$$|R(x(t)) - R(y(t))| \leq \varepsilon |x(t) - y(t)| \tag{3.10}$$

and $\varepsilon > 0$ can be made arbitrarily small if $\delta > 0$ is made small. In backward time solutions of the linearized system decay exponentially in E^{u}, and solutions decay in forward time in E^{s} so that the definition of $P(t)$ implies

$$\|\mathrm{e}^{(t-r)A} P(t-r)\| \leq C\mathrm{e}^{-\kappa|t-r|} \tag{3.11}$$

for some constant $\kappa > 0$ which can be bounded using the spectrum of A, and for some constant $C > 0$. Using (3.10) and (3.11) in (3.9) yields

$$\begin{aligned}
\|K(x) - K(y)\| &\leq \sup_{t\geq 0} \left| \int_0^\infty \mathrm{e}^{(t-r)A} P(t-r)(R(x(r)) - R(y(r)))\ \mathrm{d}r \right| \\
&\leq C \sup_{t\geq 0} \int_0^\infty \mathrm{e}^{-\kappa|t-r|} |R(x(r)) - R(y(r))|\ \mathrm{d}r \\
&\leq C\varepsilon \|x - y\| \sup_{t\geq 0} \int_0^\infty \mathrm{e}^{-\kappa|t-r|}\ \mathrm{d}r = \frac{2C\varepsilon}{\kappa} \|x - y\|.
\end{aligned}$$

For $\varepsilon < \kappa/(2C)$, K is a contraction so the Banach Fixed Point Theorem 1.8 gives the existence of $x(t)$ and hence also $W^{\mathrm{s}}_{\mathrm{loc}}(0)$ using (3.5).

`Step 2:` The next question is *differentiability*. We know the solution $\psi(t, x_{\mathrm{s}})$ is continuous in t. To show continuity $\psi \in C([0,\infty) \times \mathcal{U}, \mathbb{R}^d)$, we use the triangle inequality

$$|\psi(t,a) - \psi(s,b)| \leq |\psi(t,a) - \psi(t,b)| + |\psi(t,b) - \psi(s,b)|, \tag{3.12}$$

where the second term can be made small by continuity in time. We claim that the first term is also small. Indeed, note that

$$K(x)(t) = K_{x_{\mathrm{s}}}(x)(t)$$

is continuous as an operator in x_{s} since

$$\|K_a(x)(t) - K_b(x)(t)\| = \sup_{t\geq 0} |\mathrm{e}^{tA} a - \mathrm{e}^{tA} b| \leq C\|a - b\|.$$

To also obtain $\psi \in C^1([0,\infty) \times \mathcal{U}, \mathbb{R}^d)$, we look at partial derivatives. Differentiating the fixed-point equation (3.9), we find

$$\partial_t \psi(t,a) = A\mathrm{e}^{tA} a + \partial_t \int_0^\infty \mathrm{e}^{(t-r)A} P(t-r) R(\psi(r,a))\ \mathrm{d}r$$

which is continuous since R and P are. The other derivative $\mathrm{D}_a\psi = \eta$, if it exists, must satisfy the fixed-point equation

$$\eta(t, x_\mathrm{s}) = \mathrm{e}^{tA} P_\mathrm{s} + \int_0^\infty \mathrm{e}^{(t-r)A} P(t-r)\eta(r, x_\mathrm{s}) \cdot \nabla R(\psi(r, x_\mathrm{s}))\ \mathrm{d}r, \quad (3.13)$$

where the ∇R exists in L^1, as R is Lipschitz on $\mathcal{U}$ and hence differentiable almost everywhere by Rademacher's Theorem; furthermore, the dominated convergence theorem has been used to interchange limits. The fixed-point problem (3.13) can again be solved, now for η using the same Banach contraction mapping argument as above. Hence, $\eta(t, x_\mathrm{s})$ exists and is continuous in x_s. Since both partials exist and are continuous, we have $\psi \in C^1([0, \infty) \times \mathcal{U}, \mathbb{R}^d)$.

`Step 3:` The last question is *tangency* at $x = 0$. We evaluate (3.13) at $(t, x_\mathrm{s}) = (0, 0)$, which implies $\eta(0, 0) = P_\mathrm{s}$ since $\nabla R(\psi(r, 0)) = \nabla R(0)$ is zero almost everywhere as $R = o(|x|)$ as $|x| \to 0$. Therefore, we have

$$\mathrm{D}_{x_\mathrm{s}} h(a)|_{a=0} = P_\mathrm{u} P_\mathrm{s} = 0$$

so $E^\mathrm{s}(0)$ is indeed tangent to $W^\mathrm{s}_\mathrm{loc}(0)$. □

Differentiability of the stable/unstable manifolds can be improved to C^k if $f \in C^k$ for $k \geq 1$; in this case, even analyticity transfers, i.e., if $f \in C^\omega$ then $W^{\mathrm{s,u}}_\mathrm{loc}(x_*)$ are C^ω. The manifolds are invariant once they have been extended under the flow ϕ_t of (3.3), i.e.,

$$W^{\mathrm{s,u}}(x_*) = \bigcup_{t \in \mathcal{T}} \phi_t(W^{\mathrm{s,u}}_\mathrm{loc}(x_*)).$$

A natural analogous statement for maps holds.

Theorem 3.9 (Stable/Unstable Manifolds (for maps)). *Consider a C^1-diffeomorphism*

$$x \mapsto g(x) \qquad \text{for } x \in \mathbb{R}^d,$$

which has a hyperbolic fixed point x_* and $\mathrm{D}g(x_*)$ has k multipliers inside the unit circle and $d-k$ multipliers outside the unit circle with corresponding eigenspaces $E^\mathrm{s}(x_*)$ and $E^\mathrm{u}(x_*)$ for the linearized system. Then there exists a neighborhood $\mathcal{U}$ of x_* with local ***stable*** *and* ***unstable manifolds*** *$W^\mathrm{s}_\mathrm{loc}(x_*)$ and $W^\mathrm{u}_\mathrm{loc}(x_*)$*

$$\begin{aligned} W^\mathrm{s}_\mathrm{loc}(x_*) &= \{y \in \mathcal{U} : \phi_t(y) \to x_* \text{ as } t \to \infty \text{ and } \phi_t(y) \in \mathcal{U}\ \forall t \geq 0\}, \\ W^\mathrm{u}_\mathrm{loc}(x_*) &= \{y \in \mathcal{U} : \phi_t(y) \to x_* \text{ as } t \to -\infty \text{ and } \phi_t(y) \in \mathcal{U}\ \forall t \leq 0\}. \end{aligned}$$

Furthermore, the stable/unstable manifolds are differentiable. $W^\mathrm{s}_\mathrm{loc}(x_*)$ and $W^\mathrm{u}_\mathrm{loc}(x_*)$ are tangent to $E^\mathrm{s}(x_*)$ and $E^\mathrm{u}(x_*)$ at x_*.

The second major proof technique (**Hadamard-Perron method**) for invariant manifold theorems will be briefly illustrated in the context of Theorem 3.9 for $x_*=0$. Consider the set

$$\mathcal{B}^{\mathrm{s}}_\varepsilon := \{y \in E^{\mathrm{s}}(0) : |y| \leq \varepsilon\}.$$

Let $\mathcal{L}(L,\varepsilon)$ be the space of Lipschitz-continuous maps $h:\mathcal{B}^{\mathrm{s}}_\varepsilon \to E^{\mathrm{u}}(0)$ with Lipschitz constant L and $h(0)=0$, which is a Banach space under the supremum norm. The **graph transform** is constructed as an operator

$$G:\mathcal{L}(L,\varepsilon)\to\mathcal{L}(L,\varepsilon), \qquad G(\varphi)=\tilde{\varphi}.$$

It is defined by considering the mapping

$$\varphi \mapsto g^{-1}(\mathrm{graph}(\varphi)). \tag{3.14}$$

One then proves (nontrivial!) that the resulting set in (3.14) is again the graph of a Lipschitz map $\tilde{\varphi}$ for sufficiently small $\varepsilon>0$ over the correct domain $\mathcal{B}^{\mathrm{s}}_\varepsilon$; see Figure 10(b). Then one proves (again nontrivial!) that G is a contraction for suitable choices of (L,ε) and obtains the existence of $W^{\mathrm{s}}_{\mathrm{loc}}(0)$ as a fixed point of G.

The third major technique to construct invariant manifolds in dynamical systems is based on using the implicit function theorem on a Banach space, i.e., to construct the existence of the invariant manifold via the existence of an implicit function. This method will not be discussed here.

Example 3.10. Consider a variant of the (unforced) **Duffing equation/oscillator**

$$x''+\delta x'-x+px^3=0, \qquad x\in\mathbb{R},$$

where $p,\delta\in\mathbb{R}$ are parameters and $\delta\geq 0$. Re-writing it as a first-order system gives

$$\begin{aligned} x' &= y,\\ y' &= -\delta y + x - px^3. \end{aligned} \tag{3.15}$$

Obviously, $(x,y)=(0,0)$ is an equilibrium. If $\delta=0$, then one checks $\dim E^{\mathrm{s}}(0)=1=\dim E^{\mathrm{u}}(0)$. Therefore, there exist one-dimensional stable and unstable manifolds by Theorem 3.8. ♦

Example 3.11. Consider the linear map (sometimes called **Arnold's cat map**) defined by

$$g(x)=Ax, \qquad A=\begin{pmatrix} 2 & 1\\ 1 & 1\end{pmatrix}, \; x\in\mathbb{R}^2. \tag{3.16}$$

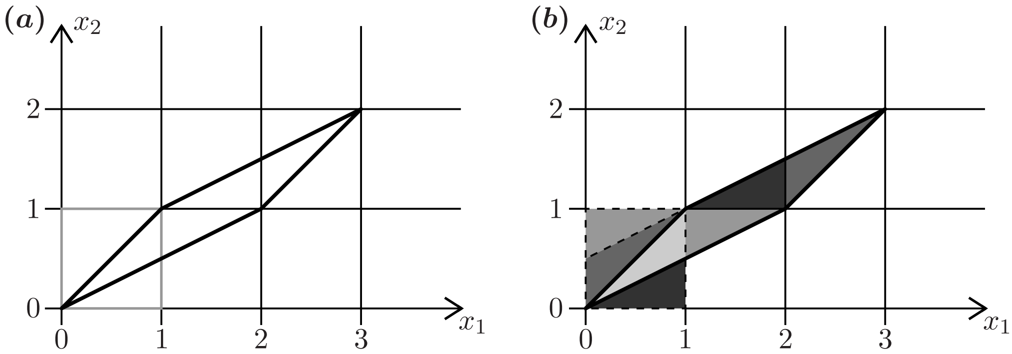

Figure 11: (a) Mapping of the fundamental domain $[0,1]^2$ (grey) via (3.16) to a polygon (black). (b) Identifying the resulting object back onto the fundamental domain.

One checks (exercise!) that g preserves the integer lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$ and induces a map on the torus $\mathcal{X} = \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$; see Figure 11. The origin 0 (recall our convention that 0 takes the right dimensions so here $0 = (0,0)$) is a hyperbolic saddle fixed point (exercise!) and hence Theorem 3.9 applies to this point. The map $g : \mathbb{T}^2 \to \mathbb{T}^2$ is the classical example for a **hyperbolic toral automorphism**, i.e., it does not only have a hyperbolic splitting at $(0,0)$ but near every point in $\mathcal{X}$ it has well-defined contracting and expanding directions. ♦

Exercises for Section 3

E3.1 (a) Prove that any smooth submanifold $\mathcal{M} \subseteq \mathbb{R}^d$ can be written locally as the graph of a smooth map. (b) Start from the coordinate-based representation of tangent vectors in (3.2) and upgrade it to a definition of tangent vectors. Then prove that your definition implies our original construction in Definition 3.4.

E3.2 Consider an invertible continuous dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$. (a) Let $x_0 \in \mathcal{X}$. Show that the ω-limit set $\omega(x_0)$ is invariant. (b) Let S be an invariant set. Show that the closure, the interior and the boundary of S are positively invariant. (c) Let $S_1, \ldots, S_n$ be invariant sets. Show that the intersection and the union of $S_1, \ldots, S_n$ are again invariant.

E3.3 Consider the system

$$\begin{aligned} x' &= y \\ y' &= x - px^3 \end{aligned} \tag{3.17}$$

with a parameter $p \in \mathbb{R}$, that arises out of Example 3.10 with $\delta = 0$. (a) Find the equilibria and check if they are hyperbolic. (b) Compute a polynomial approximation of the stable and unstable manifolds of the equilibrium point 0. (c) Exploit the special structure of (3.17) to find an explicit expression of these manifolds. What do you observe when you extend the local stable/unstable manifolds to their global counterpart?

E3.4 Let $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$g\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y^2 \\ 2y - 2x^2 + x^4y^2 \end{pmatrix}$$

and consider the dynamical system $(\mathbb{R}^2, \mathbb{N}_0, \phi_t)$ that is induced by iterating the map g. (a) Find all fixed points of g. (b) Show that $S = \{y = x^2\}$ is an invariant set. (c) Consider the fixed point 0. Is it hyperbolic? If so, compute its stable and unstable eigenspaces. (d) Conclude that $W^{\mathrm{s}}_{\mathrm{loc}}(0) = \{(x, y) \in \mathbb{R}^2 : (x, y) \in S \text{ and } |x| < \varepsilon\}$ for some $\varepsilon > 0$.

4 Equivalence

Having covered dynamical stability, local hyperbolicity and invariant manifolds near equilibria, it is natural to ask for a more abstract classification. In particular, one would hope that we dynamically could find a more standard form via well-chosen coordinates near hyperbolic equilibria. In fact, this is similar in philosophy to the rectification theorem [40, Thm. 8.3] but more challenging to prove as the geometry near hyperbolic equilibria is already quite complex. We start with the main general notions of dynamical equivalence.

Definition 4.1. A dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ is called **topologically equivalent** to a system $(\mathcal{Y}, \mathcal{T}, \psi_t)$ if there exists a homeomorphism $h : \mathcal{X} \to \mathcal{Y}$ mapping orbits from the first system to the second system preserving the direction of time. If the time parametrization is also preserved, the systems are called **conjugate**.

If $h, h^{-1} \in C^k$, $k \geq 1$, then the systems are called **C^k-equivalent** (sometimes this is called **smoothly equivalent** if the precise $k \geq 1$ does not matter for the current argument). Similarly, for a preserved time parametrization and a C^k-diffeomorphism, we have a **C^k-conjugacy**.

Proposition 4.2. *Two discrete-time maps*

$$x \mapsto f(x),\ x \in \mathbb{R}^d, \qquad y \mapsto g(y),\ y \in \mathbb{R}^d,$$

are topologically conjugate if and only if there exists a homeomorphism $h : \mathbb{R}^d \to \mathbb{R}^d$ *such that*

$$f = h^{-1} \circ g \circ h. \tag{4.1}$$

Proof. We only show that topological conjugacy implies (4.1); the converse is an easy exercise. Topological conjugacy implies that there exists a homeomorphism $h : \mathbb{R}^d \to \mathbb{R}^d$ such that

$$y = h(x) \quad \text{and} \qquad h(f(x)) = g(y).$$

Therefore, $h(f(x)) = g(y) = g(h(x))$ and since h is invertible, the result follows. □

Proposition 4.3. *Consider two continuous-time flows* ϕ_t *and* ψ_t *generated by*

$$x' = f(x), \quad x \in \mathbb{R}^d \qquad \text{and} \qquad y' = g(y), \quad y \in \mathbb{R}^d$$

with $f, g \in C^1$*. Suppose* $y = h(x)$ *for some* C^1*-diffeomorphism* $h : \mathbb{R}^d \to \mathbb{R}^d$*. Then* ϕ_t*,* ψ_t *are* C^1*-conjugate, and*

$$f(x) = (\mathrm{D}h)^{-1}(x) g(h(x)). \tag{4.2}$$

Proof. Regarding (4.2) we make the important calculation

$$g(h(x)) = g(y) = y' = \mathrm{D}h(x)x' = \mathrm{D}h(x)f(x).$$

Therefore, h maps solutions to solutions and we have $h(\phi_t(x)) = \psi_t(h(x))$. □

Definition 4.4. A dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ with steady state x_* is called **locally topologically equivalent** to a system $(\mathcal{X}, \mathcal{T}, \psi_t)$ near x_* if there exists a homeomorphism $h : \mathcal{U}(x_*) \to \mathcal{V}(y_*)$ mapping orbits from the first system to the second system preserving the direction of time for neighborhoods $\mathcal{U}(x_*), \mathcal{V}(y_*)$, where $\psi_t(y_*) = y_*$ for all $t \in \mathcal{T}$.

Theorem 4.5 (Hartman-Grobman Theorem). *Consider the ODE* $x' = f(x)$*,* $x \in \mathbb{R}^d$*,* $f \in C^1$*, with flow* ϕ_t *and suppose* x_* *is a hyperbolic equilibrium. Then* ϕ_t *is locally topologically conjugate to the flow* e^{tA} *(with* $A = \mathrm{D}f(x_*)$*) generated by the linearized system.*

We are only going to prove the key statement for maps

$$g(x) = Ax + R(x), \qquad x \in \mathbb{R}^d, g \in C^1, \text{ g is invertible,} \tag{4.3}$$

and then comment on the proof of the Hartman-Grobman Theorem; see also Figure 12.

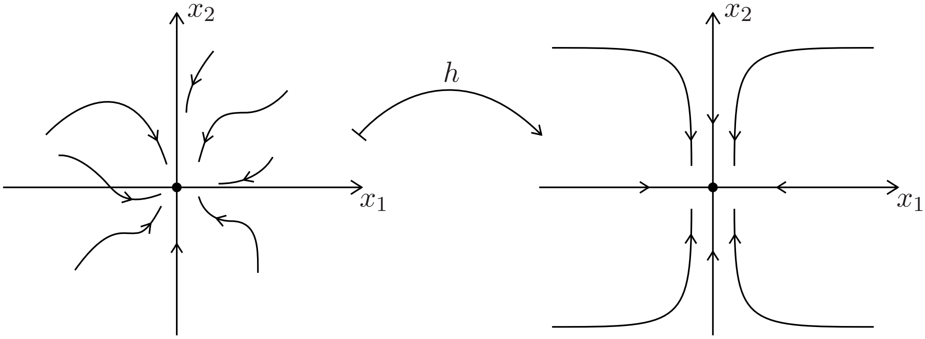

Figure 12: Generic stable node for a nonlinear system conjugated via h to a node for a linear system. For *generic* matrices with spectrum in the left-half complex plane, there exists a weakest contracting eigendirection (exercise!); see also Definition 6.9.

Theorem 4.6. *Suppose* 0 *is a hyperbolic fixed point of* (4.3). *Then there exists a unique homeomorphism* $h(x) = x + H(x)$, $H(0) = 0$, *H bounded, such that in a neighborhood* $\mathcal{U} = \mathcal{U}(0) \subset \mathbb{R}^d$ *we have*

$$h \circ A = g \circ h, \tag{4.4}$$

i.e., g is locally conjugate to A.

Proof. `Step 1:` We have to prove the *existence* of H. Suppose A has eigenvalues inside and outside the unit circle; the other cases can be treated similarly. Split A via $A_\mathrm{s} := A|_{E^\mathrm{s}(0)}$ and $A_\mathrm{u} := A|_{E^\mathrm{u}(0)}$. Select a matrix norm $\|\cdot\|$ such that

$$\alpha := \max(\|A_\mathrm{u}^{-1}\|, \|A_\mathrm{s}\|) < 1, \qquad |R(x) - R(y)| \leq \varepsilon|x - y|, \tag{4.5}$$

where we select $\varepsilon < (1-\alpha)/2$ and the inequality for R holds on $\mathcal{U}$ but we cut off R with a smooth cut-off function outside $\mathcal{U}$ so that it holds globally. The result (4.4) is equivalent to the **functional equation**

$$H(Ax) - AH(x) = R(x + H(x)) \qquad \forall x \in \mathcal{U}. \tag{4.6}$$

Consider the Banach space $C(\mathbb{R}^d, \mathbb{R}^d)$ with the supremum norm. Define two invertible linear operators

$$(KH)(x) := H(Ax), \qquad (\mathcal{A}H)(x) := AH(x),$$

where K preserves the norm (exercise!), while $\mathcal{A}$ has expanding and contracting directions. Setting $L := K - \mathcal{A}$, it follows that (4.6) can be viewed as a fixed-point problem

$$LH(x) = R(x + H(x)),$$

where we claim that L is invertible. To prove this, consider the splittings

$$C(\mathbb{R}^d,\mathbb{R}^d) = C(\mathbb{R}^d, E^{\mathrm{s}}(0)) \oplus C(\mathbb{R}^d, E^{\mathrm{u}}(0)), \quad K = K_{\mathrm{s}} \oplus K_{\mathrm{u}}, \quad L = L_{\mathrm{s}} \oplus L_{\mathrm{u}}.$$

We decompose along each direction and find

$$L_{\mathrm{u}} = K_{\mathrm{u}} - A_{\mathrm{u}} = -A_{\mathrm{u}}(\mathrm{Id} - A_{\mathrm{u}}^{-1}K_{\mathrm{u}})$$

so that $L_{\mathrm{u}}^{-1} = -(\mathrm{Id} - A_{\mathrm{u}}^{-1}K_{\mathrm{u}})^{-1}A_{\mathrm{u}}^{-1}$, where the existence of A_{u}^{-1} holds by assumption and $(\mathrm{Id} - A_{\mathrm{u}}^{-1}K_{\mathrm{u}})$ is invertible using the **Neumann series**

$$(\mathrm{Id} - A_{\mathrm{u}}^{-1}K_{\mathrm{u}})^{-1} = \sum_{k=0}^{\infty}(A_{\mathrm{u}}^{-1}K_{\mathrm{u}})^k$$

as $\|A_{\mathrm{u}}^{-1}K_{\mathrm{u}}\| \leq \alpha < 1$. Furthermore, we have $\|L_{\mathrm{u}}^{-1}\| \leq 1/(1-\alpha)$ using the geometric series. A similar argument applies to L_{s} (exercise!) so in summary we have

$$L^{-1} = (K_{\mathrm{u}} - A_{\mathrm{u}})^{-1} \oplus (K_{\mathrm{s}} - A_{\mathrm{s}})^{-1}, \qquad \|L^{-1}\| \leq \frac{2}{1-\alpha}.$$

So we may now try to solve $H(x) = L^{-1}R(x+H(x))$ for H. Since the goal is to apply Banach Fixed Point Theorem 1.8, we estimate

$$\begin{aligned} \|L^{-1}R(\mathrm{Id}+H_1) - L^{-1}R(\mathrm{Id}+H_2)\| &\leq \frac{2}{1-\alpha}\|R(\mathrm{Id}+H_1) - R(\mathrm{Id}+H_2)\| \\ &\leq \frac{2\varepsilon}{1-\alpha}\|H_1 - H_2\|. \end{aligned}$$

This implies that we have a contraction by (4.5) and H exists and is continuous.

`Step 2:` It remains to prove that $h = \mathrm{Id} + H$ is invertible. First, note that we can construct a map $\tilde{h}$ such that

$$\tilde{h}(x) = x + \tilde{H}(x), \qquad A \circ \tilde{h} = \tilde{h} \circ g. \tag{4.7}$$

This follows by exchanging the roles of A, g above and repeating the same line of argument. Therefore, using (4.4) and (4.7) we find

$$A \circ \tilde{h} \circ h = \tilde{h} \circ g \circ h = \tilde{h} \circ h \circ A. \tag{4.8}$$

One checks that, if $R(x) \equiv 0$, the homeomorphism is unique in the linear case so (4.8) implies

$$\tilde{h} \circ h = \mathrm{Id} = h \circ \tilde{h}$$

and we obtain $\tilde{h} = h^{-1}$.

`Step 3:` It remains to check that $H(0) = 0$ and that h is unique. One evaluates $Ah^{-1}(x) = h^{-1}(g(x))$ at $x = 0$, which gives the linear equation

$$Ah^{-1}(0) = h^{-1}(0) \qquad \Leftrightarrow \qquad Ah^{-1}(0) - \mathrm{Id}h^{-1}(0) = 0$$

so $h^{-1}(0)$ is an eigenvector for multiplier 1, but A has no multipliers on the unit circle by assumption. This means we must have $h^{-1}(0) = 0$. The proof of uniqueness is a (relatively easy) exercise. □

The proof of the Hartman-Grobman Theorem 4.5 uses the **time-one map**. We let ϕ_t denote the flow of $x' = f(x)$ and define

$$\phi_1(x) = \mathrm{e}^A x + R(x)$$

for some remainder R. Upon localizing the vector field near $x = 0$ with a cut-off, one checks that R is indeed small so that Theorem 4.6 applies so we get a homeomorphism h with

$$h \circ \mathrm{e}^A = \phi_1 \circ h. \tag{4.9}$$

The final step is to define

$$h_s := \phi_s \circ h \circ \mathrm{e}^{-sA}$$

and note that h_s also satisfies (4.9) (exercise!). Using the uniqueness part of Theorem 4.6 one obtains $h_s \equiv h$ completing the construction.

Exercises for Section 4

E4.1 Consider the dynamical systems generated by $x' = Ax$ and $y' = By$ for matrices $A, B \in \mathbb{R}^{d\times d}$. Show that the following three statements are equivalent: (i) The flows are linearly conjugate, i.e., the conjugating homeomorphism is a linear map. (ii) The flows are C^1-conjugate. (iii) The matrices A and B are similar, i.e., $B = S^{-1}AS$.

E4.2 Consider the two differential equations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

(a) Rewrite both equations in polar coordinates $x = r\cos(\theta)$, $y = r\sin(\theta)$. (b) Sketch the phase portraits around the origin. (c) Show that the two dynamical systems are conjugate. Are they also C^1-conjugate?

E4.3 We consider the two differential equations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Prove that the two dynamical systems are topologically conjugate using a homeomorphism of the form $h(x, y) = (x + f(y), y)$ for some $f : \mathbb{R} \to \mathbb{R}$. *Hint*: Derive and solve a (differential) equation for the unknown function f.

E4.4 Show that the two dynamical systems induced by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -\sin(x) + 3y^3 \\ x^2 - y \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -x + y(y-1) \\ \mathrm{e}^x - \mathrm{e}^y \end{pmatrix}$$

are locally conjugate around the origin.

5 Poincaré Maps

Having covered the dynamics near hyperbolic steady states quite comprehensively in the last few sections, we can proceed to periodic motion. Consider the ODE

$$x' = f(x), \qquad x \in \mathbb{R}^d,\ f \in C^1, \tag{5.1}$$

with flow ϕ_t and suppose $\Gamma = \Gamma(t)$ is a periodic orbit. The idea of a Poincaré map is to record returns to a manifold transversal to Γ; see Figure 13.

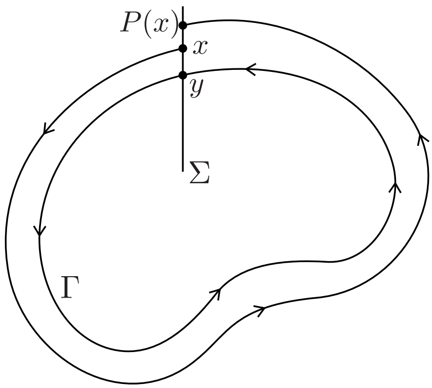

Figure 13: Sketch of the Poincaré map construction.

Theorem 5.1 (Existence/Smoothness of the Poincaré Map)**.** *Consider* (5.1), *suppose* $\Gamma(t) = \Gamma$ *has period* $T > 0$ *and* $y \in \Gamma$. *Let*

$$\Sigma := \{x \in \mathbb{R}^d : (x - y) \cdot f(y) = 0\} \tag{5.2}$$

*be a cross-section (or **Poincaré section**). Then there exists $\delta > 0$ and $\tau : \mathcal{B}(y;\delta) \to \mathbb{R}$, $\tau \in C^1$, such that*

$$\tau(y) = T \qquad \text{and} \qquad \phi_{\tau(x)}(x) \in \Sigma \quad \forall x \in \mathcal{B}(y;\delta).$$

*In summary, the **Poincaré map** $P(x) := \phi_{\tau(x)}(x)$ is locally well-defined and C^1 on $\mathcal{B}(y;\delta) \cap \Sigma$.*

Proof. The idea is to get τ via the implicit function theorem. Define the function

$$F(t,x) := (\phi_t(x) - y) \cdot f(y) \tag{5.3}$$

which is in C^1 since $\phi \in C^1$. Furthermore, $F(T,y) = 0$ by periodicity of Γ. Differentiating (5.3) in t, we find

$$\partial_t F(T,y) = f(y) \cdot \partial_t \left(\phi_t(x)\right)|_{(t,x)=(T,y)} = |f(y)|^2 \neq 0$$

since y is not an equilibrium. Therefore, the implicit function theorem provides a C^1 map $\tau : \mathcal{B}(y;\delta) \to \mathbb{R}$ such that

$$\tau(y) = T, \qquad 0 = F(\tau(x),x) = (\phi_{\tau(x)}(x) - y) \cdot f(y) \quad \Rightarrow \phi_{\tau(x)}(x) \in \Sigma,$$

which finishes the proof. $\square$

The definition (5.2) imposes that Γ intersects Σ **transversally** at y. This construction generalizes naturally if Σ is replaced by a smooth manifold $\mathcal{M}$ (exercise!). Up to a coordinate change, we may assume local coordinates $x = (x_1, \ldots, x_{d-1})$ on Σ, so that $P(0) = 0$ and consider the Jacobian

$$A = \mathrm{D}P(0) \qquad \text{with multipliers } \mu_1, \ldots, \mu_{d-1}.$$

In particular, analyzing local stability of the periodic orbit is equivalent to local stability of the fixed point via its multipliers.

Proposition 5.2. *The multipliers of A are independent of y, of the cross-section as well as of the local coordinates.*

Proof. Suppose there are two points $y, \tilde{y}$, two sections $\Sigma, \tilde{\Sigma}$, and coordinates $x, \tilde{x}$ with Poincaré maps $P, \tilde{P}$. Then, using the construction of P and $\tilde{P}$, one finds that there exists a diffeomorphism $h(x) = \tilde{x}$ such that $P, \tilde{P}$ are locally conjugate so

$$P = h^{-1} \circ \tilde{P} \circ h.$$

Differentiating and evaluating at $x = 0$ gives

$$A = B^{-1}\tilde{A}B, \qquad B := \mathrm{D}h(0), \quad \tilde{A} = \mathrm{D}\tilde{P}(0).$$

Then we calculate using the standard characteristic equation

$$\det(A - \mu\mathrm{Id}) = \det(B^{-1})\det(\tilde{A} - \mu\mathrm{Id})\det(B) = \det(\tilde{A} - \mu\mathrm{Id})$$

because $\det(B^{-1})\det(B) = 1$ by standard rules for determinants. $\square$

Example 5.3. Consider the planar system with parameter $p \in \mathbb{R}$ given by

$$\begin{aligned} x' &= -y + x(p - x^2 - y^2), \\ y' &= x + y(p - x^2 - y^2), \end{aligned}$$

which turns out to have a limit cycle for certain parameters p. The single multiplier μ can be calculated exactly (exercise! Hint: polar coordinates). However, in general cases, this is not possible analytically. ♦

To analyze the stability of a periodic orbit Γ, one may also link this to classical ODE theory. Indeed, write

$$x(t) = \Gamma(t) + u(t), \qquad 0 < |u(t)| \ll 1,$$

i.e., u is a small perturbation. Then, we have formally

$$u' = x' - \Gamma' = f(\Gamma(t) + u(t)) - f(\Gamma(t)) = A(t)u + o(|u|)$$

where $A(t) = \mathrm{D}f(\Gamma(t))$.

Definition 5.4. The ODE $u' = A(t)u$ is called the **variational equation** along Γ.

For some basic analytical background on the solution properties of linear nonautonomous systems, we refer to [40, Sec. 5].

Definition 5.5. A matrix $M(t)$ is called a **fundamental solution** of a time-dependent linear system $x' = A(t)x$ if $M(0) = \mathrm{Id}$ and $M'(t) = A(t)M(t)$ for all $t \in \mathcal{T}$.

The previous definition naturally generalizes to matrices $M = M(t, t_0)$, where the starting time is $t_0 \neq 0$ but we assume here $t_0 = 0$. If T is the period of Γ and M solves the variational equation about Γ. Then $u(T) = M(T)u(0)$ and $M(T)$ is called the **monodromy matrix**.

Theorem 5.6. *The monodromy matrix $M(T)$ has eigenvalues $1, \mu_1, \ldots, \mu_{d-1}$, where μ_j are the multipliers of the Poincaré map.*

Proof. Fix $y \in \Gamma$. First, by the **rectification theorem** from ODEs [40, Thm. 8.3], we may assume that

$$f(x) = (0, \ldots, 0, 1)^\top \in \mathbb{R}^d \quad \text{for } x \text{ near } y, \tag{5.4}$$

and take local coordinates $(x_1, \ldots, x_{d-1}, z)$, $x_d =: z \in \mathbb{R}$, such that the Poincaré map $P(x) = \phi_{\tau(x)}(x)$ is given solely in the first $(d-1)$ coordinates. We write $\phi_t(x) = \phi(t, x)$ for clarity, and differentiate

$$\begin{aligned} \mathrm{D}P(y) &= \mathrm{D}\,\phi(\tau(x), x)|_{x=y} = \partial_t \phi(T, y)\ (\nabla\tau(y))^\top + \mathrm{D}_x\phi(T, y) \\ &= f(y)\ (\nabla\tau(y))^\top + \mathrm{D}_x\phi(T, y). \end{aligned} \tag{5.5}$$

We claim that $\mathrm{D}_x\phi(T, y) = M(T)$. Indeed, we have $\mathrm{D}_x\phi(0, y) = \mathrm{Id}$ and we calculate

$$(\mathrm{D}_x\phi(t, y))' = \mathrm{D}_x f(\phi(t, y)) = \mathrm{D}f(\phi(t, y))\mathrm{D}_x\phi(t, y)$$

interchanging derivatives with respect to t and x. Therefore, equation (5.5) implies using (5.4) that

$$M(T) = \begin{pmatrix} \mathrm{D}_x P(y) & v_1 \\ v_2^\top & m \end{pmatrix} \tag{5.6}$$

where $v_{1,2} \in \mathbb{R}^{d-1}$ are some vectors and $m \in \mathbb{R}$. We claim that $v_1 = 0$ and $m = 1$, which follows if we can prove $M(T)f(y) = f(y)$, i.e., $\mu = 1$ is an eigenvalue of the monodromy matrix with eigenvector aligned with tangent vector to the periodic orbit $\Gamma(t)$. This situation can be easily drawn into Figure 13 (exercise!). We have

$$\Gamma''(t) = f(\Gamma(t))' = A(t)\Gamma'(t)$$

so $\Gamma'(t)$ solves $u' = A(t)u$. Therefore, we have

$$M(T)f(y) = M(T)\Gamma'(0) = \Gamma'(T) = \Gamma'(0) = f(y)$$

as required. The structure of the monodromy matrix in (5.6) with $v_1 = 0$ and $m = 1$ now implies the result. □

The construction of the Poincaré map naturally leads to the question, when we can realize any given map $g : \mathbb{R}^d \to \mathbb{R}^d$ as the **time-T map** for some flow ϕ_t, i.e.,

$$g(x) = \phi_T(x), \qquad \text{for some } T > 0.$$

Example 5.7. It is easy to check that $g(x) = -\frac{1}{2}x$ has no realization as the time-T for a flow generated by $x' = f(x)$ with $x \in \mathbb{R}$. ♦

Theorem 5.8 (Suspension Flow)**.** *Given any diffeomorphism $g : \mathbb{R}^d \to \mathbb{R}^d$, there exists a $(d+1)$-dimensional manifold $\mathcal{M}$ such that an ODE on $\mathcal{M}$ has g as its time-T map.*

Proof. Fix some $T > 0$ and define the manifold by

$$\mathcal{M} := \{(x,t) \in \mathbb{R}^d \times \mathbb{R} : t \in [0,T]\}/((g(x),0) \sim (x,T)).$$

Then consider the flow ϕ_t on $\mathcal{M}$ defined by $(x', t') = (0,1)$ (also called the

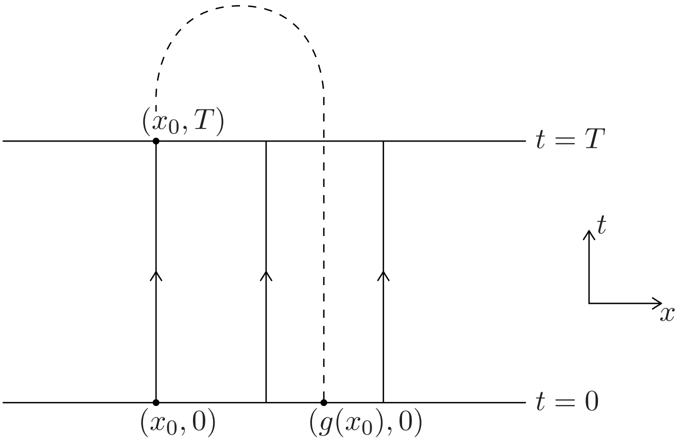

Figure 14: Illustration of the suspension flow construction.

suspension). The time-T map leaves $\{t = 0\} \subset \mathcal{M}$ invariant and induces the map g; see Figure 14. $\square$

Exercises for Section 5

E5.1 Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth function and consider the ODE $x' = f(x)$ with $x \in \mathbb{R}$. Prove that there does not exist a periodic orbit (a) by basic tools from an introductory calculus lecture, (e.g. mean-value theorem), (b) by considering $\int_0^T f(x(t))\frac{\mathrm{d}x(t)}{\mathrm{d}t}\,\mathrm{d}t$, (c) by exploiting the existence of a potential $V : \mathbb{R} \to \mathbb{R}$ such that $-\nabla V(x) = f(x)$.

E5.2 Consider the planar ODE system

$$\begin{aligned} x' &= \mu x - y - x\,(x^2 + y^2), \\ y' &= x + \mu y - y\,(x^2 + y^2), \end{aligned}$$

where $\mu > 0$ is a parameter. Find a periodic orbit and determine its stability using a Poincaré map. *Hint: Use polar coordinates.*

6 Structural Stability

Having encountered the amazing classification results one can obtain for hyperbolic equilibria, fixed points and periodic orbits, one might ask whether it is possible to be even more ambitious. It would be most elegant to analyze *all* dynamical systems of a certain class at once by restricting to those displaying *typical* dynamics. Consider a compact d-dimensional set $\mathcal{K} \subset \mathbb{R}^d$.

Definition 6.1. The function space $C^k(\mathcal{K}, \mathbb{R}^d)$ is given by all functions $f = (f_1, f_2, \ldots, f_d)^\top$ with the norm

$$\|f\|_{C^k} = \max_l \sum_{|\alpha|=0}^{k} \sup_{x\in\mathcal{K}} |\mathrm{D}^\alpha f_l(x)|,$$

where $\alpha = (\alpha_1, \ldots, \alpha_d) \in (\mathbb{N}_0)^d$ is a multi-index and $\mathrm{D}^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}$.

$C^k(\mathcal{K}, \mathbb{R})$ is a Banach space, and the definition generalizes to a compact smooth manifold $\mathcal{M}$ replacing $\mathcal{K}$, by taking the supremum over all charts. For non-compact $\mathcal{K}$ we do not obtain Banach spaces, but still metric spaces, i.e., we could set

$$d_0(f, g) := \sum_j 2^{-j} \frac{\sup_{x\in\mathcal{K}_j} |f(x) - g(x)|}{\sup_{x\in\mathcal{K}_j} |f(x) - g(x)| + 1}$$

where $\mathcal{K}_j$ is a countable family of growing compact sets, or we could set

$$d_0(f, g) := \min\left\{1, \sup_{x\in\mathcal{K}} |f(x) - g(x)|\right\}, \tag{6.1}$$

where both definitions naturally generalize to manifolds, if the manifold has a metric $d(x, y) = |x - y|$; here we shall stick to (6.1) until further notice. For elementary background regarding ODEs on manifolds we refer to [40].

Definition 6.2. Let $f \in C^1(\mathcal{M}, \mathrm{T}\mathcal{M})$ be a vector field. Then f is called **structurally stable** if there exists $\varepsilon > 0$ such that for all vector fields $g \in C^1(\mathcal{M}, \mathrm{T}\mathcal{M})$ with

$$\|f - g\|_{C^1} < \varepsilon,$$

g is topologically equivalent to f. If f is not structurally stable, it is called **structurally unstable**.

Example 6.3. Consider the unit torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ with vector field f defined by

$$\begin{aligned} x' &= \omega_1, \\ y' &= \omega_2. \end{aligned} \tag{6.2}$$

for fixed $\omega_{1,2} \in \mathbb{R}$. Then f is structurally unstable, since for $\omega_1/\omega_2 \in \mathbb{Q}$ all orbits are periodic while for $\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$, all orbits are **aperiodic**, i.e., are never closed. ♦

Definition 6.4. Given a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$, a point $x \in \mathcal{X}$ is called **nonwandering** if for any neighborhood $\mathcal{U} = \mathcal{U}(x)$ of x and for any $t_0 \in \mathcal{T}$, there exists a $t > t_0$ such that

$$\phi_t(\mathcal{U}) \cap \mathcal{U} \neq \emptyset.$$

The set of such points is called the **nonwandering set** $\mathrm{NW}(\phi)$; see Figure 15(a). All other points are called **wandering**.

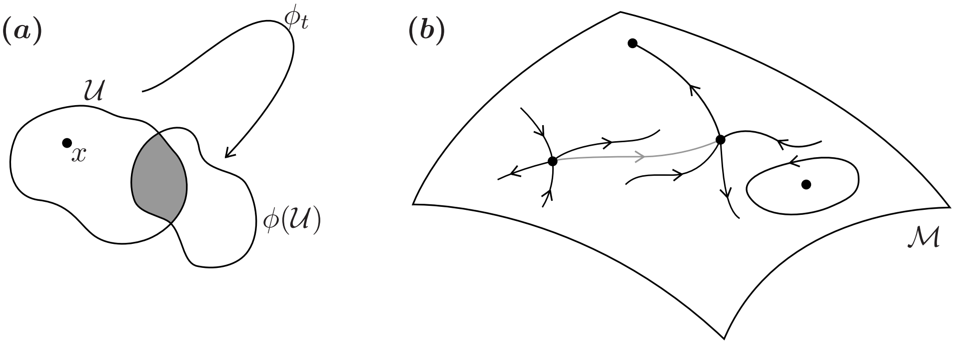

Figure 15: (a) Sketch of the idea in the definition of a nonwandering point. (b) Illustration of one aspect of Peixoto's Theorem: a saddle-to-saddle connection (grey) is structurally unstable.

Example 6.5. Consider the flow ϕ generated by (6.2), then we easily check (exercise!) that $\mathrm{NW}(\phi) = \mathbb{T}^2 = \mathcal{X}$. ♦

Theorem 6.6 (Peixoto's Theorem)**.** *A C^1-vector field f on a compact two-dimensional differentiable manifold $\mathcal{M}$ is structurally stable if and only if*

(T1) the number of periodic orbits and equilibria is finite, and each is hyperbolic;

(T2) for saddles x, y (potentially $x = y$), $W^{\mathrm{s}}(x) \cap W^{\mathrm{u}}(y) = \emptyset$ (respectively for $x = y$ we require $W^{\mathrm{s}}(x) \cap W^{\mathrm{u}}(x) = \{x\}$);

(T3) $\mathrm{NW}(\phi)$ *consists only of equilibria and periodic orbits.*

Furthermore, if $\mathcal{M}$ is orientable, the set of structurally stable vector fields is open and dense in $C^1(\mathcal{M})$.

We shall not prove Theorem 6.6 but see Figure 15(b). Peixoto's Theorem for $d \geq 3$ is false. We only have a unidirectional statement under similar assumptions.

Definition 6.7. A **Morse-Smale flow** ϕ_t is one for which

(D1) the number of periodic orbits and equilibria is finite, and each is hyperbolic;

(D2) stable and unstable manifolds only intersect transversally;

(D3) $\mathrm{NW}(\phi)$ consists only of equilibria and periodic orbits.

Theorem 6.8 (Morse-Smale Stability)**.** *Consider C^1-vector fields on a compact differentiable manifold. Then Morse-Smale systems are structurally stable.*

Definition 6.9. A property $\mathcal{P}$ of elements of a topological space $\mathcal{Z}$ is called **generic** if it holds on a countable intersection of open dense sets.

Remark: There is another related idea to define 'generic' not topologically but measure-theoretically. This concept is known under the name **prevalence**. It would lead us a bit too far away from the main lines of argument to give a detailed discussion of the two definitions and their consequences. Hence, we stick to the technically slightly easier approach via genericity.

A countable intersection of open dense sets is also called a **residual**. Peixoto's Theorem states that on compact two-dimensional orientable manifolds structural stability is a generic property. For $d \geq 3$ Morse-Smale systems are not generic.

Example 6.10. Consider the space of matrices $\mathcal{Z} = \mathbb{R}^{d\times d}$ with topology induced by any matrix norm. Then for $A \in \mathcal{Z}$ the property $\mathcal{P}$ of A being hyperbolic, i.e., A has no zero real part eigenvalues, is generic; see Exercise E6.2.

In summary, there are very strong statements on generic planar flows and their structure and structural stability. Basically, hyperbolicity plus transversality give the desired results. Yet, for all higher dimensions, major problems arise, which we are going to comprehend much better in Sections 13-15.

Exercises for Section 6

E6.1 For any of the following planar vector fields, let $\mathcal{K}$ be a sufficiently large compact set that contains all the dynamically relevant compact invariant sets in its interior. Discuss the structural stability of the induced dynamical systems on $\mathcal{K}$.

$$\text{(a)}\quad f(x,y) = \begin{pmatrix} -y \\ x \end{pmatrix} \qquad \text{(b)}\ f(x,y) = \begin{pmatrix} 2x - x^2 \\ -y + xy \end{pmatrix}$$

$$\text{(c)}\quad f(x,y) = \begin{pmatrix} x + 2y \\ -x - y \end{pmatrix} \qquad \text{(d)}\ f(x,y) = \begin{pmatrix} y \\ x - x^3 \end{pmatrix}$$

$$\text{(e)}\quad f(x,y) = \begin{pmatrix} y \\ x - x^3 - y \end{pmatrix}.$$

E6.2 Prove that the set $\{A \in \mathbb{R}^{d\times d} \mid \mathrm{Re}(\lambda) \neq 0,\ \lambda \text{ is eigenvalue for } A\}$ is an open and dense subset of $\mathbb{R}^{d\times d}$.

7 Hyperbolic Sets

In this section, we prove in a very general setting, why hyperbolicity is such a crucial property to expect robustness of dynamics and structural stability. Let $\mathcal{M}$ be a C^1 **Riemannian manifold**, i.e., there exists a **Riemannian metric**, which is a differentiable family of symmetric, bilinear, and positive definite forms

$$\langle\cdot,\cdot\rangle_x : \mathrm{T}_x\mathcal{M} \times \mathrm{T}_x\mathcal{M} \to \mathbb{R}$$

where we set $|v| := \sqrt{\langle v, v\rangle}$. For a curve $\gamma : [a,b] \to \mathcal{M}$, $\gamma \in C^1$, we define its **length** by

$$\int_a^b |\gamma'(s)|\ \mathrm{d}s$$

which induces a metric $d(x,y)$ on $\mathcal{M}$ by taking the infimum of the length over all curves connecting $x = \gamma(a)$ and $y = \gamma(b)$. Next, we define a discrete-time dynamical system via a diffeomorphism

$$g : \mathcal{M} \to \mathcal{M}. \tag{7.1}$$

Since $g \in C^1$, we have a well-defined derivative (or **tangent map**)

$$\mathrm{D}g_x : \mathrm{T}_x\mathcal{M} \to \mathrm{T}_{g(x)}\mathcal{M}, \qquad \mathrm{D}g_x(v)(w) := v(w \circ g),$$

which is just the **Jacobian** in the case $\mathcal{M} = \mathbb{R}^d$; see Figure 16(a).

Definition 7.1. A compact g-invariant set $\Lambda \subset \mathcal{M}$ is called **hyperbolic** if there exists $\lambda \in (0,1)$, $C > 0$, and families of subspaces $E^{\mathrm{s}}(x), E^{\mathrm{u}}(x) \subset \mathrm{T}_x\mathcal{M}$ such that for every $x \in \Lambda$

(D1) ("invariant") $\mathrm{D}g_x E^{\mathrm{s}}(x) = E^{\mathrm{s}}(g(x))$ and $\mathrm{D}g_x E^{\mathrm{u}}(x) = E^{\mathrm{u}}(g(x))$.

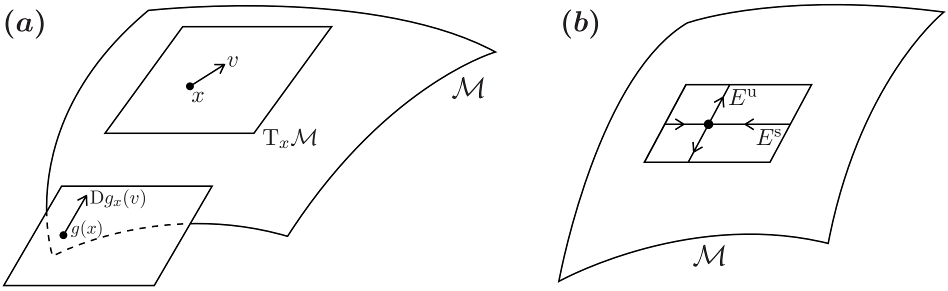

Figure 16: (a) Illustration of the derivative/tangent map on a manifold $\mathcal{M}$. (b) Sketch of a local hyperbolic splitting on $\mathcal{M}$.

(D2) ("stable") $|\mathrm{D}g_x^j v^{\mathrm{s}}| \leq C\lambda^j |v^{\mathrm{s}}|$ for every $v^{\mathrm{s}} \in E^{\mathrm{s}}(x)$ and $j \geq 0$;

(D3) ("unstable") $|\mathrm{D}g_x^{-j} v^{\mathrm{u}}| \leq C\lambda^j |v^{\mathrm{u}}|$ for every $v^{\mathrm{u}} \in E^{\mathrm{u}}(x)$ and $j \geq 0$;

(D4) ("splitting") $\mathrm{T}_x\mathcal{M} = E^{\mathrm{s}}(x) \oplus E^{\mathrm{u}}(x)$;

One easily checks (exercise!) that a hyperbolic equilibrium is just a special case of hyperbolic set. Furthermore, the **stable distribution** $E^{\mathrm{s}} := (E^{\mathrm{s}}(x))_{x\in\mathcal{M}}$ and the **unstable distribution** $E^{\mathrm{u}} := (E^{\mathrm{u}}(x))_{x\in\mathcal{M}}$ are checked (exercise!) to be continuous with respect to x; see Figure 16(b). The stable and unstable distributions are contained in the collection of all tangent spaces, which is defined in general as follows:

Definition 7.2. Suppose $\mathcal{M}$ is a C^k-manifold of dimension d. The **tangent bundle**

$$\mathrm{T}\mathcal{M} = \bigcup_{x\in\mathcal{M}} \mathrm{T}_x\mathcal{M}$$

is a C^{k-1}-manifold of dimension $2d$. If $h_\alpha = (x_1,\ldots,x_d) : \mathcal{U}_\alpha \to \mathcal{V}_\alpha$ is a coordinate chart for $\mathcal{M}$, then $(x_1,\ldots,x_d,\mathrm{d}x_1,\ldots,\mathrm{d}x_d) : \mathrm{T}\mathcal{M} \to \mathbb{R}^{2d}$ is a coordinate chart for the tangent bundle, where $\mathrm{d}x_i(v) = v(x_i)$.

Example 7.3. Consider hyperbolic toral automorphism on $\mathcal{M} = \mathbb{T}^2$, considered already in Example 3.11 defined by

$$g(x) = Ax, \qquad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Calculating eigenvalues and eigenvectors of A, one observes that Definition 7.1 is satisfied with $\Lambda = \mathcal{M}$. ♦

Definition 7.4. If for g defined (7.1), the entire manifold $\mathcal{M} = \Lambda$ is a hyperbolic set, then we call g an **Anosov diffeomorphism**.

Theorem 7.5. *Let Λ be a hyperbolic set for $g : \mathcal{M} \to \mathcal{M}$. Then there exists an open set $\mathcal{U}$ containing Λ and $\varepsilon_1 > 0$ such that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any $f : \mathcal{M} \to \mathcal{M}$ with*

$$d_0(\mathrm{D}f, \mathrm{D}g) \leq \varepsilon_1$$

and any homeomorphism $h : \mathcal{Y} \to \mathcal{Y}$ of a topological space $\mathcal{Y}$, and any continuous map $\varphi : \mathcal{Y} \to \mathcal{U}$ satisfying

$$d_0(\varphi \circ h, f \circ \varphi) \leq \delta$$

there exists a unique continuous map $\psi : \mathcal{Y} \to \mathcal{U}$ with

$$\psi \circ h = f \circ \psi \qquad \text{and} \qquad d_0(\varphi, \psi) < \varepsilon.$$

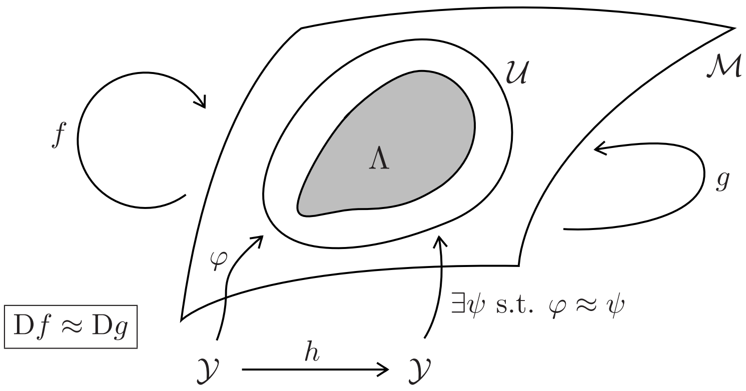

Figure 17: Illustration of Theorem 7.5.

The main conclusion of Theorem 7.5 is shown in Figure 17, which roughly states that for g, and any map f C^1-close to it, we may turn an approximate commutative diagram into an exact one due to hyperbolicity. The proof uses some basic differential geometry and the Banach Fixed Point Theorem to construct ψ, very similar to the proof of Theorem 3.8, so we shall not carry it out here.

Example 7.6. If $\mathcal{Y} = \{y\}$ is a singleton and $h(y) = y$, then Theorem 7.5 states that every map f near g satisfies

$$\psi(y) = f \circ \psi(y)$$

so f must have a fixed point. ♦

Definition 7.7. A δ-**pseudo-orbit** of a map $g : \mathcal{M} \to \mathcal{M}$ is a finite or infinite sequence $(x_k) \subset \mathcal{M}$ such that

$$d(g(x_k), x_{k+1}) \leq \delta \qquad \text{for all } k.$$

Theorem 7.8 (The Shadowing Theorem)**.** *Let Λ be a hyperbolic set of $g : \mathcal{M} \to \mathcal{M}$. Then given any $\varepsilon > 0$ there exists $\delta > 0$ such that if (x_k) is a δ-pseudo-orbit of g and $d(x_k, \Lambda) < \delta$ for all k, then there exists*

$$x \in \Lambda_\varepsilon := \{y \in \mathcal{M} : d(y, \Lambda) < \varepsilon\}$$

such that $\mathrm{Or}(x)$ ***shadows*** *the pseudo-orbit, i.e., we have*

$$d(g^k(x), x_k) < \varepsilon \qquad \textit{for any } k.$$

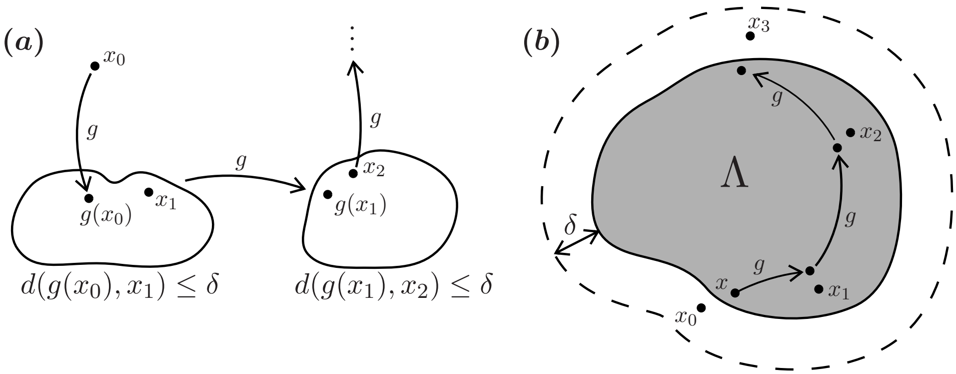

Figure 18: (a) Illustrating the definition of a pseudo-orbit, which always stays close to a true orbit. (b) Basic setting for the proof of Theorem 7.8.

Proof. The idea is to apply Theorem 7.5 so we start by choosing a neighborhood $\mathcal{U}$ satisfying the conclusions of the theorem and such that $\Lambda_\delta \subset \mathcal{U}$. If (x_k) is not defined for all $k \in \mathbb{Z}$, say

$$(x_k) = (x_{k-}, \ldots, x_{k+})$$

augment it with preimages of g for some y_- such that $d(x_{k-}, y_-) < \delta$ and by iterates/images of g for some y_+ such that $d(x_{k+}, y_+) < \delta$. This yields a bi-infinite sequence, still denoted (x_k), which is a δ-pseudo-orbit lying inside a δ-neighborhood of Λ. Define $\mathcal{Y} = (x_k)$ with the discrete topology and note that the left-shift map

$$h : \mathcal{Y} \to \mathcal{Y}, \qquad h(x_k) = x_{k+1}$$

is a homeomorphism on $\mathcal{Y}$. Now set $f = g$ in Theorem 7.5 and consider $\varphi(x_k) = x_k$ as the usual inclusion. Since we have a δ-pseudo-orbit, we must have

$$d(\varphi(h(x_k)), g(\varphi(x_k))) = d(x_{k+1}, g(x_k)) \leq \delta$$

so the assumptions are satisfied and there exists ψ with

$$\psi \circ h = g \circ \psi \qquad \text{and} \qquad d_0(\varphi, \psi) < \varepsilon.$$

Now take $x = x_0$ and observe that for any k we have

$$\begin{aligned} d(g^k(x), x_k) &= d(g^k(\varphi(x)), \varphi(x_k)) \\ d(g^k(\psi(x)), \psi(x_k)) &= d(\psi(h^k(x)), \psi(x_k)) = d(\psi(x_k), \psi(x_k)) = 0. \end{aligned}$$

Since φ is ε-close to ψ we get $d(g^k(x), x_k) < \varepsilon$ as required. □

The previous results already hint at the fact that hyperbolic sets should obey various forms of "persistence" or "robustness" to perturbations, e.g., to perturbations in the map in the C^1-topology:

Theorem 7.9 (Persistence of Hyperbolicity)**.** *Let Λ be a hyperbolic set of g. Then the following statements hold:*

(T1) There is an open set $\mathcal{U}(\Lambda)$ containing Λ and $\varepsilon_1 > 0$ such that if $\mathcal{K} \subset \mathcal{U}(\Lambda)$ is a compact invariant set of a diffeomorphism $f : \mathcal{M} \to \mathcal{M}$ with

$$d_1(f, g) < \varepsilon_1, \tag{7.2}$$

then $\mathcal{K}$ is a hyperbolic set of f.

(T2) For every open set $\mathcal{V}$ containing Λ and every $\varepsilon > 0$, there exists $\delta > 0$ such that for every $f : \mathcal{M} \to \mathcal{M}$ with $d_1(f, g) < \delta$, there exists a hyperbolic set $\mathcal{K} \subset \mathcal{V}$ of f and a homeomorphism $\psi : \mathcal{K} \to \Lambda$ such that

$$\psi \circ f|_{\mathcal{K}} = g|_{\Lambda} \circ \psi$$

and $d_0(\psi, \mathrm{Id}) < \varepsilon$.

Roughly speaking, (T1) says that hyperbolic sets persist under perturbations of the map, while (T2) means that the dynamics on the original and the perturbed hyperbolic set is topologically conjugate; cf. the Hartman-Grobman Theorem.

Proof. `Step 1:` We deal with (T1). By continuity, we may extend the distributions E^{s}_g and E^{u}_g to distributions $\tilde{E}^{\mathrm{s}}_g$ and $\tilde{E}^{\mathrm{u}}_g$ defined on $\mathcal{U}(\Lambda)$ and decompose

$$\mathrm{T}_x\mathcal{M} = \tilde{E}^{\mathrm{s}}_g(x) \oplus \tilde{E}^{\mathrm{u}}_g(x), \qquad v = v^{\mathrm{s}} + v^{\mathrm{u}}.$$

Next, consider $\alpha > 0$ to be chosen below, and define the **stable/unstable cones** by

$$\begin{aligned} K^{\mathrm{s}}_\alpha(x) &= \{v \in \mathrm{T}_x\mathcal{M} : |v^{\mathrm{u}}| \leq \alpha |v^{\mathrm{s}}|\}, \\ K^{\mathrm{u}}_\alpha(x) &= \{v \in \mathrm{T}_x\mathcal{M} : |v^{\mathrm{s}}| \leq \alpha |v^{\mathrm{u}}|\}. \end{aligned}$$

By hyperbolicity we see that there exists $\alpha > 0$ such that $\mathrm{D}g$ is contracting on $K^{\mathrm{s}}_\alpha(x)$ and expanding on $K^{\mathrm{u}}_\alpha(x)$. Since Λ is compact and the **unit tangent bundle**

$$\mathrm{T}^1\mathcal{M} = \left\{ v \in \mathrm{T}\mathcal{M} = \bigcup_{x\in\mathcal{M}} \mathrm{T}_x\mathcal{M} : |v| = 1 \right\}$$

when restricted to the hyperbolic set is also compact, we have a single $\lambda \in (0,1)$ such that

$$|\mathrm{D}g_x v| \leq \lambda |v| \quad \text{for } v \in K^{\mathrm{s}}_\alpha(x) \quad \text{and} \quad |\mathrm{D}g_x^{-1} v| \leq \lambda|v| \quad \text{for } v \in K^{\mathrm{u}}_\alpha(x)$$

for all $x \in \mathcal{U}(\Lambda)$. The same contraction/expansion statement also holds, upon potentially increasing λ but still $\lambda \in (0,1)$, for f since it is C^1-close to g by (7.2). Now one can define the subspaces

$$\begin{aligned} E^{\mathrm{s}}(x) &= \bigcap_{k\geq 0} (\mathrm{D}f^{-k})_{f^k(x)} K^{\mathrm{s}}_\alpha(f^k(x)), \\ E^{\mathrm{u}}(x) &= \bigcap_{k\geq 0} (\mathrm{D}f^{k})_{f^{-k}(x)} K^{\mathrm{u}}_\alpha(f^{-k}(x)), \end{aligned}$$

which are precisely satisfying the definitions of a hyperbolic set $\mathcal{K}$ for f.

`Step 2:` Proving (T2) is another application of Theorem 7.5. Let $\mathcal{Y} = \Lambda$ and $h = g|_\Lambda$ with the usual inclusion $\varphi : \Lambda \hookrightarrow \mathcal{M}$. Hence, we obtain a map $\psi : \Lambda \to \mathcal{M}$ such that

$$\psi \circ g|_\Lambda = f \circ \psi.$$

Let $\mathcal{K} := \psi(\Lambda)$. Now apply Theorem 7.5 again but to $\mathcal{Y} = \mathcal{K}$, $h = f|_\mathcal{K}$, and $\varphi : \mathcal{K} \hookrightarrow \mathcal{M}$ to obtain a map $\tilde{\psi} : \mathcal{K} \to \mathcal{M}$ such that

$$\tilde{\psi} \circ f|_\mathcal{K} = g|_\Lambda \circ \psi$$

and by uniqueness $\tilde{\psi} = \psi^{-1}$. The set $\mathcal{K}$ is hyperbolic by (T1). □

Corollary 7.10 (Structure of Anosov Diffeomorphisms)**.** *The set of Anosov diffeomorphisms on a given compact manifold $\mathcal{M}$ is open in the C^1-topology. Furthermore, Anosov diffeomorphisms are structurally stable.*

Proof. Openness follows directly from applying (T1) of Theorem 7.9, while structural stability follows from (T2). □

In some sense, we have pushed the concept of (uniform) hyperbolicity as far as one can go with relatively direct arguments. Next, we are going to look at cases where hyperbolicity is lost in parametric families of dynamical systems.

Exercises for Section 7

E7.1 Consider a C^1-map $f : \mathbb{R}^d \to \mathbb{R}^d$ with a hyperbolic fixed point $x_* \in \mathbb{R}^d$ and let $g \in C^1(\mathbb{R}^d, \mathbb{R}^d)$ be arbitrary. Use the implicit function theorem to prove directly (i.e., without using any previous results) that for sufficiently small ε, the perturbed map $f + \varepsilon g$ also has a hyperbolic fixed point with the same stability properties as x_*.

E7.2 Let $\mathcal{X}$ be a metric space and $g : \mathcal{X} \to \mathcal{X}$ a continuous iterated map $x_{k+1} = g(x_k)$. A periodic ε-pseudo-orbit is a finite ε-pseudo-orbit with $x_0 = x_n$ for some $n \in \mathbb{N}$. A point $x \in X$ is called **chain recurrent** if for every $\varepsilon > 0$ there is a periodic ε-pseudo-orbit from x to itself. Show that every non-wandering point is chain recurrent.

8 Local Bifurcations

Consider parameters p and the ODE

$$x' = f(x, p), \qquad x \in \mathbb{R}^d,\ p \in \mathbb{R}^m. \tag{8.1}$$

Definition 8.1. The appearance of a non-equivalent phase portrait under parameter variation is called **bifurcation**.

Example 8.2. Consider $p = 1 = d$ and the ODE

$$x' = px - x^3 = x(p - x^2). \tag{8.2}$$

For $p \leq 0$, $x_* = 0$ is the unique equilibrium point; in this case, it is globally asymptotically stable. For $p > 0$, $x_* = 0$ is unstable and two new equilibria $x_\pm = \pm\sqrt{p}$ appear, which are locally asymptotically stable; see also Figure 19(a). The bifurcation at $(x, p) = (0, 0)$ is called **pitchfork bifurcation**. It is commonly found in *symmetric* systems; note the invariance of the ODE under $x \mapsto -x$. ♦

It makes sense to slightly tweak/weaken the definition of topological equivalence for parameter-dependent systems, i.e., comparing (8.1) to

$$y' = g(y, q), \qquad y \in \mathbb{R}^d,\ q \in \mathbb{R}^m. \tag{8.3}$$

Definition 8.3. The parametrized ODEs (8.1) and (8.3) are called **(fiber) topologically equivalent** if

(D1) there exists a homeomorphism $\xi : \mathbb{R}^m \to \mathbb{R}^m$, $\xi(p) = q$;

(D2) for fixed $\xi(p) = q$, the ODEs are topologically equivalent.

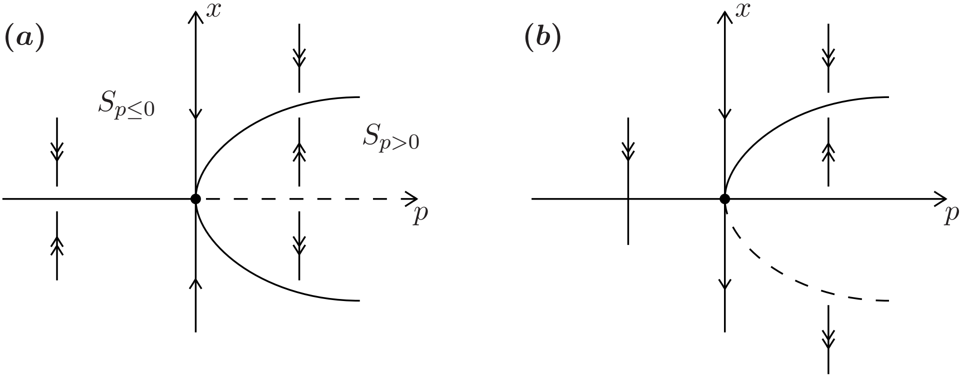

Figure 19: (a) (Supercritical) pitchfork bifurcation. (b) Fold bifurcation. For (a)-(b), unstable invariant sets, here just equilibria, are indicated by dashed lines, which is a common convention.

Fix a parameter p_0 and let $\mathcal{S}_{p_0}$ be the maximally connected subset in parameter space containing p_0 with topologically equivalent phase portraits and call $\mathcal{S}_{p_0}$ a **stratum**; see Figure 19(a).

Definition 8.4. A **bifurcation diagram** is a stratification of the parameter space ($p \in \mathbb{R}^m$) induced by topological equivalence, together with a representative phase portrait for each stratum.

Definition 8.5. An **unfolding** is a parametrized family of dynamical systems containing a bifurcation in a persistent way. The minimal number of parameters for an unfolding is called its **codimension**.

Example 8.6. Consider $m = 1 = d$ and the ODE

$$x' = p - x^2 = f(x,p). \tag{8.4}$$

For $p > 0$, there are two equilibrium points $x_\pm = \pm\sqrt{p}$. One checks by looking at the linearization

$$X' = \mathrm{D}_x f(x_\pm, p)X = -2x_\pm X = AX, \qquad X = X(t),$$

that x_+ is stable and x_- is unstable. At $p = 0$ the two equilibria coincide and disappear for $p < 0$. A single eigenvalue λ of A passes through zero at $p = 0$ and hyperbolicity of the equilibria is lost at the bifurcation point $(x, p) = (0, 0)$. The bifurcation diagram is shown in Figure 19(b). Note carefully that a very similar analysis holds for the ODE

$$x' = p + x^2 = f(x,p). \tag{8.5}$$

In both cases, we have a **fold bifurcation** (or **saddle-node bifurcation**). Furthermore, (8.4) is a codimension-one unfolding of the saddle-node bifurcation since any sufficiently small perturbation (say in C^k, $k \geq 2$) of f still exhibits the same bifurcation. ♦

The next result provides the key insight that we expect only certain "lowest-order terms" to matter at a (local) bifurcation point.

Lemma 8.7. *In a neighborhood of* $(x,p) = (0,0)$*, the ODE*

$$x' = p \pm x^2 + \mathcal{O}(x^3) \tag{8.6}$$

is locally topologically equivalent to

$$x' = p \pm x^2.$$

Remark: Recall the notation $a(x) = \mathcal{O}(b(x))$ as $x \to 0$ means $\lim_{x\to 0} \frac{|a(x)|}{|b(x)|} < \infty$. Furthermore, in (8.6) we also allow certain higher-order dependence in the parameter, e.g., px^3.

Proof. We work in a neighborhood of the origin and only consider the negative sign case $x' = p - x^2$. The other sign case follows from the coordinate transformation $(x,p) \mapsto (-x,-p)$. Re-write (8.6) as

$$y' = g(y,p) = p - y^2 + \psi(y,p), \qquad \psi = \mathcal{O}(y^3). \tag{8.7}$$

The manifold/curve of equilibria can locally be written as

$$\mathcal{M} = \{(y,p) : p = y^2 + G(y)\}, \qquad G(y) = \mathcal{O}(y^3)$$

using the implicit function theorem since $\partial_p g(0,0) = 1$. This implies there exist equilibria $y_\pm(p)$ of (8.7) close to $x_\pm(p) = \pm\sqrt{p}$; see Figure 19(b). To construct the required homeomorphism $h_p : \mathbb{R} \to \mathbb{R}$, define $h_p(x) = x$ for $p \leq 0$ and for $p > 0$ use a linear transformation

$$h_p(x) = a(p) + b(p)x, \qquad h_p(x_\pm(p)) \stackrel{!}{=} y_\pm(p),$$

where a, b can be solved for implicitly using the last two conditions/equations. Therefore, we have the required (fiber) topological equivalence of Definition 8.3. □

Theorem 8.8 (Fold Bifurcation of Flows)**.** *Any smooth one-dimensional, one-parameter ODE*

$$x' = f(x,p), \qquad x \in \mathbb{R},\ p \in \mathbb{R},$$

satisfying the conditions

(A1) $f(0,0) = 0$, $\partial_x f(0,0) = 0$,

(A2) $\partial_{xx} f(0,0) \neq 0$, $\partial_p f(0,0) \neq 0$,

is locally near $(x,p) = (0,0)$ *topologically equivalent to one of the two following ODEs*

$$y' = q \pm y^2, \tag{8.8}$$

which are also called the ***topological normal form*** *of the* ***fold bifurcation****.*

Proof. First, we expand f in a Taylor series near the origin

$$f(x,p) = f_0(p) + f_1(p)x + f_2(p)x^2 + \mathcal{O}(x^3),$$

where $f_0(0) = f(0,0) = 0$ and $f_1(0) = \partial_x f(0,0) = 0$ by (A1). For the geometric idea of the proof see Figure 20.

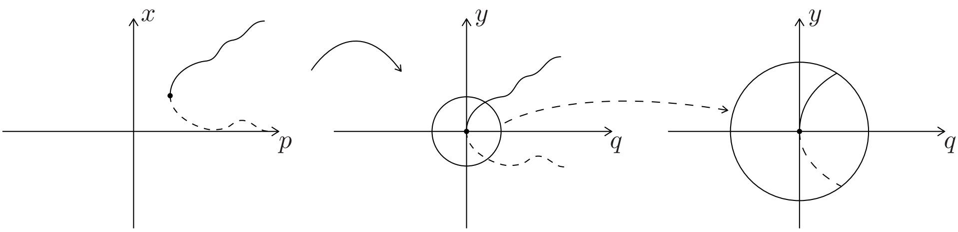

Figure 20: Transformation to normal form for a fold.

`Step 1:` Consider a linear *phase space coordinate change* $y = x + \delta(p) = x + \delta$, which yields

$$\begin{aligned} y' &= x' = f_0(p) + f_1(p)(y-\delta) + f_2(p)(y-\delta)^2 + \mathcal{O}((y-\delta)^3) \\ &= \left[f_0(p) - f_1(p)\delta + f_2(p)\delta^2 + \mathcal{O}(\delta^3)\right] + \left[f_1(p) - 2f_2(p)\delta + \mathcal{O}(\delta^2)\right] y \\ &\quad + \left[f_2(p) + \mathcal{O}(\delta)\right] y^2 + \mathcal{O}(y^3). \end{aligned}$$

The linear term in y can be dealt with requiring

$$F(p,\delta) := \left[f_1(p) - 2f_2(p)\delta + \delta^2\psi(p,\delta)\right] \stackrel{!}{=} 0,$$

where ψ is some smooth function. Note that $F(0,0) = 0$, and $\partial_\delta F(0,0) = -2f_2(0) = -2\partial_{xx} f(0,0) \neq 0$ by (A2). Therefore, the Implicit Function Theorem implies the existence of a unique smooth function $\delta = \delta(p)$ such that $F(p,\delta(p)) = 0$ locally and

$$\delta(p) = -\frac{f_1'(0)}{2f_2(0)}p + \mathcal{O}(p^2).$$

The remaining ODE is now given by

$$y' = \left[f_0'(0)p + \mathcal{O}(p^2)\right] + \left[f_2(0) + \mathcal{O}(p)\right] y^2 + \mathcal{O}(y^3). \tag{8.9}$$

`Step 2:` Consider as a *new parameter* $q = q(p)$

$$q = f_0'(0)p + p^2\mu(p),$$

for some smooth function μ. Since $q(0) = 0$, $q'(0) = f_0'(0) = \partial_p f(0,0) \neq 0$ by (A2), the Inverse Function Theorem yields a unique smooth inverse $p = p(q)$ with $p(0) = 0$. Therefore, (8.9) can now be written as

$$y' = q + b(q)y^2 + \mathcal{O}(y^3), \tag{8.10}$$

where $b(q)$ is smooth and $b(0) = f_2(0) = \partial_{xx} f(0,0) \neq 0$.

`Step 3:` Using a *scaling* $\tilde{y} = |b(q)|y$, $\tilde{q} = |b(q)|q$ in (8.10) and dropping the tildes gives

$$y' = q + sy^2 + \mathcal{O}(y^3), \qquad s = \text{sign}(b(0)) = \pm 1,$$

so that Lemma 8.7 finishes the proof. □

We interpret the conditions (A1)-(A2). In (A1), $f(0,0) = 0$ just says we have an equilibrium at a certain parameter value at a certain point (here $p = 0 = x$ but without loss of generality we may always shift coordinates in parameter and phase space). Hence, the main **bifurcation condition** is

$$\partial_x f(0,0) = 0$$

of a single eigenvalue λ of the linearization located at zero so hyperbolicity is lost. We also say that the fold bifurcation is of **codimension one** as a single parameter suffices to describe the bifurcation diagram; see Section 11 for an example of a codimension two bifurcation. For (A2), the condition

$$\partial_{xx} f(0,0) \neq 0$$

is a **non-degeneracy condition**, which ensures that the next non-trivial Taylor term x^2 does not vanish. The condition

$$\partial_p f(0,0) \neq 0,$$

guarantees (exercise!) that the eigenvalue $\lambda = \lambda(p)$ passes through the origin at non-zero speed, which is a so-called **transversality condition**; see also Figure 21(a) for a transversality condition in another context. In summary, one refers to (A2) as **genericity conditions**.

Example 8.9. Dropping the transversality condition $\partial_p f(0,0) \neq 0$ but keeping the other conditions, then one possibility is that we may have the situation

$$x' = x(p - x), \qquad x \in \mathbb{R},\ p \in \mathbb{R},$$

which has the trivial equilibrium branch $x \equiv 0$ and the equilibrium branch $x_* = p$. We find $x = 0$ is locally asymptotically stable for $p < 0$ and unstable for $p > 0$, while the reverse holds for $x_* = p$. At $p = 0$, a **trans-critical bifurcation** occurs, which nicely shows an example for **exchange-of-stability**, as can be seen from the bifurcation diagram (exercise: draw the diagram!). ♦

From standard linear algebra, it is evident that in a one-parameter family of matrices, we may not only generically encounter the situation that a *single real eigenvalue* λ passes through zero but also the case, when a *complex conjugate pair* $\lambda_{1,2} \in \mathbb{C}$, $\lambda_1 = \overline{\lambda_2}$ passes through the imaginary axis $\mathrm{i}\mathbb{R} \subset \mathbb{C}$; see Figure 21(a).

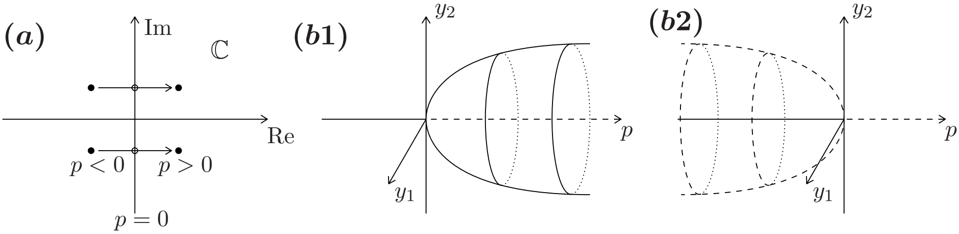

Figure 21: Hopf bifurcation. (a) Pair of transversally crossing eigenvalues. (b1) Supercritical Hopf bifurcation diagram. (b2) Subcritical Hopf bifurcation diagram.

Theorem 8.10 (Hopf Bifurcation)**.** *Any smooth* generic *two-dimensional, one-parameter ODE*

$$x' = f(x,p), \qquad x \in \mathbb{R}^2,\ p \in \mathbb{R},$$

satisfying the conditions $f(0,0) = 0$, $\mathrm{D}_x f(0,p)$ has a complex conjugate pair of eigenvalues

$$\lambda_{1,2}(p) = \mu(p) \pm \mathrm{i}\omega(p)$$

with $\mu(0) = 0$, $\omega(0) > 0$, is locally topologically equivalent to one of the two following ODEs

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} q & -1 \\ 1 & q \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \pm (y_1^2 + y_2^2) \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \tag{8.11}$$

*which are also called the **topological normal form** of the **Hopf bifurcation**.*

The Hopf and fold bifurcations are both of *codimension one*. Regarding the requirement that the setting is generic, the correct transversality condition $\mu'(0) \neq 0$ for the Hopf bifurcation is easy to guess. The non-degeneracy condition is a requirement, which guarantees non-vanishing cubic terms; see Example 8.11. The transformation to normal form is discussed in Section 9.

Example 8.11. We analyze the dynamics of (8.11) denoting the sign of the cubic term by $l_1 \in \mathbb{R} \setminus \{0\}$; it is called the **first Lyapunov coefficient**. The first Lyapunov coefficient turns out to be actually computable from the original vector field only [26] and the non-degeneracy condition for the Hopf bifurcation is $l_1 \neq 0$. Two coordinate changes are useful to analyze the Hopf normal form dynamics. First, consider the **complexification** $z = y_1 + \mathrm{i}y_2 \in \mathbb{C}$ and observe

$$\begin{aligned} z' &= y_1' + \mathrm{i}y_2' = q(y_1 + \mathrm{i}y_2) + \mathrm{i}(y_1 + \mathrm{i}y_2) + l_1(y_1 + \mathrm{i}y_2)(y_1^2 + y_2^2) \\ &= (q + \mathrm{i})z + l_1 z|z|^2, \end{aligned}$$

which is the complex Hopf normal form. Using $z = r\mathrm{e}^{\mathrm{i}\varphi}$ and

$$z' = r'\mathrm{e}^{\mathrm{i}\varphi} + \mathrm{i}r\varphi'\mathrm{e}^{\mathrm{i}\varphi} = r\mathrm{e}^{\mathrm{i}\varphi}(q + \mathrm{i} + l_1 r^2)$$

gives us the polar form

$$\begin{aligned} r' &= r(q + l_1 r^2), \\ \varphi' &= 1. \end{aligned} \tag{8.12}$$

From (8.12), we obtain the two bifurcation diagrams in Figure 21(b):

$l_1 < 0$: A **supercritical** Hopf bifurcation occurs with the transition from a stable equilibrium for $q < 0$ to an unstable equilibrium for $q > 0$ and stable periodic orbits for $q > 0$.

$l_1 > 0$: A **subcritical** Hopf bifurcation occurs with the transition from a stable equilibrium for $q < 0$ to an unstable equilibrium for $q > 0$ and unstable periodic orbits for $q < 0$.

For a Hopf bifurcation not in normal form, the first Lyapunov coefficient is computable using derivatives of the vector field up to and including order three; see [40, Thm. 14.5] for the formula. The non-degeneracy condition is then $l_1 \neq 0$. ♦

Next, we briefly consider the codimension one cases for maps

$$x_{k+1} = g(x_k, p), \qquad x_k \in \mathbb{R}^d,\ d \in \{1, 2\},\ p \in \mathbb{R}, \tag{8.13}$$

with fixed points x_* and associated linearized map $X \mapsto \mathrm{D}g(x_*, p)X = A(p)X$. Denote the multipliers of $A = A(p)$ by $\mu = \mu(p) \in \mathbb{R}$ for $d = 1$ and by $\mu_{1,2} = \mu_{1,2}(p) \in \mathbb{C}$ for $d = 2$. The condition for loss of hyperbolicity is now

$$|\mu| = 1 \quad \text{or} \quad |\mu_{1,2}| = 1 \qquad \text{i.e., multipliers lie on the unit circle.}$$

Theorem 8.12 (Fold Bifurcation of Maps)**.** *The topological normal form for a generic **fold bifurcation** occurring when $g(0,0) = 0$ and $\mu = 1$, is given by*

$$y \mapsto q + y \pm y^2, \qquad y \in \mathbb{R},\ q \in \mathbb{R}, \tag{8.14}$$

with genericity conditions $\partial_{xx} g(0,0) \neq 0$ and $\partial_p g(0,0) \neq 0$, i.e., under these conditions, the map (8.13) *is locally topologically conjugate to* (8.14)*.*

The proof is completely analogous to Theorem 8.8 and the analysis of fixed points almost identical to Example 8.6. Hence we next focus on the bifurcation, which has no direct analog for flows.

Theorem 8.13 (Flip Bifurcation)**.** *The topological normal form for a generic **flip bifurcation** (or **period-doubling bifurcation**) occurring when $g(0,0) = 0$ and $\mu = -1$, is given by*

$$y \mapsto -(1+q)y \pm y^3, \qquad y \in \mathbb{R},\ q \in \mathbb{R}, \tag{8.15}$$

with genericity conditions

$$\frac{1}{2}\left(\partial_{xx} g(0,0)\right)^2 + \frac{1}{6}\partial_{xxx} g(0,0) \neq 0 \quad \textit{and} \qquad \partial_{xp} g(0,0) \neq 0,$$

i.e., under these conditions, the map (8.13) *is locally topologically conjugate to* (8.15)*.*

Proof. Since $g(0,0) = 0$ and $\partial_x g(0,0) = -1 \neq 1$, we may apply the implicit function theorem to

$$g(x,p) - x = 0$$

to obtain a locally unique fixed point $x_* = x_*(p)$. Without loss of generality we can assume using a preliminary coordinate transformation that $x \equiv 0$ is a fixed point for $|p|$ sufficiently small. Now Taylor expansion gives

$$g(x,p) = g_1(p)x + g_2(p)x^2 + g_3(p)x^3 + \mathcal{O}(x^4),$$

where $g_1(p) = -(1 + \kappa(p))$ for some smooth function κ with $\kappa(0) = 0$ since $\mu = -1$. Using the transversality assumption

$$\kappa'(0) = \partial_{xp} g(0,0) \neq 0$$

the Inverse Function Theorem implies local invertibility and we may consider a new parameter $q = \kappa(p)$. This yields the map

$$x \mapsto \mu(q)x + a(q)x^2 + b(q)x^3 + \mathcal{O}(x^4), \tag{8.16}$$

where $\mu(q) = -(1+q)$. We consider a smooth coordinate change

$$x = y + \delta y^2, \qquad \delta = \delta(q), \tag{8.17}$$

which is locally invertible with inverse (exercise!) given by

$$y = x - \delta x^2 + 2\delta^2 x^3 + \mathcal{O}(x^4). \tag{8.18}$$

Combining (8.17)-(8.18) with (8.16) yields

$$\begin{aligned} y \mapsto\ & \mu y + (a + \delta\mu - \delta\mu^2)y^2 \\ & +(b + 2\delta a - 2\delta\mu(\delta\mu + a) + 2\delta^2\mu^3)y^3 + \mathcal{O}(y^4). \end{aligned}$$

The quadratic term is removable for sufficiently small q by setting

$$\delta(q) = \frac{a(q)}{\mu^2(q) - \mu(q)} \qquad \mu^2(0) - \mu(0) = 2 \neq 0.$$

This implies we obtain a map

$$y \mapsto \mu y + \left(b + \frac{2a^2}{\mu^2 - \mu}\right) y^3 + \mathcal{O}(y^4) = -(1+q)y + c(q)y^3 + \mathcal{O}(y^4).$$

We can compute the smooth function c at $q = 0$

$$c(0) = a^2(0) + b(0) = \left(\frac{1}{2}\partial_{xx}g(0,0)\right)^2 + \frac{1}{6}\partial_{xxx}g(0,0) \neq 0$$

by the non-degeneracy assumption. This implies we may use the scaling

$$\tilde{y} = y\sqrt{|c(q)|}$$

and dropping the tildes yields the result up to higher-order terms, which can be dropped using the same technique as in Lemma 8.7. □

Example 8.14. We analyze the normal form (8.15) with the positive sign for the cubic term. Fixed points satisfy

$$y = -(1+q)y + y^3 = y(-1 - q + y^2) = f(y,q) \tag{8.19}$$

so $y_* = 0$ is locally for q near 0 the only fixed point. It has multiplier $\mu = -1 - q$ so it is linearly stable for $q < 0$ and linearly unstable for $q > 0$.

The flip bifurcation occurs at $q = 0$. We consider the *second-iterate* map to find periodic orbits

$$\begin{aligned} f^2(y,q) &= f \circ f(y,q) = -(1+q)\left[-(1+q)y + y^3\right] + (-(1+q)y + y^3)^3 \\ &= (1+q)^2 y - \left[(1+q)(2+2q+q^2)\right] y^3 + \mathcal{O}(y^5). \end{aligned}$$

This map $y \mapsto f^2(y,q)$ has the trivial fixed point $y_* = 0$ but also two non-trivial fixed points for $q > 0$ sufficiently small

$$y_\pm = \pm\sqrt{q} + \text{higher-order terms}.$$

In fact, one checks explicitly via (8.19) that the points $y_\pm = \pm\sqrt{q}$ are stable and constitute a periodic orbit of period two; see Figure 22(a). For the normal form coefficient with negative sign see Figure 22(b)-(c). ♦

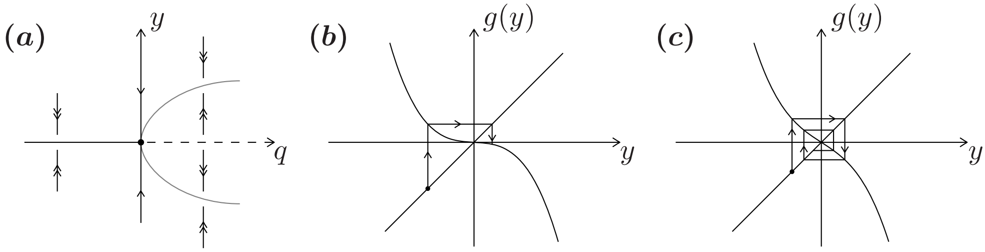

Figure 22: Flip bifurcation. (a) Bifurcation diagram of the normal form (8.15) with the positive sign for the cubic term as analyzed in Example (8.14). (b)-(c) Cobweb of the normal form (8.15) with negative coefficient for the cubic term upon changing q.

A similar, yet technically more involved treatment, can be given for the two-dimensional case

$$x_{k+1} = g(x_k, p) \qquad x_k \in \mathbb{R}^2,\ p \in \mathbb{R},$$

where a complex conjugate pair of multipliers $\mu_{1,2}(q)$ may pass through the unit circle $|\mu_{1,2}(0)| = 1$ (under the assumption $\mu_1 \neq \mu_2$). In this case, the bifurcation is called a **Neimark-Sacker bifurcation**. The Neimark-Sacker bifurcation leads to a rather complicated invariant curve, which generically bifurcates at $q = 0$.

Example 8.15. Consider the **Hénon map**

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_2 \\ p_1 - p_2 x_1 - x_2^2 \end{pmatrix} = g(x,p) \tag{8.20}$$

where $p = (p_1, p_2) \in \mathbb{R}^2$ are parameters. Fixed points x_* lying in $\{x_1 = x_2\}$ can be found solving the quadratic equation $x_2 = p_1 - p_2 x_2 - x_2^2$ so there are at most two fixed points. Local stability can be computed using the Jacobian

$$A(p) = \mathrm{D}g(x_*, p) = \begin{pmatrix} 0 & 1 \\ -p_2 & -2(x_*)_2 \end{pmatrix}.$$

One approach would be to try to calculate the multipliers explicitly and look for the relevant bifurcation conditions. It is helpful to introduce **bifurcation functions**, which encode the conditions compactly. For example, for the fold bifurcation a necessary condition is

$$\det(A(p) - \mathrm{Id}) = 0$$

while for the flip and Neimark-Sacker bifurcations we may consider as necessary conditions

$$\det(A(p) + \mathrm{Id}) = 0 \qquad \text{and} \qquad \det(A(q)) - 1 = 0.$$

Then one may work out (exercise!) the algebraic conditions for a multiplier crossing and try to check, when genericity conditions hold. ♦

In summary, the bifurcation theorems in this section allow us to check explicitly which bifurcations happen for systems in low dimensions with loss of hyperbolicity of steady states. This also justifies rigorously the more direct application of these results to concrete systems that we have emphasized in [40]. Yet, there are two more steps to go: (I) we have to put normal form transformations on a more general algebraic footing and (II) we have to reduce higher-dimensional systems to lower-dimensional ones to be able to apply our bifurcation theorems. We are going to carry out these two steps in the next two sections.

Exercises for Section 8

E8.1 Consider the one-dimensional dynamical system induced by

$$x' = x(p - x) \tag{8.21}$$

where $p \in \mathbb{R}$ is a parameter, see also Example 8.9. (a) Let $\varepsilon > 0$ and perturb the vector field of (8.21) by $\pm\varepsilon$. Analyze the bifurcation behavior of the resulting systems. (b) Show that $x' = x(p - x) + x^3$ exhibits the same bifurcation behavior as (8.21) around the origin.

E8.2 Consider the dynamical system induced by

$$x' = p \ln x + x - 1,$$

where $p \in \mathbb{R}$ is a parameter. Show that for a certain value of p, a transcritical bifurcation occurs. Find new variables y and q that bring the system into the approximate normal form of a transcritical bifurcation

$$y' = y(q - y) + \mathcal{O}(y^3).$$

E8.3 Find all bifurcations of the parametrized system given by

$$x' = -x^5 + x^3 + px, \quad p \in \mathbb{R},$$

and draw its bifurcation diagram.

E8.4 Consider the parametrized map

$$g : \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \quad g(x, p) = x\mathrm{e}^{x+p}$$

(a) Find the fixed points of $g(\cdot, p)$ depending on the parameter $p \in \mathbb{R}$. (b) When are they hyperbolic? Determine the stability of the fixed points in the hyperbolic cases. (c) When the fixed points are non-hyperbolic, bifurcations occur. Classify these bifurcations and draw the bifurcation diagram.

9 Normal Forms

We now discuss a more general complementary approach to the previous techniques to compute a normal form for an ODE $x' = f(x)$, $x \in \mathbb{R}^d$. We assume that f is a sufficiently smooth vector field.

Definition 9.1. Let $\mathcal{H}_k$ denote the space of d-component vector-valued homogeneous polynomials of degree k.

Assume $x = 0$ is an equilibrium and Taylor-expand the ODE

$$x' = Ax + f^{(2)}(x) + f^{(3)}(x) + \cdots, \qquad A = \mathrm{D}f(0), \tag{9.1}$$

where $f^{(k)} \in \mathcal{H}_k$ for $k \geq 2$. Consider a polynomial coordinate change

$$x = y + h^{(k)}(y), \qquad y \in \mathbb{R}^d,\ h^{(k)} \in \mathcal{H}_k,\ k \geq 2 \text{ fixed},$$

so that $y = x - h^{(k)}(x) + \mathcal{O}(|x|^{k+1})$. We compute the transformed ODE

$$\begin{aligned}
y' &= x' - \mathrm{D}_x h^{(k)}(x)x' + \mathcal{O}(|x|^{k+1}) \\
&= \left(\mathrm{Id} - \mathrm{D}_x h^{(k)}(x) + \mathcal{O}(|x|^k)\right)\left(Ax + \sum_{j=2}^{k} f^{(j)}(x) + \mathcal{O}(|x|^{k+1})\right) \\
&= Ay + \sum_{j=2}^{k-1} f^{(j)}(y) + \left(f^{(k)}(y) - \underbrace{\left[\mathrm{D}_y h^{(k)}(y)Ay - Ah^{(k)}(y)\right]}_{=:Lh^{(k)}(y)}\right) + \mathcal{O}(|y|^{k+1}),
\end{aligned}$$

where $L : \mathcal{H}_k \to \mathcal{H}_k$ (exercise!) is a linear operator. The terms of order less than k have remained *invariant*. We may aim to eliminate the order-k terms by solving the **homological equation**

$$Lh^{(k)} = f^{(k)}.$$

Terms in $L(\mathcal{H}_k) = \text{range}(L)$ can be eliminated, denote them by $g^{(k)}$. The remaining terms $r^{(k)} = f^{(k)} - g^{(k)}$ in a complement $\mathcal{R}_k$ to $L(\mathcal{H}_k)$ are called **resonant terms**, i.e., $\mathcal{R}_k$ contains the terms we cannot eliminate by coordinate changes. Observe carefully that the calculations above are exactly along the same strategy we have already encountered concretely in the proofs of the fold and flip bifurcation normal forms in Section 8. The general result is as follows:

Theorem 9.2 (Poincaré Normal Form)**.** *There exists a local polynomial* ***near-identity*** *coordinate change*

$$x = y + h^{(2)}(y) + h^{(3)}(y) + \cdots + h^{(k)}(y) + \mathcal{O}(|y|^{k+1}), \qquad h^{(k)} \in \mathcal{H}_k,$$

which transforms (9.1) *into*

$$y' = Ay + r^{(2)}(y) + r^{(3)}(y) + \cdots + r^{(k)}(y) + \mathcal{O}(|y|^{k+1}), \qquad r^{(k)} \in \mathcal{R}_k,$$

i.e., only resonant terms remain.

Proof. The discussion before the theorem can be applied recursively, starting with $k = 2$, then $k = 3$, and so on. $\square$

Remark: The operator L can also be viewed as an **adjoint** $\text{ad}(Ax)$ induced by Ax via the **Lie bracket** $[\cdot, \cdot]$

$$Lh = \text{ad}(Ax)(h) = [h, Ax] := (\mathrm{D}h)Ax - (\mathrm{D}Ax)h.$$

Lemma 9.3. *The complement $\mathcal{R}_k$ to $L(\mathcal{H}_k)$ is orthogonal in a scalar product $\langle \cdot, \cdot \rangle_k$ if it is chosen as*

$$\ker(L^*) = \{u \in \mathcal{H}_k : L^*u = 0\}$$

where L^* is the adjoint of L, i.e., $\langle Lu, v\rangle_k = \langle u, L^*v\rangle_k$. If there are no generalized eigenvectors for the zero eigenvalue of L we have that

$$\ker(L) = \{u \in \mathcal{H}_k : Lu = 0\}$$

and $\text{range}(L)$ *also spans $\mathcal{H}_k$, so we can also select $\mathcal{R}_k = \ker(L)$ in this case.*

Proof. Recall that the **Fredholm Alternative Theorem** says in our context that *either* $Lh = b$ has a solution $b \in \text{range}(L)$ *or* $L^*h^* = 0$ has a solution $h^* \in \mathcal{H}_k$ with $\langle h^*, b\rangle_k \neq 0$, which proves the first part. For the second part, no generalized eigenvectors imply that $\ker(L) \oplus \text{range}(L) = \mathcal{H}_k$. $\square$

In practice, we identify $\mathcal{H}_k$ using coordinates with some $\mathbb{R}^{N(k)}$ and L with a matrix $M \in \mathbb{R}^{N(k)\times N(k)}$ so that we can use the standard Euclidean inner product.

Theorem 9.4 (Hopf Normal Form)**.** *A smooth planar system* (9.1) *at a Hopf bifurcation point, i.e., the eigenvalues of A are $\pm\mathrm{i}\omega$ with $\omega > 0$, can be locally transformed into the Poincaré normal form*

$$y' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} y + (y_1^2 + y_2^2) \begin{pmatrix} p_1 & -p_2 \\ p_2 & p_1 \end{pmatrix} y + \mathcal{O}(|y|^4). \tag{9.2}$$

Remark: The Poincaré normal form (9.2) is locally topologically equivalent to the normal form (8.11) at $q = 0$. Theorem (9.4) provides the correct leading-order terms, while Theorem 8.10 *unfolds* the full bifurcation and removes the higher-order terms. In practice, one computes the Poincaré normal form first, and then studies the detailed parametric unfolding.

Proof. (of Theorem 9.4) We write $(y_1, y_2) = (x, y)$ throughout the proof for notational simplicity. Consider quadratic terms ($k = 2$), fix an (ordered) polynomial basis $\{x^2, xy, y^2\}$ of $\mathcal{H}_2$. View a vector field $h = h^{(2)}$ as a map $h : \mathbb{R}^2 \to T\mathbb{R}^2 \simeq \mathbb{R}^2$ with basis coordinates ∂_x, ∂_y; for example, write $(2xy, x^2)^\top$ as $(2xy)\partial_x + (x^2)\partial_y$. Without loss of generality (upon scaling time if necessary) we have

$$\begin{aligned} Lh &= \left(\mathrm{D}A\begin{pmatrix} x \\ y \end{pmatrix}\right) h - (\mathrm{D}h)A\begin{pmatrix} x \\ y \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} - \begin{pmatrix} \partial_x h_1 & \partial_y h_1 \\ \partial_x h_2 & \partial_y h_2 \end{pmatrix}\begin{pmatrix} -y \\ x \end{pmatrix}. \end{aligned}$$

Plugging in the basis elements, we find

$$L\begin{pmatrix} x^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2xy \\ x^2 \end{pmatrix}, \quad L\begin{pmatrix} xy \\ 0 \end{pmatrix} = \begin{pmatrix} y^2 - x^2 \\ xy \end{pmatrix}, \quad L\begin{pmatrix} y^2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2xy \\ y^2 \end{pmatrix},$$

for the first component, and for the second component

$$L\begin{pmatrix} 0 \\ x^2 \end{pmatrix} = \begin{pmatrix} -x^2 \\ 2xy \end{pmatrix}, \quad L\begin{pmatrix} 0 \\ xy \end{pmatrix} = \begin{pmatrix} -xy \\ y^2 - x^2 \end{pmatrix}, \quad L\begin{pmatrix} 0 \\ y^2 \end{pmatrix} = \begin{pmatrix} -y^2 \\ -2xy \end{pmatrix}.$$

Of course, it is a lot more convenient to view this in matrix form and identify L with some $M^{(2)} \in \mathbb{R}^{6\times 6}$. Using the coordinate ordering in $\mathbb{R}^6$ by

$$(x^2\partial_x, xy\partial_x, y^2\partial_x, x^2\partial_y, xy\partial_y, y^2\partial_y)$$

we find

$$M^{(2)} = \begin{pmatrix} 0 & -1 & 0 & -1 & 0 & 0 \\ +2 & 0 & -2 & 0 & -1 & 0 \\ 0 & +1 & 0 & 0 & 0 & -1 \\ +1 & 0 & 0 & 0 & -1 & 0 \\ 0 & +1 & 0 & +2 & 0 & -2 \\ 0 & 0 & +1 & 0 & +1 & 0 \end{pmatrix}.$$

Since $\det(M^{(2)}) = -9 \neq 0$, all quadratic terms can be eliminated. The procedure can now be carried out for $k = 3$. We use the ordering of coordinates in $\mathbb{R}^8$

$$(x^3\partial_x, x^2y\partial_x, xy^2\partial_x, y^3\partial_x, x^3\partial_y, x^2y\partial_y, xy^2\partial_y, y^3\partial_y).$$

The matrix $M^{(3)}$ of L for $k = 3$ turns out to be (lengthy exercise!)

$$M^{(3)} = \begin{pmatrix} 0 & -1 & 0 & 0 & -1 & 0 & 0 & 0 \\ +3 & 0 & -2 & 0 & 0 & -1 & 0 & 0 \\ 0 & +2 & 0 & -3 & 0 & 0 & -1 & 0 \\ 0 & 0 & +1 & 0 & 0 & 0 & 0 & -1 \\ +1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & +1 & 0 & 0 & +3 & 0 & -2 & 0 \\ 0 & 0 & +1 & 0 & 0 & +2 & 0 & -3 \\ 0 & 0 & 0 & +1 & 0 & 0 & +1 & 0 \end{pmatrix}.$$

The columns $1, 2, 3, 4, 5, 8$ are linearly independent. Using the first part of Lemma 9.3, the adjoint (or transposed) matrix $(M^{(3)})^\top$ has $\dim(\ker((M^{(3)})^\top)) = 2$ with basis vectors

$$v^* = (3, 0, 1, 0, 0, 1, 0, 3)^\top \qquad \text{and} \qquad w^* = (0, -1, 0, -3, 3, 0, 1, 0)^\top$$

so that $\mathcal{R}_3 = \operatorname{span}(v^*, w^*)$ gives resonant terms

$$r_*^{(3)} = p_1 \begin{pmatrix} 3x^3 + xy^2 \\ x^2y + 3y^3 \end{pmatrix} + p_2 \begin{pmatrix} -x^2y - 3y^3 \\ 3x^3 + xy^2 \end{pmatrix},$$

which does not match (9.2). Using the second part of Lemma 9.3, we find that $\ker(M^{(3)})$ is spanned by

$$v = (1, 0, 1, 0, 0, 1, 0, 1)^\top \qquad \text{and} \qquad w = (0, -1, 0, -1, 1, 0, 1, 0)^\top,$$

which produces the correct resonant terms in (9.2). $\square$

The last proof showed that the Poincaré normal form is *not unique* due to a choice of basis. We also verified the algebra necessary for Theorem 8.10, yet Poincaré normal forms do not yield the first Lyapunov coefficient. The method of normal forms can be immediately applied to parameter-dependent systems written in the form

$$\begin{aligned} x' &= f(x,p), \\ p' &= 0. \end{aligned}$$

Theorem 9.5 (Bogdanov-Takens Bifurcation)**.** *Any generic planar two-parameter system with an equilibrium at $x = 0$ and a linearization having a double-zero eigenvalue $\lambda_{1,2}(0) = 0$ is locally topologically equivalent to one of the following normal forms*

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 \\ q_1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ q_2 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} 0 \\ y_1^2 \pm y_1 y_2 \end{pmatrix}. \tag{9.3}$$

The algebraic structure of (9.3) can be derived from the Poincaré normal form making the ansatz

$$x' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x + f^{(2)}(x) + f^{(3)}(x) + \cdots,$$

which leads already to quite a substantial calculation. Before we discuss the dynamics of (9.3), two issues remain. We have to understand whether we are allowed to drop higher-order terms. This is easy to prove for fold and Hopf bifurcations explicitly (see Section 8), while for (9.3) we sketch the proof in Section 11. The second issue is even more fundamental. We have to reduce the dimension of a given system to make normal form computations and applying bifurcation theorems even tractable.

Exercises for Section 9

E9.1 Show that a system of the form

$$x' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x + \mathcal{O}(|x|^2)$$

can be locally transformed into the Poincaré normal form

$$y' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} y + \begin{pmatrix} 0 \\ p_1 y_1^2 + p_2 y_1 y_2 \end{pmatrix} + \mathcal{O}(|y|^3).$$

E9.2 (a) Consider the dynamical system induced by iteration of a map $g : \mathbb{R}^d \to \mathbb{R}^d$. Transfer the considerations of this section to this setting of a discrete-time dynamical system. Specifically, determine the form of the homological equation. (b) Apply (a) to eliminate as many quadratic terms as possible for the map

$$g: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + (x-y)^2 \\ 3y + x^2 + 2y^2 \end{pmatrix}$$

10 Center Manifolds

So far, we have crucially assumed to work in the *minimal dimension* for the smooth ODE

$$x' = Ax + f^{(2)}(x) + f^{(3)}(x) + \cdots = f(x), \qquad x \in \mathbb{R}^d, \ f \in C^k, \qquad (10.1)$$

where $A = \mathrm{D}f(0)$ and $x_* = 0$ is an equilibrium. Suppose the spectrum of A has a center eigenspace

$$E^{\mathrm{c}}(0) = \mathrm{span}\{v_j : Av_j = \lambda_j v_j, \mathrm{Re}(\lambda_j) = 0\},$$

which is d_{c}-dimensional. In particular, A has

d_{c} eigenvalues λ_{c} with $\mathrm{Re}(\lambda_{\mathrm{c}}) = 0$;

d_{s} eigenvalues λ_{s} with $\mathrm{Re}(\lambda_{\mathrm{s}}) < 0$;

d_{u} eigenvalues λ_{u} with $\mathrm{Re}(\lambda_{\mathrm{u}}) > 0$.

If $d_{\mathrm{c}} = 0$ the Stable/Unstable Manifold Theorem applies; see Figure 23(a) for the idea, how to deal with the additional center directions.

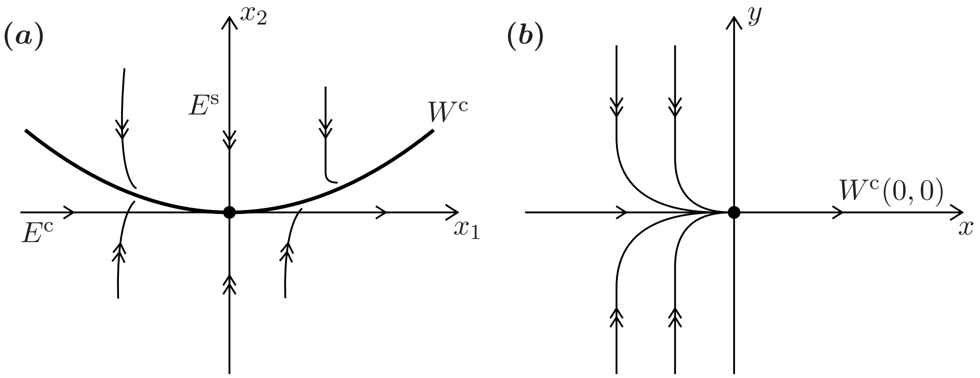

Figure 23: (a) Basic geometric idea of a center manifold W^{c}. (b) Illustration of center manifolds for (10.2) showing lack of uniqueness and smoothness. Note that the sketch is zoom of the situation near $(x, y) = (0, 0)$.

Theorem 10.1 (Center Manifold Theorem)**.** *Consider* (10.1) *with* $k \in \mathbb{N}$, *then there exists an invariant* d_c*-dimensional* C^k*-manifold* $W^{\mathrm{c}}_{\mathrm{loc}}(0)$ *locally tangent to* $E^{\mathrm{c}}(0)$*. Moreover, any trajectory staying inside a neighborhood* $\mathcal{U} = \mathcal{U}(0)$ *for* $t \geq 0$ *(*$t \leq 0$*) converges to* $W^{\mathrm{c}}_{\mathrm{loc}}(0)$ *as* $t \to +\infty$ *(*$t \to -\infty$*).*

Definition 10.2. $W^{\mathrm{c}}_{\mathrm{loc}}(0)$ is called a **center manifold**.

The proof of Theorem 10.1 can be accomplished using similar, yet more subtle, fixed-point arguments as used in the proof of the Stable/Unstable Manifold Theorem 3.8; see shall not carry out the proof here and refer to the literature in Appendix A. We shall often write $W^{\mathrm{c}}_{\mathrm{loc}}(0)$ as $W^{\mathrm{c}}(0)$ or just W^{c}.

Theorem 10.3 (Center Manifold Limitations)**.** *The following holds:*

(T1) Center manifolds need not be unique*.*

(T2) Center manifolds need not be C^ω*, even if* $f \in C^\omega$*, i.e., center manifolds are in general not analytic.*

Proof. Consider the counter-example

$$\begin{aligned} x' &= x^2, \\ y' &= -y, \end{aligned} \tag{10.2}$$

with $(x(0), y(0)) = (x_0, y_0)$. Clearly, $(0,0)$ is an equilibrium and the eigenvalues of the linearization are $\lambda = 0, -1$. The counter-example can be solved explicitly

$$x(t) = \frac{1}{\frac{1}{x_0} - t}, \qquad y(t) = y_0 \mathrm{e}^{-t}.$$

See Figure 23(b) for the phase portrait. We claim $W^{\mathrm{c}}(0)_p = \{(x,y) \in \mathbb{R}^2 : y = h_p(x)\}$ for

$$h_p(x) := \begin{cases} p \exp\left(\frac{1}{x}\right) & \text{for } x < 0, \\ 0 & \text{for } x \geq 0, \end{cases}$$

is a one-parameter family (for $p \in \mathbb{R}$) of center manifolds. Indeed, the tangent space is $\mathrm{T}W^{\mathrm{c}}_p(0) = \{y = 0\} = E^{\mathrm{c}}(0)$. Each $W^{\mathrm{c}}_p(0)$ is invariant since for p fixed, $\{y = 0, x > 0\}$ is a trajectory and for $x < 0$ we may simply solve for t in $x(t)$ above to obtain that

$$y = p\mathrm{e}^{1/x}$$

is also a trajectory for $x < 0$. Hence, $W^{\mathrm{c}}(0)$ is not unique and it is not C^ω for $p \neq 0$. $\square$

Remark: Center manifolds do not even have to be C^∞, even if f is analytic. We leave it as a (non-trivial!) exercise to check that the system

$$x_1' = -x_2x_1 - x_1^3, \quad x_2' = 0, \quad y' = -y + x_1^2$$

is one possible required counterexample.

Separating the critical and non-critical components, we may write (10.1)

$$\begin{aligned} x' &= Bx + F(x,y), \qquad & F = \mathcal{O}(x^2, xy, y^2) \\ y' &= Cy + G(x,y), \qquad & G = \mathcal{O}(x^2, xy, y^2) \end{aligned} \tag{10.3}$$

where $B \in \mathbb{R}^{d_c \times d_c}$ with $\mathrm{spec}(B) \subset \mathrm{i}\mathbb{R}$ and $C \in \mathbb{R}^{(d_s+d_u)\times(d_s+d_u)}$ with $\mathrm{spec}(C) \cap \mathrm{i}\mathbb{R} = \emptyset$.

Theorem 10.4 (Center Manifold Reduction)**.** *The ODE* (10.3) *is locally topologically equivalent near* 0 *to*

$$\begin{aligned} x' &= Bx + F(x, H(x)), \\ y' &= Cy, \end{aligned} \tag{10.4}$$

where $W^{\mathrm{c}}_{\mathrm{loc}}(0) = \{(x,y) \in \mathbb{R}^{d_c} \times \mathbb{R}^{d_s+d_u} : y = H(x)\}$ *with* $H = \mathcal{O}(x^2)$*.*

Remark: Essentially analogous center manifold results hold for maps, where the critical components in $E^{\mathrm{c}}(0)$ are associated to multipliers on the unit circle.

In practice, the most important case is $d_u = 0$, which we assume for the rest of this section. We can only apply normal form and bifurcation theory on W^{c} if we can calculate H, or at least an approximation $h : \mathbb{R}^{d_c} \to \mathbb{R}^{d_s}$. Making the ansatz $y = h(x)$ in (10.3) yields

$$y' = \mathrm{D}h(x)\; x' = \mathrm{D}h(x)\,[Bx + F(x, h(x))] = Ch(x) + G(x, h(x)).$$

Definition 10.5. The partial differential equation (PDE)

$$\mathcal{N}(h(x)) = \mathrm{D}h(x)\,[Bx + F(x, h(x))] - Ch(x) - G(x, h(x)) = 0 \tag{10.5}$$

for h with boundary conditions $h(0) = 0$, $\mathrm{D}h(0) = 0$ is called the **(center manifold) invariance equation**.

Theorem 10.6 (Approximation of Center Manifolds)**.** *If a map h can be found such that $h(0) = 0$, $\mathrm{D}h(0) = 0$, which satisfies*

$$\mathcal{N}(h(x)) = \mathcal{O}(|x|^p) \qquad \text{for some } p > 1 \text{ as } |x| \to 0,$$

i.e., the invariance equation is solved approximately, then

$$H(x) = h(x) + \mathcal{O}(|x|^p) \quad \text{as } |x| \to 0, \tag{10.6}$$

where $W^{\mathrm{c}}_{\mathrm{loc}}(0) = \{y = H(x)\}$ *is the center manifold.*

Proof. (Sketch) The proof is an extension of ideas to prove existence of invariant manifolds, so we only provide the main idea. Consider (10.3), suppose

$$H : \mathbb{R}^{d_c} \to \mathbb{R}^{d_s}, \quad H(0) = 0, \quad H \text{ locally Lipschitz},$$

is the center manifold. Let $x(t, x_0, H)$ denote the solution of

$$x' = Bx + F(x, H(x)), \qquad x(0, x_0, H) = x_0$$

as in (10.4). One now checks (non-trivial) that H can be constructed using

$$KH(x_0) = \int_{-\infty}^{0} \mathrm{e}^{-Cs} G(x(s, x_0, H), H(x(s, x_0, H)))\, \mathrm{d}s \tag{10.7}$$

and looking for a fixed point of the operator K on the space $\mathcal{K}$ of Lipschitz functions with the sup-norm; cf. proof of Theorem 3.8 and note that (10.7) essentially removes all decaying directions so only the center directions can remain. To obtain the approximation h, we study a modified fixed point problem. Consider $\theta \in C_c^1(\mathbb{R}^{d_c}, \mathbb{R}^{d_s})$ with $\theta(x) = h(x)$ for $|x|$ small, and define

$$N(x) := \mathrm{D}\theta[Bx + F(x, \theta(x))] - C\theta(x) - G(x, \theta(x))$$

and observe $N(x) = \mathcal{O}(|x|^p)$ as $|x| \to 0$. We know H is a fixed point of K and define a new operator by

$$Lw = K(w + \theta) - \theta.$$

It can be shown (non-trivial) that L is a contraction mapping on

$$\mathcal{L} := \left\{ w \in \mathcal{K} : |w(x)| \leq c|x|^p \text{ for all } x \in \mathbb{R}^{d_c} \right\}.$$

We get a fixed point w_* such that $w_* = K(w_* + \theta) - \theta$. Since $\theta(x) = h(x)$ locally and K has a fixed point H, the approximation (10.6) follows. $\square$

Hence, substituting a formal power series into the invariance equation leads to a rigorous approximation algorithm for center manifolds.

Example 10.7. Consider the planar ODE

$$\begin{aligned} x' &= xy, \\ y' &= -y + px^2, \end{aligned} \tag{10.8}$$

which has an equilibrium at 0 and is already in the standard form (10.3) with $B \equiv 0$ and $C \equiv -1$. We make a third-order ansatz

$$y = h(x) = a_2 x^2 + a_3 x^3 + \mathcal{O}(|x|^4) \qquad (\text{note: } h(0) = 0 = h'(0))$$

in (10.5) to obtain

$$\begin{aligned}
0 &= h'(x)\,[xh(x)] + h(x) - px^2 \\
&= (2a_2x + 3a_3x^2 + \mathcal{O}(|x|^4))x\left[a_2x^2 + a_3x^3 + \mathcal{O}(|x|^4)\right] \\
&\quad +(a_2 - p)x^2 + a_3x^3 + \mathcal{O}(|x|^4) \\
&= (a_2 - p)x^2 + a_3x^3 + \mathcal{O}(|x|^4).
\end{aligned}$$

So we can select $a_2 = p$, then all quadratic terms vanish. One takes $a_3 = 0$ to eliminate the cubic term to obtain

$$W^{\mathrm{c}}_{\mathrm{loc}}(0) = \{(x,y) \in \mathcal{U} = \mathcal{U}(0) : y = h(x) = px^2 + \mathcal{O}(|x|^4)\}.$$

The reduced dynamical system on the center manifold is

$$x' = px^3 + \mathcal{O}(x^5)$$

and the phase portraits for $p > 0$ and $p < 0$ are shown in Figure 24. ♦

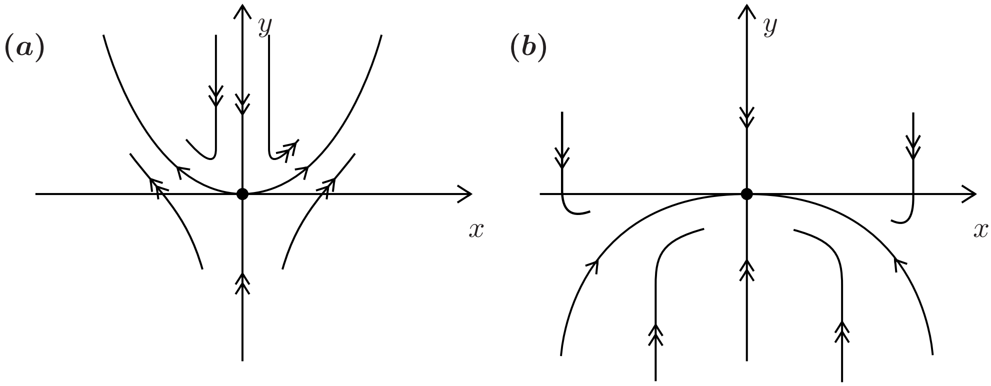

Figure 24: Sketch of the center manifolds for Example 10.7. (a) $p = p_1 > 0$. (b) $p = p_2 < 0$ with $|p_2| < p_1$.

Example 10.8. Consider the planar ODE

$$\begin{aligned}
x' &= y - 1, \\
y' &= -(y-1) + px^2 + x(y-1),
\end{aligned} \tag{10.9}$$

and note that $(0,1)$ is an equilibrium and the linearized problem

$$\begin{pmatrix} X' \\ Y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} X \\ Y \end{pmatrix} =: A\begin{pmatrix} X \\ Y \end{pmatrix}$$

at this point has eigenvalues 0 and -1. However, the system (10.9) is not in standard form. Setting $(x, y) = (\tilde{x}, \tilde{y} + 1)$, inserting into (10.9), and dropping the tildes yields

$$\begin{aligned} x' &= y, \\ y' &= -y + px^2 + xy. \end{aligned} \tag{10.10}$$

Next, we use the eigenvectors of A to diagonalize via the transformation

$$\begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = T^{-1}.$$

Inserting this coordinate change in (10.10), and dropping the tiles, yields the standard form

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ p(x+y)^2 - (x+y)y \end{pmatrix}.$$

Now one may proceed analogously to Example 10.7 to find a center manifold approximation (exercise!). ♦

Exercises for Section 10

E10.1 Consider the dynamical system induced by

$$\begin{aligned} x' &= x^2 - xy, \\ y' &= x - y + x^2 - xy + xz, \\ z' &= -3z + x^2, \end{aligned}$$

around the equilibrium point at the origin. (a) Find a linear coordinate transformation that brings the system into the form (10.3) with separated center and stable/unstable components. (b) Find an approximation of the center manifold up to cubic order. (c) Determine the stability of the equilibrium at the origin.

E10.2 Consider the following system of ordinary differential equations

$$\begin{aligned} x' &= -y + xz + 2y^4, \\ y' &= x + yz + x^2y^2, \\ z' &= -z - (x^2 + y^2) + z^2. \end{aligned}$$

(a) Find a quadratic approximation of the center manifold of the equilibrium point at the origin. (b) Use the result from (a) to determine the stability of the equilibrium and sketch the phase portrait near the origin.

E10.3 In analogy to equation (10.5), find the center manifold invariance equation for a discrete-time dynamical system induced by iteration of some map g.

E10.4 Use Exercise E10.3 to determine the stability of the origin for the map

$$g(x,y) = \begin{pmatrix} x+xy \\ \frac{1}{2}y - x^2 - xy \end{pmatrix}.$$

11 Higher Codimension

We return to our goal to unfold the Bogdanov-Takens (double-zero eigenvalue) bifurcation. At the bifurcation point $q=0$ the normal form (9.3) is

$$\begin{aligned} y_1' &= y_2, \\ y_2' &= y_1^2 \pm y_1 y_2, \end{aligned} \tag{11.1}$$

where we have dropped higher-order terms. Do third-order Taylor terms matter? Unfortunately, the situation is not as simple as for the fold in Lemma 8.7 and it helps to consider a more abstract viewpoint using the following definition.

Definition 11.1. Let $F \in C^l(\mathbb{R}^d, \mathbb{R}^d)$, $F(0)=0$, and consider $k \leq l$, $k \in \mathbb{N}$. Then the k**-jet** $J_kF(y)$ is the Taylor polynomial of F up to and including order k.

Therefore, our question is, whether the 2-jet $(y_2, y_1^2 \pm y_1 y_2)$ **determines** the topological type of the vector field, i.e., whether including higher-order jets yield vector fields locally topologically not equivalent to the 2-jet. The following example illustrates the technique we need to prove the 2-jet determinacy of (11.1).

Example 11.2. Consider the planar ODE

$$\begin{aligned} y_1' &= y_1^2 - 2y_1y_2 =: F_1(y_1,y_2), \\ y_2' &= y_2^2 - 2y_1y_2 =: F_2(y_1,y_2). \end{aligned} \tag{11.2}$$

where $0=(0,0)$ is a non-hyperbolic equilibrium point and $F=(F_1,F_2)^\top$ is its own 2-jet. Introduce polar coordinates

$$\Phi : \mathbb{S}^1 \times [0,\infty) \to \mathbb{R}^2, \qquad \Phi(\theta,r) = (r\cos\theta, r\sin\theta),$$

and then one computes the transformed vector field

$$\begin{aligned}\hat{F}(\theta,r) &= (\mathrm{D}\Phi^{-1}_{(\theta,r)} \circ F \circ \Phi)(\theta,r)\\ &= \begin{pmatrix} -\frac{\sin\theta}{r} & \frac{\cos\theta}{r}\\ \cos\theta & \sin\theta \end{pmatrix} \begin{pmatrix} r^2(\cos^2\theta - 2\cos\theta\sin\theta)\\ r^2(\sin^2\theta - 2\cos\theta\sin\theta)\end{pmatrix}\\ &= \begin{pmatrix} 3r\cos\theta\sin\theta(\sin\theta-\cos\theta)\\ \frac{1}{4}r^2(\cos\theta + 3\cos(3\theta) + \sin\theta - 3\sin(3\theta))\end{pmatrix}.\end{aligned}$$

Note that $\hat{F}$ has a circle of equilibrium points $\mathbb{S}^1 \times \{r = 0\}$. Since Φ is a diffeomorphism outside of 0, it follows that $\hat{F}$ on $\mathbb{S}^1 \times (0,\infty)$ and F on $\mathbb{R}^2 - \{0\}$ are topologically conjugate and the same holds for

$$\bar{F}(\theta,r) = \frac{1}{r}\hat{F}(\theta,r)$$

as division by any power of r just corresponds to a re-parametrization of time on orbits on the punctured plane (non-trivial exercise!). Yet, one checks easily that $\mathbb{S}^1 \times \{r = 0\}$ now contains six equilibria as shown in Figure 25.

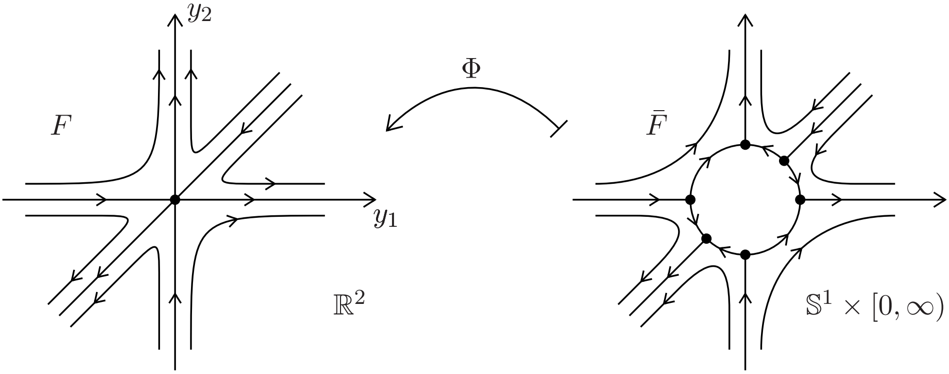

Figure 25: Polar blow-up at (0, 0) of equation (11.2).

All six equilibria are hyperbolic. The Hartman-Grobman Theorem 4.5 implies that near each of these equilibria, the 1-jet determines the vector field $\bar{F}$. Hence, inverting Φ on the punctured plane implies that F is determined by its 2-jet. ♦

Definition 11.3. The **blow-up** (or **geometric desingularization**) of a vector field $F : \mathbb{R}^d \to \mathbb{R}^d$ is a map

$$\Phi : \mathbb{S}^{d-1} \times [0,\infty) \to \mathbb{R}^d,$$

with associated l-times desingularized **blown-up vector field**

$$\bar{F}(\theta,r) = \frac{1}{r^l}(\mathrm{D}\Phi^{-1}_{(\theta,r)} \circ F \circ \Phi)(\theta,r).$$

for some $l \in \mathbb{N}$. Φ^{-1} is also known as the **blow-down map**.

Theorem 11.4. *The two-jet in* (11.1) *is determinate, i.e., k-jets with $k \geq 3$ can be dropped in any unfolding of the Bogdanov-Takens bifurcation.*

The proof of Theorem 11.4 requires three iterated blow-ups and long calculations, so we omit it here. However, the idea is precisely the same as in Example 11.2. The normal form (9.3) is a possible unfolding so it remains to analyze

$$\begin{aligned} y_1' &= y_2, \\ y_2' &= q_1 + q_2 y_1 + y_1^2 - y_1 y_2, \end{aligned} \tag{11.3}$$

where we picked the negative sign in the last term; the positive sign case works analogously.

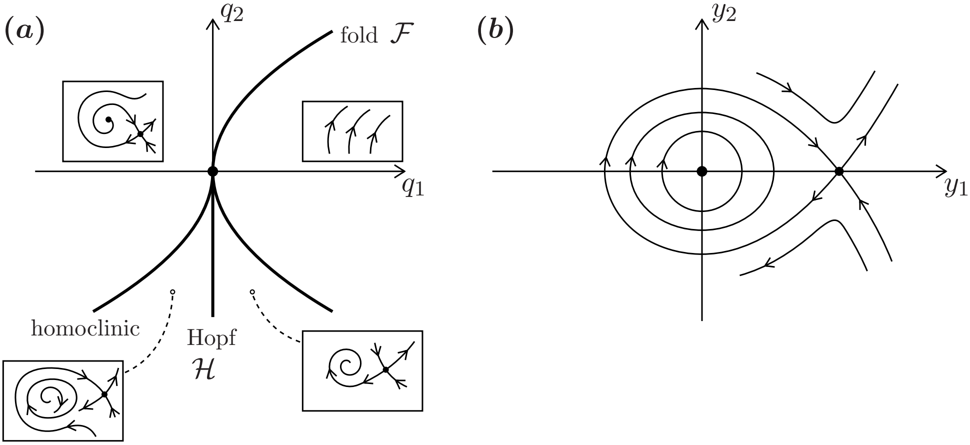

Figure 26: (a) Bifurcation diagram of the Bogdanov-Takens bifurcation normal form (11.3); for the fold and Hopf see Proposition 11.5 and for an explanation regarding homoclinic bifurcations see Section 12. (b) Phase portrait of the Hamiltonian system (11.7).

Proposition 11.5. *The ODE* (11.3) *has a curve of fold bifurcations given by*

$$\mathcal{F} = \{q \in \mathbb{R}^2 : 4q_1 - q_2^2 = 0\}, \tag{11.4}$$

and a curve of supercritical Hopf bifurcations

$$\mathcal{H} = \{q \in \mathbb{R}^2 : q_1 = 0, q_2 < 0\}$$

as shown in Figure 26(a).

Proof. Equilibria $y_*^\pm$ of (11.3) satisfy $y_2 = 0$ and $0 = q_1 + q_2 y_1 + y_1^2$ so

$$y_*^\pm = \left(-\frac{q_2 \pm \sqrt{q_2^2 - 4q_1}}{2}, 0\right) =: \left(-\frac{q_2 \pm \nu}{2}, 0\right),$$

which immediately yields the fold curve (11.4) upon inspection. One checks that crossing $\mathcal{F} \cap \{q_2 > 0\}$ generates an unstable node y_*^- and a saddle y_*^+. Crossing $\mathcal{F} \cap \{q_2 < 0\}$ yields a stable node y_*^- and a saddle y_*^+; note that this also explains the name saddle-node bifurcation as a synonym for the fold bifurcation. It is straightforward to check (exercise!) that y_*^- undergoes a Hopf bifurcation on $\mathcal{H}$ but is only a neutral saddle for $\{q \in \mathbb{R}^2 : q_1 = 0, q_2 > 0\}$. $\square$

Just moving counter-clockwise in the bifurcation diagram starting at the neutral saddle line $\{q_1 = 0, q_2 > 0\}$, we get the phase portraits in Figure 26(a) easily via Proposition 11.5 except that between $\mathcal{H}$ and $\mathcal{F} \cap \{q_2 > 0\}$ we must somehow destroy the periodic orbit generated in the Hopf bifurcation without any local bifurcations. Let us, formally, motivate what we expect. For $\nu > 0$, we translate y_*^- to the origin via $(\tilde{y}_1, \tilde{y}_2) = (y_1 - (y_*^-)_1, y_2)$. Upon dropping the tildes, this yields the ODE

$$\begin{aligned} y_1' &= y_2, \\ y_2' &= y_1(y_1 - \nu) - ((y_*^-)_1 y_2 + y_1 y_2). \end{aligned} \tag{11.5}$$

Near the bifurcation point, we have to deal with small ν so we use a **scaling**

$$Y_1 = \nu^{-1} y_1, \quad Y_2 = \nu^{-3/2} y_2, \quad t = \nu^{-1/2} T, \quad \alpha_1 := (y_*^-)_1 \nu^{-1/2}, \quad \alpha_2 := \nu^{1/2}.$$

This transformation is essentially another blow-up since it provides a zoom for ν small near the equilibrium at the origin and gives

$$\begin{aligned} Y_1' &= Y_2, \\ Y_2' &= Y_1(Y_1 - 1) - (\alpha_1 Y_2 + \alpha_2 Y_1 Y_2). \end{aligned} \tag{11.6}$$

In (11.5) the distance between the two equilibria is ν while it is precisely 1 in (11.6). Let us assume we are interested in $\alpha_{1,2} = 0$, then (11.6) becomes a Hamiltonian system (cf. Example 2.9)

$$\begin{aligned} Y_1' &= Y_2 &&= \tfrac{\partial H}{\partial Y_2}, \\ Y_2' &= Y_1(Y_1 - 1) &&= -\tfrac{\partial H}{\partial Y_1}, \end{aligned} \qquad H(Y) = \frac{Y_2^2}{2} - \frac{Y_1^2}{2} + \frac{Y_1^3}{3}. \tag{11.7}$$

The phase portrait is shown in Figure 26(b). In particular, one observes that the level set $\{H(Y) = 1/6\}$ contains an orbit asymptotic in forward and backward time to the saddle at $Y_*^+ = (1, 0)$. Like the saddle-to-saddle connections we encountered in Section 6, this situation is not structurally stable for fixed parameters but it may occur in a structurally stable way in parametrized families. This motivates us to look at certain global bifurcations.

Exercises for Section 11

E11.1 Consider the planar vector field

$$\begin{aligned} y_1' &= y_2 =: F_1(y_1,y_2),\\ y_2' &= y_1^2 + y_1y_2 =: F_2(y_1,y_2). \end{aligned} \tag{11.8}$$

In analogy to Example 11.2, calculate the blow-up via polar coordinates and show that one blow-up cannot remove the non-hyperbolicity at the origin. (Remark: In fact, one needs several successive blow-ups to do this, which turns out to be algebraically quite complex).

E11.2 Consider an ODE $x' = f(x,p)$ where $p \in \mathbb{R}^2$ are two parameters such that $f(0) = 0$ and $D_xf(0)$ has a pair of purely imaginary eigenvalues and one zero real eigenvalue. Determine a Poincaré normal form for this situation, which is also called a codimension-two **fold-Hopf bifurcation**. *Hint: This exercise is algebraically quite challenging and you might want to do the calculations with the help of a computer algebra system.*

12 Global Bifurcations

Definition 12.1. Let $(\mathcal{X},\mathcal{T},\phi_t)$ for $\mathcal{T} = \mathbb{R}$ or $\mathcal{T} = \mathbb{Z}$ be a dynamical system with steady states x_*, y_* with $x_* \neq y_*$.

(D1) An orbit $\gamma = \gamma(t)$ is called a **heteroclinic orbit** from x_* to y_* if

$$\lim_{t\to-\infty}\gamma(t) = x_* \qquad \text{and} \qquad \lim_{t\to+\infty}\gamma(t) = y_*.$$

(D2) An orbit $\gamma = \gamma(t)$ is called a **homoclinic orbit** to x_* if

$$\lim_{t\to\pm\infty}\gamma(t) = x_*.$$

Examples of heteroclinic/homoclinic orbits and the role of the stable/unstable manifolds are shown in Figure 27(a)-(b). Recall that by Peixoto's Theorem 6.6, homoclinic/heteroclinic orbits to hyperbolic equilibria are structurally unstable in $\mathbb{R}^2$ although they can be structurally stable in $\mathbb{R}^3$, e.g., consider a transversal intersection of a two-dimensional stable and a two-dimensional unstable manifold. Yet, in the following we shall be interested in the bifurcations that occur upon breaking homoclinic orbits under parameter variation for

$$x' = f(x,p), \qquad x \in \mathbb{R}^d,\ p \in \mathbb{R},\ d \in \{2,3\}, \tag{12.1}$$

where f is smooth, $f(0,p) = 0$, and $x_* = 0$ is a hyperbolic saddle point.

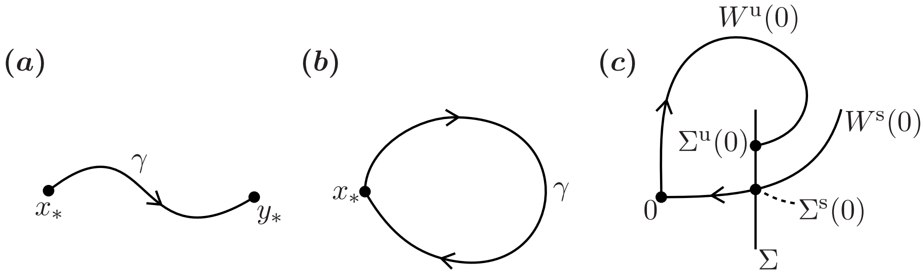

Figure 27: (a) Heteroclinic orbit. (b) Homoclinic orbit. (c) Sketch for stable/unstable manifolds of a broken homoclinic and for the construction of the split function.

Definition 12.2. Suppose (12.1) has a homoclinic orbit to $x_* = 0$ at $p = 0$. Let Σ be a transversal rectangular cross-section to $W^{\mathrm{s}}(0)$ in a sufficiently small neighborhood $\mathcal{U} = \mathcal{U}(0)$. Furthermore, we define the sets

$$\Sigma^{\mathrm{u}} := W^{\mathrm{u}}(0) \cap \Sigma \qquad \text{and} \qquad \Sigma^{\mathrm{s}} := W^{\mathrm{s}}(0) \cap \Sigma.$$

Let q be a coordinate on Σ with axis $\mathcal{Q} \subset \Sigma$ such that $\{q = 0\} = \Sigma^{\mathrm{s}}$. Then a smooth function

$$\xi(p) : \mathbb{R} \to \mathbb{R}, \qquad \xi(p) = q \text{ for } q \in \Sigma^{\mathrm{u}} \cap \mathcal{Q},$$

with $\xi(0) = 0$ is called a **split function**, or **split functional**.

Remark: There is a choice of direction for the coordinate axis for q but the following results will be independent of this choice, so there is no real need to fix a direction. If one wanted to fix a direction, a simple case in $\mathbb{R}^2$ is to use the outer unit normal direction to the closed homoclinic curve.

In particular, the split function measures a signed distance of the unstable manifold on Σ to the stable manifold; see Figure 27(c). The split function is one example of a **bifurcation function**; cf. Example (8.15).

Theorem 12.3 (Andronov-Leontovich Theorem)**.** *Consider the ODE* (12.1) *for $d = 2$ and let $\lambda_{1,2}(p)$ be the eigenvalues of $x_* = 0$ with $\lambda_1(0) < 0 < \lambda_2(0)$. Let $\gamma_0(t)$ be a homoclinic orbit to* 0*. Assume the following genericity conditions hold:*

(A1) ("non-resonance") We have $\sigma(0) := \lambda_1(0) + \lambda_2(0) \neq 0$.

(A2) ("breaking") The split function $\xi(p) = q$ satisfies $\xi'(0) \neq 0$.

Then a unique limit cycle Γ_q, lying in a neighborhood of $\gamma_0 \cup \{0\}$, bifurcates from γ_0 upon variation of p. Furthermore, the cycle is stable and exists for $\sigma(0) < 0$, while the cycle is unstable for $\sigma(0) > 0$; see Figure 28.

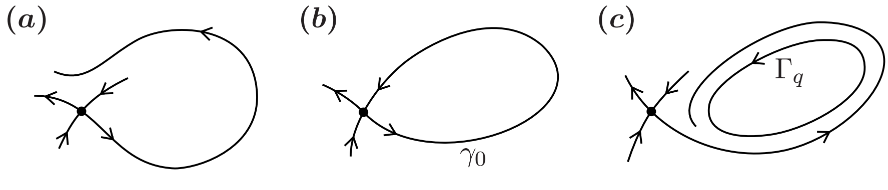

Figure 28: Illustration of Theorem 12.3 for $\sigma(0) < 0$. (a)-(c) correspond to a variation of p through the bifurcation point at $p = 0$ shown in (b).

Remark: As a particular application, Theorem 12.3 explains, what happens on the homoclinic bifurcation curve in Figure 26(a) for the Bogdanov-Takens bifurcation; calculating the curve precisely in parameter space is possible, yet very difficult.

Definition 12.4. We call $\sigma(p) := \lambda_1(p) + \lambda_2(p)$ the **saddle quantity**.

Proof. (of Theorem 12.3) `Step 1:` We drop the parameter-dependence for now for notational simplicity. We aim to *simplify* the system locally near $x_* = 0$ in a neighborhood $\mathcal{V} = \mathcal{V}(0)$. Using a linear coordinate change, we may write (12.1) without loss of generality as

$$\begin{aligned} x_1' &= \lambda_1 x_1 + f_1(x_1, x_2), \\ x_2' &= \lambda_2 x_2 + f_2(x_1, x_2). \end{aligned} \tag{12.2}$$

By the Stable/Unstable Manifold Theorem 3.8, we have

$$W^{\mathrm{s}}(0) = \{x \in \mathcal{V} : x_2 = S(x_1)\}, \qquad S(0) = 0 = S'(0), \tag{12.3}$$
$$W^{\mathrm{u}}(0) = \{x \in \mathcal{V} : x_1 = U(x_2)\}, \qquad U(0) = 0 = U'(0), \tag{12.4}$$

so that we may *rectify* the manifolds using the coordinate change

$$(y_1, y_2) = (x_1 - U(x_2), x_2 - S(x_1))$$

in (12.2); see Figure 29. Upon scaling coordinates, we obtain that in the standard rectangle $\Omega = [-1, 1] \times [-1, 1]$, the dynamics is given by

$$\begin{aligned} y_1' &= \lambda_1 y_1 + y_1 g_1(y_1, y_2), \\ y_2' &= \lambda_2 y_2 + y_2 g_2(y_1, y_2). \end{aligned} \tag{12.5}$$

where $g = (g_1, g_2) = \mathcal{O}(|y|)$.

`Step 2:` We claim that there exists a C^1-*linearization* near the saddle so that via a diffeomorphic coordinate change

$$\Phi : \Omega \to \Omega, \qquad \Phi(y_1, y_2) = (\Phi_1(y), \Phi_2(y)) =: (\xi, \zeta), \tag{12.6}$$

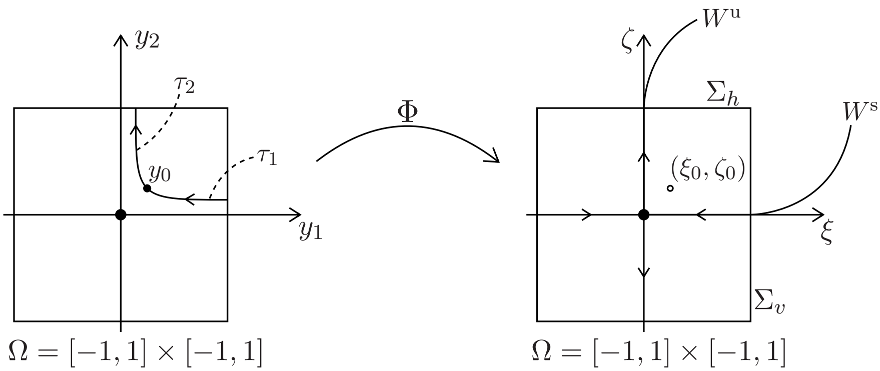

Figure 29: Local construction near the saddle to prove Theorem 12.3.

one may transform (12.5) into the form

$$\begin{aligned} \xi' &= \lambda_1 \xi, \\ \zeta' &= \lambda_2 \zeta. \end{aligned} \tag{12.7}$$

To define Φ, fix $y_0 \in \Omega$ in the first quadrant; the other quadrants work similarly. Let τ_1 and τ_2 be the absolute values of the first exit times of Ω of $y(t)$ through the right and top boundaries starting at time $t = 0$; see Figure 29. Consider the same construction for (12.7) and obtain Φ by mapping y_0 to (ξ_0, ζ_0) having the same pair of exit times; see Figure 29. Set $\Phi(0) = 0$. Φ leaves $\{y_1 = 0\} \cup \{y_2 = 0\}$ invariant. One checks (exercise!) that it is a homeomorphism, which is C^1 and invertible away from $y = 0$. We claim

$$\mathrm{D}\Phi(0) = \mathrm{Id} \tag{12.8}$$

so that Φ is indeed a diffeomorphism. This claim is difficult to check. We start by considering the linear system (12.7), which can be solved explicitly

$$\xi(t) = \xi(t_0)\mathrm{e}^{\lambda_1(t-t_0)} \qquad \text{and} \qquad \zeta(t) = \zeta(t_0)\mathrm{e}^{\lambda_2(t-t_0)}.$$

Setting $t_0 = 0$ and $\xi(0) = \xi = \Phi_1(y)$, $\zeta(0) = \zeta = \Phi_2(y)$ we must have $1 = \xi(-\tau_1) = \xi\mathrm{e}^{-\lambda_1\tau_1}$, which yields $\xi = \mathrm{e}^{\lambda_1\tau_1}$. Therefore, we differentiate $\xi = \Phi_1(y)$ to obtain

$$\mathrm{D}\Phi_1(y) = \mathrm{D}(\mathrm{e}^{\lambda_1\tau_1}) = \lambda_1\mathrm{e}^{\lambda_1\tau_1}\mathrm{D}\tau_1. \tag{12.9}$$

The solution of the nonlinear system (12.5) can be written as a mild/integral solution, inserting $-\tau_1$ gives

$$\begin{aligned} 1 \;&=\; y_1(-\tau_1) = y_1\mathrm{e}^{-\lambda_1\tau_1} + \int_0^{-\tau_1} \mathrm{e}^{\lambda_1(-\tau_1-s)} y_1(s) g_1(y(s))\ \mathrm{d}s \\ &=: \mathrm{e}^{-\lambda_1\tau_1}\left[y_1 + R(\tau_1)\right] \end{aligned}$$

Differentiating $y_1 = \mathrm{e}^{\lambda_1\tau_1} - R(\tau_1)$ gives

$$(1,0) = \lambda_1 \mathrm{e}^{\lambda_1\tau_1}\mathrm{D}\tau_1 - R'\mathrm{D}\tau_1 = \lambda_1\mathrm{e}^{\lambda_1\tau_1}\left[1 - \frac{1}{\lambda_1}\mathrm{e}^{-\lambda_1\tau_1}R'\right]\mathrm{D}\tau_1.$$

Solving for $\mathrm{D}\tau_1$, and using (12.9) leads to

$$\mathrm{D}\Phi_1(y) = \frac{1}{\left[1 - \frac{1}{\lambda_1}\mathrm{e}^{-\lambda_1\tau_1}R'\right]}(1,0). \tag{12.10}$$

By using the Leibniz integral rule we get

$$\mathrm{e}^{-\lambda_1\tau_1}R' = \mathrm{e}^{-\lambda_1\tau_1}\frac{\mathrm{d}}{\mathrm{d}\tau_1}\int_0^{-\tau_1}\mathrm{e}^{-\lambda_1 s}y_1(s)g_1(y(s))\ \mathrm{d}s = y_1(-\tau_1)g_1(y(-\tau_1))$$

As $y \to 0$, the integral solution of the nonlinear system shows that we must have $y_1 g_1(y) \to 0$ as the term is higher-order using the bound $g = \mathcal{O}(|y|)$ so $\mathrm{e}^{-\lambda_1\tau_1}R' \to 0$. Now we may conclude that $\mathrm{D}\Phi_1(y) \to (1,0)$ uniformly as $y \to 0$. Similarly, one may proceed for the derivative $\mathrm{D}\Phi_2$ to obtain (12.8). The local coordinate change induced by Φ can be extended to a global smooth coordinate change.

`Step 3:` We construct a *composition of maps*. Define two sections

$$\Sigma_v := \{(\xi,\zeta) \in \mathbb{R}^2 : \zeta \in [0,1], \xi = 1\}, \quad \Sigma_h := \{(\xi,\zeta) \in \mathbb{R}^2 : \xi \in [0,1], \zeta = 1\},$$

and first consider the local map $L : \Sigma_v \to \Sigma_h$ induced by the flow of (12.7), which can be explicitly calculated for $\zeta \in (0,1]$ as

$$\xi = L(\zeta) = \zeta^{-\frac{\lambda_1}{\lambda_2}}, \qquad L(0) := 0.$$

The global map $G : \Sigma_h \to \Sigma_v$ induced by the flow is smooth and has the general form

$$\zeta = G(\xi) = q + a\xi + \mathcal{O}(\xi^2),$$

where q is the value of the split function and $a > 0$ since orbits cannot intersect; note that we discard all points not hitting Σ_v as we are interested in invariant sets in the positive quadrant. In summary, we have constructed a Poincaré map

$$P(\zeta) := G \circ L(\zeta) = q + a\zeta^{-\frac{\lambda_1}{\lambda_2}} + \cdots .$$

For $|q|$ small, the fixed points of this map can be analyzed as shown in Figure 30 and this yields the claimed result. □

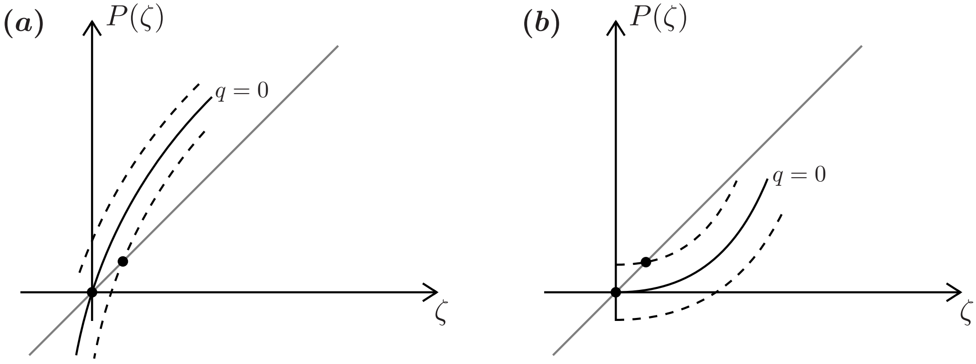

Figure 30: Fixed points of the map $P(\zeta)$ upon varying q through $q = 0$ (solid black curve); the diagonal is shown in grey. (a) An unstable nonzero fixed point (periodic orbit) occurs. (b) A non-zero stable fixed point (periodic orbit) occurs.

The C^1-linearization, which we investigated "by-hand" in Step 2, actually works in $\mathbb{R}^2$ without any nonresonance conditions. However, for general dimensions it is a special case of a very general class of theorems, which guarantees C^k-conjugacy in the absence of resonances. We state one result for maps but similar results hold for flows.

Theorem 12.5 (Sternberg Linearization)**.** *Suppose $g \in C^\infty(\mathbb{R}^d, \mathbb{R}^d)$ is a map with fixed point x_* and $\mathrm{D}g(x_*) = diag(\lambda)$ with eigenvalues $\{\lambda_j\}_{j=1}^d$ and the **nonresonance condition***

$$\lambda_j \neq \lambda^k \qquad \text{for all multiindices } k \in (\mathbb{N}_0)^d,\ |k| \geq 2,$$

is satisfied, then g is locally C^∞-conjugate to its linear part.

Theorems analogous to the Andronov-Leontovich Theorem also hold for saddle points *with real eigenvalues* for $d > 2$, e.g., by using center-manifold-type arguments along the entire homoclinic orbit. However, saddles with *complex eigenvalues*, e.g., a saddle-focus in $\mathbb{R}^3$, can be fundamentally different!

Theorem 12.6 (Shilnikov Homoclinic Orbit)**.** *Consider the ODE* (12.1) *for $d = 3$ with a saddle-focus $x_* = 0$ with $\mathrm{Re}(\lambda_{2,3}(0)) < 0 < \lambda_1(0)$. Let $\gamma_0(t)$ be a homoclinic orbit to 0. Assume the following genericity conditions hold:*

(A1) ("Shilnikov condition") $\sigma(0) := \lambda_1(0) + \mathrm{Re}(\lambda_{2,3}(0)) > 0$.

(A2) ("nonresonance") $\lambda_2(0) \neq \lambda_3(0)$.

Then for all $|p|$ sufficiently small there exist infinitely many saddle limit cycles in a neighborhood of $\gamma_0 \cup \{0\}$.

Proof. We only construct the geometry of the Poincaré map. The final result is going to follow from results in Section 13. Without loss of generality we may assume in a neighborhood $\mathcal{U} = \mathcal{U}(0)$ that

$$W^{\mathrm{s}}_{\mathrm{loc}}(0) = \{x \in \mathcal{U}(0) : x_1 = 0\},$$
$$W^{\mathrm{u}}_{\mathrm{loc}}(0) = \{x \in \mathcal{U}(0) : x_2 = 0, x_3 = 0\}.$$

Introducing two-dimensional cross-sections $\Sigma_v = \{x \in \mathcal{U} : x_1 = c_v\}$ and $\Sigma_h = \{x \in \mathcal{U} : x_2 = c_h\}$ for constants $c_v, c_h > 0$ sufficiently small, we may consider the Poincaré map as a composition

$$P = G \circ L : \Sigma_h \cap \{x_1 > 0\} \to \Sigma_h \cap \{x_1 > 0\},$$

where $G : \Sigma_v \to \Sigma_h$ and $L : \Sigma_h \to \Sigma_v$ are induced by the flow of (12.1). Let $\lambda_{2,3} = \alpha \pm \omega \mathrm{i}$ with $\omega > 0$ and $\lambda_1 = \lambda$, where the parameter-dependence is implicit. Regarding L, we use a Sternberg-type linearization to obtain

$$\begin{aligned} \zeta' &= \lambda\zeta, \\ \xi_1' &= \alpha\xi_1 - \omega\xi_2, \\ \xi_2' &= \omega\xi_1 + \alpha\xi_2, \end{aligned} \tag{12.11}$$

which can be solved explicitly for L (exercise!) mapping the rectangle $\Sigma_h \cap \{x_1 > 0\}$ to a solid spiral $\mathcal{Z} \subset \Sigma_v$; see also Figure 31.

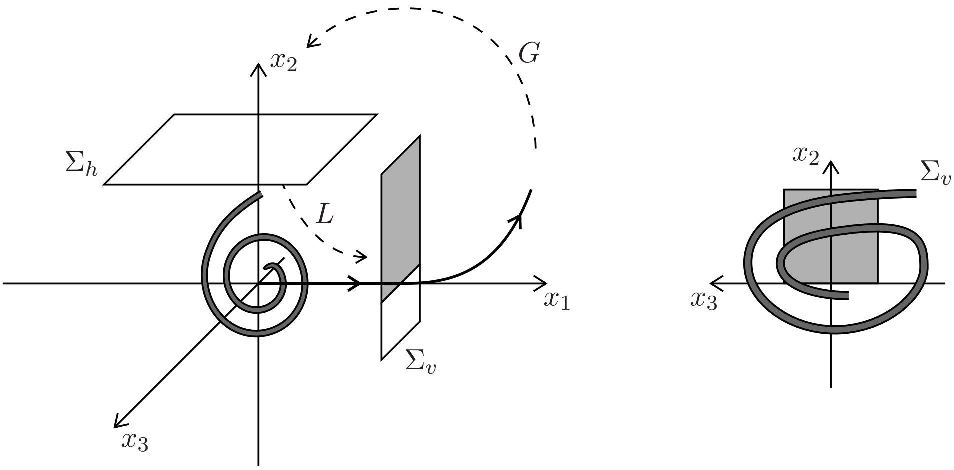

Figure 31: Sketch of Shilnikov homoclinic bifurcation and its proof. Phase space on the left and return map $G \circ L : \Sigma_v \to \Sigma_v$ on the right.

The global map G can be given to lowest-order by Taylor expansion as in the proof of the Andronov-Leontovich Theorem 12.3 but it is far more complicated algebraically, so we argue on geometric grounds. By the Shilnikov condition, G maps the spiral outside of the initial rectangle back to Σ_v as shown in Figure 31. $\square$

Example 12.7. Homoclinic orbits also appear naturally in many PDEs. Consider the FitzHugh-Nagumo equation

$$\begin{aligned} \partial_t u &= \partial_{xx} u + f(u) - v, \\ \partial_t v &= \varepsilon(u - v), \end{aligned} \tag{12.12}$$

where $f(u) = u(1-u)(u-p)$, $(x,t) \in \mathbb{R} \times [0,\infty)$, $p \in (0, 1/2)$ and ε are parameters. Suppose we restrict to (respectively search for) **travelling wave** solutions of the form

$$u(x,t) = u(x - ct) =: u(s), v = v(x - ct) =: v(s),$$

with **wave speed** $c \in \mathbb{R}$. Using the chain rule we have

$$\partial_x u = \frac{\mathrm{d}u}{\mathrm{d}s} = \dot{u}, \qquad \partial_t u = -c\frac{\mathrm{d}u}{\mathrm{d}s} = -c\dot{u},$$

and similarly for v. Upon setting $\dot{u} = w$, this yields the three-dimensional ODE

$$\begin{aligned} \dot{u} &= w, \\ \dot{w} &= v - cw - f(u), \\ \dot{v} &= \tfrac{\varepsilon}{c}(v - u). \end{aligned} \tag{12.13}$$

One checks that there is a unique equilibrium at 0 for (12.13). For the case $\varepsilon = 0$, the plane $\mathcal{P} = \{v = 0\}$ is invariant and contains the global equilibrium. The flow on $\mathcal{P}$ is

$$\begin{aligned} \dot{u} &= w, \\ \dot{w} &= -cw - f(u), \end{aligned} \tag{12.14}$$

which is a *Hamiltonian system* for $c = 0$ and using the Hamiltonian, one finds a homoclinic orbit; see Exercise E12.2. For $\varepsilon > 0$, one can also construct a substantially more complicated homoclinic orbit to the origin (the proof is highly nontrivial). However, it is relatively straightforward to check that there exist parameter values (c, a) such that 0 is a saddle-focus and the Shilnikov condition holds for the eigenvalues. ♦

Exercises for Section 12

E12.1 We consider the system

$$\begin{aligned} x' &= -x + 2y + x^2, \\ y' &= (2-\alpha)x - y - 3x^2 + \frac{3}{2}xy, \end{aligned}$$

with a parameter $\alpha \in \mathbb{R}$. (a) Determine the eigenvalues and eigenspaces of the linearization at the origin for $\alpha = 0$. (b) Show that for $\alpha = 0$ the equation $x^2(1-x) - y^2 = 0$ for $x \geq 0$ describes a homoclinic orbit. (c) Suppose we have a split function $\xi(\alpha)$ with $\xi'(0) < 0$ for a section on the x-axis around $x = 1$ with the same orientation as the axis, i.e., the unstable manifold breaks 'inwards' when we increase α. Show that under variation of the parameter α a limit cycle bifurcates from the homoclinic orbit. When does it exist, for $\alpha > 0$ or $\alpha < 0$? Sketch the phase portraits for the three cases $\alpha < 0$, $\alpha = 0$ and $\alpha > 0$.

E12.2 Consider the FitzHugh-Nagumo ODE system (12.13). (a) Prove that there exists a unique equilibrium point and calculate its stability properties. (b) For $\varepsilon = 0$ and $c = 0$, show that (12.14) is a Hamiltonian system and that it has a homoclinic orbit. (c) Again for $\varepsilon = 0$, consider (12.13) and observe that then (v, c, p) are parameters. Find parameters so that the planar vector field for (u, w) has a heteroclinic orbit; see also [40, Sec. 15-16] for a bit more regarding systems with small parameters.

13 The Smale Horseshoe

The last map in the proof of Shilnikov's Theorem 12.6, leads one naturally to consider maps of the rectangle to itself, which have a horseshoe geometry as shown in Figure 32.

Consider the unit square $\mathcal{Y} = [0,1] \times [0,1]$. Define a map g by (I) linearly stretching $\mathcal{Y}$ vertically by a factor $\mu > 2$ and linearly compressing horizontally by a factor $\lambda \in (0, 1/2)$ and (II) folding the resulting rectangle back so that $g(\mathcal{Y}) \cap \mathcal{Y}$ consists of two disjoint vertical strips $\mathcal{V}_{0,1}$. In particular, the folded portion lies outside $\mathcal{Y}$ so that on $\mathcal{Y} \cap g^{-1}(\mathcal{Y})$ the map is linear. The pre-image $g^{-1}(g(\mathcal{Y}) \cap \mathcal{Y})$ consists of two horizontal strips $\mathcal{H}_{0,1}$; see Figure 32. Now iterate this map

$$x_{k+1} = g(x_k), \;\; g : \mathcal{X} \to \mathcal{X}, \qquad \mathcal{X} := \{x_0 : x_k \in \mathcal{Y} \text{ for all } k \in \mathbb{Z}\}$$

which is called the **Smale horseshoe map**, i.e., we only consider points which remain in the unit square under iteration. Observe that $\mathcal{Y} \cap g(\mathcal{Y}) \cap$

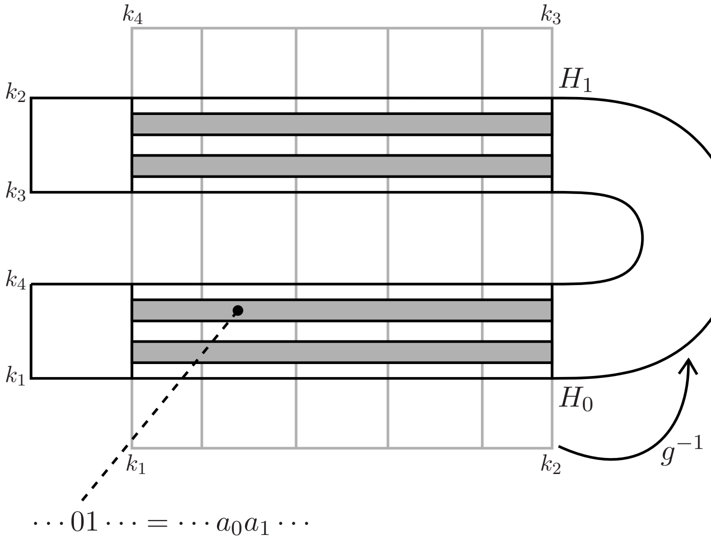

Figure 32: Inverse g^{-1} of the standard horseshoe map g (exercise: draw a suitable sketch for g). The unit square $\mathcal{Y}$ is stretched horizontally and compressed vertically, then folded into a horseshoe shape, and finally pasted back onto the rectangle. The four solid grey horizontal strips form the set of points $g^{-2}(\mathcal{Y} \cap g(\mathcal{Y}) \cap g^2(\mathcal{Y}))$ staying in $\mathcal{Y}$. The corners k_j of $\mathcal{Y}$ are also marked under the transformation g^{-1}. We also indicate the two vertical strips in $\mathcal{Y}$ of the forward image $g(\mathcal{Y}) \cap \mathcal{Y}$ and we show the symbolic coding via the horizontal strips of one point chosen as an example.

$\cdots \cap g^n(\mathcal{Y})$ is the union of 2^n disjoint vertical strips while $\mathcal{Y} \cap g^{-1}(\mathcal{Y}) \cap \cdots \cap g^{-n}(\mathcal{Y})$ is the union of 2^n disjoint horizontal strips. The Jacobian of g is constant and given on $\mathcal{H}_{0,1}$ by

$$\mathrm{D}g = \begin{pmatrix} \pm\lambda & 0 \\ 0 & \pm\mu \end{pmatrix} \tag{13.1}$$

for $+$ on $\mathcal{H}_0$ and $-$ on $\mathcal{H}_1$. Most points leave $\mathcal{Y}$ under iteration. The remaining points $\mathcal{X}$ form a Cantor set (a compact, perfect, totally disconnected set).

Let Σ_2 be the space of bi-infinite sequences $a = \{a_j\}_{j=-\infty}^{\infty}$ with elements from $\{0, 1\}$ with the metric

$$d(a, b) = \sum_{j=-\infty}^{\infty} |a_j - b_j| 2^{-|j|}$$

and also recall the (left-)shift $\sigma : \Sigma_2 \to \Sigma_2$ defined by

$$\sigma(a)_j = a_{j+1}.$$

Theorem 13.1 (Horseshoe Symbolic Dynamics). *$g : \mathcal{X} \to \mathcal{X}$ is topologically conjugate to $\sigma : \Sigma_2 \to \Sigma_2$.*

Proof. Let $x \in \mathcal{X}$, we construct the conjugacy $\phi : \mathcal{X} \to \Sigma_2$ by

$$\phi(x) = \{a_j\}_{j=-\infty}^{\infty} \quad \text{with } g^j(x) \in \mathcal{H}_{a_j}.$$

See also Figure 32 for an illustration of the labeling of points in $\mathcal{X}$. By *construction*, we must have

$$\sigma(\phi(x))_j = \sigma(a)_j = a_{j+1} = \phi(g(x))_j.$$

Let $R(b)$ be the rectangle of all x having a finite central sequence $b = b_{-m} \cdots b_n$, i.e., $g^j(x) \in \mathcal{H}_{b_j}$. The rectangle $R(b)$ has height $\mu^{-(n+1)}$ and width λ^m and is obtained by the intersection of a horizontal and a vertical strip. Since the diameter of $R(b)$ converges to 0 as $m, n \to \infty$, injectivity and continuity of ϕ follow. Lastly, we claim $R(b)$ is non-empty for any given $b = b_{-m} \cdots b_n$ so that ϕ is surjective; indeed, we have that $R(b_0 \cdots b_n)$ is a horizontal strip mapped by g vertically across $\mathcal{Y}$ so that it intersects each horizontal strip, which implies $R(b_0 \cdots b_{n+1})$ is non-empty. A similar argument holds for backward iteration and $R(b_{-m} \cdots b_{-1})$. $\square$

Corollary 13.2 (Horseshoe/Shift Dynamics). *The following properties of the horseshoe map $g : \mathcal{X} \to \mathcal{X}$ hold:*

(T1) $\mathcal{X}$ contains a countable infinite set of saddle-type periodic orbits.

(T2) $\mathcal{X}$ contains an uncountable set of non-periodic orbits.

(T3) $\mathcal{X}$ contains a dense orbit.

(T4) $\mathcal{X}$ is a hyperbolic set.

Proof. The existence of orbits as prescribed in (T1)-(T3) follows from the conjugacy to the shift map in Theorem 13.1. (T4) and the saddle-type contraction/expansion follow by looking at (13.1). $\square$

In particular, Theorem 13.2(T4) implies in combination with the persistence of hyperbolic sets Theorem 7.5 that horseshoe dynamics is robust under perturbations. This observation and Theorem 13.2(T1) almost finish the proof of Shilnikov's Theorem 12.6. What we still lack is to generalize the horseshoe construction to *curved* structures instead of rectangles for points $(x, y) \in \mathcal{Y} \subset \mathbb{R}^2$.

Definition 13.3. A **vertical curve** $x = v(y)$ is a curve for which

$$0 \leq v(y) \leq 1, \qquad |v(y_l) - v(y_r)| \leq \rho |y_l - y_r| \quad \text{for } 0 \leq y_l \leq y_r \leq 1$$

and some $\rho \in (0,1)$. Similarly, a **horizontal curve** $y = h(x)$ is a curve for which

$$0 \leq h(x) \leq 1, \qquad |h(x_l) - h(x_r)| \leq \rho |x_l - x_r| \quad \text{for } 0 \leq x_l \leq x_r \leq 1.$$

Given two non-intersecting vertical curves $v_1(y) < v_2(y)$, we may define a **vertical strip** by

$$\mathcal{V} := \{(x,y) \in \mathbb{R}^2 : x \in [v_1(y), v_2(y)], y \in [0,1]\}$$

and similarly we may define a **horizontal strip**

$$\mathcal{H} := \{(x,y) \in \mathbb{R}^2 : y \in [h_1(x), h_2(x)], x \in [0,1]\}$$

for two non-intersecting horizontal curves $h_1(x) < h_2(x)$.

Theorem 13.4 (Generalized Horseshoe Map)**.** *Let $\mathcal{I} = \{0, 1, 2, \ldots, N-1\}$ and let $\mathcal{H}_i, \mathcal{V}_i$ for $i \in \mathcal{I}$ be disjoint horizontal and vertical strips. Consider $g \in C^1(\mathcal{Y}, \mathbb{R}^2)$ with $g(\mathcal{H}_i) = \mathcal{V}_i$. For $\mu \in (0, 1/2)$ define the (cone-)bundles*

$$\mathcal{K}^{\mathrm{u}} := \left\{ (x,y) \in \bigcup_{i \in \mathcal{I}} \mathcal{V}_i : |x| < \mu |y| \right\},$$

$$\mathcal{K}^{\mathrm{s}} := \left\{ (x,y) \in \bigcup_{i \in \mathcal{I}} \mathcal{H}_i : |y| < \mu |x| \right\}.$$

and suppose g satisfies

(A1) ("invariance") $\mathrm{D}g(\mathcal{K}^{\mathrm{u}}) \subset \mathcal{K}^{\mathrm{u}}$, $(\mathrm{D}g)^{-1}(\mathcal{K}^{\mathrm{s}}) \subset \mathcal{K}^{\mathrm{s}}$*;*

(A2) ("expansion/contraction") $|(\mathrm{D}g_p(x,y))_2| \geq \frac{1}{\mu}|y|$, $|(\mathrm{D}g_p^{-1}(x,y))_1| \geq \frac{1}{\mu}|x|$ *uniformly for $p \in \mathcal{Y}$;*

and $|\det(\mathrm{D}g)|, |\det(\mathrm{D}g^{-1})| \leq \frac{1}{2}\mu^{-2}$*. Then*

(T1) g has a hyperbolic invariant set $\mathcal{X}$.

(T2) On $\mathcal{X}$, g is topologically conjugate to a shift map on the set of bi-infinite sequences Σ_N on N symbols.

The proof of Theorem 13.4 proceeds exactly along the same lines as the proof of Theorem 13.1 for the standard horseshoe, so we omit it here. Instead, we proceed to a principle to link the geometry of global orbits in an iterated map to the appearance of a horseshoe.

Theorem 13.5 (Smale-Birkhoff Homoclinic Theorem)**.** *Consider a diffeomorphism $g : \mathbb{R}^d \to \mathbb{R}^d$ such that 0 is a hyperbolic saddle fixed point. Suppose there exists $q \neq 0$ such that $W^{\mathrm{s}}(0)$ and $W^{\mathrm{u}}(0)$ intersect transversally at q. Then there exists an invariant set $\mathcal{X}$ and $\kappa \in \mathbb{Z}$ such that g^κ is topologically conjugate to the Smale horseshoe map.*

Remark: It is easy to prove (exercise!) that **transverse homoclinic points**, such as q above, cannot occur for the stable/unstable manifolds in a flow. Yet, they may obviously appear in a suitable Poincaré map.

To prove Theorem 13.5, we need a preliminary result, which is of independent interest in many other situations:

Lemma 13.6 (λ-Lemma (or Inclination Lemma))**.** *Let 0 be a hyperbolic fixed point of a diffeomorphism $g : \mathbb{R}^d \to \mathbb{R}^d$ with $\dim(W^{\mathrm{u}}(0)) = k$ and $\dim(W^{\mathrm{s}}(0)) = l$. Let $\mathcal{M}$ be a k-dimensional C^1-submanifold intersecting $W^{\mathrm{s}}(0)$ transversally at $q \in W^{\mathrm{s}}(0)$. Then for any $r > 0$, and any $\delta > 0$, there exists n_0, a k-dimensional disk $\mathcal{D} \subset \mathcal{M}$, and a point $x \in W^{\mathrm{u}}(0)$ such that for all $n \geq n_0$ we have*

$$d_1(\mathcal{B}(x, r) \cap W^{\mathrm{u}}(0), g^n(\mathcal{D})) < \delta,$$

i.e., all images of some disk in $\mathcal{M}$ are eventually C^1-close to a ball inside the unstable manifold; see Figure 33.

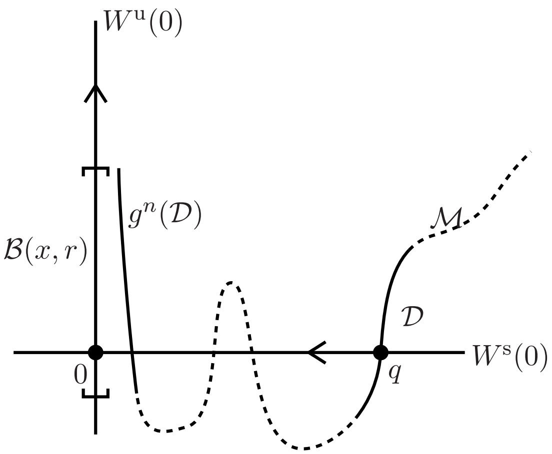

Figure 33: Illustration of the λ-Lemma (or Inclination Lemma) for the case when $\mathcal{M}$ is also an invariant manifold of the dynamics, which is a case commonly encountered in practice.

Proof. First, note that iterating g and using the invariance of $W^{\mathrm{s}}(0)$, implies that if $g^n(\mathcal{M})$ intersects $W^{\mathrm{s}}(0)$ transversally for any $n \geq 0$. Hence, we may focus on a neighborhood of 0 upon picking a submanifold $\mathcal{D} \subset \mathcal{M}$. Furthermore, it is easy to check that the result holds for C^0, i.e.,

$$d_0(\mathcal{B}(x,r) \cap W^{\mathrm{u}}(0), g^n(\mathcal{D})) < \delta,$$

so the problem is to control the tangent spaces. Without loss of generality, we write the map locally in a neighborhood $\mathcal{V} = \mathcal{V}^{\mathrm{s}} \times \mathcal{V}^{\mathrm{u}}$ near 0 as

$$g(x) = g(x_{\mathrm{s}}, x_{\mathrm{u}}) = (A^{\mathrm{s}} x_{\mathrm{s}} + \phi_{\mathrm{s}}(x), A^{\mathrm{u}} x_{\mathrm{u}} + \phi_{\mathrm{u}}(x))^\top$$

where $(x_{\mathrm{s}}, x_{\mathrm{u}}) \in \mathcal{V}^{\mathrm{s}} \times \mathcal{V}^{\mathrm{u}}$ and

$$|A^{\mathrm{s}}| \leq a < 1, \quad |(A^{\mathrm{u}})^{-1}| \leq a < 1, \quad \phi_{\mathrm{s}}(x) = o(|x|) = \phi_{\mathrm{u}}(x).$$

In the following calculations, it can be shown (non-trivial exercise!) that carrying around the higher-order terms $\phi_{\mathrm{u}}(x)$, $\phi_{\mathrm{s}}(x)$ does not affect the result so we shall simply drop them here. Let $v_0 = (v_0^{\mathrm{s}}, v_0^{\mathrm{u}}) \in \mathrm{T}_q\mathcal{D}$, which has slope $|\lambda_0| := |v_0^{\mathrm{s}}|/|v_0^{\mathrm{u}}|$, which is non-singular since $|v_0^{\mathrm{u}}| \neq 0$ due to transversality at q. Then define the iterates

$$q_1 = g(q),\ v_1 = \mathrm{D}g_q(v_0), \quad \cdots \quad q_n = g^n(q),\ v_n = \mathrm{D}g_{q_{n-1}}(v_{n-1}).$$

One then computes

$$\lambda_1 = \frac{|v_1^{\mathrm{s}}|}{|v_1^{\mathrm{u}}|} = \frac{|A^{\mathrm{s}} v_0^{\mathrm{s}}|}{|A^{\mathrm{u}} v_0^{\mathrm{u}}|} \leq a^2 \lambda_0 < a\lambda_0,$$

where we stated the last obvious inequality as we are going to lose one factor of a in the general case with higher-order terms due to cross-terms. Similarly, one finds

$$\frac{|v_1^{\mathrm{u}}|}{|v_0^{\mathrm{u}}|} > a^{-1}.$$

Therefore, for any given $\delta > 0$ and $r > 0$, there exists n_0 such that for all $n \geq n_0$, we find

$$\lambda_n \leq \delta \qquad \text{and} \qquad \frac{|v_n^{\mathrm{u}}|}{|v_0^{\mathrm{u}}|} \geq r,$$

which implies the result. □

Proof. (of Theorem 13.5; sketch) We present only the main geometric idea in $\mathbb{R}^2$. We may again assume that the stable and unstable manifolds $W^{\mathrm{s}}(0)$ and $W^{\mathrm{u}}(0)$ are the (x_1, x_2)-coordinate axes in a neighborhood $\mathcal{V} = \mathcal{V}(0)$; cf. (12.3)-(12.4). Since $q \in W^{\mathrm{s}}(0) \cap W^{\mathrm{u}}(0)$, it follows that there exists n_0 such that

$$g^n(q) \in \mathcal{V} \quad \text{and} \quad g^{-n}(q) \in \mathcal{V} \quad \text{for all } n \geq n_0.$$

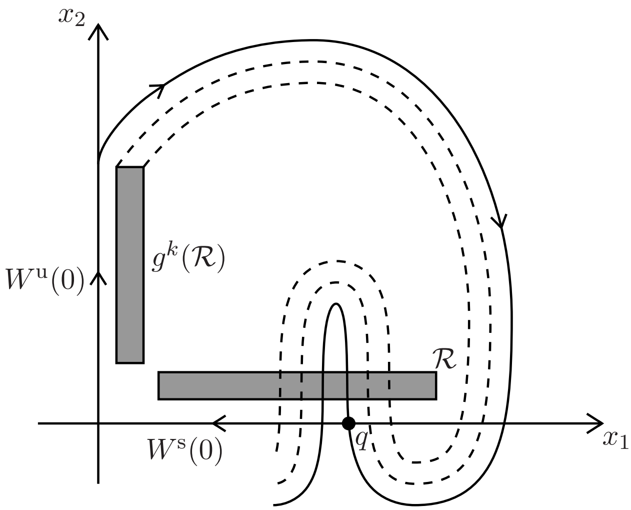

Figure 34: Geometric sketch for the Smale-Birkhoff Theorem 13.5.

Now pick a thin rectangle $\mathcal{R}$ with long horizontal sides parallel to the x_1-axis but separated from $W^{\mathrm{s}}(0)$; see Figure 34.

The vertical boundary segments are chosen as straight lines such that we may find iterates $g^{n_{\mathrm{s}}}(q)$ and $g^{n_{\mathrm{s}}+1}(q)$ with $n_{\mathrm{s}} \geq n_0$ close to the horizontal boundary of $\mathcal{R}$. By the λ-Lemma 13.6, the vertical and horizontal boundary segments of $\mathcal{R}$ are transformed to vertical and horizontal curves for $g^k(\mathcal{R})$, which stretches as a thin rectangle parallel to $W^{\mathrm{u}}(0)$. Also for this image, we can pick k to match precisely two iterates $g^{-n_{\mathrm{u}}}(q)$ and $g^{-n_{\mathrm{u}}+1}(q)$ with $n_{\mathrm{u}} \geq n_0$ close to the vertical boundary of $\mathcal{R}$; see Figure 34. The λ-Lemma 13.6 guarantees that pre-images of vertical and horizontal boundary segments are transformed to vertical and horizontal curves. This construction has the required horseshoe geometry viewing $\mathcal{R}$ as our basic domain for the horseshoe analogous to the unit rectangle above. What remains to be checked is that suitable contraction and expansion conditions hold as prescribed in Theorem 13.4, which is essentially evident (difficult exercise!) since we may always also ensure by adjusting k that g^k is a map with sufficiently strong contraction and expansion as 0 is a saddle point. $\square$

Exercises for Section 13

E13.1 On the unit square we consider the **baker's map**

$$g : [0,1)^2 \to [0,1)^2 \,, \quad g(x,y) = \begin{cases} \left(2x,\ \frac{1}{2}y\right) & \text{if } x \in [0, \frac{1}{2}) \,, \\ \left(2x-1,\ \frac{1}{2}y + \frac{1}{2}\right) & \text{if } x \in [\frac{1}{2}, 1) \,. \end{cases}$$

Remark: This map represents a baker handling dough. He rolls out the dough to twice the width but half the height. The flat dough is then cut into two pieces and the right half is placed on top of the other. (a) For the two sets $A = [0, \frac{1}{2}) \times [0, 1)$ and $B = [\frac{1}{2}, 1) \times [0, 1)$ determine the images $g(S), g^2(S), g^3(S), \ldots$ and preimages $g^{-1}(S), g^{-2}(S), g^{-3}(S), \ldots$ with $S \in \{A, B\}$. (b) Take an initial condition $(x, y) \in [0, 1)^2$ and write it in its binary representation. What can you observe when you apply the map g?

E13.2 Show that the left-shift map $\sigma : \Sigma_2 \to \Sigma_2$ has (a) a countable number of periodic points which are dense, (b) a dense forward orbit, (c) an uncountable set of non-periodic points.

14 Melnikov's Method

The Smale-Birkhoff Homoclinic Theorem 13.5 provides a criterion how to prove the existence of a horseshoe. However, verifying its main assumption regarding the existence of transverse homoclinic points is very difficult. In this section, we provide one special case, where an analytical approach is possible. Consider the ODE

$$x' = f(x) + \varepsilon g(x, t), \qquad x \in \mathbb{R}^2, \tag{14.1}$$

where f, g are smooth, g is periodic in t with minimal period $T > 0$, and $f = (\partial_{x_2} H, -\partial_{x_1} H)^\top$ is Hamiltonian. The key assumption is that the unperturbed system for $\varepsilon = 0$ contains a homoclinic orbit $\gamma = \gamma(t)$ to a hyperbolic saddle point p, i.e., for $\varepsilon = 0$ we have

$$\lim_{t \to \pm\infty} \gamma(t) = p, \qquad \mathrm{D}f(p) \text{ has eigenvalues } \lambda_1 < 0 < \lambda_2.$$

The relevant two-dimensional map is obtained as a Poincaré map

$$P_\varepsilon^{t_0} : \Sigma^{t_0} \to \Sigma^{t_0}, \qquad \Sigma^{t_0} = \{(x, t) | t = t_0 \in [0, T]\} \subset \mathbb{R}^2 \times \mathbb{S}^1,$$

for the three-dimensional autonomous version of (14.1) given by

$$\begin{aligned} x' &= f(x) + \varepsilon g(x, t), \\ t' &= 1, \end{aligned} \tag{14.2}$$

and $\mathbb{S}^1 = \mathbb{R}/(T\mathbb{Z})$; see Figure 35.

Lemma 14.1. *For $\varepsilon > 0$ sufficiently small, the Poincaré map has a locally unique hyperbolic saddle point $p_\varepsilon^{t_0} = p + \mathcal{O}(\varepsilon)$.*

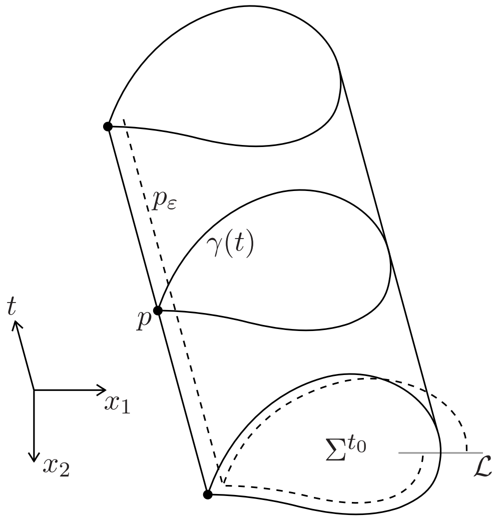

Figure 35: Phase space for the system (14.2). The solid curves illustrate the situation in the limit $\varepsilon = 0$, while dashed curves show the breaking of the family of homoclinic orbits.

Proof. This just follows from the implicit function theorem applied to $F := \mathrm{Id} - P_\varepsilon^{t_0}$ and using that p is hyperbolic. Alternatively, one may just appeal to Theorem 7.9 for a proof. $\square$

We may view the family of saddle-points as a hyperbolic periodic orbit $\Gamma_\varepsilon = \Gamma_\varepsilon(t) = p + \mathcal{O}(\varepsilon)$.

Lemma 14.2. *For $\varepsilon > 0$ sufficiently small, $W^{\mathrm{s}}_{\mathrm{loc}}(\Gamma_\varepsilon)$ and $W^{\mathrm{u}}_{\mathrm{loc}}(\Gamma_\varepsilon)$ are C^k-close (for some $k \geq 1$) to $W^{\mathrm{s}}_{\mathrm{loc}}(\Gamma_0)$ and $W^{\mathrm{u}}_{\mathrm{loc}}(\Gamma_0)$. Furthermore, orbits $\gamma^{\mathrm{s}}_\varepsilon(t;t_0)$, $\gamma^{\mathrm{u}}_\varepsilon(t;t_0)$ lying in $W^{\mathrm{s}}(\Gamma_\varepsilon)$ and $W^{\mathrm{u}}(\Gamma_\varepsilon)$ starting at time t_0 (i.e., in Σ^{t_0}) can be uniformly approximated as*

$$\begin{aligned} \gamma^{\mathrm{s}}_\varepsilon(t;t_0) &= \gamma(t-t_0) + \gamma^{\mathrm{s}}_1(t;t_0)\varepsilon + \mathcal{O}(\varepsilon^2), \qquad && t \in [t_0, \infty), \\ \gamma^{\mathrm{u}}_\varepsilon(t;t_0) &= \gamma(t-t_0) + \gamma^{\mathrm{u}}_1(t;t_0)\varepsilon + \mathcal{O}(\varepsilon^2), \qquad && t \in (-\infty, t_0]. \end{aligned} \tag{14.3}$$

Proof. Existence and closeness follows from the stable/unstable manifold theorem. Pick a sufficiently small neighborhood $\mathcal{U} = \mathcal{U}(p)$ so that the stable and unstable manifolds are $\mathcal{O}(\varepsilon)$-close. Consider an orbit $\gamma^{\mathrm{s}}_\varepsilon(t;t_0)$ in $W^{\mathrm{s}}(\Gamma_\varepsilon)$ starting outside $\mathcal{U}$ in a neighborhood of $\gamma(0)$; see Figure 35. By Gronwall's Lemma, we can follow this orbit until it reaches $\partial\mathcal{U}$, say at time $t_1 > t_0$, so that

$$|\gamma^{\mathrm{s}}_\varepsilon(t;t_0) - \gamma(t-t_0)| = \mathcal{O}(\varepsilon) \qquad \text{for } t \in [t_0, t_1].$$

Since $\gamma_\varepsilon^{\mathrm{s}}(t;t_0)$ remains in $\mathcal{U}$ for any $t > t_1$, the C^k-closeness of the stable manifolds implies

$$|\gamma_\varepsilon^{\mathrm{s}}(t;t_0) - \gamma(t-t_0)| = \mathcal{O}(\varepsilon) \qquad \text{for } t \in (t_1,\infty).$$

So the first equation in (14.3) follows. For the second one, the same argument applies following an orbit backward in time. □

In fact, one even has an equation for the first-order approximation using a variational equation (cf. Definition 5.4) along γ, i.e.,

$$(\gamma_1^{\mathrm{s}})'(t;t_0) = \mathrm{D}f(\gamma(t-t_0))\gamma_1^{\mathrm{s}}(t;t_0) + g(\gamma(t-t_0),t) \tag{14.4}$$

for $t \geq t_0$, and similarly for $t \leq t_0$ in the case of approximating $\gamma_1^{\mathrm{u}}(t;t_0)$. We would like to study the distance between $W^{\mathrm{u}}(p_\varepsilon^{t_0})$ and $W^{\mathrm{s}}(p_\varepsilon^{t_0})$ on Σ^{t_0} as shown in Figure 35. As before, consider $\gamma(0)$ and define a one-dimensional cross-section/line $\mathcal{L}$ via the normal vector

$$(-f_2(\gamma(0)), f_1(\gamma(0)))^\top \in \mathbb{R}^2$$

to $\gamma(t)$ at $\gamma(0)$. Let $\gamma_\varepsilon^{\mathrm{s}}(t_0) := \gamma_\varepsilon^{\mathrm{s}}(t_0;t_0) \in W^{\mathrm{s}}(p_\varepsilon^{t_0})$ and $\gamma_\varepsilon^{\mathrm{u}}(t_0) := \gamma_\varepsilon^{\mathrm{u}}(t_0;t_0) \in W^{\mathrm{u}}(p_\varepsilon^{t_0})$ be the unique points in $\mathcal{L}$ closest to $\gamma(0)$. Define the (signed) distance by

$$d(t_0) := \gamma_\varepsilon^{\mathrm{u}}(t_0) - \gamma_\varepsilon^{\mathrm{s}}(t_0).$$

Lemma 14.3. *The function $d(t_0)$ can be uniformly approximated by*

$$d(t_0) = \varepsilon \frac{f(\gamma(0)) \wedge (\gamma_1^{\mathrm{u}}(t_0) - \gamma_1^{\mathrm{s}}(t_0))}{|f(\gamma(0))|} + \mathcal{O}(\varepsilon^2). \tag{14.5}$$

Proof. By Lemma 14.2, we may approximate the difference $\gamma_\varepsilon^{\mathrm{u}}(t_0) - \gamma_\varepsilon^{\mathrm{s}}(t_0)$ to first order by $\gamma_1^{\mathrm{u}}(t_0) - \gamma_1^{\mathrm{s}}(t_0)$. Furthermore, recall the definition of the wedge product $a \wedge b = a_1 b_2 - b_1 a_2$, which yields that $f \wedge (\gamma_1^{\mathrm{u}} - \gamma_1^{\mathrm{s}})$ is the projection of $\gamma_1^{\mathrm{u}} - \gamma_1^{\mathrm{s}}$ onto $\mathcal{L}$ and that $|f(\gamma(0))|$ is a normalization factor; see Figure 35. □

Formula (14.5) suggests to study just the numerator in the $\mathcal{O}(\varepsilon)$-term to study intersections of the stable/unstable manifolds. The following definition turns out to be the correct one to just calculate the intersections from given data.

Definition 14.4. Define the **Melnikov function** by

$$M(t_0) = \int_{-\infty}^{\infty} f(\gamma(t-t_0)) \wedge g(\gamma(t-t_0),t)\ \mathrm{d}t.$$

Theorem 14.5 (Melnikov Zeros for Homoclinic Orbits). *If M has simple zeros and is independent of ε, then $W^{\mathrm{s}}(p_\varepsilon^{t_0})$ and $W^{\mathrm{u}}(p_\varepsilon^{t_0})$ intersect transversally. If M has no zeros, then $W^{\mathrm{s}}(p_\varepsilon^{t_0}) \cap W^{\mathrm{u}}(p_\varepsilon^{t_0}) = \emptyset$.*

Proof. Consider the time-dependent distance function

$$\begin{aligned}\delta(t;t_0) &:= f(\gamma(t-t_0)) \wedge (\gamma_1^{\mathrm{u}}(t;t_0) - \gamma_1^{\mathrm{s}}(t;t_0))\\ &= f(\gamma(t-t_0)) \wedge \gamma_1^{\mathrm{u}}(t;t_0) - f(\gamma(t-t_0)) \wedge \gamma_1^{\mathrm{s}}(t;t_0)\\ &=: \delta^{\mathrm{u}}(t;t_0) - \delta^{\mathrm{s}}(t;t_0).\end{aligned}$$

By Lemma 14.3, we have $d(t_0) = \varepsilon\delta(t_0;t_0)/|f(\gamma(0))| + \mathcal{O}(\varepsilon^2)$. The trick is to find ODEs for $\delta^{\mathrm{u}}(t;t_0)$ and $\delta^{\mathrm{s}}(t;t_0)$ and solve them by integration. We start with the stable part and differentiate

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta^{\mathrm{s}}(t;t_0) = \mathrm{D}f(\gamma(t-t_0))\gamma'(t-t_0) \wedge \gamma_1^{\mathrm{s}}(t;t_0) + f(\gamma(t-t_0)) \wedge (\gamma_1^{\mathrm{s}})'(t;t_0).$$

Using the variational equation (14.4), $\gamma' = f(\gamma)$, and omitting the arguments $(t-t_0)$ and $(t;t_0)$ for notational simplicity leads to

$$\begin{aligned}(\delta^{\mathrm{s}})' &= \mathrm{D}f(\gamma)f(\gamma) \wedge \gamma_1^{\mathrm{s}} + f(\gamma) \wedge (\mathrm{D}f(\gamma)(\gamma_1^{\mathrm{s}}) + g(\gamma,t))\\ &= \mathrm{Tr}(\mathrm{D}f(\gamma))\ \delta^{\mathrm{s}} + f(\gamma) \wedge g(\gamma,t).\end{aligned}$$

However, $\mathrm{Tr}(\mathrm{D}f(\gamma)) = 0$ as the vector field is Hamiltonian and smooth (in fact: C^2 would suffice here). Integrating the ODE for δ^{s} from t_0 to ∞ yields

$$\delta^{\mathrm{s}}(\infty;t_0) - \delta^{\mathrm{s}}(t_0;t_0) = \int_{t_0}^{\infty} f(\gamma(t-t_0)) \wedge g(\gamma(t-t_0),t)\ \mathrm{d}t. \tag{14.6}$$

By Lemma 14.2, $\gamma_1^{\mathrm{s}}(t;t_0)$ is bounded for $t \geq t_0$ so that we may compute

$$\delta^{\mathrm{s}}(\infty;t_0) = \lim_{t\to+\infty} f(\gamma(t-t_0)) \wedge \gamma_1^{\mathrm{s}}(t;t_0) = f(p) \wedge \lim_{t\to+\infty}\gamma_1^{\mathrm{s}}(t;t_0) = 0,$$

so (14.6) gives a formula for $\delta^{\mathrm{s}}(t_0;t_0)$. Similarly, one may prove

$$\delta^{\mathrm{u}}(t_0;t_0) = \int_{-\infty}^{t_0} f(\gamma(t-t_0)) \wedge g(\gamma(t-t_0),t)\ \mathrm{d}t, \tag{14.7}$$

so that adding (14.6) and (14.7) and employing Lemma 14.3 gives us

$$d(t_0) = \varepsilon\frac{M(t_0)}{|f(\gamma(0))|} + \mathcal{O}(\varepsilon^2), \tag{14.8}$$

where we note that $|f(\gamma(0))| = \mathcal{O}(1)$ as $\varepsilon \to 0$. If $M(t_0)$ is independent of ε, the first term in (14.8) dominates. Suppose $M(t_0)$ has zeros, then there exists a time t_* such that $\gamma_\varepsilon^{\mathrm{s}}(t_*) = \gamma_\varepsilon^{\mathrm{u}}(t_*)$, which corresponds to an intersection in a homoclinic point in the stable/unstable manifolds of the saddle point of the Poincaré map, i.e.,

$$W^{\mathrm{s}}(p_\varepsilon^{t_*}) \cap W^{\mathrm{u}}(p_\varepsilon^{t_*}) \neq \emptyset.$$

Since Poincaré maps for $t_0 \in [0, T]$ are topologically equivalent (exercise!), $W^{\mathrm{s}}(p_\varepsilon^{t_0})$ and $W^{\mathrm{u}}(p_\varepsilon^{t_0})$ must intersect for all $t_0 \in [0, T]$; see Figure 35. If the zeros of $M(t_0)$ are simple, then the intersection points are transversal homoclinic points. In the case, when M has no zeros, the manifolds $W^{\mathrm{s}}(p_\varepsilon^{t_0})$ and $W^{\mathrm{u}}(p_\varepsilon^{t_0})$ do not intersect for any t_0. $\square$

Example 14.6. Consider the periodically-forced **Duffing equation**

$$\begin{aligned} x' &= y, \\ y' &= x - x^3 + \varepsilon\alpha\cos(t) - \varepsilon y, \end{aligned} \tag{14.9}$$

where α is a parameter. The system (14.9) is Hamiltonian for $\varepsilon = 0$ with Hamiltonian function

$$H(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

There is a hyperbolic saddle point at $p = (0, 0)$ and the level set $\{H = 0\}$ consists of two homoclinic orbits $\gamma_\pm$; see Figure 36.

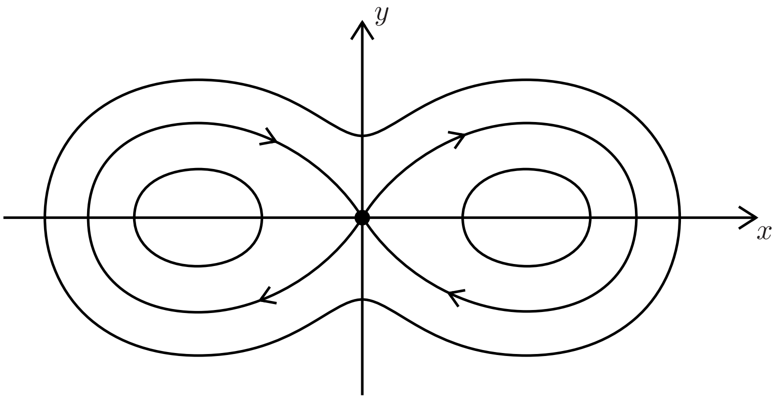

Figure 36: Phase portrait showing the double-homoclinic loop in the unforced (i.e., $\varepsilon = 0$) Duffing equation (14.9).

In this case, one may even explicitly calculate the homoclinic orbits and check that

$$\gamma_+(t) = \left(\sqrt{2}\,\operatorname{sech}(t), -\sqrt{2}\,\operatorname{sech}(t)\,\tanh(t)\right), \qquad \gamma_-(t) = -\gamma_+(t).$$

Therefore, we can give an explicit formula for the Melnikov function. We focus on $\gamma_+(t)$ since the calculation is similar for $\gamma_-(t)$ by symmetry. Let γ_+^y denote the second component of γ_+, then we have

$$M(t_0) = \int_{-\infty}^{\infty} \gamma_+^y(t)\left(\alpha\cos(t+t_0) - \gamma_+^y(t)\right)\,\mathrm{d}t.$$

After plugging in the explicit formula for $\gamma_+^y(t)$, the resulting integral can actually be evaluated (non-trivial exercise!) and one obtains

$$M(t_0) = -\frac{4}{3} + \sqrt{2}\alpha\pi\ \mathrm{sech}\left(\frac{\pi}{2}\right)\sin(t_0).$$

Therefore, after quite a bit of algebraic manipulation (exercise!), one finds that there exists an amplitude $\alpha > 0$ of the forcing such that $M(t_0)$ has simple zeros. Theorem 14.5 applies and shows the existence of a transverse homoclinic point. Therefore, the Smale-Birkhoff Theorem can be used to show that the return map of the Duffing equation for suitable forcing contains horseshoe dynamics. ♦

Exercises for Section 14

E14.1 (a) Verify the equation

$$\mathrm{D}f(\gamma)f(\gamma)\wedge\gamma_1^{\mathrm{s}}+f(\gamma)\wedge(\mathrm{D}f(\gamma)\gamma_1^{\mathrm{s}}+g(\gamma,t)) = \mathrm{Tr}(\mathrm{D}f(\gamma))\delta^{\mathrm{s}}+f(\gamma)\wedge g(\gamma,t),$$

that was used in the proof of Theorem 14.5. (b) Assume that g does not depend on time and let $A\subset\mathbb{R}^2$ be the set enclosed by the homoclinic orbit γ. Show that

$$M(t_0) = s\int_A \mathrm{Tr}(\mathrm{D}g(x))\,\mathrm{d}x \quad \text{with}\ \ s=1 \text{ or } s=-1.$$

E14.2 You may assume the following theorem without proof: Consider the ODE

$$x' = f(x)+\varepsilon g(x,p), \quad p\in\mathbb{R}^m,\ x=x(t)\in\mathbb{R}^2,\ f,g\in C^1.$$

Assume for $\varepsilon = 0$ there exists a one-parameter family of periodic orbits $\gamma_\alpha(t)$ of period T_α with a parameter $\alpha\in\mathbb{R}$. If there exists a point $(p_0,\alpha_0)\in\mathbb{R}^{m+1}$ such that the **Melnikov function**

$$M(p,\alpha) := \int_0^{T_\alpha} f(\gamma_\alpha(t))\wedge g(\gamma_\alpha(t),p)\,\mathrm{d}t$$

satisfies $M(p_0,\alpha_0)=0$ and $\partial_\alpha M(p_0,\alpha_0)\neq 0$ then for all sufficiently small $\varepsilon\neq 0$ and for $p=p_0$, there exists a unique hyperbolic limit cycle in an $\mathcal{O}(\varepsilon)$-neighborhood of the cycle $\gamma_{\alpha_0}(t)$. Now consider a variant of the **Liènard equation**

$$\begin{aligned} x' &= y-\varepsilon(p_1x+p_2x^2+p_3x^3),\\ y' &= -x. \end{aligned}$$

Using the last theorem, verify under which conditions this Liènard equation has a locally unique limit cycle for $\varepsilon\neq 0$ sufficiently small.

15 Chaos

The Smale horseshoe is a key example of chaotic dynamics. Yet, one may also define the concept of chaos more abstractly. The next definition works for general metric spaces $\mathcal{X}$ but for didactic simplicity, we shall make the usual assumption that $\mathcal{X}\subseteq\mathbb{R}^d$.

Definition 15.1. A dynamical system $(\mathcal{X},\mathcal{T},\phi_t)$ has **sensitive dependence on initial conditions at** x_0 if there exists $r>0$ such that for any $\delta>0$, there exists y_0 with $|x_0-y_0|<\delta$ and a time $T>0$ such that

$$|\phi_T(x_0)-\phi_T(y_0)|\geq r.$$

If $\mathcal{S}$ is a set then the system exhibits **sensitive dependence on initial conditions on** $\mathcal{S}$ if the previous definition applies to any $x_0\in\mathcal{S}$.

Example 15.2. Consider the dynamical system on $\mathcal{X}=[0,1]$ with $\mathcal{T}=\mathbb{N}_0$ given by the **doubling map**

$$g(x)=2x \pmod 1, \tag{15.1}$$

which can be taken as one of the simplest examples, where we can expect sensitive dependence on initial conditions; see Figure 37(a). ♦

Theorem 15.3 (Sensitive Dependence for the Doubling Map)**.** *The doubling map* (15.1) *has sensitive dependence on initial conditions on* $\mathcal{X}=[0,1]$.

Proof. We are going to show that g is **expansive** with constant $\frac14$, i.e., for any $x_0\neq y_0$, there exists k such that

$$|g^k(x_0)-g^k(y_0)|\geq\frac{1}{4},$$

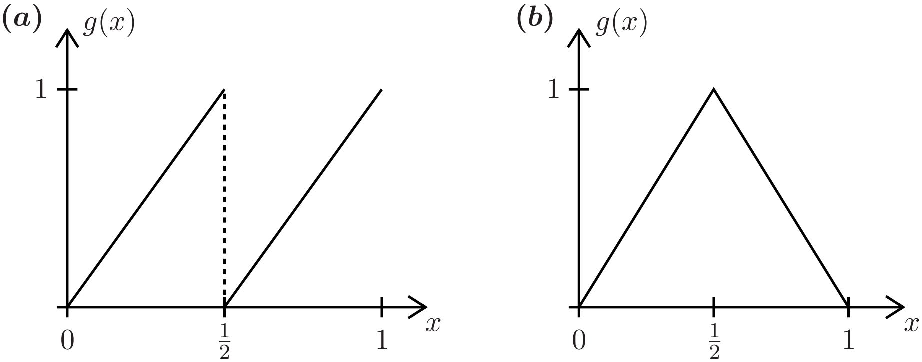

Figure 37: (a) Sketch of the doubling map 15.1. (b) Sketch of the tent map (15.3) for $p = 2$.

which immediately implies the result. Consider two points x, y such that $x < y$ and $y - x < \frac{1}{4}$. We claim that

$$|g(y) - g(x)| \geq 2|y - x|. \tag{15.2}$$

If x, y are in the same interval, i.e., $x, y \in [0, 1/2)$ or $x, y \in [1/2, 1)$ then we immediately have (15.2). If x, y are in different intervals and $y > x$, then the doubling map can be written as

$$g(x) = 2x, \qquad g(y) = 2y - 1.$$

This observation yields again (15.2) since

$$|g(y) - g(x)| = |2y - 1 - 2x| = |1 - 2(y - x)| \geq 1 - 2|y - x| \geq 2|y - x|$$

due to $|y - x| \leq 1/4$. To finish the proof, we argue by contradiction. Suppose $x_0 < y_0$ and $|g^k(y_0) - g^k(x_0)| \leq 1/4$ for all $k \in \mathbb{N}_0$, then by (15.2) we have

$$|g^k(y_0) - g^k(x_0)| \geq 2^k |x_0 - y_0|,$$

and selecting k sufficiently large yields the desired result. □

The logistic map is another example, where one may prove sensitive dependence for certain parameters. The calculations are easier in the following simplification.

Example 15.4. Consider the (family of) **tent maps**

$$g(x; p) = \begin{cases} px & \text{if } x \in [0, 1/2) \\ p(1 - x) & \text{if } x \in [1/2, 1] \end{cases} \tag{15.3}$$

for $p \in (1, 2]$. For example, it is relatively easy to check (exercise!) that the classical tent map $g(x; 2)$ has sensitive dependence on initial conditions. ♦

In addition to sensitive dependence, chaos should also contain some notion of spreading through various parts of phase space.

Definition 15.5. A dynamical system is called **topologically transitive** if there exists a point $x \in \mathcal{X}$, whose forward orbit is dense in $\mathcal{X}$.

Theorem 15.6 (Tent Map Topological Transitivity)**.** *The tent map* (15.3) *is topologically transitive for* $p = 2$*.*

Remark: Similar results and proofs also work for other parameter values and various other classes of dynamical systems; see Section 16.

Proof. We use **symbolic dynamics**, similar to the horseshoe analysis. Let $\Sigma_2^+ = \{s = s_0 s_1 \cdots : s_j \in \{0,1\}\}$ be the space of infinite one-sided sequences of two symbols. Define the **itinerary map** $h : [0,1] \to \Sigma_2^+$ by

$$h(x)_j = \begin{cases} 0 & \text{if } g^j(x) \in [0,1/2] =: \mathcal{I}_0, \\ 1 & \text{if } g^j(x) \in [1/2,1] =: \mathcal{I}_1. \end{cases}$$

We exclude all points x, which are going to eventually hit the point $x = 1/2$, which is not going to alter our argument regarding topological transitivity. Then note that h satisfies $h(g(x)) = \sigma(h(x))$, where σ is the left-shift map, i.e., $\sigma(s)_j = s_{j+1}$ for $j \in \mathbb{N}_0$. We may associate subsets of $[0,1]$ with subsets of symbols via

$$\mathcal{I}_{s_0 s_1 \cdots s_n} := \{x \in [0,1] : g^j(x) \in \mathcal{I}_{s_j} \text{ for } j \in \{0,1,\ldots,n\}\}.$$

Observe that each $\mathcal{I}_{s_0 s_1 \cdots s_n}$ is an interval and has length $2^{-(n+1)}$. The sequence of compact intervals

$$\mathcal{I}_{s_0} \supset \mathcal{I}_{s_0 s_1} \supset \mathcal{I}_{s_0 s_1 s_2} \supset \cdots$$

has a non-empty intersection by Cantor's Intersection Theorem and one easily checks that the intersection is a single point x_0 so that we may define $H(s) = x_0$. One finds $H \circ h(x_0) = x_0$ but observes that H is surjective yet not injective (see remark below), so it is only a **semi-conjugacy** $H \circ \sigma(s) = g \circ H(s)$. In any case, having this semi-conjugacy, checking topological transitivity is very easy: let s^* be a sequence constructed by concatenating all possible symbol sequences of length 1, length 2, length 3 and so on. Define $x^* = H(s^*)$ and observe that eventually $g^j(x^*) \in \mathcal{I}_{s_0 s_1 \cdots s_n}$ for any given $s_0 s_1 \cdots s_n$. Since the length of $\mathcal{I}_{s_0 s_1 \cdots s_n}$ tends to zero as $n \to \infty$, density of the forward orbit of x^* follows. $\square$

Remark: The map $H : \Sigma_2^+ \to [0,1]$ can also be viewed as sending a symbol sequence to the **binary form** of a real number by

$$H(s) = \sum_{j=0}^{\infty} \frac{s_j}{2^{j+1}}.$$

It is not injective since, e.g., $H(1,0,0,0,\cdots) = \frac{1}{2} = H(0,1,1,1,\ldots)$.

Definition 15.7. An invariant set $\mathcal{S} \subseteq \mathcal{X}$ is called **chaotic** if it has sensitive dependence on initial conditions and $\phi_t|_{\mathcal{S}}$ is topologically transitive.

Proving the following results is lengthy, yet no new techniques are needed.

Theorem 15.8 (Chaos in Interval Maps)**.** $\mathcal{S} = [0,1]$ *is chaotic for the tent map* (15.3) *with* $p = 2$*, for the doubling map* (15.1)*, and for the logistic map* $x \mapsto 4x(1-x)$*.*

Theorem 15.9 (Chaotic Horseshoe)**.** *The horseshoe map is chaotic on* $\mathcal{X}$*, where* $\mathcal{X} \subset [0,1] \times [0,1]$ *is the maximal invariant Cantor set as constructed in Section 13.*

Unfortunately, testing for chaos in higher-dimensional maps

$$x_{k+1} = g(x_k),\ g : \mathbb{R}^d \to \mathbb{R}^d, \qquad x_k \in \mathbb{R}^d$$

for $d \geq 2$ turns out to be difficult; the same problem occurs for flows in dimension $d \geq 3$. The following concept is a good practical indicator.

Definition 15.10. Suppose $g \in C^2$, fix $x_0 \in \mathbb{R}^d$, define the matrix

$$A_k = \mathrm{D}g^k_{x_0} = \mathrm{D}g_{x_{k-1}} \cdots \mathrm{D}g_{x_1}\mathrm{D}g_{x_0} \in \mathbb{R}^{d\times d} \tag{15.4}$$

and consider $(A_k)^\top A_k$ with eigenvalues $(\lambda_j^{(k)})^2$ for $j \in \{1,2,\ldots,d\}$ ordered decreasingly (upon taking the positive square-root of each eigenvalue) by

$$\lambda_1^{(k)} \geq \lambda_2^{(k)} \geq \cdots \geq \lambda_d^{(k)}.$$

Define the j-th **Lyapunov exponent** l_j at x_0 by

$$l_j(x_0) = \lim_{k\to\infty} \frac{1}{k} \ln\left(\lambda_j^{(k)}\right) \tag{15.5}$$

if this limit exists.

One may interpret, or define and then prove (15.5) afterwards, Lyapunov exponents as deformation rates of the unit sphere centered at x_0 under the linearized dynamics. In particular, looking at $A_k(\mathbb{S}^{d-1})$ and the resulting axes of the ellipsoid one; see Figure 38.

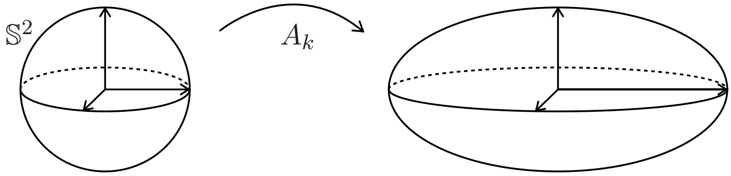

Figure 38: Lyapunov exponents can be viewed as a measure for the deformation of the axes of a unit sphere into the axes of an ellipsoid.

Theorem 15.11 (Liouville Formula)**.** *Suppose $l_j(x_0)$ exist for all j and $|\det \mathrm{D}g_p|$ is the same for every p, then*

$$\sum_{j=1}^{d} l_j(x_0) = \ln(|\det \mathrm{D}g_{x_0}|).$$

Proof. The determinant is the product of the eigenvalues so we have

$$\prod_{j=1}^{d} (\lambda_j^{(k)})^2 = \det(\mathrm{D}g_{x_0}^k)^2 \stackrel{(15.4)}{=} |\det \mathrm{D}g_{x_0}|^{2k}.$$

By standard logarithm rules we have

$$\begin{aligned}\ln|\det \mathrm{D}g_{x_0}| &= \frac{1}{2k}\ln|\det \mathrm{D}g_{x_0}|^{2k} = \lim_{k\to\infty}\frac{1}{2k}\ln\left(\prod_{j=1}^{d}(\lambda_j^{(k)})^2\right)\\ &= \lim_{k\to\infty}\frac{1}{k}\sum_{j=1}^{d}\ln\left(\lambda_j^{(k)}\right) = \sum_{j=1}^{d} l_j(x_0),\end{aligned}$$

and the result is proven. □

Example 15.12. For the Smale horseshoe, the Jacobian is independent of x_0 and given by (13.1) with $\lambda \in (0, \frac{1}{2})$, $\mu > 2$ so that we have

$$l_1 = \lim_{k\to\infty}\frac{1}{k}\ln \mu^k = \ln \mu > 0, \qquad l_2 = \ln \lambda < 0, \tag{15.6}$$

which leads to the general observation that Lyapunov exponents are directly related to multipliers for cases where $\mathrm{D}g_{x_0}$ is independent of x_0. In fact, one may prove (non-trivial!) that Lyapunov exponents are independent of the starting point for certain classes of invariant sets. ♦

Example 15.13. Consider the standard tent map $g(x; 2)$ from (15.3). Except at the single point $x = \frac{1}{2}$, we have $l_1(x_0) = \ln(2) > 0$. ♦

The last two examples indicate that a positive Lyapunov exponent should be a necessary condition for chaos; similarly, one can also characterize the situation for flows and make a definition of Lyapunov exponents (exercise!) such that again a *positive Lyapunov exponent* is a first indicator for chaos in the context of nonlinear ODEs.

Exercises for Section 15

E15.1 Let $\mathcal{X}$ be infinite and suppose that the map $g : \mathcal{X} \to \mathcal{X}$ is topologically transitive and the set of periodic points is dense in $\mathcal{X}$. Prove that g has sensitive dependence on initial conditions.

E15.2 Let $(\mathcal{X}, \mathcal{T}, \phi_t)$ and $(\mathcal{Y}, \mathcal{T}, \psi_t)$ be two dynamical systems that are topologically conjugate. (a) Show that topological transitivity is preserved under topological conjugacy, i.e., $(\mathcal{X}, \mathcal{T}, \phi_t)$ is topologically transitive if and only if $(\mathcal{Y}, \mathcal{T}, \psi_t)$ is topologically transitive. (b) Now let $\mathcal{X}$ and $\mathcal{Y}$ be closed intervals (or more general: compact sets). Show that $(\mathcal{X}, \mathcal{T}, \phi_t)$ has sensitive dependence on initial conditions on $\mathcal{X}$ if and only if $(\mathcal{Y}, \mathcal{T}, \psi_t)$ has sensitive dependence on initial conditions on $\mathcal{Y}$. *Hint: Use uniform continuity.*

E15.3 On the circle $\mathbb{S}^1$, represented by the unit interval with endpoints identified, consider the **circle rotation**

$$g : [0, 1) \to [0, 1), \quad x \mapsto x + \alpha \mod 1,$$

for some fixed parameter $\alpha \in \mathbb{R}$. (a) Show that the system is topologically transitive if and only if $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. (b) Is the system chaotic?

E15.4 This exercise illustrates **synchronization of chaos**. Let $g_p(x) \colon [0, 1] \to [0, 1]$ be defined as

$$g_p(x) = \begin{cases} px & \text{if } x \in [0, 1/2), \\ p(x - 1/2) & \text{if } x \in [1/2, 1], \end{cases}$$

where $p \in (1, 2]$ is a parameter. (a) Consider the dynamical system that is given by the iteration of g_p. Calculate its Lyapunov exponent. *Disclaimer: g_p is not in C^2, but you may take Definition 15.10 anyways. Technically, you can define a set $S = \{x \in [0, 1] : g_p^n(x) \notin \{\frac{1}{2}, 1\}$ for all $n \in \mathbb{N}_0\}$ and restrict g_p to S.* (b) Now, consider the two-dimensional system

$$\begin{aligned} x_{n+1} &= g_p(x_n) + a(g_p(y_n) - g_p(x_n)), \\ y_{n+1} &= g_p(y_n) + a(g_p(x_n) - g_p(y_n)), \end{aligned} \tag{15.7}$$

with $a \in [0,1]$. Show that the unit square $[0,1]^2$ is invariant and calculate both Lyapunov exponents for this system. (c) Let $(z_n)_{n\in\mathbb{N}}$ be a sequence with $z_n \in [0,1]$ such that $z_{n+1} = g_p(z_n)$. Show that $(z_n, z_n)^\top$ is a solution of (15.7). Further, determine the transverse stability of this solution by making the ansatz

$$\begin{aligned} x_n &= z_n + \alpha_n, \\ y_n &= z_n - \alpha_n, \end{aligned}$$

to determine an iteration procedure for α_n when $|\alpha_n| \ll 1$. You may assume that locally $g_p(x+h) = g_p(x) + g_p'(x)h$.

16 One-Dimensional Maps

The tent map, the logistic map, and the doubling map have served as excellent examples to understand chaotic dynamics. In the case of continuous maps, we can do even better and understand their *orbit structure*.

Theorem 16.1 (Li-Yorke Theorem; "period-three implies chaos")**.** *Suppose $g : [a,b] \to \mathbb{R}$ with $a < b$, $g \in C^0$, has a periodic orbit with minimal period three. Then g has periodic orbits of all periods.*

For a proof using transition graphs and symbolic dynamics, see [40]. In fact, one may get a lot finer information on the appearance of different families of periodic orbits.

Definition 16.2. The **Sharkovskii ordering** $\lhd$ on $\mathbb{N}$ is defined by

$$\begin{gathered} 1 \lhd 2 \lhd 2^2 \lhd 2^3 \lhd \cdots \lhd 2^n \lhd 2^{n+1} \cdots \\ \cdots \lhd 9 \cdot 2^{n+1} \lhd 7 \cdot 2^{n+1} \lhd 5 \cdot 2^{n+1} \lhd 3 \cdot 2^{n+1} \lhd \cdots \\ \cdots \lhd 9 \cdot 2^n \lhd 7 \cdot 2^n \lhd 5 \cdot 2^n \lhd 3 \cdot 2^n \lhd \cdots \lhd 9 \lhd 7 \lhd 5 \lhd 3. \end{gathered}$$

Theorem 16.3 (Sharkovskii Theorem)**.** *Suppose $g : [a,b] \to \mathbb{R}$, $g \in C^0$, has a periodic orbit with minimal period n. Then g has periodic orbits of all periods $k \lhd n$.*

Example 16.4. Consider the tent map (cf. Example 15.4), which can also be defined as

$$x_{k+1} = p \min\{x_k, 1 - x_k\} = g(x_k; p) = g(x_k), \qquad p \in (0,2]. \tag{16.1}$$

Varying the parameter p, we get the result shown in Figure 39(a).

As an example, consider $p = 2$, then we get

$$g(2/7) = 4/7, \qquad g(4/7) = 6/7, \qquad g(6/7) = 2/7,$$

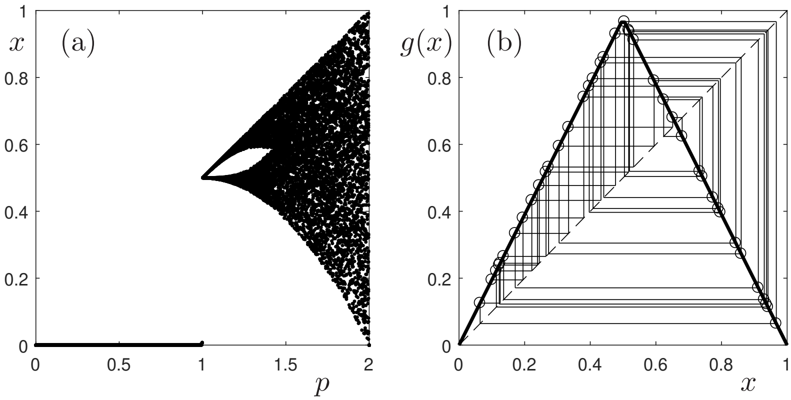

Figure 39: (a) Bifurcation diagram of the tent map (16.1) obtained by direct simulation for a 500 point equally spaced mesh of the parameter space $p \in (0, 2]$. Transients have been removed (here: first 500 iterations), then 40 iterates are plotted for each value of p. (b) Illustration of the cobweb construction for $p = 1.95$.

so that Sharkovskii's Theorem and the Li-Yorke Theorem apply. One can prove that period three orbits do exist for large ranges of parameter values as indicated already by Figure 39(a). ♦

The classical tent map (for $p = 2$) shows strong similarity to the classical logistic map $y_{k+1} = 4y_k(1 - y_k)$ (exercise: simulate the logistic map!).

Proposition 16.5. *The tent map $x_{k+1} = 2\min\{x_k, 1 - x_k\} = g(x_k)$ and the logistic map $y_{k+1} = 4y_k(1 - y_k) = f(y_k)$ are conjugate.*

Proof. We claim $h(x) = (1 - \cos(\pi x))/2$ is a conjugacy. Clearly, h is a diffeomorphism on $[0, 1]$. Suppose $x \in [0, 1/2]$. Then we calculate

$$\begin{aligned} f(h(x)) &= 4\left(\frac{1-\cos(\pi x)}{2}\right)\left(\frac{1+\cos(\pi x)}{2}\right) = 1 - \cos^2(\pi x) \\ &= \sin^2(\pi x) = \frac{1-\cos(2\pi x)}{2} = \frac{1-\cos(\pi g(x))}{2} = h(g(x)), \end{aligned}$$

The case $x \in (1/2, 1]$ is similar (exercise!) and the result follows. □

Although there may be many periodic orbits, two important questions remain: (a) how are new periodic orbits generated, and (b) what is their stability? We start with question (b) and return to (a) in Section 17. We need some preliminary techniques.

Definition 16.6. Let $\mathcal{I}$ be an interval in $\mathbb{R}$ and suppose $g \in C^3(\mathcal{I}, \mathbb{R})$. Define the **Schwarzian derivative** of g by

$$Sg(x) := \frac{g'''(x)}{g'(x)} - \frac{3}{2}\left(\frac{g''(x)}{g'(x)}\right)^2 \tag{16.2}$$

For an isolated critical point x of g we define $Sg(x) = \lim_{y\to x} Sg(y)$.

Lemma 16.7 (Minimum Principle)**.** *Suppose $g \in C^3(\mathcal{I}, \mathcal{I})$ with $g'(x) \neq 0$ for all $x \in \mathcal{I}$. If $Sg(x) < 0$ for all $x \in \mathcal{I}$, then $|g'(x)|$ does not attain a local minimum in the interior of $\mathcal{I}$.*

Proof. Suppose z is a critical point of g', which implies $g''(z) = 0$. Since $Sg < 0$, we have by (16.2) that $g'''(z)/g'(z) < 0$. Therefore, $g'''(z)$ and $g'(z)$ have opposite signs. If $g'(z) < 0$, it follows that $g'''(z) > 0$ so z is a local minimum of g', so z is also a local maximum of $|g'|$ using $g'(z) < 0$. In the other possible case we have $g'''(z) < 0$ and $g'(z) > 0$, which yields that z is a local maximum of g' and hence also of $|g'|$. Since g' does not vanish in $\mathcal{I}$ by assumption, the result follows. $\square$

Theorem 16.8 (Singer's Theorem)**.** *Let $\mathcal{I}$ be a closed (potentially unbounded) interval and consider $g \in C^3(\mathcal{I}, \mathcal{I})$. If $Sg < 0$ and g has n critical points then g has at most $n+2$ locally asymptotically stable periodic orbits.*

Proof. The idea is to check that if there are too many attracting periodic orbits, then the regions they attract must overlap. Let z be a periodic point of period m and denote by $\mathcal{V}$ the maximal attracting neighborhood of z, i.e., $g^{mn}(y) \to z$ as $n \to \infty$ for all $y \in \mathcal{V}$ and $W(z)$ is the maximal interval inside $\mathcal{V}$ (remark: $\mathcal{V}$ is also called the **basin of attraction** of z). Note we have $g^m(W(z)) \subseteq W(z)$ by construction. Suppose

$$W(z) = (a, b) \quad \text{for } a < b \text{ and } \quad W(z) \cap \partial\mathcal{I} = \emptyset.$$

Under this assumption, we claim that g^m must have a critical point in $W(z)$. Since $W(z)$ is maximal, the map g^m must preserve the endpoints a, b of $W(z)$. There are several cases.

`Case 1:` Suppose $g^m(a) = g^m(b)$, then g^m has a maximum or minimum in $\mathcal{I}$ by the intermediate value theorem and therefore also a critical point.

`Case 2:` Suppose $g^m(a) \neq g^m(b)$ then g^m permutes a, b. Suppose $g^m(a) = a$ and $g^m(b) = b$. Then $(g^m)' \geq 1$ for all points in $\partial\mathcal{V}$ since otherwise a or b would be a locally stable fixed point for g^m, whose basin of attraction overlaps $\mathcal{V}$. By the minimum principle given in Lemma 16.7, we have that if g^m has no critical points in $\mathcal{V}$, then $(g^m)' > 1$ on $\mathcal{V}$. This contradicts $g^m(W(z)) \subseteq W(z)$.

Combining the two cases, we may now conclude that g^m has a critical point $p \in W(z)$. Using the chain rule, this implies that one of the points $p, g(p), \ldots g^{m-1}(p)$ is a critical point of g. So either $W(z)$ is unbounded, or it intersects $\partial\mathcal{I}$, or there is a critical point of g, whose orbit meets $W(z)$. Since there are at most n critical points by assumption and two boundary points of $W(z)$, the maximum number of locally attracting periodic orbits is $n + 2$. □

Example 16.9. Consider the logistic map $g(x) = px(1-x)$. Then $Sg(x) = -6/(1-2x)^2$ for $x \neq 1/2$ and $Sg(1/2) = -\infty$ at the single critical point. Since $Sg < 0$ for $x \in \mathcal{I} = [0,1]$, it follows that the logistic map has at most three locally asymptotically stable periodic orbits. ♦

Singer's Theorem 16.8 motivates us that we should look closer at different notions of attractivity and the concept of an attractor, which is going to be considered in Section 18.

Exercises for Section 16

E16.1 Consider the logistic map $g(x) = px(1-x)$ on the unit interval $[0,1]$ with $p \in (1,3)$. In this case, all initial conditions $x_0 \in (0,1)$ converge to the fixed point $\frac{p-1}{p}$. (a) Compute the Lyapunov exponent $l(x_0)$ for $x_0 \in (0,1)$. (b) Plot a bifurcation diagram for the logistic map, analogously to Figure 39(a), for $p \in [0,4]$. (c) Numerically compute the Lyapunov exponent of the logistic map for all $p \in [0,4]$. Compare your results with the bifurcation diagram.

E16.2 Let $g \in C^0([0,1],[0,1])$ be a unimodal map and let $x_{\max}$ be its unique local point of maximum. Prove the following: (a) If $g(x_{\max}) \leq x_{\max}$ then all solutions tend to fixed points, which lie in $[0, g(x_{\max})]$. (b) If $g(x_{\max}) > x_{\max}$ then there is a unique fixed point $x_0 \in \big(x_{\max}, g(x_{\max})\big)$. (c) If $g(x_{\max}) > x_{\max}$ then orbits either tend to fixed points in $[0, g^2(x_{\max})]$ or are attracted into $[g^2(x_{\max}), g(x_{\max})]$. (d) If g has a fixed point $x_0 \in \big(x_{\max}, g(x_{\max})\big)$ then there are $x_{-1} \in (0, x_{\max})$ and $x_{-2} \in (x_0, 1)$ such that $g(x_{-2}) = x_{-1}$ and $g(x_{-1}) = x_0$.

17 Renormalization

Renormalization is a well-established tool in dynamical systems theory. Here we illustrate it in two contexts, the second one concerns the question about generating sequences upon parameter variation for periodic orbits in unimodal maps as raised in Section 16.

Definition 17.1. Let $(\mathcal{X}, \mathcal{T}, \phi_t)$ be a dynamical system. A **renormalization group (RG)** is a group $\mathcal{R}$ together with an induced group action by $\mathcal{R}$ on the dynamical system.

Definition 17.1 is *not classical* but convenient here; in fact, it is difficult to give a very general definition of renormalization. As we shall see, $\mathcal{R}$ is often required to have connections to **scaling** and/or **self-similarity**. The action of $\mathcal{R}$ can change the following: the phase space $\mathcal{X}$, the phase space $\mathcal{X}$ viewed as space of initial conditions, the time domain $\mathcal{T}$, and the flow ϕ_t. The action on ϕ_t is often defined by renormalizing an underlying map g or a vector field f. Of course, a parameter space $\mathcal{P}$ can be renormalized as well since we can always view $\mathcal{X}_\mathcal{P} := \mathcal{X} \times \mathcal{P}$ as the phase space with $\phi_t|_\mathcal{P} = \text{Id}$.

Example 17.2. Consider the fold normal form for a map (cf. Theorem 8.12)

$$y \mapsto q + y - y^2 = g(y; q) = g(y), \qquad y \in \mathbb{R},\ q \in \mathbb{R}. \tag{17.1}$$

Assume $q < 0$ is small, so that (17.1) has no fixed points, yet is close to bifurcation. Iterates passing from $y > 0$ to $y < 0$ need a long time as the map is small near $y = 0$. This effect is known as **critical slowing down** (or **intermittency**). We want to formally determine the number of iterates $N(q)$ to pass near zero asymptotically as $q \nearrow 0$, $y \to 0$. Consider the second iterate map

$$\begin{aligned} g^2(y;q) = (g \circ g)(y;q) &= q + (q + y - y^2) - (q + y - y^2)^2 \\ &= (2-q)q + (1-2q)y - (2-2q)y^2 + 2y^3 - y^4 \\ &= 2q + y - 2y^2 + \mathcal{O}(q^2, qy, qy^2, y^3) \approx 2q + y - 2y^2. \end{aligned}$$

If $N(q)$ iterates are required for g, then approximately $N(q)/2$ iterates are required for g^2 for the same passage near $y = 0$. Let $R_\alpha(y) := \alpha y =: Y$ and $S_\beta := \beta q =: Q$ be scalings. For $\alpha = 2$ and $\beta = 4$ define a transformation

$$R_2 g^2((R_2)^{-1}Y, (S_4)^{-1}Q) = 2\left[g^2(Y/2; Q/4)\right] \approx Q + Y - Y^2,$$

i.e., the scaling has brought back the mapping, at least approximately, into the (self-)similar fold normal form; cf. to scaling and blow-up methods in Section 11. However, this means we must have

$$\frac{N(q)}{2} \approx N(4q) \qquad \Rightarrow N(q) = \mathcal{O}(q^{-1/2}) \quad \text{as } q \nearrow 0$$

giving our desired passage-time asymptotic approximation. $(\mathbb{R}^+, \cdot) =: \mathcal{G}$ is a group so we write the RG action on the space of continuous one-parameter interval maps $\mathcal{Z} := C^0(\mathcal{I} \times \mathbb{R}, \mathcal{I} \times \mathbb{R})$ as

$$\mathcal{G} \times \mathcal{G} \times \mathcal{Z} \overset{\mathcal{R}}{\mapsto} \mathcal{Z}, \qquad \mathcal{R}(g) = R_\alpha g(R_\alpha^{-1} \cdot, S_\beta^{-1} \cdot),\ g \in \mathcal{Z}.$$

More abstractly, instead of guessing the scales $\alpha = 2$, $\beta = 4$, and using approximations, one may study exact solutions to the **functional equation**

$$(\mathcal{D}^{(2)}g)(y;p) := g^2(y;p) = R_\alpha g(R_\alpha^{-1}y, S_\beta^{-1}q),$$

which is an operator equation for g involving the **doubling operator** $\mathcal{D}^{(2)}$. This is very similar to functional iteration schemes we used to prove existence of invariant manifolds in Sections 3 and 10. ♦

Example 17.2 explains the notions of renormalization, scale, and self-similarity. The example also crucially used the normal form structure of the fold to determine the scales α and β.

Example 17.3. Consider the logistic map (1.4) again

$$x_{k+1} = px_k(1 - x_k) = g(x_k; p) = g(x_k), \qquad p \in (0, 4]. \tag{17.2}$$

One checks the following (exercise!): g has a stable fixed point $\gamma_0 = (p-1)/p$ for $p \in (1, 3] = (p_0, p_1]$ with multiplier $g'(\gamma_0) = 2 - p$. A flip (or period-doubling) bifurcation occurs at p_1 and there is a stable two-cycle $\gamma_1 = \{\gamma_{11}, \gamma_{12}\}$ for $p \in (3, 1 + \sqrt{6}] = (p_1, p_2]$ with multiplier

$$\frac{\partial g^2}{\partial x}(\gamma_{11}) = \frac{\partial g}{\partial x}(\gamma_{11})\frac{\partial g}{\partial x}(\gamma_{12}) = 4 + 2p - p^2. \tag{17.3}$$

Numerically, one easily obtains that the two-cycle bifurcates in a further flip bifurcation at some $p_2 > p_1$, the resulting four-cycle γ_2 bifurcates again at some $p_3 > p_2$, leading to a **period-doubling sequence** (or **period-doubling cascade**) of bifurcation points $\{p_n\}_{n=0}^{\infty}$ as shown in Figure 40; the cascade is also known as the **period-doubling route-to-chaos**. ♦

To analyze the structure of the period-doubling cascade it helps to consider special cycles.

Definition 17.4. A cycle (or fixed point) of an iterated map is called **superstable** if all its multipliers are zero.

Example 17.5. Continuing Example 17.3, we find γ_0 is superstable for $p = P_0 = 2$ and γ_1 is superstable for $p = P_1 = 1 + \sqrt{5}$ (exercise!). Numerics indicates the existence of a sequence of superstable cycles at parameters $\{P_n\}_{n=0}^{\infty}$; see Figure 40. Generalizing the idea from (17.3) in using the chain rule, we get for the 2^n-cycle, $n \in \mathbb{N}_0$, that

$$\frac{\partial g^{2^n}}{\partial x}(\gamma_{nj}, P_n) = 0 \quad \Leftrightarrow \quad \exists j : \gamma_{nj} = \frac{1}{2} =: x_{\max} \quad \Rightarrow g^{2^n}(x_{\max}, P_n) = x_{\max},$$

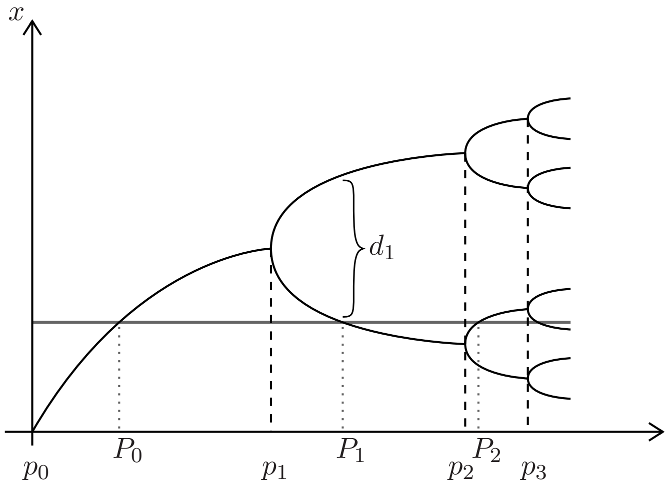

Figure 40: Sketch of the bifurcation diagram of the logistic map (17.2) with bifurcation points at p_j and superstable cycles at P_j; see also Examples 17.3, 17.5, and 17.7.

i.e., the local maximum of g at $x_{\max}$ has to belong to any superstable cycle. Therefore, a superstable cycle occurs precisely at intersection of the line $\{x = x_{\max}, p \in (0,4]\}$ with the bifurcation diagram in Figure 40. Now define

$$d_n = g^{2^{n-1}}(x_{\max}, P_n) - x_{\max}, \tag{17.4}$$

which is actually the closest point on γ_n to the maximum at $x_{\max}$ (non-trivial exercise!); see Figure 40. ♦

The previous construction does not depend upon the precise shape of the logistic map. It can be carried out for large classes of maps:

Definition 17.6. Let $\mathcal{I} = [0,1]$. Then $g \in C^0(\mathcal{I},\mathcal{I})$ is called **unimodal** if $g(0) = 0 = g(1)$ and g has a unique local maximum $x_{\max} \in (0,1)$. If in addition, g has negative Schwarzian derivative on $\mathcal{I}$, it is called **S-unimodal**.

Example 17.7. Continuing Example 17.5, Feigenbaum made several numerical (and partially analytical) observations. First, there exist limits

$$p_\infty := \lim_{n\to\infty} p_n, \qquad \lim_{n\to\infty} \frac{p_n - p_{n-1}}{p_{n+1} - p_n} = \delta_{\mathrm{F}} \approx 4.6692\cdots,$$

where δ_{F} is actually **universal**, i.e., the constant did not seem to depend on the S-unimodal map he considered. Second, the constant is linked to

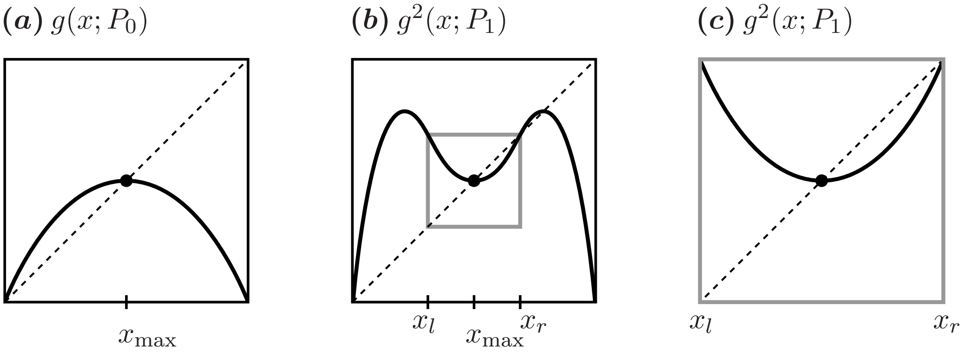

Figure 41: Renormalization of a unimodal map g. (a) map g; (b) second-iterate map $g^2 = g \circ g$; (c) zoom into (b) near the fixed point. Note that (c) is self-similar up to reflection and translation to the original map (a).

superstable cycles via the result

$$P_n = p_\infty - \kappa_1(\delta_\mathrm{F})^{-n} + o((\delta_\mathrm{F})^{-n}) \qquad \text{as } n \to \infty,$$

where $\kappa_1 > 0$ is not universal, i.e., depends upon the particular map. Third, even for the phase space measure d_n defined in (17.4) one finds a scaling law

$$d_n = \kappa_2(\alpha_\mathrm{F})^{-n}, \quad \alpha_\mathrm{F} \approx -2.5029\cdots, \qquad \text{as } n \to \infty,$$

where $\kappa_2 > 0$ is not universal but α_F is. The constants $\delta_\mathrm{F}, \alpha_\mathrm{F}$ are known as **Feigenbaum constants**. The key argument to establish Feigenbaum's universality observations is again renormalization. Consider Figure 41, and observe that the graphs of

$$g(x, P_0) \quad \text{and} \quad g^2(x, P_1)$$

look similar once we zoom into a certain region around $x = x_{\max}$ and reflect the graphs shown in Figure 41. In fact, $x_{\max}$ is a superstable fixed point of both maps. ⧫

Example 17.7 suggests for a general S-unimodal map $g = g(x;p)$ with superstable cycle sequence P_n to consider the renormalization groups $(\mathbb{R}^-, \cdot)$, $(\mathbb{R}, +)$ with actions R_α and T_β given by multiplication by α and translation by β. More precisely, we consider the self-similar approximation

$$g(x, P_0) \approx \alpha g^2\left(\frac{x}{\alpha}; P_1\right), \qquad P_1 = P_0 + \beta_0, \tag{17.5}$$

i.e., we claim the second iterate is, upon conjugacy by scaling induced by R_α and shifting by T_{β_0} for β_0 chosen to yield the next superstable parameter value; cf. Example 17.2. Upon iterating (17.5), we get

$$g(x, P_0) \approx \alpha^n g^{2^n}\left(\frac{x}{\alpha^n}; P_n\right), \tag{17.6}$$

so if we start at $P_\infty =: P_n$, i.e., at the *onset of chaos*, we get the functional equation for just the first step as

$$g(x) = R_\alpha(\mathcal{D}^{(2)}g)(R_{1/\alpha}x; P_\infty) = R_\alpha g^2(R_{1/\alpha}x; P_\infty). \tag{17.7}$$

One has to augment (17.7) with boundary conditions to solve it. Upon translating *a priori* to the origin and adjusting the scale of x we see that $g'(0) = 0$ and $g(0) = 1$ are reasonable conditions. The fixed point problem for g turns out to be solvable (very difficult proof!) if and only if $\alpha = \alpha_{\mathrm{F}}$, thereby determining α_{F}.

Remark: Linearizing the operator on the right-hand side of (17.7) using a Fréchet derivative on a suitable function space shows that the fixed point g has one unstable eigenvalue, which is precisely given by δ_{F}. Hence, the renormalization group and the functional equation (17.7) indeed contain full information about the universal Feigenbaum constants.

Exercises for Section 17

E17.1 Show that the logistic map f, and the tent map g, given by

$$f, g : [0,1] \to [0,1]\,,\ f(x) = 4x(1-x)\,,\ g(x) = \begin{cases} 2x, & x \in [0, \frac{1}{2}], \\ 2(1-x), & x \in [\frac{1}{2}, 1], \end{cases}$$

are topologically conjugate via the conjugacy $h(x) = \sin^2(\frac{1}{2}\pi x) = \frac{1}{2}(1 - \cos(\pi x))$.

E17.2 Consider the ODEs

$$x' = p + f(x), \quad x = x(t) \in \mathbb{R},\ p \in \mathbb{R}.$$

For $x \geq 0$, we consider three cases (I) $f(x) = \sqrt{x}$, (II) $f(x) = x$, (III) $f(x) = x^2$, and then extend these vector fields symmetrically to an even function for $x \in \mathbb{R}$. (a) Prove that (I)-(III) all have topologically fold bifurcations in the sense that upon variation of p, two equilibrium points collide and disappear. (b) What can you say regarding critical slowing down for the cases (I)-(III)?

18 Attractors

Having seen that the full orbit structure of one-dimensional maps can be rather involved (Sections 16-17) we now study the attracting parts of phase space in general dynamical systems more abstractly. In this section, we always consider a C^0 dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$, which is well-defined in forward time, i.e., for all $t \geq 0$, e.g., $\mathcal{T} = [0, \infty)$ for flows or $\mathcal{T} = \mathbb{N}_0$ for maps.

Definition 18.1. A compact set $\mathcal{U}$ is called a **trapping region** provided that $\phi_t(\mathcal{U}) \subset \text{int}(\mathcal{U})$ for all $t > 0$.

Example 18.2. Consider the **Lorenz system**, classically written as

$$\begin{aligned} x_1' &= \sigma(x_2 - x_1), \\ x_2' &= \rho x_1 - x_2 - x_1 x_3, \\ x_3' &= x_1 x_2 - \beta x_3, \end{aligned} \tag{18.1}$$

with parameters $\sigma, \rho, \beta > 0$. It is easy to prove [40] that the compact ellipsoid $\mathcal{E} = \mathcal{U}$ with boundary defined by

$$\sigma x_1^2 + x_2^2 + \beta \left(x_3 - \frac{\rho + \sigma}{2} \right)^2 = \frac{\beta(\rho+\sigma)^2}{4} \tag{18.2}$$

is a trapping region for (18.1). ♦

Definition 18.3. A set $\mathcal{A}$ is called an **attracting set** if there exists a trapping region $\mathcal{U}$ such that

$$\mathcal{A} = \bigcap_{t \geq 0} \phi_t(\mathcal{U}), \tag{18.3}$$

see also Figure 42(a). A set $\mathcal{A}$ is called an **attractor** if it is an attracting set and it is **indecomposable**, i.e., if $\tilde{\mathcal{A}} \neq \emptyset$ is an attracting set and $\tilde{\mathcal{A}} \subseteq \mathcal{A}$ then $\tilde{\mathcal{A}} = \mathcal{A}$.

A trapping region $\mathcal{U}$ for an attracting set $\mathcal{A}$ is then also called an **isolating neighborhood** of $\mathcal{A}$. A **repelling set** is an attracting set for the time-reversed dynamical system and similarly one defines via time reversal a **repeller**.

Theorem 18.4. *Consider a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$. Let $\mathcal{V}$ be a trapping region with attracting set $\mathcal{A}$. Then the following hold:*

(T1) $\mathcal{A}$ is closed. If $\mathcal{T} = \mathbb{R}$ or $\mathcal{T} = \mathbb{Z}$, then $\mathcal{A}$ is invariant;

(T2) if $x_0 \in \mathcal{V}$ then $\omega(x_0) \subseteq \mathcal{A}$;

(T3) if $x_* \in \mathcal{A}$ is a hyperbolic steady state and the dynamical system is also defined in backward time, then $W^{\mathrm{u}}(x_*) \subseteq \mathcal{A}$.

Proof. For (T1), one easily sees that $\mathcal{A}$ is closed since it is the intersection of closed sets. Furthermore, if $p \in \mathcal{A}$, then $p \in \phi_t(\mathcal{V})$ for all $t > 0$. One can re-write the definition (18.3) of $\mathcal{A}$ as

$$\mathcal{A} = \bigcap_{t \geq -s} \phi_{t+s}(\mathcal{V}) = \bigcap_{t \geq -s} \phi_t(\mathcal{V})$$

since $\mathcal{V}$ is a trapping region so that $\phi_{-s}(p) \in \mathcal{A}$. (T2) is basically immediate from the definitions (exercise!). Regarding (T3), since $x_* \in \mathcal{A}$ it follows that $x_* \in \text{int}(\mathcal{V})$ by the definition of a trapping region, which has to be mapped into its *interior* via ϕ_t. This implies that the local unstable manifold $W^{\text{u}}_{\text{loc}}(x_*)$ is contained in $\mathcal{V}$ for a sufficiently small neighborhood near x_*. But then we may simply use the definition of the (global) unstable manifold

$$\begin{aligned} W^{\text{u}}(x_*) &= \bigcup_{t\geq 0} \phi_t(W^{\text{u}}_{\text{loc}}(x_*)) = \bigcap_{s\geq 0}\bigcup_{t\geq 0} \phi_s \circ \phi_t \circ \underbrace{\phi_{-s}(W^{\text{u}}_{\text{loc}}(x_*))}_{\subset W^{\text{u}}_{\text{loc}}(x_*)\subset \mathcal{V}} \\ &\subset \bigcap_{s\geq 0}\bigcup_{t\geq 0} \phi_s \circ \phi_t(\mathcal{V}) = \bigcap_{s\geq 0} \phi_s(\mathcal{V}) = \mathcal{A} \end{aligned}$$

to finish the proof. $\square$

Although Definition 18.3 is very natural, i.e., just taking the forward intersection of sets under the dynamics, it may not be appropriate for all situations as the next example shows.

Example 18.5. Consider the circle $\mathbb{S}^1 = [0,1]/(0 \sim 1)$ and let $\mathcal{X} = (0,\infty) \times \mathbb{S}^1$ be the punctured plane. Consider the vector field

$$\begin{aligned} r' &= 1 - r, \\ \theta' &= -\left(\theta - \tfrac{1}{4}\right)\left(\theta - \tfrac{3}{4}\right) \end{aligned} \tag{18.4}$$

in polar coordinates. Clearly, there is a trapping region, say $\mathcal{U} = \mathbb{S}^1 \times [1/2, 3/2]$ containing the unit circle and there are two equilibrium points $(1, 1/4)$ and $(1, 3/4)$. One checks that $\mathcal{A} = \{(r,\theta) \in \mathcal{X} : r = 1\}$ is an attractor but all initial conditions except for $(1, 3/4)$ are attracted to $(1, 1/4)$. Therefore, it might be more natural to use a definition to make $(1, 1/4)$ the only attractor for the given trapping region $\mathcal{U}$; see also Figure 42(b). ♦

Definition 18.6. Let $\mathcal{A}$ be a closed invariant non-empty set. The **basin of attraction** is

$$\text{BA}(\mathcal{A}) := \{x_0 \in \mathcal{X} : \omega(x_0) \subset \mathcal{A}\}.$$

Definition 18.7. Let $\mathcal{A}$ be a closed invariant set and suppose μ is a given measure on $\mathcal{X}$. $\mathcal{A}$ is called a **Milnor attractor** if

$$\mu(\text{BA}(\mathcal{A})) > 0$$

and there is no smaller set $\tilde{\mathcal{A}} \subset \mathcal{A}$ such that $\mu(\text{BA}(\mathcal{A}) \setminus \text{BA}(\tilde{\mathcal{A}})) = 0$.

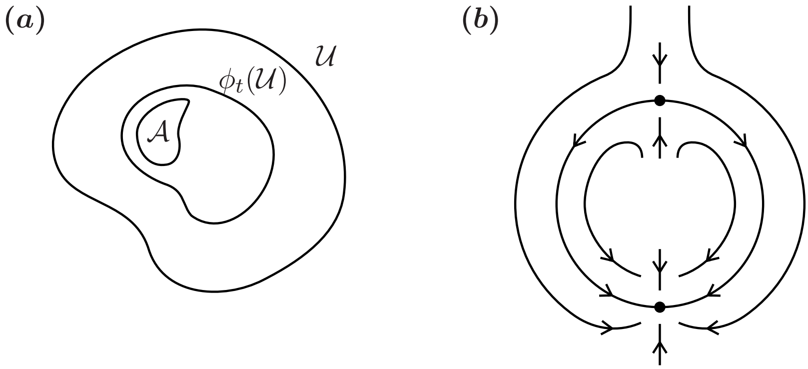

Figure 42: (a) General idea of an attractor $\mathcal{A}$ inside a set $\mathcal{U}$, i.e., the flow ϕ_t contracts everything towards/onto $\mathcal{A}$ for $t > 0$. (b) Illustration of a key example to understand the different attractor definitions.

This definition fixes some intuition problems but also creates new ones.

Example 18.8. Regarding the ODE (18.4), we see that $(1, 1/4)$ is its Milnor attractor. However, if we look at

$$x' = -x^2, \qquad x = x(t) \in \mathbb{R},$$

then $x_* = 0$ is a Milnor attractor, despite the fact that it is (nonlinearly) unstable as an equilibrium point. ⧫

Here, we shall simply use the term attractor from Definition 18.3 from now on but warn the reader that there are variations of attractor definitions in use. As expected from Sections 13-17, the topology/geometry of attractors can be extremely intricate.

Example 18.9. Consider the solid torus $\mathbb{S}^1 \times \mathbb{D}^2$, where $\mathbb{D}^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ is the unit disc in $\mathbb{R}^2$. For $\lambda \in (0, 1/2)$, define the map

$$g(\theta, x, y) := \left(2\theta, \lambda x + \frac{1}{2}\cos(2\pi\theta), \lambda y + \frac{1}{2}\sin(2\pi\theta)\right), \qquad (18.5)$$

which stretches the solid torus in the $\mathbb{S}^1$-direction by a factor of 2 and compresses in the direction of the disk $\mathbb{D}^2$ by a factor λ and wraps the resulting torus twice back inside the original one as shown in Figure 43.

The forward iteration of the map g yields tori stacked inside themselves as shown on a cross-section in Figure 43. One may prove (exercise!) that

$$\mathcal{A} = \bigcap_{k=0}^{\infty} g^k(\mathbb{S}^1 \times \mathbb{D}^2)$$

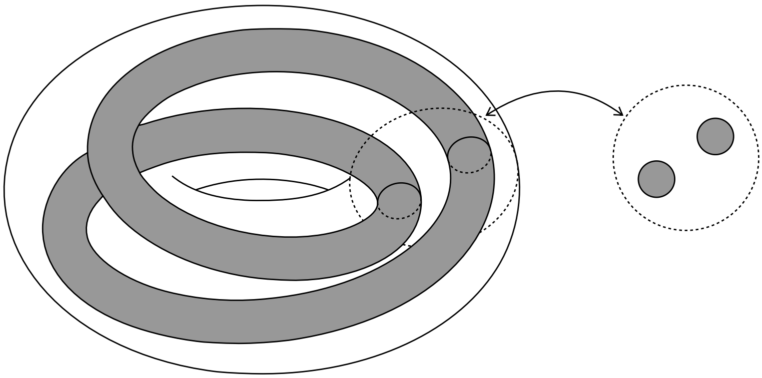

Figure 43: Illustration of the Smale-Williams solenoid showing one forward iteration from the solid torus to the double-wrapped torus of the map (18.5). On the right-hand side a cross-section is shown.

is a non-empty attractor and is the product of a Cantor set with an interval. The set $\mathcal{A}$ is called the **Smale-Williams solenoid**. ♦

Example 18.10. Continuing the Lorenz equations in Example 18.2 with trapping region $\mathcal{U}$, we immediately obtain an attractor $\mathcal{A}$ via Definition 18.3. For the so-called classical Lorenz parameter values $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, an illustration of the attractor can be found in [40]. ♦

Definition 18.11. Let $\mathcal{A}$ be an attractor. Then $\mathcal{A}$ is called

(D1) **chaotic attractor** if $\mathcal{A}$ is also a chaotic set (see Definition 15.7);

(D2) **hyperbolic attractor** if $\mathcal{A}$ is also a hyperbolic set; (see Definition 7.1);

(D3) **strange attractor** if $\mathcal{A}$ is a chaotic but not a hyperbolic set.

One checks relatively easily that the Smale-Williams solenoid is a hyperbolic attractor and with a bit more work that it is chaotic. Although it may be evident geometrically, it is highly non-trivial to show that the Lorenz attractor is indeed a strange attractor. In fact, the next result shows already some of the difficulties.

Theorem 18.12. *The Lorenz attractor has volume zero.*

Proof. Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be the vector field associated to the ODE (18.1). Let $\mathcal{E}$ be the trapping region/ellipsoid and define

$$\mathcal{E}(t) := \{\phi_t(x_0) : x_0 \in \mathcal{E}\}.$$

Let $V(t)$ be the classical (Riemannian) volume of $\mathcal{E}(t)$ as a set inside $\mathbb{R}^3$. Then we have

$$\frac{\mathrm{d}}{\mathrm{d}t}V(t) = \frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{E}(t)} \mathrm{d}x = \int_{\partial\mathcal{E}(t)} \vec{n}\cdot f(x)\ \mathrm{d}A,$$

where $\mathrm{d}A$ is the surface area form for $\partial\mathcal{E}(t)$ and $\vec{n}$ is the associated unit normal vector. Hence, the Divergence Theorem implies

$$\frac{\mathrm{d}}{\mathrm{d}t}V(t) = \int_{\mathcal{E}(t)} \nabla\cdot f(x)\ \mathrm{d}x. \tag{18.6}$$

The divergence can easily be calculated from (18.1) and we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}V(t) = -(\sigma+1+\beta)V(t), \tag{18.7}$$

which implies the result by solving (18.7). $\square$

Remark: The change-of-volume formula (18.6) under a vector field f is valid in far more generality if f and $\mathcal{E}(t)$ are smooth in $\mathbb{R}^d$. It is then proved using Stokes' Theorem instead of the Divergence Theorem.

For the Smale-Williams solenoid, one may prove that it has three-dimensional volume equal to zero. These observations motivate an alternative definition for a strange attractor $\mathcal{A}$, i.e., that $\mathcal{A}$ should be chaotic and $\mathcal{A}$ should satisfy a certain "dimension requirement". To make this precise, we need a suitable definition of dimension.

Definition 18.13. Let $\mathcal{S}$ be a subset in $\mathbb{R}^d$. Let $N(r,\mathcal{S})$ be the minimum number of unit cubes/boxes of side length r needed to cover $\mathcal{S}$. Then define the **box-counting dimension** by

$$\dim_B(\mathcal{S}) := \lim_{r\to 0}\frac{\ln N(r,\mathcal{S})}{\ln(1/r)}.$$

For example, we can cover the unit square $[0,1]^2$ by $N=2^{2j}$ boxes of side length $r=2^{-j}$, so taking the limit $j\to\infty$ yields that $\dim_B([0,1]^2)=2$.

Example 18.14. To show that $\dim_B(\mathcal{S})$ does capture non-integer values, consider the classical middle-third Cantor set. It is obtained by removing the middle third $(1/3,2/3)$ from $[0,1]$, then removing the middle thirds from $\mathcal{K}_1=[0,1/3]\cup[2/3,1]$ to create

$$\mathcal{K}_2 = \left[0,\frac{1}{9}\right]\cup\left[\frac{2}{9},\frac{1}{3}\right]\cup\left[\frac{2}{3},\frac{7}{9}\right]\cup\left[\frac{8}{9},1\right],$$

and then repeating the process iteratively to create $\mathcal{K}_j$ and the resulting Cantor set

$$\mathcal{K} = \bigcap_{j\geq 0} \mathcal{K}_j.$$

At level j for $\mathcal{K}_j$, we need 2^j boxes/intervals of length $(1/3)^j$. Hence, we obtain

$$\dim_{\mathrm{B}}(\mathcal{K}) = \lim_{j\to\infty} \frac{\ln(2^j)}{\ln(3^j)} = \frac{\ln 2}{\ln 3},$$

which is approximately equal to 0.6309, i.e., certainly not an integer. ♦

Recall that we already encountered Cantor sets as invariant sets for dynamics in the case of the Smale-Williams solenoid and the Smale horseshoe. In addition to box-counting dimension, there are several other notions, which we shall not discuss here, such as **Lyapunov dimension**, **Hausdorff dimension**, and **correlation dimension**, which can also capture non-integer dimensions. An alternative definition of a **strange attractor** is a chaotic attractor, which is also a **fractal set**, i.e., of non-integer dimension. Yet, here we shall always use the more standard Definition 18.11.

Exercises for Section 18

E18.1 Consider the system

$$\begin{aligned} x' &= x - x^3, \\ y' &= -y. \end{aligned}$$

(a) Show that the set $Q = [-2,2] \times [-1,1] \subset \mathbb{R}^2$ is a trapping region. Determine the corresponding attracting set. (b) Find an attractor of the system. Is it also a Milnor attractor?

E18.2 Let $(\mathcal{X}, \Sigma, \mu)$ be a measure space and consider a continuous flow ϕ_t on $\mathcal{X}$. A compact invariant set $\mathcal{A} \subset \mathcal{X}$ is called a **weak Milnor attractor** if $\mu\big(\mathrm{BA}(\mathcal{A})\big) > 0$. It is clear that a Milnor attractor is also a weak Milnor attractor. Prove the following: (a) If a weak Milnor attractor $\mathcal{A}$ exists, then the set $\mathcal{A}_m$ defined by

$$\mathcal{A}_m = \bigcap_{E =_0 \mathrm{BA}(\mathcal{A})} \Omega(E), \qquad \text{where} \qquad \Omega(E) = \overline{\bigcup_{x\in E} \omega(x)},$$

satisfies $\mathcal{A}_m \subset \mathcal{A}$ and $\mathcal{A}_m$ is a Milnor attractor. The symbol $E =_0 F$ means that E and F coincide except for a set of measure zero. (b) If two weak attractors contain a single common Milnor attractor, they do not need to contain each other (find a counterexample). (c) If

$\mathcal{X} \subset \mathbb{R}^d$ is compact with Lebesgue measure $0 < \mathrm{Leb}(\mathcal{X}) < \infty$, and $\mathcal{A}$ is a weak Milnor attractor for ϕ_t with respect to the Lebesgue measure, then for all $\varepsilon > 0$ there exists a closed set $\mathcal{U}_\varepsilon \subset \mathrm{BA}(\mathcal{A})$ with $\mathrm{Leb}(\mathcal{U}_\varepsilon) > \mathrm{Leb}(\mathrm{BA}(\mathcal{A})) - \varepsilon$ that is uniformly attracted to $\mathcal{A}$, namely

$$\lim_{t\to\infty} d(\phi_t(\mathcal{U}_\varepsilon), \mathcal{A}) = 0,$$

where d is the Hausdorff semidistance induced by the Euclidean distance.

19 Conley's Theorem

In Section 18, we studied dynamics converging to attractors. Within an attractor, extremely complicated recurrent dynamics can occur, e.g., chaos. In this section we prove that dynamical systems generally consist of these two parts: a "gradient-like" part and a "recurrent" part. Gradient-like dynamics basically means decreasing along level sets of a Lyapunov function as studied in Section 2; cf. Definition 2.3.

Definition 19.1. Let $g : \mathcal{X} \to \mathcal{X}$ be a map on a metric space $(\mathcal{X}, d)$. A point $x \in \mathcal{X}$ is called **chain recurrent** if for any $\delta > 0$, there exists a finite sequence of points $\{x_k\}_{k=0}^n$ with $x_0 = x = x_n$ such that

$$d(g(x_k), x_{k+1}) < \delta \quad \text{for all } k \in \{0, 1, \dots, n-1\}. \tag{19.1}$$

The set of chain recurrent points is denoted by $\mathrm{CR}(g)$.

The condition (19.1) is also referred to as the existence of a δ**-chain** from x to x, or otherwise put that there exists a finite δ-pseudo-orbit from x to x for any $\delta > 0$; see also Definition 7.7. Chain recurrence is a weaker notion than non-wandering, i.e., $\mathrm{NW}(g) \subseteq \mathrm{CR}(g)$; cf. Definition 6.4.

Definition 19.2. Two points $x, y \in \mathrm{CR}(g)$ are called **chain equivalent** if for any $\delta > 0$, there exist δ-chains from x to y, and from y to x.

One checks (exercise!) that chain equivalence is an equivalence relation which subdivides $\mathrm{CR}(g)$ into **chain classes**. We can now state Conley's Theorem, which we are going to expose in the context of maps; see also Figure TODO.

Theorem 19.3 (Conley's Theorem[1])**.** *Let $g : \mathcal{X} \to \mathcal{X}$ be a homeomorphism on a compact metric space $(\mathcal{X}, d)$. Then there exists $L \in C(\mathcal{X}, \mathbb{R})$ such that:*

[1] Sometimes Conley's Theorem is called "Fundamental Theorem of Dynamical Systems". This naming convention is an over-emphasis and should be avoided. The theorem is abstractly very elegant and conceptually helpful, yet it lacks true practical applicability.

(T1) $x \notin \mathrm{CR}(g)$ *implies* $L(g(x)) < L(x)$.

(T2) *For any* $x, y \in \mathrm{CR}(g)$, *we have* $L(x) = L(y)$ *if and only if* x *and* y *are chain equivalent.*

(T3) $L(\mathrm{CR}(g))$ *is a compact nowhere dense subset of* $\mathbb{R}$.

We are going to prepare the proof in several steps, starting with the relation of Lyapunov functions to an attractor $\mathcal{A}$ for g with isolating neighborhood $\mathcal{U}$. Define the **dual repeller**

$$\mathcal{A}^* := \bigcap_{k \geq 0} g^{-k}(\mathcal{X} - \bar{\mathcal{U}}).$$

Lemma 19.4. *There exists* $L \in C(\mathcal{X}, [0,1])$ *such that* $L|_{\mathcal{A}^*} = 1$, $L|_{\mathcal{A}} = 0$, *and* $L(x) \in (0,1)$ *for all* $x \notin \mathcal{A} \cup \mathcal{A}^*$. *Furthermore,* $L(g(x)) < L(x)$ *for any* $x \notin \mathcal{A} \cup \mathcal{A}^*$.

Proof. Consider an isolating neighborhood $\mathcal{U}$ of $\mathcal{A}$. Since g maps $\mathcal{U}$ properly into its interior, there exists a function $\alpha : \mathcal{X} \to [0,1]$ such that

$$\alpha(\mathcal{X} \setminus \mathcal{U}) = 1, \quad \alpha(g(\bar{\mathcal{U}})) = 0, \quad \alpha(x) \in (0,1) \ \forall x \in \mathcal{U} \setminus g(\bar{\mathcal{U}}).$$

Let $x \notin \mathcal{A} \cup \mathcal{A}^*$, which implies that $\mathrm{Or}(x)$ intersects $\mathcal{U} \setminus g(\bar{\mathcal{U}})$ precisely once. Hence, evaluating α on $\mathrm{Or}(x)$ yields a non-increasing sequence starting with ones and eventually ending with all zeros. Denote this sequence by $\alpha_k(x) = \alpha(g^k(x))$. Take another sequence a_k with $\sum_{k=-\infty}^{\infty} a_k = 1$, assume $(a_k) \in l^1(\mathbb{Z})$, and define the Lyapunov function via the linear combination

$$L(x) := \sum_{k=-\infty}^{\infty} a_k \alpha_k(x).$$

Since each $\alpha_k(x)$ is non-increasing along orbits, so is $L(x)$. By construction we have $L|_{\mathcal{A}^*} \equiv 1$ and $L|_{\mathcal{A}} \equiv 0$. Since the $\alpha_k(x)$ change from one to zero, we find $L(x) \in (0,1)$. Furthermore, for each x, there exists at least one k such that $\alpha_k(g(x)) < \alpha_k(x)$, so we can conclude $L(g(x)) < L(x)$ for any $x \notin \mathcal{A} \cup \mathcal{A}^*$. $\square$

Next, we have a closer look at the relation between attractors, their dual repellers and the chain recurrent set. It is easy to check (exercise!) that the compactness of $\mathcal{X}$ implies that g can only have a countable number of attractors, say $\{\mathcal{A}_i\}_{i=1}^{\infty}$.

Lemma 19.5. $\mathrm{CR}(g) = \bigcap_{i=1}^{\infty} (\mathcal{A}_i \cup \mathcal{A}_i^*)$.

Proof. `Step 1:` "$\subset$". This inclusion is equivalent to proving that $x \notin \mathcal{A} \cup \mathcal{A}^*$ for an attractor $\mathcal{A}$ implies $x \notin \mathrm{CR}(g)$. Fix an isolating neighborhood $\mathcal{U}$ of $\mathcal{A}$. Observe that every orbit in $\mathcal{X} \setminus (\mathcal{A} \cup \mathcal{A}^*)$ meets $\mathcal{U} - g(\mathcal{U})$ exactly once, so we can assume $x \in \mathcal{U} - g(\mathcal{U})$ without loss of generality. Take $\delta_0 > 0$ so that any δ_0-chain $x_0 = x, x_1, x_2$ must have $x_2 \in g^2(\mathcal{U})$. Furthermore, we observe that there exists a small $\delta_1 > 0$ such that no δ_1-chain can go from $g^2(\mathcal{U})$ to $\mathcal{X} \setminus g(\mathcal{U})$; see Figure TODO. Now define $\delta := \min\{\delta_0, \delta_1\}$. If there would exist a δ-chain from x to x, then $x_2 = x$ must lie in $g^2(\mathcal{U})$ and its complement simultaneously, which is impossible so $x \notin \mathrm{CR}(g)$.

`Step 2:` "$\supset$". Consider any $x \in \mathcal{X}$, $\delta > 0$, and define the open set

$$\mathrm{Ch}(x, \delta) := \{y \in \mathcal{X} : \exists \text{ a } \delta\text{-chain from } x \text{ to } y\}.$$

By construction we have $g(\mathrm{Ch}(x, \delta)) \subset \mathrm{Ch}(x, \delta)$ as the $\delta/2$-neighborhood of $g(\mathrm{Ch}(x, \delta))$ is contained in $\mathrm{Ch}(x, \delta)$. In particular, $\mathrm{Ch}(x, \delta)$ is a trapping region. As above, we re-formulate the inclusion condition and consider $x \notin \mathrm{CR}(g)$. Then there exists δ_0 such that $x \notin \mathrm{Ch}(x, \delta_0)$ but trivially we have $g(x) \in \mathrm{Ch}(x, \delta_0)$. Now let $\mathcal{A}$ be an attractor with isolating neighborhood $\mathrm{Ch}(x, \delta_0)$ and dual repeller $\mathcal{A}^*$. By the previous arguments, we have $x \notin \mathcal{A} \cup \mathcal{A}^*$, which finishes the proof. $\square$

The last lemma allows us to give a nice characterization of chain classes.

Lemma 19.6. *If $x, y \in \mathrm{CR}(g)$, then x and y are in the same chain class if and only if for all i we have either $x, y \in \mathcal{A}_i$ or $x, y \in \mathcal{A}_i^*$.*

Proof. We argue using proof by contraposition. If x, y are not in the same chain class, there exists $\delta_0 > 0$ such that either $y \notin \mathrm{Ch}(x, \delta_0)$ or $x \notin \mathrm{Ch}(y, \delta_0)$. Without loss of generality, we consider $y \notin \mathrm{Ch}(x, \delta_0)$. Let $\mathcal{A}_i$ be the attractor with trapping region $\mathrm{Ch}(x, \delta_0)$ and with dual repeller $\mathcal{A}_i^*$. By Lemma 19.5, it follows that $x, y \in \mathcal{A}_i \cup \mathcal{A}_i^*$. This entails $x \in \mathcal{A}_i$ and $y \in \mathcal{A}_i^*$ and yields the first implication. For the reverse implication, suppose for some i, we have $x \in \mathcal{A}_i$ and $y \in \mathcal{A}_i^*$ but then we can find a $\delta_0 > 0$ so that no δ_0-chain can go from x to y. This implies the contrapositive statement for the second implication. $\square$

Proof. (of Theorem 19.3) We start with (T1). For each $i \in \mathbb{N}$, Lemma 19.4 implies the existence of a function $L_i : \mathcal{X} \to \mathbb{R}$ such that $L_i^{-1}(1) = \mathcal{A}_i^*$ and $L_i^{-1}(0) = \mathcal{A}_i$, and L_i is strictly decreasing along orbits in $\mathcal{X} \setminus (A_i \cup A_i^*)$. The trick is now to define

$$L : \mathcal{X} \to \mathbb{R}, \qquad L(x) := \sum_{i=1}^{\infty} \frac{2L_i(x)}{3^i}.$$

Clearly, L is continuous by continuity of each L_i. To show that L is actually decreasing outside of $\mathrm{CR}(g)$, take any $x \notin \mathrm{CR}(g)$. By Lemma 19.5, there exists some i such that $x \notin \mathcal{A}_i \cup \mathcal{A}_i^*$. This entails $L(g(x)) < L(x)$ and finishes (T1). Regarding (T2), if x, y are chain equivalent then this trivially yields $L(x) = L(y)$. Conversely, take $x, y \in \mathrm{CR}(g)$ and suppose $L(x) = L(y)$. Observe that $2L_i(x)$ is the i-th digit of the ternary expansion of $L(x)$, so $L_i(x) = L_i(y)$ for every i. Therefore, $x, y \in \mathcal{A}_i$ or $x, y \in \mathcal{A}_i^*$, and by Lemma 19.6 we get that x, y are in the same chain class. Having finished (T2), let us proceed to (T3). If $x \in \mathrm{CR}(g)$, we must have $x \in \mathcal{A}_i \cup \mathcal{A}_i^*$ for every i. This implies that $L_i(x) = 0$ or $L_i(x) = 1$ for every i. Therefore, the ternary expansion of $L(x)$ only contains the digits 0 or 2, which implies that $L(x)$ is in the middle-third Cantor set; cf. Example 18.14. Therefore, $L(\mathrm{CR}(g))$ is contained in, or equal to, the middle-third Cantor set and is hence compact and nowhere dense. $\square$

Exercises for Section 19

E19.1 Consider the discrete dynamical systems on $\mathbb{S}^1$ induced by the following maps

$$f(\theta) = \theta + \alpha \cos^2(\theta), \quad g(\theta) = \theta + \alpha \cos(\theta), \qquad \text{with } 0 < \alpha < 1/2.$$

Identify the chain recurrent points for the two dynamical systems.

E19.2 Let $\mathcal{X}$ be a compact metric space and $g : \mathcal{X} \to \mathcal{X}$ a continuous map inducing a discrete dynamical system with evolution operator ϕ. Prove that: (a) For any $x \in \mathcal{X}$, $\omega(x)$ is **chain transitive**, i.e., given any two points $y_1, y_2 \in \omega(x)$ there is an ε-pseudo-orbit connecting them, for any $\varepsilon > 0$. Specifically, what does this say about $\omega(x)$ and $\mathrm{CR}(\phi)$? (b) If a point lands in $\mathrm{CR}(\phi)$, its forward orbit remains within $\mathrm{CR}(\phi)$ for all forward time. More specifically, for any $x \in \mathrm{CR}(\phi)$, $g^n(x)$ lies in the same chain class as x for all $n \geq 0$. (c) Each chain class in $\mathrm{CR}(\phi)$ is invariant, and $\mathrm{CR}(\phi)$ is invariant.

20 Invariant Measures

Having covered precise trajectory results for dynamics, we have found that there are systems (such as chaotic ones), where it may not be practical to look at individual trajectories. Hence, it is beneficial to also consider a statistical perspective. Before we start with dynamics, we begin with a brief introduction (and/or refresher) to elementary definitions from measure theory. Basically, measures provide a natural generalization of classical

(Riemannian) length, area, volume, etc. to broad classes of sets. Readers familiar with measure theory basics can skip ahead to Definition 20.6.

Definition 20.1. Let $\mathcal{X}$ be a non-empty set. A **σ-algebra** $\mathcal{F}$ on $\mathcal{X}$ is a non-empty collection of subsets of $\mathcal{X}$, which is closed under complements and under countable unions, i.e.,

$$E_j \in \mathcal{F} \ \forall j \in \mathbb{N} \ \Rightarrow \ \bigcup_{j=1}^{\infty} E_j \in \mathcal{F} \qquad \text{and} \qquad E \in \mathcal{F} \ \Rightarrow E^c \in \mathcal{F}.$$

A **sub-σ-algebra** $\tilde{\mathcal{F}}$ of $\mathcal{F}$ is a σ-algebra such that $\tilde{\mathcal{F}} \subset \mathcal{F}$.

As a concrete example, consider a topological space $\mathcal{X}$. Let $\mathrm{Bor}_{\mathcal{X}}$ be the smallest σ-algebra containing all open sets, which is called the **Borel σ-algebra**. Clearly, $\mathrm{Bor}_{\mathcal{X}}$ contains all open sets, all closed sets, as well as unions and intersections of these sets.

Definition 20.2. Let $\mathcal{X}$ be a set with σ-algebra $\mathcal{F}$. A **measure** is a function $\mu : \mathcal{F} \to [0, \infty]$ such that $\mu(\emptyset) = 0$ and for a disjoint sequence of sets $\{E_j\}_{j \in \mathbb{N}}$ in $\mathcal{F}$ we have

$$\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j). \tag{20.1}$$

The property (20.1) is called **countable additivity**. Furthermore, $(\mathcal{X}, \mathcal{F}, \mu)$ is called a **measure space** and elements in $\mathcal{F}$ are called **measurable sets**. A map between measure spaces is called **measurable** if it maps pre-images of measurable sets to measurable sets.

Quite often, we want to discard sets from consideration, which have zero measure, the **null sets**. A property holding for all $x \in \mathcal{X}$ except on a null set, is said to hold **almost everywhere** or just **a.e.**.

Probably the simplest example of a measure on a set $\mathcal{X}$, with σ-algebra just given by the **power set** $\mathcal{P}(\mathcal{X}) = \mathcal{F}$, is the **Dirac measure** δ_{x_0} at a point x_0 defined by $\delta_{x_0}(E) = 1$ if $x_0 \in E$ and $\delta_{x_0}(E) = 0$ if $x_0 \notin E$.

Definition 20.3. A measure μ on $(\mathcal{X}, \mathcal{F})$ with $\mu(\mathcal{X}) = 1$ is called a **probability measure**.

Example 20.4. For probability measures, it can help to think of $\mathcal{X}$ as a "sample space" and $\mathcal{F}$ as the "collection of events". For example, for a six-sided fair die we would set $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\mathcal{X})$, and want that $\mu(\{j\}) = 1/6$ for $j = 1, 2, 3, 4, 5, 6$. ♦

Example 20.5. From an analytical perspective, it is more intuitive to think of the length, area and volume. The most commonly used measure on $\mathbb{R}$, employed as a generalization of the length of an interval $l((a,b)) := b-a$, is the **Lebesgue measure**

$$\mu_{\text{Leb}}(E) := \inf\left\{\sum_{j=1}^{\infty} l(\mathcal{I}_j) : \{\mathcal{I}_j\}_{j\in\mathbb{N}} \text{ open intervals with } E \subseteq \bigcup_{j=1}^{\infty} \mathcal{I}_j\right\},$$

which can easily be generalized to $\mathbb{R}^d$. ♦

Although it takes effort, it is relatively straightforward to build an integration theory for a measure μ, e.g., integrating a **simple function** built as sum of indicator functions is defined as

$$\int_{\mathcal{X}} \sum_{j=1}^{n} a_j \mathbf{1}_{E_j}(x)\ \mathrm{d}\mu(x) := \sum_{j=1}^{n} a_j \mu(E_j), \qquad a_j \in (0,\infty),\ E_j \in \mathcal{F}.$$

Approximating more complicated functions by simple functions, one can extend the integral to a sufficiently large class of measurable functions. For a measure μ on a space $(\mathcal{X},\mathcal{F})$, we denote by $L^p(\mathcal{X},\mathcal{F},\mu)$ (or sometimes just L^p) for $p \in [1,\infty)$ the **L^p-space** of all measurable functions $f : \mathcal{X} \to \mathbb{R}$ such that

$$\|f\|_{L^p} := \left(\int_{\mathcal{X}} |f(x)|^p\ \mathrm{d}\mu(x)\right)^{1/p} < \infty.$$

L^∞ will denote the space of measurable functions with finite essential supremum. In what follows, we shall use some basic tools and results of measure theory, and thereby also learn more technical details along the way.

Coming back to dynamics, we have established the basic theory of hyperbolic, non-hyperbolic and chaotic dynamical systems. Chaotic dynamics is one motivation to consider measure-theoretic tools in dynamics. Measure theory needs less smoothness/regularity and chaotic structures are frequently helpful to obtain sufficiently nice measure-theoretic dynamics. We start by developing some basic tools for topological spaces $\mathcal{X}$ endowed with their Borel σ-algebra.

Definition 20.6. Let $(\mathcal{X}_1,\mathcal{F}_1,\mu_1)$ and $(\mathcal{X}_2,\mathcal{F}_2,\mu_2)$ be measure spaces with Borel probability measures μ_1 and μ_2. A measurable map $G : \mathcal{X}_1 \to \mathcal{X}_2$ is called **measure-preserving** if

$$\mu_2(\mathcal{B}_2) = \mu_1(G^{-1}(\mathcal{B}_2)) \qquad \forall \mathcal{B}_2 \in \mathcal{F}_2.$$

Definition 20.7. Let $(\mathcal{X},\mathcal{F},\mu)$ be a measure space with Borel probability measure μ. Then μ is called an **invariant measure** with respect to $g : \mathcal{X} \to \mathcal{X}$ if g is measure-preserving; we also say μ is g-invariant.

Certain iterated maps as well as flows (via a time-t map) might generate measure-preserving maps (or measure-preserving transformations). Note that sets/spaces of finite measure can always be normalized to have measure one, so just looking at probability measures is not a major restriction.

Example 20.8. Consider the circle $\mathbb{S}^1 = \mathbb{R}/\mathbb{Z}$ with Lebesgue measure μ_{Leb} induced from $\mathbb{R}$, which is a probability measure in this context since $\mu([0,1]) = 1$. As in Example 1.18, consider the rotation

$$x_{k+1} = g(x_k) := (x_k + r) \mod 1,$$

for $r \in \mathbb{R}$. Since the map just shifts any measurable set and potentially decomposes it, one easily checks that it is also a measure-preserving transformation. ♦

An invariant measure can be thought of as describing the distribution of a *typical* trajectory of the system.

Theorem 20.9 (Poincaré Recurrence)**.** *Let $(\mathcal{X}, \mathcal{F}, \mu)$ be a probability space. Suppose $g : \mathcal{X} \to \mathcal{X}$ is measure-preserving and fix any $\mathcal{B}$ with $\mu(\mathcal{B}) > 0$. Then almost every point in $\mathcal{B}$ returns to $\mathcal{B}$ infinitely often under forward iteration of g.*

Proof. For $N \geq 0$, define

$$\mathcal{B}_N := \bigcup_{n=N}^{\infty} g^{-n}(\mathcal{B}) \qquad \text{and} \qquad \mathcal{B}_* := \bigcap_{N=0}^{\infty} \mathcal{B}_N$$

so that $\mathcal{B}_*$ collects all points in $\mathcal{X}$, which enter $\mathcal{B}$ infinitely often under forward iteration of g. So $\mathcal{B} \cap \mathcal{B}_*$ consists of all the points $x \in \mathcal{B}$, which re-enter $\mathcal{B}$ infinitely often under forward iteration, i.e., there exists a sequence $0 < n_1 < n_2 < \cdots$ such that for all $i \in \mathbb{N}$

$$g^{n_i}(x) \in \mathcal{B} \qquad \text{and} \qquad g^{n_i}(x) \in \mathcal{B} \cap \mathcal{B}_*,$$

where the second conclusion follows since $g^{n_i - n_j}(g^{n_j}(x)) \in \mathcal{B}$ for all $j \in \mathbb{N}$; cf. the proof of Theorem 18.4 (T3). Next, observe that the set $\mathcal{B} \cap \mathcal{B}_*$ could have small measure but we would like to show that $\mu(\mathcal{B} \cap \mathcal{B}_*) = \mu(\mathcal{B})$. Indeed, we see that $g^{-1}(\mathcal{B}_N) = \mathcal{B}_{N+1}$ implies $\mu(\mathcal{B}_N) = \mu(\mathcal{B}_{N+1})$ as g is measure-preserving. Upon iterating the argument, this yields $\mu(\mathcal{B}_N) = \mu(\mathcal{B}_0)$ for all N. Furthermore, we observe

$$\mathcal{B}_0 \supseteq \mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \cdots \quad \Rightarrow \quad \mu\left(\bigcap_{N=0}^{\infty} \mathcal{B}_N\right) = \mu(\mathcal{B}_0),$$

which leads to the conclusion $\mu(\mathcal{B} \cap \mathcal{B}_*) = \mu(\mathcal{B} \cap \mathcal{B}_0)$. However, upon using $\mathcal{B} \subset \mathcal{B}_0$, we obtain $\mu(\mathcal{B} \cap \mathcal{B}_0) = \mu(\mathcal{B})$, which shows $\mu(\mathcal{B} \cap \mathcal{B}_*) = \mu(\mathcal{B})$ and finishes the proof. □

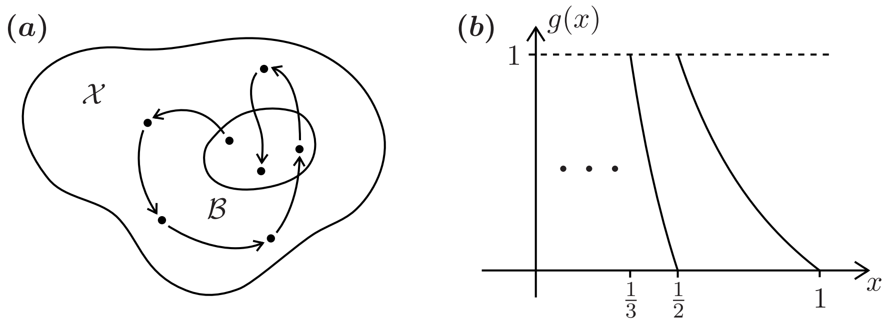

Figure 44: (a) Illustration of Poincaré recurrence as stated in Theorem 20.9. (b) Sketch of the Gauss transformation (20.7) from Example 20.11.

Although it might be nice to have abstract invariant measures, there is, a priori, no reason to expect that we may be able to find them. The next result provides a promising partial answer to this question.

Theorem 20.10 (Krylov-Bogulyubov)**.** *Let $\mathcal{X}$ be a compact metric space and $g : \mathcal{X} \to \mathcal{X}$ be a continuous map. Then there exists a g-invariant Borel probability measure on $\mathcal{X}$.*

Proof. Fix any $x \in \mathcal{X}$ and consider a continuous function $f : \mathcal{X} \to \mathbb{R}$ (which we also call an **observable**). Now define

$$S_f^n(x) := \frac{1}{n} \sum_{j=0}^{n-1} f(g^j(x)). \tag{20.2}$$

Let $\mathcal{F}^*$ be a countable dense subset in $C(\mathcal{X}, \mathbb{R}) = C(\mathcal{X})$. For any given f, the sequence $\{S_f^n(x)\}_{n=1}^{\infty}$ is bounded since $\mathcal{X}$ is compact. We claim that since $\mathcal{F}^*$ is countable, there exists a sequence $n_j \to \infty$ such that the limit

$$\lim_{j \to \infty} S_f^{n_j}(x) =: S_f^{\infty}(x) \tag{20.3}$$

exists for all $f \in \mathcal{F}^*$. To prove the claim (20.3) one uses a standard **diagonal argument**. Label the elements of $\mathcal{F}^*$ by f_1, f_2, etc. and first use that $\{S_{f_1}^m(x)\}_{m=1}^{\infty}$ is bounded to extract a subsequence $m_1(k)$ such that $S_{f_1}^{m_1(k)}(x)$ converges as $k \to \infty$. Out of this subsequence extract a further subsequence $m_2(k)$ such that $S_{f_2}^{m_2(k)}(x)$ converges and continue this process. Finally, pick the diagonal sequence by setting $n_j = m_j(j)$ and verify that (20.3) holds. Next, we use density of $\mathcal{F}^*$, which implies that for any $h \in C(\mathcal{X})$ and any $\varepsilon > 0$, there exists $f \in \mathcal{F}^*$ such that

$$\max_{y \in \mathcal{X}} |h(y) - f(y)| < \varepsilon. \tag{20.4}$$

A direct triangle inequality step now yields that for sufficiently large j, we may combine (20.3) and (20.4) to obtain

$$\left|S_h^{n_j}(x) - S_f^{\infty}(x)\right| \leq S_{|f-h|}^{n_j}(x) + \left|S_f^{n_j}(x) - S_f^{\infty}(x)\right| \leq 2\varepsilon.$$

Hence, $S_h^{n_j}(x)$ is a Cauchy sequence and the limit $S_h^{\infty}(x)$ exists for every $h \in C(\mathcal{X})$ and defines a bounded linear functional

$$L_x(h) := S_h^{\infty}(x), \tag{20.5}$$

which is also positive in the sense that $h \geq 0$ implies $L_x(h) \geq 0$ by the construction (20.2), and we have $\|L_x\| = 1$. The **Riesz Representation Theorem** implies that for a bounded positive linear functional on $C(\mathcal{X})$, there exists a measure μ_x such that

$$L_x(h) = \int_{\mathcal{X}} h(y)\, \mathrm{d}\mu_x(y). \tag{20.6}$$

However, we may simply calculate

$$\left|S_h^{n_j}(g(x)) - S_h^{n_j}(x)\right| = \frac{1}{n_j}\left|h(g^{n_j}(x)) - h(x)\right|,$$

which implies $S_h^{\infty}(g(x)) = S_h^{\infty}(x)$, which can be combined with (20.5) and (20.6) to conclude that μ_x is invariant. □

Example 20.11. Consider $\mathcal{X} = [0,1)$ and the **Gauss transformation**

$$g : \mathcal{X} \to \mathcal{X} \qquad g(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1/x - [1/x] & \text{if } x \neq 0, \end{cases} \tag{20.7}$$

where $[1/x]$ denotes the greatest integer less or equal to $1/x$. The Gauss transformation maps each interval $(1/(n+1), 1/n]$ for $n \in \mathbb{N}$ continuously and monotonically onto $[0,1)$ with discontinuities at each $1/n$; see Figure 44. The Gauss transformation is related to the **continued fraction expansion**

$$x = [a_1, a_2, a_3, \ldots] = \cfrac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}, \qquad a_j \in \mathbb{N},$$

of $x \in \mathcal{X}$. Indeed, one checks if $g^{n-1}(x) \neq 0$, then we have $a_j = \left[1/g^{j-1}(x)\right]$ for $j \leq n$. Furthermore, one observes that

$$g(x) = [a_2, a_3, \ldots]$$

so that a bit more work shows the Gauss transformation conjugates to the left-shift map on integer-valued infinite-sequences (exercise!). There exists an invariant measure, the **Gauss measure**

$$\mu((a,b)) = \frac{1}{\ln 2}\int_a^b \frac{1}{1+y}\,\mathrm{d}y = (\ln 2)^{-1}\ln\frac{1+b}{1+a}. \tag{20.8}$$

Remark: One should not confuse the Gauss measure with **Gaussian measures** given on $\mathbb{R}^d$ by

$$\frac{1}{\sqrt{(2\pi)^d\det(Q)}}\exp\left(-\frac{1}{2}(x-m)^\top Q^{-1}(x-m)\right)\,\mathrm{d}x$$

with positive definite covariance matrix $Q\in\mathbb{R}^{d\times d}$ and mean vector $m\in\mathbb{R}^d$.

To prove that (20.8) is invariant for (20.7), note that the pre-image of an interval (a,b) consists of infinitely many intervals. If we restrict to $(1/(n+1),1/n)$, then the pre-image is $(1/(n+b),1/(n+a))$. Now one calculates

$$\begin{aligned}\mu(g^{-1}((a,b))) &= \mu\left(\bigcup_{n=1}^{\infty}\left(\frac{1}{n+b},\frac{1}{n+a}\right)\right)\\ &= \frac{1}{\ln 2}\sum_{n=1}^{\infty}\ln\left(\frac{1+a+n}{a+n}\cdot\frac{b+n}{1+b+n}\right) = \mu((a,b))\end{aligned}$$

so μ is indeed a g-invariant measure. ♦

We have seen in the proof of Theorem 20.10 that it is often helpful to work with **observables** $f:\mathcal{X}\to\mathbb{R}$, which is emphasized again by the next result.

Lemma 20.12. *$g:\mathcal{X}\to\mathcal{X}$ is measure-preserving for the measure μ if and only if for each $f\in L^1(\mathcal{X},\mathcal{F},\mu)$ we have*

$$\int_{\mathcal{X}} f(y)\,\mathrm{d}\mu(y) = \int_{\mathcal{X}} f(g(y))\,\mathrm{d}\mu(y).$$

Proving Lemma 20.12 is a direct application (exercise!) of using the definitions and applying the Monotone Convergence Theorem[2].

[2]**Monotone Convergence Theorem**: For a measure space $(\mathcal{X},\mathcal{F},\mu)$ consider a sequence of measurable functions $\{f_j:\mathcal{X}\to[0,\infty]\}$ such that $f_j\leq f_{j+1}$ for all j and $f=\lim_{j\to\infty}f_j$, then $\int_{\mathcal{X}} f(x)\,\mathrm{d}\mu(x)=\lim_{j\to\infty}\int_{\mathcal{X}} f_j(x)\,\mathrm{d}\mu(x)$.

Exercises for Section 20

E20.1 Show that the measure μ given by the density

$$h(x) = \frac{1}{\pi\sqrt{x(1-x)}}$$

with respect to the Lebesgue measure on $[0,1]$ is an invariant measure of the logistic map $g : [0,1] \to [0,1]$, $x \mapsto 4x(1-x)$.

E20.2 Consider the doubling map

$$g : [0,1] \to [0,1], \quad g(x) = 2x \mod 1$$

Find three invariant measures of g. *Hint: Dirac measures.*

E20.3 Consider the map

$$g : [0,1] \to [0,1], \quad g(x) = \begin{cases} 2x & \text{if } x \in [0, \frac{1}{2}] \\ x - \frac{1}{2} & \text{if } x \in (\frac{1}{2}, 1] \end{cases}.$$

Find an invariant measure of g that is given by a density with respect to the Lebesgue measure. *Hint: Look for a piecewise constant density.*

E20.4 Prove the following statement: Let $g : \mathcal{X} \to \mathcal{X}$ be a homeomorphism of a compact metric space. Suppose μ is an g-invariant measure on $\mathcal{X}$ satisfying $\mu(\mathcal{X}) = 1$ and $\mu(\mathcal{U}) > 0$ for every non-empty open set $\mathcal{U} \subset \mathcal{X}$ and suppose that $\mathcal{X}$ is connected. Then for any $x, y \in \mathcal{X}$ and any $\varepsilon > 0$, there is an ε-chain from x to y.

21 Ergodicity

Of course, it may happen that a measure-preserving transformation decomposes into several invariant subsets. However, then it makes sense to restrict to each subset and study indecomposable objects; see Figure 45(a).

Definition 21.1. Let $(\mathcal{X}, \mathcal{F}, \mu)$ be a probability space. A measure-preserving transformation/map $g : \mathcal{X} \to \mathcal{X}$ is called **ergodic** if the only invariant measurable sets have zero or full measure, i.e., $g^{-1}(\mathcal{A}) = \mathcal{A}$ implies $\mu(\mathcal{A}) = 0$ or $\mu(\mathcal{A}) = 1$.

Lemma 21.2. *A map $g : \mathcal{X} \to \mathcal{X}$ is ergodic if and only if for every measurable $f : \mathcal{X} \to \mathbb{R}$ with $f \circ g = f$ almost everywhere (a.e.) implies f constant a.e..*

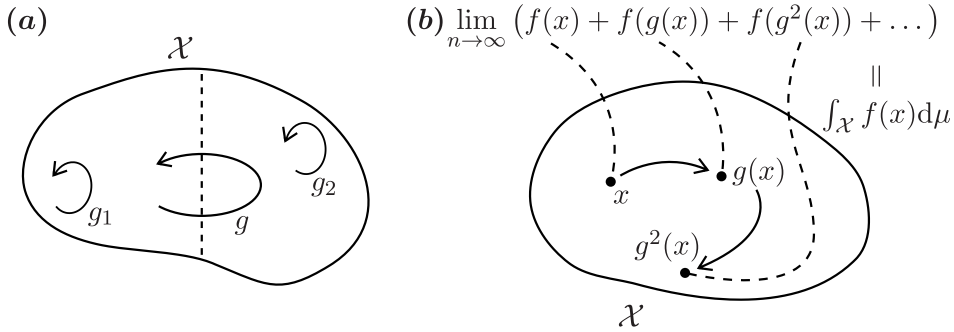

Figure 45: (a) Illustration of the notion of ergodicity: either we can split the phase space into two invariant regions so that g restricted to the regions leaves them invariant, or this is not possible and we are ergodic. (b) Illustration of the Birkhoff Ergodic Theorem 21.7.

We shall not prove Lemma 21.2, which is a (non-trivial!) exercise. However, one could also simply take the result as an alternative definition of ergodicity.

Example 21.3. Consider $\mathcal{X} = \mathbb{S}^1 \simeq [0,1]/(0 \sim 1)$ and the circle rotation map given by

$$x_{k+1} = g(x_k) := (x_k + r) \mod 1. \tag{21.1}$$

We claim that g is ergodic with respect to Lebesgue measure on $\mathbb{S}^1$ if and only if r is irrational. It suffices by a slight extension of Lemma 21.2 (exercise!) to check that if $f \in L^2(\mathcal{X}, \mathrm{Bor}_{\mathcal{X}}, \mu) = L^2$ is g-invariant, then f is constant. Consider the **Fourier series** of f

$$\sum_{n=-\infty}^{\infty} a_n \mathrm{e}^{2\pi \mathrm{i} n x} = f(x) \qquad \text{in } L^2.$$

Since we also have

$$\sum_{n=-\infty}^{\infty} a_n \mathrm{e}^{2\pi \mathrm{i} n(x+r)} = f(g(x)) = f(x) \qquad \text{in } L^2.$$

by invariance of f, the uniqueness of Fourier coefficients a_n implies $a_n = a_n \mathrm{e}^{2\pi i n r}$. Since r is irrational, $\mathrm{e}^{2\pi i n r} \neq 1$ so $a_n = 0$ for all $n \neq 0$, which means that f is constant. Therefore, irrational r implies ergodicity for (21.1). The converse is equally straightforward to show. ♦

Ergodicity implies very nice recurrent properties. To uncover these results, we need some preparation.

Definition 21.4. Let f be a measurable real-valued function on $(\mathcal{X}, \mathcal{F}, \mu)$ and define the **Koopman operator** $\mathcal{L}_g$ by

$$(\mathcal{L}_g f)(x) = f(g(x)).$$

One may study $\mathcal{L}_g$ on various function spaces. As an important case, consider $L^1 = L^1(\mathcal{X}, \mathcal{F}, \mu)$, suppose as usual that g is measure-preserving, and observe that

$$\mathcal{L}_g L^1 \subseteq L^1, \qquad \|\mathcal{L}_g f\|_{L^1} = \int_{\mathcal{X}} f(g(x))\ \mathrm{d}\mu(x) = \int_{\mathcal{X}} f(y)\ \mathrm{d}\mu(y) = \|f\|_{L^1}.$$

One easily checks that $\mathcal{L}_g$ is a **positive operator**, i.e., $f \geq 0$ implies $\mathcal{L}_g f \geq 0$.

Theorem 21.5 (Maximal Ergodic Theorem)**.** *Consider the Koopman operator $\mathcal{L}_g =: \mathcal{L}$ on L^1, fix any $f \in L^1$, and define (for $n \geq 1$) the following*

$$f_0 := 0, \quad f_n := f + \mathcal{L}f + \mathcal{L}^2 f + \cdots + \mathcal{L}^{n-1} f \qquad F_N := \max_{0 \leq n \leq N} f_n \geq 0.$$

Then it follows that $\int_{\{x: F_N(x) > 0\}} f(x)\ \mathrm{d}\mu(x) \geq 0$.

Proof. First, note that $F_N \in L^1$ and $F_N \geq f_n$ by construction. Since the Koopman operator is positive, we find $\mathcal{L}F_N \geq \mathcal{L}f_n$ so that using the definition of f_n, it follows that $\mathcal{L}F_N + f \geq f_{n+1}$. Therefore, a direct estimate yields

$$\mathcal{L}F_N(x) + f(x) \geq \max_{1 \leq n \leq N} f_n(x) \overset{\text{if } F_N(x) > 0}{=} \max_{0 \leq n \leq N} f_n(x) = F_N(x).$$

This provides the elegant estimate $f \geq F_N - \mathcal{L}F_N$ on $\mathcal{A} := \{x \in \mathcal{X} : F_N(x) > 0\}$, which we can just integrate over $\mathcal{A}$ to obtain

$$\int_{\mathcal{A}} f\ \mathrm{d}\mu \geq \int_{\mathcal{A}} F_N\ \mathrm{d}\mu - \int_{\mathcal{A}} \mathcal{L}F_N\ \mathrm{d}\mu = \int_{\mathcal{X}} F_N\ \mathrm{d}\mu - \int_{\mathcal{A}} \mathcal{L}F_N\ \mathrm{d}\mu \tag{21.2}$$

where one uses $F_N = 0$ on $\mathcal{X} \setminus \mathcal{A}$ for the first term. The second term on the right-hand side in (21.2) can be estimated from below since $F_N \geq 0$ implies $\mathcal{L}F_N \geq 0$, which gives

$$\int_{\mathcal{A}} f\ \mathrm{d}\mu \geq \int_{\mathcal{X}} F_N\ \mathrm{d}\mu - \int_{\mathcal{X}} \mathcal{L}F_N\ \mathrm{d}\mu = 0, \tag{21.3}$$

where the last equality is a consequence of g being measure-preserving. $\square$

Corollary 21.6. *Suppose $g : \mathcal{X} \to \mathcal{X}$ is measure-preserving, $h \in L^1(\mathcal{X}, \mathcal{F}, \mu)$, and define*

$$\mathcal{B}_\alpha := \left\{ x \in \mathcal{X} : \sup_{n \geq 1} \frac{1}{n} \sum_{j=0}^{n-1} h(g^j(x)) > \alpha \right\}. \tag{21.4}$$

If $g^{-1}\mathcal{A} = \mathcal{A}$ then $\int_{\mathcal{B}_\alpha \cap \mathcal{A}} h\ \mathrm{d}\mu \geq \alpha \mu(\mathcal{B}_\alpha \cap \mathcal{A})$.

Proof. Just restrict g to $\mathcal{A}$ so that we can simply apply Theorem 21.5 with $\mathcal{A} = \mathcal{X}$ and $f = h - \alpha$ to obtain the result. $\square$

Theorem 21.7 (Birkhoff Ergodic Theorem)**.** *Let $(\mathcal{X}, \mathcal{F}, \mu)$ be a probability space with measure-preserving transformation $g : \mathcal{X} \to \mathcal{X}$ and fix any $f \in L^1 = L^1(\mathcal{X}, \mathcal{F}, \mu)$. The following statements hold for almost every $x \in \mathcal{X}$:*

(T1) There exists $f^* \in L^1$ such that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f(g^j(x)) = f^*(x). \tag{21.5}$$

(T2) $f^* \circ g = f^*$ and $\int_{\mathcal{X}} f \, \mathrm{d}\mu = \int_{\mathcal{X}} f^* \, \mathrm{d}\mu$.

(T3) If g is ergodic, then $f^* \equiv \int_{\mathcal{X}} f \, \mathrm{d}\mu$ so that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f(g^j(x)) = \int_{\mathcal{X}} f \, \mathrm{d}\mu, \tag{21.6}$$

*i.e., **time-average** equals **space-average**.*

Proof. We start with (T1), which is actually the most difficult/crucial part. We define

$$f^*(x) := \limsup_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f(g^j(x)) \quad \text{and} \quad f_*(x) := \liminf_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f(g^j(x))$$

and observe that $f^* \circ g = f^*$ and $f_* \circ g = f_*$. We would like to show that (I) $f^* = f_*$ so that the limit (21.5) exists and (II) that $f^* \in L^1$. For $\alpha, \beta \in \mathbb{R}$, define $\mathcal{E}_{\alpha,\beta} = \{x \in \mathcal{X} : f_*(x) < \beta \text{ and } \alpha < f^*(x)\}$. Since

$$\{x \in \mathcal{X} : f_*(x) < f^*(x)\} = \bigcup_{\alpha,\beta\in\mathbb{Q},\beta<\alpha} \mathcal{E}_{\alpha,\beta}$$

it suffices to prove $\mu(\mathcal{E}_{\alpha,\beta}) = 0$ for $\beta < \alpha$ to establish (I). Observe that $g^{-1}\mathcal{E}_{\alpha,\beta} = \mathcal{E}_{\alpha,\beta}$, define $\mathcal{B}_\alpha$ as in (21.4) and note that $\mathcal{B}_\alpha \cap \mathcal{E}_{\alpha,\beta} = \mathcal{E}_{\alpha,\beta}$. Hence, Corollary 21.6 applies to yield

$$\int_{\mathcal{E}_{\alpha,\beta}} f \, \mathrm{d}\mu = \int_{\mathcal{E}_{\alpha,\beta}\cap\mathcal{B}_\alpha} f \, \mathrm{d}\mu \geq \alpha\mu(\mathcal{E}_{\alpha,\beta} \cap \mathcal{B}_\alpha) = \alpha\mu(\mathcal{E}_{\alpha,\beta}). \tag{21.7}$$

Replacing f, α, β by $-f, -\beta, -\alpha$ it is easy to check (exercise!) that we also obtain the inequality

$$\int_{\mathcal{E}_{\alpha,\beta}} f \, \mathrm{d}\mu \leq \beta\mu(\mathcal{E}_{\alpha,\beta}). \tag{21.8}$$

Combining (21.7)-(21.8) gives

$$\alpha\mu(\mathcal{E}_{\alpha,\beta}) \leq \beta\mu(\mathcal{E}_{\alpha,\beta}) \quad \Rightarrow \quad \mu(\mathcal{E}_{\alpha,\beta}) = 0 \text{ if } \beta < \alpha.$$

This shows (I) and to check (II) let $h_n(x) := \left|\frac{1}{n}\sum_{j=0}^{n-1} f(g^j(x))\right|$ so Fatou's Lemma[3] implies $f^* \in L^1$ since

$$\int_{\mathcal{X}} \liminf_{n\to\infty} h_n \ \mathrm{d}\mu \leq \liminf_{n\to\infty} \frac{1}{n}\sum_{j=0}^{n-1}\int_{\mathcal{X}} |f \circ g^j| \ \mathrm{d}\mu = \int_{\mathcal{X}} |f| \ \mathrm{d}\mu.$$

The first statement in (T2), i.e., $f^* = f^* \circ g$ is now easy to check. For the second part, let $\mathcal{D}_k^n := \{x \in \mathcal{X} : k/n \leq f^*(x) \leq (k+1)/n\}$, which are sets to decompose f^* at "scale n" for $k \in \mathbb{Z}$, $n \in \mathbb{N}$. One may select $\varepsilon > 0$ sufficiently small so that $\mathcal{D}_k^n \cap \mathcal{B}_{(k/n)-\varepsilon} = \mathcal{D}_k^n$, which makes Corollary 21.6 applicable again

$$\int_{\mathcal{D}_k^n} f \ \mathrm{d}\mu \geq \left(\frac{k}{n} - \varepsilon\right)\mu(\mathcal{D}_k^n) \quad \Rightarrow \quad \int_{\mathcal{D}_k^n} f \ \mathrm{d}\mu \geq \frac{k}{n}\mu(\mathcal{D}_k^n).$$

Using this inequality and the definition of $\mathcal{D}_k^n$ leads to

$$\int_{\mathcal{D}_k^n} f^* \ \mathrm{d}\mu \leq \frac{k+1}{n}\mu(\mathcal{D}_k^n) \leq \frac{1}{n}\mu(\mathcal{D}_k^n) + \int_{\mathcal{D}_k^n} f \ \mathrm{d}\mu.$$

Now one sums over k to conclude $\int_{\mathcal{X}} f^* \ \mathrm{d}\mu \leq \frac{1}{n} + \int_{\mathcal{X}} f \ \mathrm{d}\mu$ and taking the limit $n \to \infty$ implies $\int_{\mathcal{X}} f^* \ \mathrm{d}\mu \leq \int_{\mathcal{X}} f \ \mathrm{d}\mu$. Applying this to $-f$ gives $\int_{\mathcal{X}}(-f)^*\mathrm{d}\mu \leq \int_{\mathcal{X}}(-f) \ \mathrm{d}\mu$ but since $(-f)^* = -f_*$ by construction we have $\int_{\mathcal{X}} f_* \ \mathrm{d}\mu \geq \int_{\mathcal{X}} f \ \mathrm{d}\mu$. Combining this with the previous inequality we have

$$\int_{\mathcal{X}} f_* \ \mathrm{d}\mu \geq \int_{\mathcal{X}} f \ \mathrm{d}\mu \geq \int_{\mathcal{X}} f^* \ \mathrm{d}\mu$$

so that using $f^* = f_*$ finishes the proof of (T2). For (T3), f^* is constant almost everywhere by ergodicity due to Lemma 21.2 so the result follows by combining (T1) and (T2). □

The next result is a nice consequence of the Birkhoff Ergodic Theorem.

[3]**Fatou's Lemma**: For a measure space $(\mathcal{X}, \mathcal{F}, \mu)$ consider a sequence of measurable functions $\{f_j : \mathcal{X} \to [0,\infty]\}$, then $\int_{\mathcal{X}} \liminf_{j\to\infty} f_j(x) \ \mathrm{d}\mu(x) \leq \liminf_{j\to\infty}\int_{\mathcal{X}} f_j(x) \ \mathrm{d}\mu(x)$.

Theorem 21.8 (L^p Ergodic Theorem of Von Neumann)**.** *Fix $p \in [1, \infty)$, suppose $g : \mathcal{X} \to \mathcal{X}$ is measure-preserving for a probability space $(\mathcal{X}, \mathcal{F}, \mu)$. Consider $f \in L^p = L^p(\mathcal{X}, \mathcal{F}, \mu)$ and the (n-term) average*

$$(S_n f)(x) := \frac{1}{n} \sum_{j=0}^{n-1} f(g^j(x))$$

then there exists $f^* \in L^p$ s.t. $f^* \circ g = f^*$ and $\lim_{n\to\infty} \|S_n f - f^*\|_{L^p} = 0$.

Proof. Fix $\varepsilon > 0$. The proof is a standard $\varepsilon/3$**-argument** to check that $S_n f$ is a Cauchy sequence so we provide only the main steps. Consider $h \in L^\infty$ such that $\|h-f\|_{L^p} \leq \varepsilon/3$. Since $\mathcal{X}$ is a probability space, it follows that $h \in L^p$. Therefore, the Birkhoff Ergodic Theorem implies that $S_n h$ converges in L^p as the limit h^* is again bounded and hence is in L^p. This yields the existence of a sufficiently large $n \in \mathbb{N}$ such that $\|S_n h - S_{n+k} h\|_{L^p} \leq \varepsilon/3$. Hence, the triangle inequality and $\|S_n f\|_{L^p} \leq \|f\|_{L^p}$ give

$$\begin{aligned} \|S_n f - S_{n+k} f\|_{L^p} &\leq \|S_n f - S_n h\|_{L^p} + \|S_n h - S_{n+k} h\|_{L^p} \\ &\qquad + \|S_{n+k} h - S_{n+k} f\|_{L^p} = \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

and the existence of $f^* \in L^p$ follows, which is easily checked to be also g-invariant. $\square$

Exercises for Section 21

E21.1 Prove Lemma 21.2. *Hint: Consider the set $\mathcal{A}_c = \{x \in \mathcal{X}\,;\ f(x) \geq c\}$ for some $c \in \mathbb{R}$.*

E21.2 On the torus $\mathbb{T}^2$, represented by the unit square $[0,1]^2$ with opposite sides identified, we consider the map

$$g : \mathbb{T}^2 \to \mathbb{T}^2, \quad g(x,y) = (2x + y \bmod 1, x + y \bmod 1),$$

which we already encountered in Example 3.11. Show that g is ergodic with respect to the Lebesgue measure. *Hint: Use a Fourier series and Exercise E21.1.*

22 Mixing

The next result hints at a crucial consequence of ergodicity, when we are interested in return probabilities.

Theorem 22.1 (Limit of Cesàro Means). *Let $(\mathcal{X}, \mathcal{F}, \mu)$ be a probability space and suppose $g : \mathcal{X} \to \mathcal{X}$ is a measure-preserving map. Then g is ergodic if and only if for all $\mathcal{A}, \mathcal{B} \in \mathcal{F}$ we have*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \mu(g^{-j}(\mathcal{A}) \cap \mathcal{B}) = \mu(\mathcal{A})\mu(\mathcal{B}). \tag{22.1}$$

Proof. Suppose g is ergodic, then let $f = \mathbf{1}_{\mathcal{A}}$ be the indicator function of $\mathcal{A}$ in the Birkhoff Ergodic Theorem 21.7 so that the space average is just $\mu(\mathcal{A})$. Hence, we find

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \mathbf{1}_{\mathcal{A}}(g^j(x)) = \mu(\mathcal{A}).$$

Multiplying the last equality by $\mathbf{1}_{\mathcal{B}}$, integrating, and applying the Dominated Convergence Theorem[4] gives

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \mu(g^{-j}(\mathcal{A}) \cap \mathcal{B}) = \mu(\mathcal{A})\mu(\mathcal{B}).$$

The converse of our result assumes (22.1). Suppose $\mathcal{S}$ is any invariant set, $g^{-1}\mathcal{S} = \mathcal{S}$, and let $\mathcal{A} = \mathcal{S} = \mathcal{B}$. Then (22.1) implies $\mu(\mathcal{S}) = \mu(\mathcal{S})^2$, so $\mu(\mathcal{S}) = 0$ or $\mu(\mathcal{S}) = 1$, which just means ergodicity holds. □

We know by Theorem 22.1 that the sets $\mu(g^{-j}(\mathcal{A}) \cap \mathcal{B})$ become asymptotically independent "on average" for ergodic maps. Hence, it is very natural to ask whether we can do better than just average properties.

Definition 22.2. Let g be a measure-preserving transformation of a probability space $(\mathcal{X}, \mathcal{F}, \mu)$.

(D1) g is called **weak-mixing** if for all $\mathcal{A}, \mathcal{B} \in \mathcal{F}$ we have

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \left|\mu(g^{-j}(\mathcal{A}) \cap \mathcal{B}) - \mu(\mathcal{A})\mu(\mathcal{B})\right| = 0. \tag{22.2}$$

(D2) g is called **strong-mixing** (or just **mixing**) if for all $\mathcal{A}, \mathcal{B} \in \mathcal{F}$ we have

$$\lim_{n\to\infty} \mu(g^{-n}(\mathcal{A}) \cap \mathcal{B}) = \mu(\mathcal{A})\mu(\mathcal{B}). \tag{22.3}$$

[4] **Dominated Convergence Theorem**: For a measure space $(\mathcal{X}, \mathcal{F}, \mu)$ consider a sequence $\{f_j\}$ in L^1 such that $f_j \to f$ a.e. as $j \to \infty$ and there exists $g \in L^1$ such that $|f_j| \leq g$ a.e. for all j, then $\int_{\mathcal{X}} f(x)\,\mathrm{d}\mu(x) = \lim_{j\to\infty} \int_{\mathcal{X}} f_j(x)\,\mathrm{d}\mu(x)$.

Mixing properties entail the **decay of correlations**. The next result is relatively straightforward to prove (exercise!).

Proposition 22.3. *Strong-mixing $\Rightarrow$ weak-mixing $\Rightarrow$ ergodic.*

The converse implications in Proposition 22.3 are not true in general; see e.g. Example 22.10.

Example 22.4. Consider $\mathcal{X} = \mathbb{S}^1 \simeq [0,1]/(0 \sim 1)$ and the family of **expanding circle endomorphisms** given by

$$x_{k+1} = g(x_k) := \alpha x_k \mod 1, \qquad \alpha \in \mathbb{Z},\ |\alpha| > 1. \tag{22.4}$$

One checks that Lebesgue measure μ is g-invariant. Next, consider $\alpha = 2$ and two intervals $\mathcal{A} = [p/2^i, (p+1)/2^i]$ and $\mathcal{B} = [q/2^j, (q+1)/2^j]$, where we may fix any p, q so that $\mathcal{A}, \mathcal{B} \subseteq [0,1]$. Then $g^{-1}(\mathcal{B})$ is the union of 2 uniformly spaced intervals of length $1/2^{j+1}$ and similarly $g^{-n}(\mathcal{B})$ is the union of 2^n uniformly spaced intervals of length $1/2^{j+n}$. Hence, for $n > i$, we have that $\mathcal{A} \cap g^{-n}(\mathcal{B})$ consists of 2^{n-i} intervals of length $1/2^{n+j}$ and we calculate

$$\mu(\mathcal{A} \cap g^{-n}(\mathcal{B})) = 2^{n-i}(1/2^{n+j}) = 1/2^{i+j} = \mu(\mathcal{A})\mu(\mathcal{B})$$

so that an approximation procedure shows that (22.4) is strong-mixing for $\alpha = 2$. ♦

The considerations in the last example can easily be turned into a more general proof of the following result.

Proposition 22.5. *Expanding circle endomorphisms are strong-mixing.*

For more general maps, proving mixing and/or working with properties directly becomes almost impossible. However, an operator-theoretic viewpoint provides a nice characterization.

Definition 22.6. Let $L^p(\mathcal{X}, \mathcal{F}, \mu) =: L^p$ for $p \in [1, \infty]$. View the Koopman operator as $\mathcal{L}_g = \mathcal{L} : L^\infty \to L^\infty$ and let $\mathcal{P}_g = \mathcal{P} : L^1 \to L^1$ be its adjoint defined by requiring that

$$\langle \mathcal{P}_g f, h \rangle := \int_{\mathcal{X}} (\mathcal{P}_g f) h \, \mathrm{d}\mu \stackrel{!}{=} \int_{\mathcal{X}} f(\mathcal{L}_g h) \, \mathrm{d}\mu = \langle f, \mathcal{L}_g h \rangle. \tag{22.5}$$

for $f \in L^1$ and $h \in L^\infty$. The operator $\mathcal{P}_g$ is called the **Perron-Frobenius (or transfer) operator**.

Theorem 22.7 (Operator-Characterization of Mixing)**.** *Let $(\mathcal{X},\mathcal{F},\mu)$ be a probability space and $g:\mathcal{X}\to\mathcal{X}$ be measure-preserving. Then g is strong-mixing if and only if*

$$\lim_{n\to\infty}\langle \mathcal{P}^n f,h\rangle = \langle f,1\rangle\langle 1,h\rangle, \qquad f\in L^1,\ h\in L^\infty, \tag{22.6}$$

where we used the natural abbreviation $\mathcal{P}=\mathcal{P}_g$ for the Perron-Frobenius operator.

Proof. We start by showing that strong-mixing yields (22.6). The mixing condition (22.3) can be written in integral form as

$$\lim_{n\to\infty}\int_{\mathcal{X}}\mathbf{1}_{\mathcal{A}}(x)\mathbf{1}_{\mathcal{B}}(g^n(x))\ \mathrm{d}\mu(x)=\int_{\mathcal{X}}\mathbf{1}_{\mathcal{A}}(x)\ \mathrm{d}\mu(x)\int_{\mathcal{X}}\mathbf{1}_{\mathcal{B}}(x)\ \mathrm{d}\mu(x)$$

so that using the definition of the Koopman and Perron-Frobenius operators gives

$$\langle \mathbf{1}_{\mathcal{A}},1\rangle\langle 1,\mathbf{1}_{\mathcal{B}}\rangle=\lim_{n\to\infty}\langle \mathbf{1}_{\mathcal{A}},\mathcal{L}^n\mathbf{1}_{\mathcal{B}}\rangle=\lim_{n\to\infty}\langle \mathcal{P}^n\mathbf{1}_{\mathcal{A}},\mathbf{1}_{\mathcal{B}}\rangle \tag{22.7}$$

so the desired equality holds for indicator functions, and by linearity also for linear combinations of indicator functions, which are just the standard simple functions. We approximate $h\in L^\infty$ by simple functions h_k such that $h_k\to h$, and we approximate $f\in L^1$ by simple functions such that $f_k\to f$. Then we have

$$\begin{aligned}|\langle \mathcal{P}^n f,h\rangle-\langle f,1\rangle\langle 1,h\rangle| \leq\ & |\langle \mathcal{P}^n f,h\rangle-\langle \mathcal{P}^n f_k,h_k\rangle|\\ &+|\langle \mathcal{P}^n f_k,h_k\rangle-\langle f_k,1\rangle\langle 1,h_k\rangle|\\ &+|\langle f_k,1\rangle\langle 1,h_k\rangle-\langle f,1\rangle\langle 1,h\rangle|\end{aligned} \tag{22.8}$$

so that a standard $\varepsilon/3$-argument yields the result as all three terms on the right-hand side of (22.8) can be made small. The converse of the result is easier since if (22.6) holds then setting $f=\mathbf{1}_{\mathcal{A}}$ and $h=\mathbf{1}_{\mathcal{B}}$ gives strong-mixing. $\square$

The last results naturally point into the direction of using *invariants* of the Koopman operator $\mathcal{L}$ (respectively the Perron-Frobenius operator $\mathcal{P}$) to characterize/classify measure-preserving transformations. Observe that we can view the Koopman operator also as an operator $\mathcal{L}_g=\mathcal{L}:L^2\to L^2$. In this context, it is as a unitary operator since it preserves the L^2 inner product. Hence, if $\mathcal{L}f=\lambda f$ for some $\lambda\in\mathbb{C}$, i.e., λ is an **eigenvalue** and f is an **eigenfunction**, then $\lambda\in\{z\in\mathbb{C}:|z|=1\}$.

Definition 22.8. Consider a probability space $(\mathcal{X},\mathcal{F},\mu)$ and let $L^2=L^2(\mathcal{X},\mathcal{F},\mu)$. The Koopman operator $\mathcal{L}=\mathcal{L}_g:L^2\to L^2$, defined by $\mathcal{L}f(x)=f(g(x))$, has:

(D1) **pure-point spectrum** if the eigenfunctions of $\mathcal{L}$ span L^2;

(D2) **continuous spectrum** if 1 is the only eigenvalue.

Remark: Of course, many linear operators have *both* types of spectra, i.e., some eigenvalues and then some remaining part consisting of continuous/essential spectrum. This is highly relevant for the dynamical systems analysis of PDEs [39].

The next result illustrates that one can characterize a lot of dynamical effects of $g : \mathcal{X} \to \mathcal{X}$ purely by looking at the spectrum of $\mathcal{L}_g = \mathcal{L}$.

Theorem 22.9 (Spectral Characterization)**.** *The following results hold:*

(T1) g is ergodic if and only if 1 is a simple eigenvalue of $\mathcal{L}$.

(T2) If g is ergodic, then every eigenvalue of $\mathcal{L}$ is simple.

(T3) Suppose g is invertible with measurable inverse, then g is weak-mixing if and only if $\mathcal{L}_g$ has continuous spectrum.

Proof. (T1) is easy to check (exercise!). Regarding (T2), if g is ergodic, then every eigenfunction is constant a.e., so that if there are two eigenfunctions f_1 and f_2 with the same eigenvalue then f_1/f_2 is also an eigenfunction with eigenvalue 1 so f_1/f_2 is also constant a.e., which proves the result. For (T3), g weak-mixing implying continuous spectrum is relatively easy (exercise!). The converse is difficult, i.e., we now assume that $\mathcal{L}_g$ has continuous spectrum. An equivalent characterization of weak-mixing is

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} |\langle \mathcal{L}^j f, f\rangle - \langle f, 1\rangle\langle 1, f\rangle|^2 = 0 \qquad \text{for } f \in L^2.$$

For a constant f, this is obvious, so we only need to check it in the orthogonal complement to the constants in L^2, i.e., when $\langle f, 1\rangle = 0$. Therefore, we have to prove

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} |\langle \mathcal{L}^j f, f\rangle|^2 = 0 \qquad \text{for } f \in L^2. \tag{22.9}$$

Since $\mathcal{L}$ is a unitary operator on the Hilbert space L^2, we may apply the **Spectral Theorem**, which guarantees for every $f \in L^2$ the existence of a unique finite Borel measure μ_f on the spectrum $\mathcal{K}$ such that

$$\langle \mathcal{L}^j f, f\rangle = \int_{\mathcal{K}} \lambda^j \, \mathrm{d}\mu_f(\lambda). \tag{22.10}$$

Furthermore, since $\mathcal{L}$ has only continuous spectrum, the Spectral Theorem guarantees that μ_f has no atoms[5]. Combining (22.9) and (22.10) means we just have to prove

$$\lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}\left|\int_{\mathcal{K}}\lambda^j\ \mathrm{d}\mu_f(\lambda)\right|^2 = 0 \qquad \text{for } f\in L^2. \tag{22.11}$$

Since for points λ in the spectrum we have $\bar{\lambda}=\lambda^{-1}$, we may re-write the sum in (22.11) as

$$\begin{aligned}\frac{1}{n}\sum_{j=0}^{n-1}|\int_{\mathcal{K}}\lambda^j\mathrm{d}\mu_f(\lambda)|^2 &= \frac{1}{n}\sum_{j=0}^{n-1}\left(\int_{\mathcal{K}}\lambda^j\ \mathrm{d}\mu_f(\lambda)\ \int_{\mathcal{K}}\lambda^{-j}\ \mathrm{d}\mu_f(\lambda)\right)\\ &= \frac{1}{n}\sum_{j=0}^{n-1}\int_{\mathcal{K}\times\mathcal{K}}(\lambda\bar{\tau})^j\ \mathrm{d}(\mu_f\times\mu_f)(\lambda,\tau)\\ &= \int_{\mathcal{K}\times\mathcal{K}}\frac{1}{n}\sum_{j=0}^{n-1}(\lambda\bar{\tau})^j\ \mathrm{d}(\mu_f\times\mu_f)(\lambda,\tau) \qquad (22.12)\end{aligned}$$

where the second equality step follows from a re-labelling of the second variable and using Fubini's Theorem[6]; essentially, the last argument is a classical case of *doubling variables.* For points off the diagonal in $\mathcal{K}\times\mathcal{K}$, we get

$$\lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}(\lambda\bar{\tau})^j = \lim_{n\to\infty}\frac{1}{n}\left(\frac{1-(\lambda\bar{\tau})^n}{1-(\lambda\bar{\tau})}\right) = 0. \tag{22.13}$$

Since μ_f is non-atomic, the diagonal has measure zero, the limit in (22.13) holds a.e., and the Dominated Convergence Theorem can be applied to (22.12) to get the result. $\square$

Example 22.10. Consider the circle rotation $g(x) = x+r \pmod 1$ for $x\in\mathbb{S}^1=\mathbb{R}/\mathbb{Z}$. For any $n\in\mathbb{Z}$, the function $f_n=\exp(2\pi\mathrm{i}nx)$ is an eigenfunction of $\mathcal{L}$ since

$$\mathcal{L}f_n(x) = f_n(g(x)) = \mathrm{e}^{2\pi\mathrm{i}nr}\mathrm{e}^{2\pi\mathrm{i}nx} = \mathrm{e}^{2\pi\mathrm{i}nr}f_n(x).$$

If $r\in\mathbb{R}\setminus\mathbb{Q}$, then the eigenfunctions span L^2 and Theorem 22.9(T3) implies that the irrational circle rotation is not weak-mixing. However, it is shown in Example 21.3 that irrational circle rotations are ergodic; cf. Proposition 22.3. ♦

[5]For a measure space $(\mathcal{X},\mathcal{F},\mu)$, a set $A\in\mathcal{F}$ is called an **atom** if $\mu(A)>0$ and any measurable subset $B\subset A$ with $\mu(B)<\mu(A)$ satisfies $\mu(B)=0$.

[6]**Fubini's Theorem**: Let $(\mathcal{X}_i,\mathcal{F}_i,\mu_i)$ for $i\in\{1,2\}$ be σ-finite measure spaces and let f be a non-negative measurable function on $\mathcal{X}_1\times\mathcal{X}_2$, then $\int_{\mathcal{X}_1}\int_{\mathcal{X}_2}f(x,y)\ \mathrm{d}\mu_1(x)\ \mathrm{d}\mu_2(y) = \int_{\mathcal{X}_2}\int_{\mathcal{X}_1}f(x,y)\ \mathrm{d}\mu_2(y)\ \mathrm{d}\mu_1(x)$.

Exercises for Section 22

E22.1 Let $(\mathcal{X}, \mathcal{B}, \mu)$ be a probability space and $g : \mathcal{X} \to \mathcal{X}$ a μ-preserving map. Prove that g is ergodic if and only if

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \langle \mathcal{P}^k f, h\rangle \;=\; \langle f, 1\rangle\langle 1, h\rangle \quad \text{for all} f \in L^1, h \in L^\infty.$$

E22.2 Prove Proposition 22.3.

E22.3 Let $(\mathcal{X}, \mathcal{F}, \mu)$ be a probability space and suppose that $g : \mathcal{X} \to \mathcal{X}$ preserves the measure μ. Show that g is strong-mixing if and only if

$$\lim_{n\to\infty} \mu(\mathcal{A} \cap g^{-n}(\mathcal{A})) = \mu(\mathcal{A})^2 \;\text{ for all }\; \mathcal{A} \in \mathcal{F}.$$

Hint: Use Theorem 22.7 and start with $f = \mathbf{1}_{\mathcal{A}} \circ g^k$ *and* $h = \mathbf{1}_{\mathcal{A}} \circ g^m$.

E22.4 (a) Let $g : \mathcal{X} \to \mathcal{X}$ be an isometry of a compact metric space $(\mathcal{X}, d)$. Prove that g is not mixing for any invariant Borel measure whose support is not a single point. (b) Now apply the last result in the context of circle rotations.

23 Entropy

Spectral properties provide one possible way to distinguish two dynamical systems. Hence, a natural question to ask is, which other useful **invariants** may be defined? Borrowing the concept of **entropy** from statistical physics yields one very helpful invariant. There are two related approaches to entropy: a measure-theoretic and a topological one. We are going to consider measure-theoretic entropy for a measure-preserving transformation

$$g : \mathcal{X} \to \mathcal{X}$$

on a probability space $(\mathcal{X}, \mathcal{F}, \mu)$.

Definition 23.1. A (finite) **partition** $\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_m\}$ of $(\mathcal{X}, \mathcal{F}, \mu)$ is a disjoint collection of elements of $\mathcal{F}$, whose union is $\mathcal{X}$. The **join** (or **common refinement**) $\mathcal{A} \vee \mathcal{B}$ of two partitions $\mathcal{A}$ and $\mathcal{B}$ is the partition into intersections $\mathcal{A}_k \cap \mathcal{B}_n$.

Usually we shall assume that a partition $\mathcal{A}$ is directly related to a sub-(σ-)algebra, i.e., by considering all elements in $\mathcal{F}$ which can be written as unions of elements of $\mathcal{A}$. Similarly we may relate a given sub-algebra of $\mathcal{F}$ to a partition (exercise!). Furthermore, we write $\mathcal{A} \subseteq \mathcal{B}$ to mean that $\mathcal{B}$ is a **refinement** (or finer partition), i.e., elements of $\mathcal{A}$ can be constructed via unions of elements of $\mathcal{B}$.

Definition 23.2. For a partition $\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_m\}$ define the **entropy**

$$H(\mathcal{A}) := -\sum_{j=1}^{m} \mu(\mathcal{A}_j) \ln \mu(\mathcal{A}_j)$$

Entropy has a natural interpretation as average information of the elements of a partition. Indeed, let $I(\mathcal{C}) := -\ln \mu(\mathcal{C})$ be the **information** of the event (or measurable set) $\mathcal{C}$, then the information is a non-negative and decreasing function, i.e., observing a lower probability event gives more information. Furthermore, the trivial event $\mathcal{X}$ has $I(\mathcal{X}) = 0$ and for independent events, say $\mu(\mathcal{C}_1 \cap \mathcal{C}_2) = \mu(\mathcal{C}_1)\mu(\mathcal{C}_2)$, we have

$$I(\mathcal{C}_1 \cap \mathcal{C}_2) = I(\mathcal{C}_1) + I(\mathcal{C}_2), \tag{23.1}$$

so information is additive; in fact, using a logarithm (up to a constant) is the only choice if one requires the last observations from a definition of information.

Definition 23.3. Consider a measure-preserving transformation $g : \mathcal{X} \to \mathcal{X}$ and a partition $\mathcal{A}$. Define the **entropy** of g with respect to $\mathcal{A}$ as

$$h(g, \mathcal{A}) := \lim_{n \to +\infty} \frac{1}{n} H\left(\bigvee_{j=0}^{n-1} g^{-j} \mathcal{A} \right) \tag{23.2}$$

where $g^{-j}\mathcal{A} = \{g^{-j}\mathcal{A}_1, \ldots, g^{-j}\mathcal{A}_m\}$. Finally, define the **entropy** of g by taking the supremum over all finite partitions

$$h(g) := \sup_{\mathcal{A} \text{ finite}} h(g, \mathcal{A}).$$

It can be shown (non-trivial exercise!) that the limit in (23.2) actually exists. The next result is expected (exercise!) since we can just insert a conjugating map in the last definition without too much trouble.

Theorem 23.4 (Entropy as a Conjugacy Invariant)**.** *Two measure-preserving transformations, which are topologically conjugate under a measure-preserving conjugacy, have the same entropy.*

The main problem is that entropy is still almost impossible to calculate from the definition of $h(g)$ for concrete examples. A technical tool is needed to "sort" the information into smaller pieces. It is a natural concept to condition upon previous knowledge, i.e., for measurable sets $\mathcal{C}_1, \mathcal{C}_2$ with $\mu(\mathcal{C}_2) > 0$ define the **conditional probability** as

$$\mu(\mathcal{C}_1 | \mathcal{C}_2) := \frac{\mu(\mathcal{C}_1 \cap \mathcal{C}_2)}{\mu(\mathcal{C}_2)}.$$

Hence, we can now average the usual entropy conditionally over another partition.

Definition 23.5. For two partitions $\mathcal{A}, \mathcal{B}$ define the **conditional entropy**

$$H(\mathcal{A}|\mathcal{B}) := -\sum_{i,j} \mu(\mathcal{B}_j)\mu(\mathcal{A}_i|\mathcal{B}_j) \ln \mu(\mathcal{A}_i|\mathcal{B}_j).$$

Theorem 23.6 (Properties of Conditional Entropy)**.** *If $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are finite sub-algebras of $\mathcal{F}$, then we have*

(T1c) $H(\mathcal{A} \vee \mathcal{B}) = H(\mathcal{A}) + H(\mathcal{B}|\mathcal{A})$;

(T2c) $\mathcal{A} \subseteq \mathcal{B} \Rightarrow H(\mathcal{A}) \leq H(\mathcal{B})$;

(T3c) $\mathcal{C} \subseteq \mathcal{B} \Rightarrow H(\mathcal{A}|\mathcal{C}) \geq H(\mathcal{A}|\mathcal{B})$;

(T4c) $H(\mathcal{A} \vee \mathcal{B}|\mathcal{C}) \leq H(\mathcal{A}|\mathcal{C}) + H(\mathcal{B}|\mathcal{C})$;

(T5c) $H(g^{-1}\mathcal{A}|g^{-1}\mathcal{B}) = H(\mathcal{A}|\mathcal{B})$ *and* $H(g^{-1}\mathcal{A}) = H(\mathcal{A})$.

Remark: The properties are all quite intuitive, e.g., (T1c) claims that the entropy ("information") contained in the join of $\mathcal{A}, \mathcal{B}$ is the entropy of $\mathcal{A}$ plus the additional information in $\mathcal{B}$.

Proof. Most of the properties can be proved by (quite lengthy) definition-chasing. However, let us prove (T3c) to see, how the structure of the entropy is used. Consider the function $\psi : [0, \infty) \to \mathbb{R}$ defined by $\psi(x) := x \ln x$ with the obvious limit point definition $\psi(0) = 0$. Observe that ψ is *strictly convex* so that, for fixed i, j, we have

$$\psi\left(\sum_k \frac{\mu(\mathcal{B}_k \cap \mathcal{C}_j)}{\mu(\mathcal{C}_j)} \frac{\mu(\mathcal{A}_i \cap \mathcal{B}_k)}{\mu(\mathcal{B}_k)}\right) \leq \sum_k \frac{\mu(\mathcal{B}_k \cap \mathcal{C}_j)}{\mu(\mathcal{C}_j)} \psi\left(\frac{\mu(\mathcal{A}_i \cap \mathcal{B}_k)}{\mu(\mathcal{B}_k)}\right).$$

Since $\mathcal{C} \subseteq \mathcal{B}$, the left-hand side of the last inequality simplifies to

$$\psi\left(\frac{\mu(\mathcal{A}_i \cap \mathcal{C}_j)}{\mu(\mathcal{C}_j)}\right) = \frac{\mu(\mathcal{A}_i \cap \mathcal{C}_j)}{\mu(\mathcal{C}_j)} \ln \frac{\mu(\mathcal{A}_i \cap \mathcal{C}_j)}{\mu(\mathcal{C}_j)}.$$

Using these last two results, multiplying both sides by $\mu(\mathcal{C}_j)$ and summing over the indices i, j we get

$$\begin{aligned}
\sum_{i,j} \mu(\mathcal{A}_i \cap \mathcal{C}_j) \ln \frac{\mu(\mathcal{A}_i \cap \mathcal{C}_j)}{\mu(\mathcal{C}_j)} &\leq \sum_{i,j,k} \mu(\mathcal{B}_k \cap \mathcal{C}_j) \frac{\mu(\mathcal{A}_i \cap \mathcal{B}_k)}{\mu(\mathcal{B}_k)} \ln \frac{\mu(\mathcal{A}_i \cap \mathcal{B}_k)}{\mu(\mathcal{B}_k)} \\
&= \sum_{i,k} \mu(\mathcal{B}_k) \frac{\mu(\mathcal{A}_i \cap \mathcal{B}_k)}{\mu(\mathcal{B}_k)} \ln \frac{\mu(\mathcal{A}_i \cap \mathcal{B}_k)}{\mu(\mathcal{B}_k)},
\end{aligned}$$

which just means $-H(\mathcal{A}|\mathcal{C}) \leq -H(\mathcal{A}|\mathcal{B})$ and (T3c) follows. ☐

Several properties of Theorem 23.6 have natural consequences for the entropy of a transformation g.

Theorem 23.7 (Properties of Entropy)**.** *The following hold:*

(T1) $h(g,\mathcal{A}) \le H(\mathcal{A})$*;*

(T2) $h(g,\mathcal{A}\vee\mathcal{B}) \le h(g,\mathcal{A}) + h(g,\mathcal{B})$*;*

(T3) $h(g,\mathcal{A}) \le h(g,\mathcal{B}) + H(\mathcal{A}|\mathcal{B})$*;*

(T4) if g *is invertible with measurable inverse, then* $h(g,\mathcal{A}) = h\left(g, \bigvee_{j=-k}^{k} g^j\mathcal{A}\right)$.

Proof. As above, proofs are lengthy but not conceptually difficult. Let us just show that, how the results of Theorem 23.6 lead to (T3). We want to use the definition in (23.2) and start with the estimate

$$\begin{aligned} H\left(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{A}\right) &\overset{\text{(T2c)}}{\le} H\left(\left(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{A}\right)\vee\left(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{B}\right)\right) \\ &\overset{\text{(T1c)}}{=} H\left(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{B}\right) + H\left(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{A}\,\middle|\,\bigvee_{j=0}^{n-1} g^{-j}\mathcal{B}\right). \end{aligned}$$

The second term on the right can be be estimated from above as follows

$$\begin{aligned} H\left(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{A}\,\middle|\,\bigvee_{j=0}^{n-1} g^{-j}\mathcal{B}\right) &\overset{\text{(T4c)}}{\le} \sum_{j=0}^{n-1} H\left(g^{-j}\mathcal{A}\,\middle|\,\bigvee_{k=0}^{n-1} g^{-k}\mathcal{B}\right) \\ &\overset{\text{(T3c)}}{\le} \sum_{j=0}^{n-1} H(g^{-j}\mathcal{A}|g^{-j}\mathcal{B}) \\ &\overset{\text{(T5c)}}{=} nH(\mathcal{A}|\mathcal{B}). \end{aligned}$$

Hence, combining this estimate with the previous one, dividing by n, and taking the limit $n\to\infty$ yields the result. □

Finally we can introduce a crucial theorem, how we may calculate entropy in certain cases as it allows us to *select* certain sub-algebras respectively certain types of partitions.

Theorem 23.8 (Kolmogorov-Sinai)**.** *Suppose* g *is a measure-preserving invertible transformation on* $(\mathcal{X},\mathcal{F},\mu)$*. Let* $\mathcal{A}$ *be a finite sub-algebra of* $\mathcal{F}$ *such that*

$$\lim_{n\to\infty} \bigvee_{j=-n}^{n} g^j\mathcal{A} =: \lim_{n\to\infty} \mathcal{A}^n = \mathcal{F},$$

i.e., $\mathcal{A}$ *is* ***generating****, then we may conclude* $h(g) = h(g,\mathcal{A})$.

Proof. The idea of the proof is essentially an approximation procedure involving multiple steps with error terms in the form of conditional entropies, which have to be controlled. So let us first find the error term.

`Step 1:` (error term identification) To conclude our result, it is going to suffice from the definition of $h(g)$ to consider any finite partition $\mathcal{B} \subseteq \mathcal{F}$ and prove that $h(g, \mathcal{B}) \leq h(g, \mathcal{A})$. Using Theorem 23.7, we get the key estimate

$$\begin{aligned} h(g, \mathcal{B}) &\overset{(T3)}{\leq} h(g, \mathcal{A}^n) + H(\mathcal{B}|\mathcal{A}^n) \\ &\overset{(T4)}{=} h(g, \mathcal{A}) + H(\mathcal{B}|\mathcal{A}^n). \end{aligned}$$

So if we could show that $\lim_{n\to\infty} H(\mathcal{B}|\mathcal{A}^n) = 0$ for a generating partition $\mathcal{A}$, then the result would follow.

`Step 2:` (partition approximation) Recall the **symmetric difference** between two sets $\mathcal{C}_1, \mathcal{C}_2$

$$\mathcal{C}_1 \triangle \mathcal{C}_2 := (\mathcal{C}_1 \setminus \mathcal{C}_2) \cup (\mathcal{C}_2 \setminus \mathcal{C}_1)\,.$$

We claim that given $\varepsilon > 0$ and two finite equal-length partitions $\mathcal{D}, \mathcal{B}$ we have the existence of a $\delta > 0$ such that

$$d(\mathcal{D}, \mathcal{B}) := \sum_j \mu(\mathcal{D}_j \triangle \mathcal{B}_j) < \delta \quad \Rightarrow \quad \rho(\mathcal{D}, \mathcal{B}) := H(\mathcal{D}|\mathcal{B}) + H(\mathcal{B}|\mathcal{D}) < \varepsilon, \tag{23.3}$$

where ρ is also known as the **Rokhlin metric**. Consider a new partition $\mathcal{C}$ constructed by sets $\mathcal{D}_i \cap \mathcal{B}_j$ $(i \neq j)$ and sets $\bigcup_j (\mathcal{D}_j \cap \mathcal{B}_j)$. One notices $\mathcal{D} \vee \mathcal{B} = \mathcal{D} \vee \mathcal{C}$ and $\mathcal{D} \vee \mathcal{B} = \mathcal{B} \vee \mathcal{C}$; we focus on the last equality, the other case follows by a similar argument. Since for $i \neq j$, we have $\mathcal{D}_i \cap \mathcal{B}_j \subset \bigcup_j \mathcal{D}_j \triangle \mathcal{B}_j$, our assumption in (23.3) implies

$$\mu(\mathcal{D}_i \cap \mathcal{B}_j) < \delta \text{ for } i \neq j, \quad \text{and} \quad \mu\left(\bigcup_j (\mathcal{D}_j \cap \mathcal{B}_j)\right) > 1 - \delta.$$

This implies we can calculate an entropy estimate

$$H(\mathcal{C}) \leq \kappa\delta \ln \delta - (1 - \delta)\ln(1 - \delta) < \frac{\varepsilon}{2} \qquad \text{for some } \kappa > 1$$

where the last estimate follows by making δ sufficiently small. Therefore, we find

$$H(\mathcal{B}) + H(\mathcal{D}|\mathcal{B}) = H(\mathcal{D} \vee \mathcal{B}) = H(\mathcal{B} \vee \mathcal{C}) \leq H(\mathcal{B}) + H(\mathcal{C}) < H(\mathcal{B}) + \frac{\varepsilon}{2},$$

which implies $H(\mathcal{D}|\mathcal{B}) < \frac{\varepsilon}{2}$. By the same line of argument, one also gets $H(\mathcal{B}|\mathcal{D}) < \frac{\varepsilon}{2}$ and (23.3) follows. Otherwise said, if the partitions do not differ too much in the sense of symmetric differences, the Rokhlin metric is small, i.e., there is a uniformly continuous inclusion map from the metric space of partitions under the metric d to the metric ρ.

`Step 3:` (sub-algebra approximation) Recall that $\mathcal{A}^n$ is a sequence of finite subalgebras generating $\mathcal{F}$ eventually. Let us take a finite subalgebra $\mathcal{B}$ of $\mathcal{F}$. We claim that we can find a finite algebra $\mathcal{D}$ contained in $\mathcal{A}^\infty$ such that

$$H(\mathcal{D}|\mathcal{B}) + H(\mathcal{B}|\mathcal{D}) \leq \varepsilon, \tag{23.4}$$

which is going to follow once we have verified the symmetric difference condition in (23.3). Note that the statement (23.4) is already very close to, what we want, i.e., it gives us control over arbitrary subalgebras just using our generating sequence $\mathcal{A}^n$. To prove (23.4), first let us select $(\mathcal{A}^\infty)_j$ such that $\mu(\mathcal{B}_j \cap (\mathcal{A}^\infty)_j) < \tilde{\delta}$ for some $\tilde{\delta} > 0$, which can be chosen sufficiently small in the end of the argument since $\mathcal{A}^\infty$ does generate $\mathcal{F}$ by assumption. Then we set

$$\mathcal{D}_j := (\mathcal{A}^\infty)_j \setminus \bigcup_{i \neq j} ((\mathcal{A}^\infty)_i \cap (\mathcal{A}^\infty)_j),$$

which gives us a finite partition of the same length as $\mathcal{B}$. Then we add one more set $\mathcal{D}_* := \mathcal{X} \setminus \bigcup_j \mathcal{D}_j$ and let $\mathcal{D} = \{\mathcal{D}_1, \ldots, \mathcal{D}_j, \ldots, \mathcal{D}_*\}$ be our partition. Then it is possible to check (exercise!) that $\mathcal{D}$ indeed satisfies (23.4) as we can select $\tilde{\delta}$ sufficiently small.

`Step 4:` (limit of the approximation) Finally, we can tackle the error term $H(\mathcal{B}|\mathcal{A}^n)$. By (23.4), we can find $\mathcal{D}$ such that

$$H(\mathcal{B}|\mathcal{D}) < \varepsilon$$

but $\mathcal{D}$ is completely contained in $\mathcal{A}^{n_0}$ for some n_0 as $\mathcal{D}$ is finite. For $n \geq n_0$, we finally get by Theorem 23.6

$$H(\mathcal{B}|\mathcal{A}^n) \overset{(T3c)}{\leq} H(\mathcal{B}|\mathcal{A}^{n_0}) \overset{(T3c)}{\leq} H(\mathcal{B}|\mathcal{D}) < \varepsilon$$

and since $\varepsilon > 0$ was arbitrary we may take the limit $n \to \infty$ to conclude that the error term derived in the first step vanishes. $\square$

With the Kolmogorov-Sinai Theorem in hand, we are finally ready to use the entropy as a conjugacy invariant to prove that certain dynamical systems cannot be topologically conjugate.

Example 23.9. Consider a vector $p = (p_0, \dots, p_{k-1})$ with $\sum_j p_j = 1$. Let $\mathcal{X} = \Sigma_k$ be the space of bi-infinite sequences of symbols from $\{0, 1, \dots, k-1\}$. A suitable sigma-algebra $\mathcal{F}$ is generated by the **cylinder sets**

$$\mathcal{C}_\alpha := \{x \in \mathcal{X} : x_i = \alpha_i, -m \leq i \leq n\},$$

where α is a given finite sequence of symbols and the associated measure μ is just

$$\mu(\mathcal{C}_\alpha) := \prod_{i=-m}^{n} p_{\alpha_i},$$

which associates a probability of occurrence to a sequence using the given vector p. The dynamics will be given just by the usual left-shift map

$$g(x;p)_n := x_{n+1}, \qquad g : (\mathcal{X}, \mathcal{F}, \mu) \to (\mathcal{X}, \mathcal{F}, \mu),$$

which is a measure-preserving transformation. If we want to compute the entropy via the Kolmogorov-Sinai Theorem 23.8, this requires the choice of a helpful generating subalgebra. One natural option is

$$\mathcal{A}_j := \{x : x_0 = j\}, \quad \mathcal{A} = (\mathcal{A}_0, \dots, \mathcal{A}_{k-1})$$

since the sets are simple and one verifies quite quickly that we can obtain all cylinder sets upon using shift dynamics so that

$$\bigvee_{j=-\infty}^{\infty} g^j \mathcal{A} = \mathcal{F}.$$

Observe that the typical elements of $\mathcal{A}^n$ have measures that are easy to calculate

$$\begin{aligned} \mu\left(\mathcal{A}_{j_0} \cap g^{-1}\mathcal{A}_{j_1} \cap \cdots \cap g^{-(n-1)}\mathcal{A}_{j_{n-1}}\right) &= \mu\left(\{x : x_0 = j_0, \dots, x_{n-1} = j_{n-1}\}\right) \\ &= p_{j_0} \cdots p_{j_{n-1}}. \end{aligned}$$

Hence, we can also determine entropy of the generating subalgebra by summation

$$\begin{aligned} H(\mathcal{A}^n) &= -\sum (p_{j_0} \cdots p_{j_{n-1}}) \ln(p_{j_0} \cdots p_{j_{n-1}}) \\ &= -\sum_{j_0, \dots, j_{n-1}=0}^{k-1} (p_{j_0} \cdots p_{j_{n-1}}) \left(\ln p_{j_0} + \cdots + \ln p_{j_{n-1}}\right) \\ &= -n \sum_{j=0}^{k-1} p_j \ln p_j \end{aligned}$$

and the Kolmogorov-Sinai Theorem 23.8 yields

$$h(g) = h(g, \mathcal{A}) = \lim_{n\to\infty} \frac{1}{n} H(\mathcal{A}^n) = -\sum_{j=0}^{k-1} p_j \ln p_j.$$

Therefore, the Bernoulli shift on two symbols with $p = (\frac{1}{2}, \frac{1}{2})$ and the Bernoulli shift on three symbols with $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ are not topologically conjugate by Theorem 23.4 since their entropies are $\ln 2$ and $\ln 3$ respectively. ♦

The last results on entropy show that it is actually quite difficult to *coarse-grain* dynamical systems, i.e., to extract useful low-dimensional invariants, which can be used to partition systems into different conjugacy classes.

Exercises for Section 23

E23.1 Let $\mathcal{A}, \mathcal{B}$ be partitions of $(\mathcal{X}, \mathcal{F}, \mu)$ and let $g : \mathcal{X} \to \mathcal{X}$ be a measure preserving map. Show that (a) $h(g, \mathcal{A}) \leq h(g, \mathcal{B})$ if $\mathcal{A} \subseteq \mathcal{B}$. (b) $h(g^n, \bigvee_{i=0}^{n-1} g^{-i}\mathcal{A}) = n \cdot h(g, \mathcal{A})$ for every $n \in \mathbb{N}$. (c) $h(g^n) = n \cdot h(g)$ for every $n \in \mathbb{N}$. (d) $h(g^{-n}) = n \cdot h(g)$ for every $n \in \mathbb{N}$ if g is invertible.

E23.2 Calculate the entropy of a rational rotation $g : \mathbb{S}^1 \to \mathbb{S}^1$ given by $g(x) = (x + r) \mod 1$ for $r \in \mathbb{Q}$.

A Remarks on Literature

For researchers in differential equations and dynamical systems, all the material in this book can be considered as somewhat standard, although it could be argued that not every technique is equally deeply ingrained in different sub-communities of the area. In any case, let me stress the obvious fact that this book does not contain novelties or even any fundamentally novel proofs or calculations. In this book, I have just tried to really distill known results from other sources down to the absolute essentials. The idea is just that we have a common ground that we can rely upon to proceed to the real innovation of initial reduction and then tree-like topical coverage in the (Tx) book series as discussed in the preface. Nevertheless, it is clear that readers who might want to have advanced reading on basic ODEs and nonlinear dynamics, or just a different historical presentation perspective, should get some additional literature guidance. This is precisely what this part of the appendix aims to provide.

In the appendix of (F1), I have already provided some guidance on the basics of the field. Here I am going to focus more on the graduate-level book literature, and how you can use it to complement or extend (F2) after you have mastered all the material above. In particular, I am going to assume from now on that you have understood (F1)-(F2) and are reading this appendix to learn even more, e.g., regarding topics for which there is (not yet!) a (Tx) book available at the time of first publication of (F1)-(F2).

First, the books should be mentioned that are similar in style, level and topic to (F2). A dynamical systems view on ODEs can also be found in [14, 45, 68, 69], while [4, 60] focus on maps. Books geared towards introducing pure-mathematics techniques and areas in dynamics are [9, 67]. Books directed towards bifurcation theory are the very detailed [42], the classic [26], and a variant built up via increasing dimensions [27]. A link from an introduction including bifurcations to more topics geared a bit towards Hamiltonian systems, Melnikov methods and related topics can be found in [52, 57, 74]. An introduction with a more detailed link to topological methods is [33]. Finally, there is the rather encyclopedic approach in [34], which works in much more detail through low-dimensional classification theory and the various uses of hyperbolicity.

Sources generalizing the idea of hyperbolicity in far more technical detail are [23, 29, 56, 72]. As a source for more on structural stability, good starting points are [5, 71]. A book specializing in the Hopf bifurcation is [50], planar bifurcations are covered in [16], while [13] focuses entirely on center manifolds, normal techniques are detailed in [11, 54, 55], the

Lorenz equations are covered in [66], and global bifurcations are featured in [30, 31, 73]. Numerical bifurcation computations are treated in [3, 19, 36, 38, 41, 42, 61, 65]. A link from bifurcations to catastrophe theory can be obtained from [6, 20, 25, 48, 59, 64], while links to functional-analytic approaches such as Lyapunov-Schmidt are treated in [15, 32, 37, 39, 63, 62]. Books centered more around chaotic dynamics are [10, 28, 44, 47, 73]. A very detailed introduction to mechanical and Hamiltonian systems can be found in [2]. The books [17, 53] are much more advanced sources on iterated maps, showing that one can come quite close to full classifications in cases of low dimension. For the link between numerical methods as discrete dynamical systems, we refer to [1]. For the various renormalization group variants, there seems to be no coherent mathematical introduction, but for the physical origins the reader may want to consider [7, 8, 12, 24, 46, 75]. For ergodic theory some much more detailed starting points are [18, 21, 22, 35, 49, 58, 70] while applications of the Koopman operator are covered in [43, 51].

Needless to say, there are many other topics, which we have only very barely touched upon here. Examples are infinite-dimensional dynamics, PDE dynamics, random/stochastic dynamics, networks, multiscale problems, non-autonomous/control systems, delay equations, complex iterated maps, symmetric systems, dynamics on groups, non-smoothness and differential equations with constraints, just to name a few more topics. Literature lists for parts of these topics will be provided within the forthcoming (Tx) volumes of this series. In addition, many very applied disciplines have developed (or re-invented) their own variants of studying nonlinear dynamics. Evidently it would be impossible to even attempt a more detailed literature survey along these directions at this point as dynamics is necessarily encountered universally across scientific domains.

B An Outlook: From History to Future

It may need quite some time of reflection to understand why it is crucial to understand nonlinear dynamics. At first, the subject could appear either as a collection of examples from applications, or alternatively as being too abstract due to the vast number of mathematical disciplines needed to study it rigorously. On both accounts, there are appealing revelations to be made regarding historical developments.

Virtually every emerging science or engineering topic we are surrounded by has core dynamical components. Examples range from classical mechanics, statistical physics, quantum mechanics, biological dynamics, economic systems, algorithms / machine learning, up to many areas that we have not even discovered yet as deeply as we could or should have. One may wonder, why every emerging area must acquire a dynamical flavor? The main reason is that complexity needs assembly and emergence, and these processes are inherently structured as dynamical, i.e., they evolve until a certain complex structure has been reached. Hence, it is a reasonable prediction that the tools and techniques in this book are going to age very well from the perspective of practical importance. In fact, the potential is high that they are going to elevated even further with every new application field demand. In fact, it is often really entertaining to see new applications to present interesting patterns that eventually seem to slot almost miraculously and seamlessly into the generic theory of nonlinear dynamics. One can even save substantial amounts of time and hassle in scientific discovery if one has a good picture how the generic landscape of nonlinear dynamics phenomena is shaped. At least, we have made a small first step towards this goal here.

This perspective might satisfy anyone, who has some interest in the world outside of scientific abstraction. From a philosophical standpoint, it is still a perfectly valid challenge to alter the viewpoint completely. What if we do not see dynamics as an area to describe nonlinear phenomena we encounter? Why is it relevant from a pure basic research view? One, among many, compelling reasons is that the tools and techniques of nonlinear dynamics are very constructive at translating intuition into abstraction. The evidence is rapidly increasing that even some of the challenging mathematical conjectures can be attacked using dynamics and differential equations, and that the dynamical proofs carry fascinating simplicity and elegance. Some famous examples are the solution of the Poincaré conjecture using the dynamics of infinite-dimensional evolution equations, the ergodic theory behind the Green-Tao Theorem on arithmetic progressions, or the use of renormalization in the resolution of the Ten Martini problem. None of these problems really have a current immediate practical use (and they might never have) but their proofs are effectively de-constructions of the difficulties to more manageable blocks that are then re-assembled via dynamical tools.

In summary, there seems to be really good evidence that dynamics tools are useful, no matter what your interests are. Yet, it has to be acknowledged that the results presented in the first two volumes (F1)-(F2) of this book series can only be a starting point. In fact, again drawing upon historical evidence, one usually has to make additional assumptions on the given dynamics to obtain mathematically deeper as well as more applicable results. This is precisely what the follow-up volumes (Tx) are for and I hope to see you there soon.

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